

codex-windows / 019f2174-3bd9-7ab0-81bc-75cd4f5f4b2b

Metadata

```
- Source: `codex-windows`  
- Kind: `codex`  
- Source file: `C:\Users\avidu\codex\archived_sessions\rollout-2026-07-02T11-41-33-019f2174-3bd9-7ab0-81bc-75cd4f5f4b2b.jsonl`  
- SHA-256: `6b6511f7ee4925a17b6f30e88b378bd45a2726bc2d374d56256cf8abf2c9962d`  
- Source modified: `2026-07-02T06:11:33+00:00`  
- Imported at: `2026-07-05T16:48:16+00:00`  
- cli_version: `0.142.5`  
- cwd: `C:\Users\avidu\Projects\badminton-highlight-indexer`  
- model_provider: `openai`  
- session_id: `019f2174-3bd9-7ab0-81bc-75cd4f5f4b2b`  
- source: `vscode`
```

Transcript

1. user (2026-07-02T06:11:33.468Z)

? Picking up KhelSutra after the UX + runnable-tool session. State: the bucket?reel CUJ is fully UI-driven (B-P2, with a live screencast in the promo bucket), the SPA is trilingual (en+kn+hi), plain-language copy shipped (A1), and the 7-language expansion is owner-approved (#58).

? Read first (the "use the docs" rule): memory next-session-ux-localization.md (your full handoff ? you-are-here, ordered next steps, ramp-up kit, open owner decisions), then khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md §7, LOCAL_APPLIANCE_ARCHITECTURE.md §9 (T2/T4/T11) + §4, and MULTILINGUAL_EXPANSION.md.

? Clean slate: khelsutra master has #55?#60 merged; engine master has #284/#289; 0 of my PRs open.

? The engine/serving track (Cloud-Run M5?M12) is a CONCURRENT session ? do NOT touch COMPUTE_DECOUPLED_SERVING/, backend/compute/, or the engine DOC_STATUS/NEXT_STEPS. Shared checkout:
branch from origin/master, re-verify HEAD before commit, stage explicit paths.

Mission (in order; ONE PR per item, merge on green + rising coverage):

1. Finish the SPA UX A-items ? A4 (mobile ?44px touch targets + visible tooltips) ? A5 (friendly engine-error mapping on the BUILD path: reuse web/src/lib/errors.ts::classifyError, add a shared errors.* i18n namespace) ? A8 (background-processing reassurance + icon+text status chips).
2. B3 ? make the site/ marketing pages trilingual via a shared t()/data-i18n shim. Recommended: hoist catalogs to repo-root /locales/ + a tiny site prebuild copy (B3 ticket in the handoff); negotiation must match the SPA (localStorage khelsutra.lang ? ?lang= ? navigator ? en).
3. ? NORTH STAR (the goal ? runnable tool for alpha users): T4 ? in-UI file UPLOAD (wire the SPA to the EXISTING engine chunked-upload endpoints in backend/api/uploads.py; alpha users have a file, not a gs:// URI). T2 ? host SPA+engine behind Cloudflare Access on the CPU droplet (168.144.126.176) with Gemini/CPU (NOT GPU, NOT M12); rotate the GEMINI_API_KEY; reuse site/scripts/provision-vm.sh + PORTABILITY.md. T11 ? retention sweep.

Parallel line (owner-approved #58, owner-paced): MULTILINGUAL_EXPANSION ? add th/id/da/es/zh-Hans/

ja/ko. Phase A (es/da/id) is actionable now (es needs the 3-form ICU plural {one,many,other}). Phases B/C need owner answers (Q2 zh-Hans/Hant, Q3 CJK offline strategy, Q4 native reviewers).

Don't redo: the CUJ, B-P2, A1, the doc-hygiene, #58. A2 (WhatsApp/social) is DROPPED (separate scope) ? reel stays pure-download. Open owner decisions are in the handoff §3.

Ultracode's on ? fan out research/verification with workflows, verify against the docs, keep tight
one-item PRs, merge the good ones on green. Let's make the appliance point-and-go for alpha. ?

2. assistant (2026-07-02T06:11:33.468Z)

I'll ramp up properly first ? per the repo-freshness rule, pull the repos, then read the handoff and docs before doing any work. Let me start in parallel.

3. assistant (2026-07-02T06:11:33.468Z)

```
[external_agent_tool_call: Bash]
description: Pull and status khelsutra repo
command: cd /c/Users/avidu/Projects/khelsutra && echo "=== KHELSUTRA ===" && git status --short
&& echo "--- branch ---" && git branch --show-current && echo "--- pull ---" && git pull --ff-
only 2>&1 | head -20 && echo "--- recent ---" && git log --oneline -8
[/external_agent_tool_call]
```

4. assistant (2026-07-02T06:11:33.468Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\next-session-ux-localization.md
[/external_agent_tool_call]
```

5. assistant (2026-07-02T06:11:33.468Z)

```
[external_agent_tool_result]
=== KHELSUTRA ===
?? .claude/
?? site/access.db
--- branch ---
master
--- pull ---
Already up to date.
--- recent ---
ccdcfcd feat(web): A1 ? plain-language copy pass on the en catalog (de-jargon) (#60)
a20002f docs: propose Multilingual Expansion ? 7 new locales (th/id/da/es/zh-Hans/ja/ko) (#58)
a0475bc docs: hygiene refresh ? sync trackers to shipped reality (#59)
7750afc docs(dry-run): Path-B live e2e DONE ? in-UI bucket?reel CUJ + screencast (DoD §6 box 3
?) (#57)
4525c5d feat(web): B-P2 ? in-UI bucket ingest (point at a gs:// video ? process ? load) (#56)
d0461fd docs: correct stale "bucket ingest is net-new" claims (B-P1 already shipped in M2/#280)
(#55)
2421557 docs(dry-run): record Path-A local result (en+hi reel downloaded) + the engine fix it
surfaced (#54)
113587f docs: set the dry-run CUJ as a tracked goal (UI drives a bucket video -> reel download,
multilingual) (#53)
[/external_agent_tool_result]
```

6. assistant (2026-07-02T06:11:33.468Z)

```
[external_agent_tool_result]
1      ---
2      name: next-session-ux-localization
3      description: "? START HERE for the khelsutra UX/localization + runnable-tool track.
Where we are + ordered next steps (SPA UX A-items ? B3 ? runnable-tool North Star) + the
approved 7-language line + open owner decisions + ramp-up kit. Refreshed 2026-06-21. NOTE: the
engine/serving track has its OWN handoff in session-handoff.md ? don't conflate."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: 577fd4df-ed3f-43d3-854d-d63d732e2e07
8      ---
```

```

9
10      # Handoff ? khelsutra UX / localization / runnable-tool (refreshed 2026-06-21)
11
12      > This is the **khelsutra (web/ SPA + site/)** product track. The **engine/serving**
track (Cloud-Run M5?M12, a CONCURRENT session) has its own `session-handoff.md` ? **don't touch
its files** (`COMPUTE_DECOUPLED_SERVING`, `backend/compute/`, engine `DOC_STATUS`/`NEXT_STEPS`).
13
14      ## 1. You are here
15      The **bucket?reel CUJ is fully UI-driven + proven** (B-P2 shipped + live screencast),
the **SPA is trilingual (en+kn+hi)**, **plain-language copy (A1) is merged**, and the
**7-language expansion is owner-approved** (#58). The **North Star = make the tool
runnable/shareable for alpha users from the UI** ? a hosted SPA+engine behind edge auth. **It is
NOT GPU-gated and NOT M12** ? cheapest path = the existing **CPU droplet + Gemini**.
16
17      **Merged this session (all green):** engine
[#284](https://github.com/Khelsutra/badminton-highlight-indexer/pull/284)(gs:// guard) .
[#289](https://github.com/Khelsutra/badminton-highlight-indexer/pull/289)(compile-from-bucket
fix) . khelsutra [#55](https://github.com/avidullu/khelsutra/pull/55)(stale-doc) .
[#56](https://github.com/avidullu/khelsutra/pull/56)(B-P2 ingest) .
[#57](https://github.com/avidullu/khelsutra/pull/57)(DoD box3) .
[#59](https://github.com/avidullu/khelsutra/pull/59)(doc-hygiene) .
[#60](https://github.com/avidullu/khelsutra/pull/60)(A1 copy). Owner-merged
[#58](https://github.com/avidullu/khelsutra/pull/58)(MULTILINGUAL_EXPANSION). Screencast:
`gs://khelsutra-rally-corpus/promo/khelsutra-bucket-cuj-screencast.mp4`.
18
19      ## 2. Next steps ? ordered (one PR per item, off `origin/master`, merge on green +
rising coverage)
20      **A) SPA UX A-items, then B3** (A1 ?; A2 DROPPED ? social/sharing is a separate
scope):*
21      - **A4 ? mobile touch targets + visible tooltip text.** `web/src/components/{QueryBar,Ra
llyGrid,SelectionFooter,IngestPanel,CompileResult,LanguageSwitcher}.tsx` + `index.css`: ?44px
tap targets (`min-h-11/py), enlarge the tiny query parsed-pill + sample chips, surface `title`=
tooltips as visible text on phones. (M)
22      - **A5 ? friendly engine-error mapping (build path).** Reuse
`web/src/lib/errors.ts::classifyError` (built for ingest) on `App.tsx::onBuild` (currently shows
raw ApiError for 5xx/network) ? localized message in `CompileResult`. Add a **shared `errors.**
namespace** (recommend unifying IngestPanel onto it). (S?M)
23      - **A8 ? background-processing reassurance + status as icon+text chips** (never icon-
only) in IngestPanel progress + build state. (S?M)
24      - **B3 ? `site/` trilingual via a shared `t()` shim.** New ~70-LOC vanilla
`t(key,vars)`/`data-i18n` shim (`site/public/i18n.js`) over `index/capture/own-model.html` + a
switcher + `site/server.js` Accept-Language negotiation; extend
`site/src/{server,assets}.test.js` + a shim unit test. **DECISION (recommended by the spec
scan): hoist catalogs to repo-root `/locales/{en,kn,hi}/common.json` as the single source of
truth, re-point the SPA's 3 imports (or a Vite `@locales` alias), and a tiny `site` prebuild
copies them to `site/public/locales/` + a drift-guard test** (site is dependency-free/no-build
today; this is the only new build step). Negotiation must match the SPA exactly (localStorage
`khelsutra.lang` ? `?lang=` ? navigator ? en; en-fallback never a raw key). (M)
25
26      **B) ? NORTH STAR ? runnable tool for alpha users (the crucial fresh-context work the
owner paused for):**
27      - **T4 ? in-UI ...
[/external_agent_tool_result]

```

7. assistant (2026-07-02T06:11:33.469Z)

Clean slate confirmed (master, up to date, my last PRs merged). Let me read the key docs and current-state to finish ramping up.

8. assistant (2026-07-02T06:11:33.469Z)

```

[/external_agent_tool_call: Read]
file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md
[/external_agent_tool_call]

```

9. assistant (2026-07-02T06:11:33.469Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\kheIsutra\docs\FRIENDLY_AND_MULTILINGUAL.md
[/external_agent_tool_call]
```

10. assistant (2026-07-02T06:11:33.469Z)

```
[external_agent_tool_result]
<system-reminder>[Truncated: PARTIAL view ? showing lines 1-32 of 111 total (50742 tokens, cap
25000). Call Read with offset=33 limit=32 for the next page, or Grep to find a specific section.
Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>
```

```
1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     > **Scope of this file = LIVE only:** the current handoff + *today's* checkpoints + open
PRs/decisions. Keep it read-first-short (?1 page). Older checkpoints are archived in [[current-
state-archive]] ? when today's checkpoints age out (next clean-slate day), move them there,
don't let them pile up here.
13
14     ***? NEXT SESSION FOCUS (2026-06-21, after the big localization+dry-run session):** SPA
is fully **en+kn+hi (DONE)**; a real **UI?engine?reel?download CUJ ran (en+hi)**; a **critical
engine bug fixed** (#281). *** NEXT BUILD (owner-picked): (1) Path B ? point at a bucket video
IN THE UI.** ? **B-P1 VERIFIED ALREADY DONE ? NO ENGINE CODE (do NOT re-scope it as a
feature):** `POST /api/process` has accepted `gs://` URIs since **M2/#280** (submit
resolves the URI + skips the local-exists-404 for non-local backends `main.py:283`; the worker
fetches to scratch `acquire_local_input main.py:159` ? same hash?validate?segment?DB pipeline ?
queryable API-DB `video_id`); **TESTED**
(`tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches` + hosted-
mode-reject / unknown-scheme-400 / CLI variants). Confirmed by a **3-skeptic verification
workflow (0 refuted, gaps:[])**. **Scope pin:** holds in DEFAULT single-user mode (hosted-mode
`allowed_video_root` correctly **400s** a bucket URI ? by design). Stale "local-path-only / net-
new" docs **CORRECTED ? kheIsutra [#55]** (DRY_RUN_CUJ $1/2/4/6/8 + LOCAL_APPLIANCE + README).
**? So Path B = ONLY B-P2 = the SPA ingest/progress screen** (enter a `gs://` URI ? `POST
/api/process` ? poll `/api/jobs` ? load `result.video_id`; back-half already works) ? **100%
kheIsutra `web/` SPA + i18n (en/kn/hi), ZERO engine code.** Ready B-P2 sketch: `DRY_RUN_CUJ.md`
$2 (new `web/src/{api/types,api/client,hooks/useIngestJob,components/IngestPanel}.tsx` +
`App.tsx` `videoId?state handoff + `ingest.*` keys). See [[lesson-verify-engine-surface-before-
scoping]]. ** (2) Droplet deploy (opt 3):** deploy the FIXED engine to `168.144.126.176` (un-
provisioned, 2-core/3.8GB) ? re-run CUJ on Linux. **STILL OPEN (localization DoD):** B3 (`site/`
marketing still English-only) + native-speaker review of the AI-drafted kn/hi. **Ramp-up:**
`kheIsutra/docs/DRY_RUN_CUJ.md` + `FRIENDLY_AND_MULTILINGUAL.md` $7 + the dry-run checkpoint
below. **Clean slate (at session start):** both repos on master, 0 open PRs, no worktrees/VMS.
kheIsutra master `2421557` · engine master `2afd54f` (? origin/master `cee4baa`, #283 M4
squashed).
15
16     **Checkpoint:** 2026-06-21 (ENGINE SERVING M5+ ? OWNER BATCH-AUTHORIZED the independent
path). Owner redirected: "first surface the owner-gated decisions so I can take them + unblock
max independent PRs." Ran a 6-agent grounded gate-map **workflow** (`wf_4849400f`) over README
$2/$7/$8 + code ? a decision menu splitting every M5?M12 gate into **(a) default-OFF code I can
merge now** vs **(b) thin owner residual**. **Owner answers (AskUserQuestion):** ** (1) BLANKET
YES** ? write+merge ALL default-OFF serving code (M5,M6,M8,M9-auth,M9-consent-
cols,M11-plumbing,M12), one isolated PR each; *** SPEND OK up to $100/?10k** for due-diligence +
**production-grade testing/verification** (so the real M6/M7 cloud runs are ALSO unblocked
within the cap ? must respect it, DELETE resources after, confirm cmd+cost be...
```

[/external_agent_tool_result]

11. assistant (2026-07-02T06:11:33.469Z)

[external_agent_tool_result]

```
1      # Friendly & Multilingual ? non-tech-savvy UX + Hindi/Kannada localization
2
3      > **Status:** `DRAFT` ? owner-gated (implementation starts only on explicit approval) .
**Owner:** Avi Dullu . **Created:** 2026-06-21 . **Last updated:** 2026-06-21
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE).
5      > **Tracking anchors:** $7 progress tracker is the source of truth; indexed in
`docs/README.md`; thread in `NEXT_STEPS.md`; pointer in engine `docs/DOC_STATUS.md`; memory
`current-state` + `next-session-ux-localization`.
6      > **Relation to existing docs:** this is the **implementation project** for two engine
design docs ? it *adapts & supersedes-in-practice* `badminton-highlight-
indexer/docs/I18N_PLAN.md` (localization) and the UX half of `~/docs/UI_PROPOSAL.md` (non-tech-
savvy console). Those stay the canonical *design*; this doc carries the *tracked execution*
against today's surfaces.
7      > **Honesty note:** each claim tagged `[verified]` (checked against code this session),
`[researched]` (needs sign-off), or `[design]` (proposed). Not legal advice ? the OFL font
question (D-FONT) is owner/counsel-gated.
8
9      ---
10
11     ## 0. TL;DR
12     A coach or parent in Bangalore should land on KhelSutra, feel at home, set up their
camera with confidence, and get highlights ? **in Kannada or Hindi**, with **zero hand-
holding**. This project isolates that goal into one tracked workstream-pair: ** (A) ** make the UI
super friendly for non-tech-savvy users, and ** (B) ** localize to Hindi + Kannada. Both surfaces
(`web/` React SPA + `site/` vanilla marketing) have **zero i18n today** `[verified]`, so the
foundation ? shared `locales/{en,kn,hi}` catalogs + a negotiation contract + CI parity ? is the
real deliverable; translations follow. **The single most important caveat:** translations are
**human-reviewed** (Gemini drafts are *inference only, never training* ? the standing
guardrail), and Kannada/Hindi rendering is **blocked on the OFL font decision (D-FONT)**. The
cheapest unblocked high-impact first slice is the **SPA i18n scaffold + language switcher (en-
only)** ? it locks the shared catalog/negotiation contract and adds the render-test safety net
the SPA currently lacks.
13
14     ## 1. Problem & goal
15     **Why now.** The public site (kheelsutra.guru) is live, clean, honest, and tested. The
PMF beachhead is **Bangalore badminton academies**; many coaches, parents, and players are far
more comfortable in **Kannada** than English, and most are **not highly tech-savvy** `[verified
? design intent]` (`UI_PROPOSAL.md:25,28`, `I18N_PLAN.md:10-13`, `PMF_MARKET_RESEARCH.md:116`,
`KHELSUTRA_PRODUCT_OVERVIEW.md:13/50/61`). An English-only, knob-heavy product adds friction at
the exact point of first contact.
16
17     **Concrete outcome ("good" looks like):** a Kannada/Hindi-comfortable, phone-first, non-
technical coach/parent can ? without help ? (1) understand what the product does and how to
record reliably, (2) request access / use the console, (3) pick rallies and get a reel, and (4)
share it on WhatsApp, **entirely in their language**.
18
19     **Read to get current:** this doc's $3?$4; the engine design `I18N_PLAN.md` (canonical
i18n design) + `UI_PROPOSAL.md` $2 (personas, the 6 alpha screens, "teach recording *before*
upload"); `VIDEO_RECORDING_GUIDELINES.md`; this repo's `PER_RALLY_SELECTION.md` (the honest SPA
scope) + `REQUEST_ACCESS_FORM.md`.
20
21     ## 2. Decisions locked (structural ? owner directive 2026-06-21)
22     | # | Decision | Source / date | Implication |
23     |---|-----|-----|-----|
24     | D1 | **Isolate as ONE tracked project** spanning both workstreams (A: UX, B: kn/hi
l10n) | owner, 2026-06-21 | This doc; $7 = source of truth; archived on DoD |
25     | D2 | **Home = `kheelsutra/docs/`** (the implementation repo) | owner-gated tracked-doc
convention | `web/` + `site/` both live here; $7 tracks kheelsutra PRs; engine P2 backend-l10n is
a cross-repo deferred row |
```

26 | D3 | **Adapt ...
[/external_agent_tool_result]

12. assistant (2026-07-02T06:11:33.469Z)

I've ramped up. Key context absorbed:

- **A1 already merged (#60)** ? though the \$7 tracker still shows A1 as ? (stale; I'll fix that).
- Mission order: **A4 ? A5 ? A8 ? B3 ? North Star (T4/T2/T11)** , plus the owner-approved 7-language line in parallel.
- Shared checkout discipline: branch from `origin/master`, don't touch the serving track.

Let me get oriented on the actual `web/` code surface before scoping. Let me look at the structure and key files.

13. assistant (2026-07-02T06:11:33.469Z)

```
[external_agent_tool_call: Bash]
description: Map web/src and site file structure
command: echo "=== web/src tree ===" && find web/src -type f | sort && echo "" && echo "===
locales ===" && find web/src -path '*locales*' -o -name '*.json' 2>/dev/null | grep -i locale |
sort && echo "" && echo "=== site tree (public html + server) ===" && find site -maxdepth 2
-type f \( -name '*.html' -o -name '*.js' -o -name '*.json' \) | grep -v node_modules | sort
[/external_agent_tool_call]
```

14. assistant (2026-07-02T06:11:33.469Z)

```
[external_agent_tool_result]
=== web/src tree ===
web/src/App.tsx
web/src/api/client.test.ts
web/src/api/client.ts
web/src/api/types.ts
web/src/components/CompileResult.tsx
web/src/components/IngestPanel.test.tsx
web/src/components/IngestPanel.tsx
web/src/components/LanguageSwitcher.tsx
web/src/components/QueryBar.examples.test.ts
web/src/components/QueryBar.tsx
web/src/components/RallyGrid.tsx
web/src/components/SelectionFooter.tsx
web/src/hooks/useIngestJob.test.ts
web/src/hooks/useIngestJob.ts
web/src/hooks/useRallySelection.ts
web/src/i18n/hygiene.test.ts
web/src/i18n/i18n.test.tsx
web/src/i18n/index.ts
web/src/i18n/locales/en/common.json
web/src/i18n/locales/hi/common.json
web/src/i18n/locales/kn/common.json
web/src/index.css
web/src/lib/errors.test.ts
web/src/lib/errors.ts
web/src/lib/floor.test.ts
web/src/lib/floor.ts
web/src/lib/format.test.ts
web/src/lib/format.ts
web/src/lib/log.ts
web/src/lib/query.test.ts
web/src/lib/query.ts
web/src/lib/selection.test.ts
web/src/lib/selection.ts
web/src/main.tsx
web/src/test/setup.ts
```

```
=== locales ===
web/src/i18n/locales
web/src/i18n/locales/en
web/src/i18n/locales/en/common.json
web/src/i18n/locales/hi
web/src/i18n/locales/hi/common.json
web/src/i18n/locales/kn
web/src/i18n/locales/kn/common.json

=== site tree (public html + server) ===
site/package-lock.json
site/package.json
site/public/capture.html
site/public/flyer-hi.html
site/public/flyer-kn.html
site/public/flyer.html
site/public/index.html
site/public/own-model.html
site/server.js
site/src/assets.test.js
site/src/deploy-config.test.js
site/src/index.js
site/src/index.test.js
site/src/notify.js
site/src/server.test.js
site/src/store.js
site/src/ux.dom.test.js
[/external_agent_tool_result]
```

15. assistant (2026-07-02T06:11:33.469Z)

Now let me read the key files I'll be working with for the A-items. Reading the components, errors lib, CSS, and the en catalog in parallel.

16. assistant (2026-07-02T06:11:33.469Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\kheIsutra\web\src\App.tsx
[/external_agent_tool_call]
```

17. assistant (2026-07-02T06:11:33.469Z)

```
[external_agent_tool_result]
1      import { useEffect, useMemo, useState } from "react";
2      import { useTranslation, Trans } from "react-i18next";
3      import { useRallySelection } from "../hooks/useRallySelection";
4      import { RallyGrid } from "../components/RallyGrid";
5      import { QueryBar } from "../components/QueryBar";
6      import { SelectionFooter } from "../components/SelectionFooter";
7      import { LanguageSwitcher } from "../components/LanguageSwitcher";
8      import { CompileResult, type BuildState } from "../components/CompileResult";
9      import { IngestPanel } from "../components/IngestPanel";
10     import { runQuery, type ParsedQuery } from "../lib/query";
11     import { floorRisk } from "../lib/floor";
12     import { reelFilename } from "../lib/format";
13     import { parseDetail } from "../lib/errors";
14     import { log, newCorrelationId } from "../lib/log";
15     import { ApiError, compileReel, getConfig, highlightUrl, listRallies, minShotCount }
from "../api/client";
16     import type { EngineConfig, RallySegment } from "../api/types";
17
18     /** One compact, structured description of how a query was understood (for the trace
log). */
19     function parseTrace(q: ParsedQuery) {
```

```

20     return {
21       raw: q.raw,
22       recognized: q.recognized,
23       filters: q.filters.map((f) => `${f.field}${f.op}${f.value}`),
24       sort: q.sort ? `${q.sort.key}:${q.sort.dir}` : null,
25       limit: q.limit ?? null,
26       unavailable: q.unavailable.map((u) => `${u.kind}:${u.token}`),
27     };
28   }
29
30   // Demo data shown when no ?video=<id> is supplied. Shape matches the verified
play_segments
31   // contract; only honest fields are surfaced. A real session loads from POST /api/query.
32   const DEMO_RALLIES: RallySegment[] = [
33     { id: 1, start_time: 12, end_time: 20, duration: 8, server: null, receiver: null,
metadata: { shot_count: 9, shot_count_confidence: "High" } },
34     { id: 2, start_time: 41, end_time: 47, duration: 6, server: null, receiver: null,
metadata: { shot_count: 5 } },
35     { id: 3, start_time: 63, end_time: 84, duration: 21, server: null, receiver: null,
metadata: { shot_count: 19, shot_count_confidence: "Medium" } },
36     { id: 4, start_time: 95, end_time: 101, duration: 6, server: null, receiver: null,
metadata: {} },
37     { id: 5, start_time: 130, end_time: 142, duration: 12, server: null, receiver: null,
metadata: { shot_count: 11 } },
38     { id: 6, start_time: 158, end_time: 187, duration: 29, server: null, receiver: null,
metadata: { shot_count: 24, shot_count_confidence: "High" } },
39   ];
40
41   type LoadState =
42     | { status: "demo" }
43     | { status: "loading" }
44     | { status: "loaded"; count: number }
45     | { status: "error"; error: string };
46
47   export default function App() {
48     const { t } = useTranslation();
49     // Job-tethered (D10): a finished job's video_id arrives via ?video=<id> ? seeded from
the URL,
50     // but now also SETTABLE so the in-UI ingest panel (B-P2) can hand off a freshly-
processed video
51     // without a reload (it also rewrites ?video= so a refresh is stable).
52     const [videoId, setVideoId] = useState<string | null>(
53       () => new URLSearchParams(window.location.search).get("video"),
54     );
55
56     const [rallies, setRallies] = useState<RallySegment[]>(videoId ? [] : DEMO_RALLIES);
57     const [order, setOrder] = useState<RallySegment[]>(videoId ? [] : DEMO_RALLIES);
58     const [loadState, setLoadState] = useState<LoadState>(videoId ? { status: "loading" }
: { status: "demo" });
59     const [reelName, setReelName] = useState("highlights");
60     const [build, setBuild] = useState<BuildState>({ status: "idle" });
61
62     // Display order is separate from selection: a query can re-sort the cards while the
shared
63     // selection set (sel) stays the single source of truth for what lands in the reel.
64     const sel = useRallySelection(rallies, "all");
65
66     // Load real rallies for a job's video (POST /api/query). Default ALL-selected (D8).
67     useEffect(() => {
68       if (!videoId) return;
69       const cid = newCorrelationId();
70       lo...
[/external_agent_tool_result]

```



```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\web\src\lib\errors.ts
[/external_agent_tool_call]
```

19. assistant (2026-07-02T06:11:33.470Z)

```
[external_agent_tool_result]
1      // Error helpers shared by the ingest flow and the build flow. Kept i18n-FREE: callers
map an
2      // IngestErrorCode to a localized string (so hooks stay testable without a translation
context),
3      // while the engine's own `detail` text passes through verbatim (it's the most specific
message).
4      import { ApiError } from "../api/client";
5
6      /** Pull FastAPI's {"detail": "..."} message out of an error body, else return the raw
body. */
7      export function parseDetail(body: string): string {
8          try {
9              const j = JSON.parse(body) as { detail?: unknown };
10             if (j && typeof j.detail === "string") return j.detail;
11         } catch {
12             /* not JSON ? fall through to the raw body */
13         }
14         return body;
15     }
16
17     /** Stable, localizable failure categories surfaced by the ingest flow.
18     * `engine` carries the engine's own message in `detail`; the rest are localized by
code. */
19     export type IngestErrorCode = "invalidUri" | "networkError" | "emptyResult" | "timeout"
| "engine";
20
21     export interface IngestError {
22         code: IngestErrorCode;
23         /** Present for `engine` ? the verbatim FastAPI detail / status; rendered as-is, not
translated. */
24         detail?: string;
25     }
26
27     /** Classify a thrown client error into a stable code. A network throw (ApiError with no
`status`)
28     * is the common "engine unreachable" case ? its own friendly code; any HTTP error
surfaces the
29     * engine's `detail` (e.g. an unsupported-scheme 400, a not-found 404). */
30     export function classifyError(err: unknown): IngestError {
31         if (err instanceof ApiError) {
32             if (err.status === undefined) return { code: "networkError" }; // fetch threw ?
engine truly unreachable
33             if (err.body) {
34                 const detail = parseDetail(err.body);
35                 if (detail) return { code: "engine", detail }; // a real engine message (e.g. a
400 unsupported-scheme)
36             }
37             // A bare 5xx with no body is almost always the engine being down / a gateway that
can't reach it
38             // (e.g. the dev proxy when :8000 is off) ? for a non-technical user a friendly "is
it running?"
39             // beats a cryptic "HTTP 500". A 4xx with no body keeps the status (it's a client-
side mistake).
40             if (err.status >= 500) return { code: "networkError" };
41             return { code: "engine", detail: `HTTP ${err.status}` };
42         }
43         return { code: "engine", detail: String(err) };
44     }
```

45

[/external_agent_tool_result]

20. assistant (2026-07-02T06:11:33.470Z)

[external_agent_tool_call: Read]

file: C:\Users\avidu\Projects\khelsutra\web\src\components\CompileResult.tsx

[/external_agent_tool_call]

21. assistant (2026-07-02T06:11:33.470Z)

[external_agent_tool_result]

```
1      // The compiled-reel result (PER_RALLY_SELECTION.md P5): once Build ? POST /api/compile
succeeds,
2      // preview the reel inline (<video> over GET /api/highlights/...) and offer a download.
The engine
3      // returns a server-local filepath, NOT a URL (main.py:394), so the SPA builds the URL
itself. On a
4      // failure (e.g. the min_shot_count 400) we show the engine's reason ? loud, never
swallowed.
5
6      import { useTranslation } from "react-i18next";
7
8      export type BuildState =
9      | { status: "idle" }
10     | { status: "building" }
11     | { status: "done"; url: string; filename: string }
12     | { status: "error"; error: string };
13
14     interface Props {
15         build: BuildState;
16         onDismiss: () => void;
17     }
18
19     export function CompileResult({ build, onDismiss }: Props) {
20         const { t } = useTranslation();
21         if (build.status === "idle" || build.status === "building") return null;
22
23         return (
24             <section
25                 data-testid="compile-result"
26                 className={
27                     "mt-6 rounded-xl border p-4 " +
28                     (build.status === "error" ? "border-red-200 bg-red-50" : "border-green/30 bg-
green/5")
29             }
30             >
31                 <div className="flex items-start justify-between gap-3">
32                     <h2 className="font-semibold text-ink">
33                         {build.status === "done" ? t("result.ready") : t("result.failed")}
34                     </h2>
35                     <button
36                         type="button"
37                         onClick={onDismiss}
38                         aria-label={t("result.dismiss")}
39                         className="rounded-full px-2 text-ink/50 hover:text-ink"
40                     >
41                         ?
42                     </button>
43                 </div>
44
45                 {build.status === "done" ? (
46                     <div className="mt-3">
47                         {/* eslint-disable-next-line jsx-a11y/media-has-caption ? a stitched rally
reel has no captions */}
```

```

48         <video
49             data-testid="reel-video"
50             src={build.url}
51             controls
52             playsInline
53             className="w-full max-w-xl rounded-lg bg-black"
54         />
55         <a
56             href={build.url}
57             download={build.filename}
58             className="mt-3 inline-block rounded-full bg-green px-4 py-2 text-sm font-
semibold text-[#04140D]"
59         >
60             {t("result.download", { filename: build.filename })}
61         </a>
62     </div>
63 ) : (
64     <p className="mt-2 text-sm text-red-800" data-testid="compile-error">
65         {build.error}
66     </p>
67 )}
68 </section>
69 );
70 }
71
[/external_agent_tool_result]

```

22. assistant (2026-07-02T06:11:33.470Z)

Now let me read the remaining components and the CSS + en catalog.

23. assistant (2026-07-02T06:11:33.470Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khealsutra\web\src\components\QueryBar.tsx
[/external_agent_tool_call]

```

24. assistant (2026-07-02T06:11:33.470Z)

```

[external_agent_tool_result]
1      // The fast-entry query bar (PER_RALLY_SELECTION.md P3). A deterministic, OFFLINE parser
turns
2      // plain text ("top 15 longest", "over 10s") into a selection over the SAME shared set
the
3      // checkboxes use. It is honest by construction: a live "parsed-pill" shows exactly how
the
4      // text was understood, and concepts with no data behind them (shot-type / player /
score) are
5      // greyed out as "not available yet" ? never silently faked.
6      import { useMemo, useState } from "react";
7      import type { FormEvent } from "react";
8      import { useTranslation, Trans } from "react-i18next";
9      import { parseQuery, summarize } from "../lib/query";
10     import type { ParsedQuery } from "../lib/query";
11
12     interface Props {
13         /** Apply a parsed query to the rally set (App runs runQuery + writes the shared
selection).
14         * `source` distinguishes a typed-and-submitted query from an example-chip tap (for
the trace). */
15         onApply: (q: ParsedQuery, source: "typed" | "example") => void;
16     }
17
18     // Each example is a real, parseable query ? tapping one fills the box AND applies it.

```

```

Led by rally
19 // LENGTH (the trusted axis); shot-count filtering exists but is experimental, so it's
not featured
20 // here (a test pins that every example parses to a recognized, fully-available query).
21 export const EXAMPLES = ["over 10s", "top 15 longest", "shortest first", "first 10"];
22
23 const CHIP = "rounded-full px-2.5 py-0.5 text-xs font-medium";
24
25 export function QueryBar({ onApply }: Props) {
26   const { t } = useTranslation();
27   const [value, setValue] = useState("");
28   const parsed = useMemo(() => parseQuery(value), [value]);
29   const clauses = summarize(parsed);
30   const typed = value.trim().length > 0;
31
32   const applyText = (text: string) => {
33     setValue(text);
34     onApply(parseQuery(text), "example");
35   };
36   const submit = (e: FormEvent) => {
37     e.preventDefault();
38     if (parsed.recognized) onApply(parsed, "typed");
39   };
40
41   return (
42     <div>
43       <form onSubmit={submit} className="flex flex-wrap items-center gap-2">
44         <input
45           type="text"
46           value={value}
47           onChange={(e) => setValue(e.target.value)}
48           placeholder={t("query.placeholder")}
49           aria-label={t("query.ariaLabel")}
50           className="min-w-[16rem] flex-1 rounded-xl border border-black/15 bg-white
px-4 py-2 text-sm text-ink outline-none focus:border-green"
51         />
52         <button
53           type="submit"
54           disabled={!parsed.recognized}
55           className="rounded-xl bg-ink px-4 py-2 text-sm font-semibold text-white
disabled:opacity-40"
56         >
57           {t("query.apply")}
58         </button>
59       </form>
60
61       {/* Live interpretation ? transparent + hand-correctable (no hidden NL guessing).
*/}
62       <div aria-live="polite" data-testid="query-pill" className="mt-2 flex flex-wrap
items-center gap-1.5">
63         {clauses.map((c) => (
64           <span key={c} className={CHIP + " bg-green/15 text-green"}>
65             {c}
66           </span>
67         ))}
68         {parsed.unavailable.map((u) => (
69           <span
70             key={u.token}
71             title={t("query.notAvailableTitle", { token: u.token, kind: u.kind })}
72             className={CHIP + " bg-black/5 text-ink/45"}
73           >
74             <span className="line-through">{u.token}</span> · {t("query.notAvailable")}
75           </span>
76         ))}
77         {typed && !parsed.recognized && parsed.unavailable.length === 0 && (

```

```

78         <span className="text-xs text-ink/50">{t("query.unreadable")}</span>
79     )}
80     {!typed && <span className="text-xs text-ink/50">{t("query.hint")}</span>}
81 </div>
82
83     <div className="mt-3 flex flex-wrap gap-2 text-sm">
84         <span className="text-ink/40">{t("query.try")}</span>
85         {EXAMPLES.map((ex) => (
86             <button
87                 ...
88             )}
89     </div>
90 </external_agent_tool_result>

```

25. assistant (2026-07-02T06:11:33.470Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\RallyGrid.tsx
[/external_agent_tool_call]

```

26. assistant (2026-07-02T06:11:33.470Z)

```

[external_agent_tool_result]
1    // The selectable rally grid ? one card per play_segment, HONEST fields only:
2    // ordinal (by start order), mm:ss start, and rally LENGTH (the trusted axis). A shot-
count chip
3    // is shown ONLY when present, and clearly flagged EXPERIMENTAL ? shot detection is
still under
4    // development and not yet reliable, so length is what we lead with (owner 2026-06-20).
5    // Never renders server/receiver/events (motion-only fabrications,
PER_RALLY_SELECTION.md §4).
6    import { useTranslation } from "react-i18next";
7    import type { RallySegment } from "../api/types";
8    import { mmss } from "../lib/format";
9
10   interface Props {
11       rallies: RallySegment[];
12       isSelected: (id: number) => boolean;
13       onToggle: (id: number) => void;
14       /** Stable chronological label for a rally (by start_time), so re-sorting via a query
15        * doesn't renumber the cards. Falls back to display position when absent. */
16       ordinalOf?: (id: number) => number;
17   }
18
19   export function RallyGrid({ rallies, isSelected, onToggle, ordinalOf }: Props) {
20       const { t } = useTranslation();
21       if (rallies.length === 0) {
22           return <p className="text-ink/60 text-sm">{t("grid.empty")}</p>;
23       }
24       return (
25           <ul className="grid grid-cols-[repeat(auto-fill,minmax(150px,1fr))] gap-3 list-none
p-0 m-0">
26               {rallies.map((r, i) => {
27                   const selected = isSelected(r.id);
28                   const n = ordinalOf ? ordinalOf(r.id) : i + 1;
29                   const shot = r.metadata?.shot_count;
30                   const conf = r.metadata?.shot_count_confidence;
31                   return (
32                       <li key={r.id}>
33                           <button
34                               type="button"
35                               onClick={() => onToggle(r.id)}
36                               aria-pressed={selected}
37                               aria-label={t(selected ? "grid.inReel" : "grid.notInReel", { n })}
38                               className={
39                                   "w-full text-left rounded-xl border p-3 transition " +
40                                   (selected

```

```

41         ? "border-green bg-green/10"
42         : "border-black/10 bg-white hover:border-green/60")
43     }
44     >
45     <div className="flex items-center justify-between">
46         <span className="font-semibold text-ink">{t("grid.rally", { n })}</span>
47         <span
48             aria-hidden="true"
49             className={
50                 "inline-flex h-5 w-5 items-center justify-center rounded-full text-
xs " +
51                 (selected ? "bg-green text-[#04140D]" : "border border-black/20
text-transparent")
52             }
53         >
54             ?
55         </span>
56     </div>
57     <div className="mt-1 text-sm text-ink/70">
58         {mmss(r.start_time)} <span className="text-
ink/40">{t("grid.start")}</span> . <strong className="text-ink/90">{mmss(r.duration)}</strong>
59         <span className="text-ink/40">{t("grid.long")}</span>
60     </div>
61     {typeof shot === "number" && (
62         <div
63             title={t("grid.shotsTitle", {
64                 conf: conf && conf !== "Unknown" ? t("grid.shotsConfidence", { conf
65                 }) : "",
66             })}
67             className="mt-2 inline-block rounded-full bg-amber-50 px-2 py-0.5
text-xs text-amber-700"
68         >
69             {t("grid.shots", { count: shot })}
70         </div>
71     )}
72     </button>
73 </li>
74 </ul>
75 );
76 }
77
[/external_agent_tool_result]

```

27. assistant (2026-07-02T06:11:33.470Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\kheIsutra\web\src\components\SelectionFooter.tsx
[/external_agent_tool_call]

```

28. assistant (2026-07-02T06:11:33.470Z)

```

[external_agent_tool_result]
1     // Sticky compile footer (PER_RALLY_SELECTION.md P4): live selection summary, a reel-
name input,
2     // the Build button, and ? above all ? an HONEST min_shot_count warning. The engine
silently drops
3     // selected rallies below stitching.min_shot_count; we surface that BEFORE Build so a
rally never
4     // vanishes without a reason (D7). We warn, we do NOT override the floor (D11).
5     import { useTranslation, Trans } from "react-i18next";
6     import { mmss } from "../lib/format";
7     import type { FloorRisk } from "../lib/floor";
8

```

```

9      interface Props {
10         count: number;
11         totalSec: number;
12         reelName: string;
13         onReelNameChange: (v: string) => void;
14         onBuild: () => void;
15         /** A compile is in flight (Build wired in P5). */
16         building?: boolean;
17         /** Non-null when some selected rallies sit below the engine floor. */
18         warning: FloorRisk | null;
19     }
20
21     export function SelectionFooter({ count, totalSec, reelName, onReelNameChange, onBuild,
building, warning }: Props) {
22         const { t } = useTranslation();
23         const buildDisabled = count === 0 || building || warning?.allDropped === true;
24
25         return (
26             <footer className="fixed inset-x-0 bottom-0 border-t border-black/10 bg-white/95
backdrop-blur">
27                 {warning && (
28                     <div
29                         role="status"
30                         data-testid="floor-warning"
31                         className={
32                             "border-b px-6 py-2 text-sm " +
33                             (warning.allDropped
34                                 ? "border-red-200 bg-red-50 text-red-800"
35                                 : "border-amber-200 bg-amber-50 text-amber-900")
36                         }
37                     >
38                         {warning.allDropped ? (
39                             count === 1 ? (
40                                 <Trans i18nKey="footer.floorOne" values={{ min: warning.minShotCount }}
components={{ b: <strong /> }} />
41                             ) : (
42                                 <Trans i18nKey="footer.floorAll" values={{ count, min:
warning.minShotCount }} components={{ b: <strong /> }} />
43                             )
44                             ) : (
45                                 <>
46                                     <strong>{t("footer.floorSomeLead", { count: warning.atRisk.length
})}</strong>{" "}
47                                     <Trans i18nKey="footer.floorSomeRest" values={{ min: warning.minShotCount
}} components={{ code: <code /> }} />
48                                 </>
49                             )}
50                                 </div>
51                             )}
52
53                             <div className="mx-auto flex max-w-5xl flex-wrap items-center gap-3 px-6 py-3">
54                                 <span className="text-sm text-ink/80" data-testid="selection-summary">
55                                     <strong className="text-ink">{t("footer.count", { count })}</strong>
56                                     {t("footer.summaryRest", { total: mmss(totalSec) })}
57                                 </span>
58
59                                 <label className="ml-auto flex items-center gap-2 text-sm text-ink/60">
60                                     <span className="hidden sm:inline">{t("footer.reelName")}</span>
61                                     <input
62                                         type="text"
63                                         value={reelName}
64                                         onChange={(e) => onReelNameChange(e.target.value)}
65                                         placeholder={t("footer.reelNamePlaceholder")}
66                                         aria-label={t("footer.reelName")}
67                                         className="w-40 rounded-lg border border-black/15 bg-white px-3 py-1.5 text-

```

```

ink outline-none focus:border-green"
67         />
68     </label>
69
70     <button
71         type="button"
72         onClick={onBuild}
73         disabled={buildDisabled}
74         className="rounded-full bg-green px-5 py-2 text-sm font-semibold
text-[#04140D] disabled:opacity-40"
75     >
76         {building ? t("footer.building") : t("footer.build", { count })}
77     </button>
78 </div>
79 </footer>
80 );
81 }
82
[/external_agent_tool_result]

```

29. assistant (2026-07-02T06:11:33.471Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx
[/external_agent_tool_call]

```

30. assistant (2026-07-02T06:11:33.471Z)

```

[external_agent_tool_result]
1     // The front-half ingest screen (appliance S1/S2, tracker B-P2). Point the tool at a
video ? a
2     // `gs://` bucket URI or a path on the engine host ? and it submits to POST
/api/process, shows
3     // live progress while polling GET /api/jobs/{id}, and hands the resulting video_id to
the existing
4     // rally-selection back-half (App lifts ?video=<id>). Honest by construction: the engine
reports
5     // coarse phase WORDS (not a %), and a success-shaped failure (done but no rallies) is
shown, never
6     // silently treated as a load.
7     import { useState } from "react";
8     import type { FormEvent } from "react";
9     import { useTranslation } from "react-i18next";
10    import { useIngestJob } from "../hooks/useIngestJob";
11
12    interface Props {
13        /** Called with the engine's video_id once a submitted job finishes successfully. */
14        onLoaded: (videoId: string) => void;
15    }
16
17    export function IngestPanel({ onLoaded }: Props) {
18        const { t } = useTranslation();
19        const [uri, setUri] = useState("");
20        const { phase, submit, cancel, reset, busy } = useIngestJob(onLoaded);
21
22        const onSubmit = (e: FormEvent) => {
23            e.preventDefault();
24            submit(uri);
25        };
26
27        // `engine` failures carry the engine's own message (a 400 scheme/path detail, a
worker error);
28        // every other code maps to a friendly localized line ? the network case especially,
since this
29        // is the first screen a fresh user hits and "is the engine running?" must read

```



```

plainly.
30     const errorMsg =
31       phase.status === "error"
32       ? phase.error.code === "engine"
33       ? (phase.error.detail ?? t("ingest.failed", { error: "" })))
34       : t(`ingest.${phase.error.code}`)
35       : null;
36
37     return (
38       <section className="mt-6 rounded-2xl border border-black/10 bg-white p-5" data-
testid="ingest-panel">
39         <h2 className="text-lg font-bold">{t("ingest.title")}</h2>
40         <p className="mt-1 max-w-2xl text-sm text-ink/60">{t("ingest.intro")}</p>
41
42         <form onSubmit={onSubmit} className="mt-3 flex flex-wrap items-center gap-2">
43           <input
44             type="text"
45             value={uri}
46             onChange={(e) => setUri(e.target.value)}
47             placeholder={t("ingest.uriPlaceholder")}
48             aria-label={t("ingest.uriAriaLabel")}
49             disabled={busy}
50             className="min-w-[18rem] flex-1 rounded-xl border border-black/15 bg-white
px-4 py-2 text-sm text-ink outline-none focus:border-green disabled:opacity-50"
51           />
52           <button
53             type="submit"
54             disabled={busy}
55             className="rounded-xl bg-ink px-4 py-2 text-sm font-semibold text-white
disabled:opacity-40"
56           >
57             {phase.status === "submitting" ? t("ingest.submitting") : t("ingest.submit")}
58           </button>
59           {busy && (
60             <button
61               type="button"
62               onClick={cancel}
63               className="rounded-xl border border-black/15 px-4 py-2 text-sm hover:border-
green"
64             >
65               {t("ingest.cancel")}
66             </button>
67           )}
68         </form>
69
70         <p className="mt-2 text-xs text-ink/40">{t("ingest.hint")}</p>
71
72         {/* Live status (aria-live so screen readers hear progress). Indeterminate: the
engine reports
73         phase words ("hashing", "segmenting"), not a percentage ? so a spinner + the
phase word. */}
74         <div aria-live="polite" className="mt-3 text-sm" data-testid="ingest-status">
75           {busy && (
76             <p className="flex items-center gap-2 text-ink/70">
77               <Spinner />
78               <span>
79                 {t("ingest.working")}
80                 {phase.status === "polling" && phase.progress
81                   ? ` · ${t("ingest.progress", { progress: phase.progress })}`
82                   : ""}
83               </span>
84             </p>
85           )}
86           {phase.status === "error" && (
87             <div className="rounded-lg b...

```

```
[/external_agent_tool_result]
```

31. assistant (2026-07-02T06:11:33.471Z)

```
[external_agent_tool_call: Read]
```

```
file: C:\Users\avidu\Projects\kheIsutra\web\src\components\LanguageSwitcher.tsx
```

```
[/external_agent_tool_call]
```

32. assistant (2026-07-02T06:11:33.471Z)

```
[external_agent_tool_result]
```

```
1      import { useTranslation } from "react-i18next";
2      import { SUPPORTED_LOCALES, DEFAULT_LOCALE, type Locale } from "../i18n";
3
4      // Each locale's label is its own autonym, shown in its own script. Kannada renders
crisply once the
5      // bundled Noto Sans Kannada font lands (tracker B4); most devices already ship a system
Kannada font.
6      const LANG_LABEL_KEY: Record<Locale, string> = {
7        en: "lang.en",
8        kn: "lang.kn",
9        hi: "lang.hi",
10     };
11
12     /**
13      * A persistent language switcher. Selecting a locale calls i18next's changeLanguage,
which the
14      * detector caches to localStorage (LANG_STORAGE_KEY) ? so the choice is the top-
priority signal on
15      * the next visit (negotiation: persisted pref wins). Icon-free + non-color active state
via
16      * aria-pressed keeps it usable for non-tech-savvy users (FRIENDLY_AND_MULTILINGUAL §4).
17      */
18     export function LanguageSwitcher() {
19       const { t, i18n } = useTranslation();
20       // Highlight the user's CHOSEN locale (i18n.language), not the resolved-for-content
one:
21       // resolvedLanguage falls back to `en` when a catalog is missing, so picking Kannada
(content
22       // still English until the kn catalog lands, B5) must still show Kannada as selected.
Compare on
23       // the base subtag so a detected "en-IN" highlights "en".
24       const active = (i18n.language || DEFAULT_LOCALE).split("-")[0];
25
26       return (
27         <nav aria-label={t("lang.label")} className="flex items-center gap-1">
28           {SUPPORTED_LOCALES.map((loc) => {
29             const isActive = active === loc;
30             return (
31               <button
32                 key={loc}
33                 type="button"
34                 lang={loc}
35                 aria-pressed={isActive}
36                 onClick={() => void i18n.changeLanguage(loc)}
37                 className={[
38                   "rounded-full px-3 py-1.5 text-sm font-semibold transition",
39                   isActive ? "bg-white text-ink" : "text-white/70 hover:text-white",
40                 ].join(" ")}
41               >
42                 {t(LANG_LABEL_KEY[loc])}
43               </button>
44             );
45           })}
46       </nav>
```

```
47     );
48 }
49
[/external_agent_tool_result]
```

33. assistant (2026-07-02T06:11:33.471Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khealsutra\web\src\i18n\locales\en\common.json
[/external_agent_tool_call]
```

34. assistant (2026-07-02T06:11:33.471Z)

```
[external_agent_tool_result]
1  {
2    "app": {
3      "badge": "pick your rallies · alpha",
4      "title": "Pick the rallies for your reel",
5      "intro": "Filter by <b>rally length</b> ? e.g. <i>?top 15 longest?</i> ? then fine-
6        tune by tapping cards. Everything starts <b>selected</b>; just drop the ones you don't want."
7    },
8    "lang": {
9      "label": "Language",
10     "en": "English",
11     "kn": "?????",
12     "hi": "?????"
13   },
14   "bulk": {
15     "selectAll": "Select all",
16     "clear": "Clear",
17     "invert": "Invert"
18   },
19   "mode": {
20     "demo": "These are sample rallies ? process a video above to work on a real match.",
21     "loading": "Finding the rallies in your video?",
22     "error": "Couldn't load this video's rallies: {error}",
23     "loaded": "Found <b>{count}</b> rallies in your video."
24   },
25   "query": {
26     "placeholder": "e.g. over 10s · top 15 longest · shortest first",
27     "arialLabel": "Filter rallies by length, count or order",
28     "apply": "Apply",
29     "notAvailable": "not available yet",
30     "notAvailableTitle": "?{token}? needs {kind} detection ? not available yet",
31     "unreadable": "Couldn't read that ? try ?over 10s? or ?top 15 longest?.",
32     "hint": "Type over 10s for rallies longer than 10 seconds ? or tap a sample below.",
33     "try": "Sample queries:",
34     "support": "Filtering by <b>rally length</b> is fully supported. Shot-count is
35     <x>experimental</x> (under development)."
36   },
37   "grid": {
38     "empty": "No rallies to show yet.",
39     "rally": "Rally {n}",
40     "inReel": "Rally {n}, in reel",
41     "notInReel": "Rally {n}, not in reel",
42     "start": "start",
43     "long": "long",
44     "shots": "~{count} shots · experimental",
45     "shotsTitle": "Shot count is experimental (under development) and may be
46     inaccurate{conf}. Filter by rally length instead.",
47     "shotsConfidence": " ? confidence: {conf}"
48   },
49   "footer": {
50     "count": "{count, plural, one {# rally} other {# rallies}}",
51     "summaryRest": "selected · {total} total",
52   }
53 }
```

```

49     "reelName": "Reel name",
50     "reelNamePlaceholder": "my-highlights",
51     "build": "Build reel ({count})",
52     "building": "Building?",
53     "floorOne": "<b>Your only selected rally</b> has fewer than {min} shots ? it would
be skipped, so the build would fail. Pick a longer rally.",
54     "floorAll": "<b>All {count} selected rallies</b> have fewer than {min} shots ?
they'd all be skipped, so the build would fail. Pick longer rallies.",
55     "floorSomeLead": "{count, plural, one {# rally} other {# rallies}}",
56     "floorSomeRest": "with fewer than {min} shots will be skipped when building (they're
too short).",
57 },
58 "result": {
59     "ready": "Your reel is ready",
60     "failed": "Build failed",
61     "dismiss": "Dismiss",
62     "download": "Download {filename}"
63 },
64 "build": {
65     "noVideo": "These are sample rallies ? process a real video first to build a reel."
66 },
67 "ingest": {
68     "title": "Process a video",
69     "intro": "Point KhelSutra at a video and it finds the rallies for you ? this can
take a few minutes.",
70     "uriPlaceholder": "gs://your-bucket/match.mp4",
71     "uriAriaLabel": "Video location ? a gs:// bucket URL or a path on the engine host",
72     "hint": "Paste a cloud-storage link (gs://?), or a path to a video on the machine
running KhelSutra.",
73     "submit": "Process video",
74     "submitting": "Submitting?",
75     "cancel": "Cancel",
76     "working": "Processing your video?",
77     "progress": "{progress}",
78     "retry": "Try again",
79     "failed": "Couldn't process the video: {error}",
80     "invalidUri": "Enter a video location first ? a gs:// link or a path on the machine
running KhelSutra.",
81     "networkError": "Can't reach KhelSutra. Is it running?",
82     "emptyResult": "Processing finished but found no rallies in this video. Try a
clearer match recording.",
83     "timeout": "This is taking longer ..."
[/external_agent_tool_result]

```

35. assistant (2026-07-02T06:11:33.471Z)

Now the CSS, i18n index, and the hygiene/key-parity tests (to understand the CI guards I must keep green).

36. assistant (2026-07-02T06:11:33.471Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\web\src\index.css
[/external_agent_tool_call]

```

37. assistant (2026-07-02T06:11:33.471Z)

```

[external_agent_tool_result]
1     @import "tailwindcss";
2
3     /* Brand tokens (mirror the marketing site palette). Tailwind v4 generates utilities
4        from these ? e.g. bg-paper, text-ink, bg-ink, text-lime, text-green. */
5     @theme {
6         --color-ink: #0c261f;

```

```

7      --color-ink2: #123a2e;
8      --color-green: #19b36b;
9      --color-lime: #c7f2b0;
10     --color-paper: #f6f8f4;
11   }
12
13   /* Bundled OFL Noto fonts (B4 / D7): Latin uses the system stack; Kannada & Devanagari
codepoints
14     resolve to the self-hosted Noto faces (unicode-range-scoped), so kn/hi render crisply
with no
15     Google Fonts runtime dependency. Order matters ? the Latin stack is matched first for
Latin. */
16   body {
17     font-family:
18       ui-sans-serif, system-ui, -apple-system, "Segoe UI", Roboto, Helvetica, Arial,
19       "Noto Sans Kannada Variable", "Noto Sans Devanagari Variable", sans-serif;
20   }
21
[/external_agent_tool_result]

```

38. assistant (2026-07-02T06:11:33.471Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\index.ts
[/external_agent_tool_call]

```

39. assistant (2026-07-02T06:11:33.471Z)

```

[external_agent_tool_result]
1      // i18n scaffold (tracker B1) ? the shared localization foundation for KhelSutra Studio.
2      //
3      // Per docs/FRIENDLY_AND_MULTILINGUAL.md + the engine's I18N_PLAN.md:
4      //   - keys, never prose (t("app.title"), enforced later by a no-bare-strings lint, B2);
5      //   - ICU MessageFormat for interpolation/plurals (Kannada/Hindi plural rules live in
the catalog,
6      //     not the code ? replaces the SPA's hand-rolled `count > 1 ? "s" : ""`);
7      //   - one negotiation order shared with the marketing site (B3);
8      //   - always-fallback to `en` so a missing translation degrades to English, never a raw
key.
9      import i18n from "i18next";
10     import { initReactI18next } from "react-i18next";
11     import LanguageDetector from "i18next-browser-languagedetector";
12     import ICU from "i18next-icu";
13     import enCommon from "../locales/en/common.json";
14     import knCommon from "../locales/kn/common.json";
15     import hiCommon from "../locales/hi/common.json";
16
17     // Supported locales (D6): `en` + `kn` + `hi`. Anything else falls back to `en`.
18     export const SUPPORTED_LOCALES = ["en", "kn", "hi"] as const;
19     export type Locale = (typeof SUPPORTED_LOCALES)[number];
20     export const DEFAULT_LOCALE: Locale = "en";
21
22     // ONE persisted-preference key, shared with the marketing site's t() shim (D9 / tracker
B3), so a
23     // visitor's language choice carries across both surfaces.
24     export const LANG_STORAGE_KEY = "khelsutra.lang";
25
26     // Catalogs are co-located here (D9). `kn` + `hi` shipped (B5/B6, AI-drafted ? native
review
27     // pending). A missing key in any locale still falls back to `en` ? the by-design
invariant.
28     const resources = {
29       en: { common: enCommon },
30       kn: { common: knCommon },
31       hi: { common: hiCommon },

```

```

32     } as const;
33
34     void i18n
35         .use(ICU)
36         .use(LanguageDetector)
37         .use(initReactI18next)
38         .init({
39             resources,
40             fallbackLng: DEFAULT_LOCALE,
41             supportedLngs: [...SUPPORTED_LOCALES],
42             nonExplicitSupportedLngs: true, // en-IN -> en
43             defaultNS: "common",
44             ns: ["common"],
45             interpolation: { escapeValue: false }, // React already escapes
46             returnNull: false,
47             // Negotiation order (I18N_PLAN §2.3): persisted pref -> ?lang= -> browser -> en.
48             detection: {
49                 order: ["localStorage", "querystring", "navigator"],
50                 lookupLocalStorage: LANG_STORAGE_KEY,
51                 lookupQuerystring: "lang",
52                 caches: ["localStorage"],
53             },
54             react: { useSuspense: false },
55         });
56
57     // Keep <html lang> in sync with the active locale (screen readers + SEO use it).
58     // Reflects the resolved content language, so an `en` fallback for an absent locale still reads as `en`.
59     function syncDocumentLang(lng: string) {
60         if (typeof document !== "undefined") {
61             document.documentElement.lang = (i18n.resolvedLanguage || lng ||
62             DEFAULT_LOCALE).split("-")[0];
63         }
64     }
65     i18n.on("languageChanged", syncDocumentLang);
66     syncDocumentLang(i18n.language ?? DEFAULT_LOCALE);
67
68     export default i18n;
69
70     [/external_agent_tool_result]

```

40. assistant (2026-07-02T06:11:33.471Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\hygiene.test.ts
[/external_agent_tool_call]

```

41. assistant (2026-07-02T06:11:33.472Z)

```

[external_agent_tool_result]
1     // CI guardrails that keep i18n "by design" (FRIENDLY_AND_MULTILINGUAL.md §4):
2     // 1. KEY-PARITY ? every locale catalog has EXACTLY the `en` key set (the
3     //    extensibility proof:
4     //    drop a stub locale and this lists every gap). Trivially green while only `en`
5     //    exists; it
6     //    starts enforcing the moment a kn/hi catalog lands (B5/B6).
7     // 2. NO-BARE-STRINGS (attribute level, v1) ? user-facing component attributes
8     //    (placeholder /
9     //    aria-label / title) must route through t(), never a hardcoded literal. A curated
10    start per
11    the doc; an AST-level sweep of JSX text can extend it later.
12    import { describe, it, expect } from "vitest";
13    import { readdirSync, readFileSync } from "node:fs";
14    import { join } from "node:path";

```

```

11
12     const LOCALES_DIR = join(import.meta.dirname, "locales");
13     const SRC_DIR = join(import.meta.dirname, "..");
14
15     function flatten(obj: Record<string, unknown>, prefix = ""): string[] {
16         const keys: string[] = [];
17         for (const [k, v] of Object.entries(obj)) {
18             const key = prefix ? `${prefix}.${k}` : k;
19             if (v && typeof v === "object" && !Array.isArray(v)) keys.push(...flatten(v as
Record<string, unknown>, key));
20             else keys.push(key);
21         }
22         return keys.sort();
23     }
24
25     const loadKeys = (locale: string): string[] =>
26         flatten(JSON.parse(readFileSync(join(LOCALES_DIR, locale, "common.json"), "utf8")) as
Record<string, unknown>);
27
28     describe("i18n key-parity ? every locale has EXACTLY the en keys", () => {
29         const enKeys = loadKeys("en");
30         const others = readdirSync(LOCALES_DIR, { withFileTypes: true })
31             .filter((d) => d.isDirectory() && d.name !== "en")
32             .map((d) => d.name);
33
34         it("en is the non-empty source catalog", () => {
35             expect(enKeys.length).toBeGreaterThan(20);
36         });
37
38         for (const loc of others) {
39             it(`${loc} matches en exactly (no missing, no extra keys)`, () => {
40                 const keys = loadKeys(loc);
41                 const missing = enKeys.filter((k) => !keys.includes(k));
42                 const extra = keys.filter((k) => !enKeys.includes(k));
43                 expect({ locale: loc, missing, extra }).toEqual({ locale: loc, missing: [], extra:
[] });
44             });
45         }
46     });
47
48     describe("no-bare-strings ? user-facing component attributes go through t()", () => {
49         const FILES = [
50             "App.tsx",
51             "components/QueryBar.tsx",
52             "components/RallyGrid.tsx",
53             "components/SelectionFooter.tsx",
54             "components/CompileResult.tsx",
55             "components/LanguageSwitcher.tsx",
56             "components/IngestPanel.tsx",
57         ];
58         // A hardcoded placeholder / aria-label / title with letters is a bare-string
59         regression ? it
60             // must be an expression ({t(...)}), not a quoted literal.
61             const BARE_ATTR = /\b(?:placeholder|aria-label|title)\s*=\s*"[\^"]*[A-Za-z]{2}[\^"]*" /g;
62
63         for (const f of FILES) {
64             it(`${f}: no hardcoded placeholder/aria-label/title`, () => {
65                 const src = readFileSync(join(SRC_DIR, f), "utf8");
66                 expect(src.match(BARE_ATTR) ?? []).toEqual([]);
67             });
68         }
69     });
70
71     describe("Indic fonts (B4) ? self-hosted OFL Noto, no runtime CDN", () => {
72         it("main.tsx bundles the OFL Noto Kannada + Devanagari fonts (self-hosted, not Google

```

```

Fonts)", () => {
72     const main = readFileSync(join(SRC_DIR, "main.tsx"), "utf8");
73     expect(main).toMatch(/@fontsource-variable\/noto-sans-kannada/);
74     expect(main).toMatch(/@fontsource-variable\/noto-sans-devanagari/);
75 });
76 it("the font stack falls back to Noto for Kannada + Devanagari glyphs", () => {
77     const css = readFileSync(join(SRC_DIR, "index.css"), "utf8");
78     expect(css).toMatch(/Noto Sans Kannada Variable/);
79     expect(css).toMatch(/Noto Sans Devanagari Variable/);
80 });
81 });
82
83 // The Studio SPA must render correctly on an AIR-GAPPED appliance (GPU/CPU/RAM, no
internet) ?
84 // every asset (fonts, styles, scripts) is bundled local, nothing pulled from a CDN...
[/external_agent_tool_result]

```

42. assistant (2026-07-02T06:11:33.472Z)

I now have complete context on the `web/` surface. Let me set up the work. First, mark the chapter and cut the A4 branch from `origin/master` (shared-checkout discipline), and load the task-tracking tools.

43. assistant (2026-07-02T06:11:33.472Z)

```

[external_agent_tool_call: mcp_ccd_session__mark_chapter]
input: {"title":"A4 ? mobile touch targets","summary":"Ramp-up done; starting the SPA UX A-items
(A4 first) per the runnable-tool mission."}
[/external_agent_tool_call]

```

44. assistant (2026-07-02T06:11:33.472Z)

```

[external_agent_tool_result]
Chapter marked: "A4 ? mobile touch targets". Continue your current work.
[/external_agent_tool_result]

```

45. assistant (2026-07-02T06:11:33.472Z)

```

[external_agent_tool_call: ToolSearch]
input: {"query":"select:TaskCreate,TaskUpdate,TaskList","max_results":5}
[/external_agent_tool_call]

```

46. assistant (2026-07-02T06:11:33.472Z)

```

[external_agent_tool_result]
[/external_agent_tool_result]

```

47. assistant (2026-07-02T06:11:33.472Z)

```

[external_agent_tool_call: Bash]
description: Cut A4 branch from origin/master
command: git fetch origin --quiet && git checkout -b feat/a4-mobile-touch-targets origin/master
2>&1 | tail -3 && echo "--- HEAD ---" && git log --oneline -1 && echo "--- branch ---" && git
branch --show-current
[/external_agent_tool_call]

```

48. assistant (2026-07-02T06:11:33.472Z)

```

[external_agent_tool_result]
branch 'feat/a4-mobile-touch-targets' set up to track 'origin/master'.
Switched to a new branch 'feat/a4-mobile-touch-targets'
--- HEAD ---
ccdcfd feat(web): A1 ? plain-language copy pass on the en catalog (de-jargon) (#60)
--- branch ---

```



```
feat/a4-mobile-touch-targets  
[/external_agent_tool_result]
```

49. assistant (2026-07-02T06:11:33.472Z)

Let me create the mission task list for visibility, then implement A4.

50. assistant (2026-07-02T06:11:33.472Z)

```
[external_agent_tool_call: TaskCreate]  
description: ?44px tap targets across  
AppBar/RallyGrid/SelectionFooter/IngestPanel/LanguageSwitcher/CompileResult/App bulk; enlarge  
parsed-pill + sample chips; surface title= tooltip text visibly for phone (no-hover) users. One  
PR, merge on green + rising coverage.  
[/external_agent_tool_call]
```

51. assistant (2026-07-02T06:11:33.472Z)

```
[external_agent_tool_result]  
Task #1 created successfully: A4 ? SPA mobile touch targets + visible tooltips  
[/external_agent_tool_result]
```

52. assistant (2026-07-02T06:11:33.472Z)

```
[external_agent_tool_call: TaskCreate]  
description: Reuse lib/errors.ts::classifyError on App.tsx::onBuild (currently shows raw  
ApiError detail/String(err)) ? localized message in CompileResult. Add a shared errors.* i18n  
namespace; unify IngestPanel onto it.  
[/external_agent_tool_call]
```

53. assistant (2026-07-02T06:11:33.472Z)

```
[external_agent_tool_result]  
Task #2 created successfully: A5 ? friendly engine-error mapping on the BUILD path  
[/external_agent_tool_result]
```

54. assistant (2026-07-02T06:11:33.472Z)

```
[external_agent_tool_call: TaskCreate]  
description: IngestPanel progress + build state: reassure on slow uplinks; status as icon+text  
chips (never icon-only).  
[/external_agent_tool_call]
```

55. assistant (2026-07-02T06:11:33.472Z)

```
[external_agent_tool_result]  
Task #3 created successfully: A8 ? background-processing reassurance + icon+text status chips  
[/external_agent_tool_result]
```

56. assistant (2026-07-02T06:11:33.473Z)

```
[external_agent_tool_call: TaskCreate]  
description: Hoist catalogs to repo-root /locales/ (single source), site prebuild copy + drift-  
guard; vanilla t()/data-i18n shim over index/capture/own-model.html + switcher + server.js  
Accept-Language negotiation matching the SPA order.  
[/external_agent_tool_call]
```

57. assistant (2026-07-02T06:11:33.473Z)

```
[external_agent_tool_result]  
Task #4 created successfully: B3 ? site/ marketing trilingual via shared t() shim  
[/external_agent_tool_result]
```

58. assistant (2026-07-02T06:11:33.473Z)

```
[external_agent_tool_call: TaskCreate]
description: Wire SPA to engine chunked-upload endpoints (backend/api/uploads.py
init|part|complete|DELETE) + file picker in IngestPanel ? feed server path to /api/process ?
existing poll?load?compile?download. New upload helpers in api/client.ts + upload mode.
[/external_agent_tool_call]
```

59. assistant (2026-07-02T06:11:33.473Z)

```
[external_agent_tool_result]
Task #5 created successfully: T4 ? in-UI local file UPLOAD (North Star keystone)
[/external_agent_tool_result]
```

60. assistant (2026-07-02T06:11:33.473Z)

```
[external_agent_tool_call: TaskCreate]
description: Serve built SPA + reverse-proxy /api?engine behind Cloudflare Access on CPU droplet
168.144.126.176, Gemini/CPU. Owner-gated bits: CF Access setup + GEMINI_API_KEY rotation. Prep
provision scripts/docs.
[/external_agent_tool_call]
```

61. assistant (2026-07-02T06:11:33.473Z)

```
[external_agent_tool_result]
Task #6 created successfully: T2 ? droplet appliance shell (hosted + shareable)
[/external_agent_tool_result]
```

62. assistant (2026-07-02T06:11:33.473Z)

```
[external_agent_tool_call: TaskCreate]
description: Retention sweep before non-owner exposure. Owner sets the window.
[/external_agent_tool_call]
```

63. assistant (2026-07-02T06:11:33.473Z)

```
[external_agent_tool_result]
Task #7 created successfully: T11 ? retention sweep (output/ + uploads/)
[/external_agent_tool_result]
```

64. assistant (2026-07-02T06:11:33.473Z)

```
[external_agent_tool_call: TaskCreate]
description: Add es/da/id catalogs (Latin, catalog-only); es needs 3-form ICU plural
{one,many,other}. Phases B/C owner-gated.
[/external_agent_tool_call]
```

65. assistant (2026-07-02T06:11:33.473Z)

```
[external_agent_tool_result]
Task #8 created successfully: MULTILINGUAL_EXPANSION Phase A (es/da/id) ? owner-paced
[/external_agent_tool_result]
```

66. assistant (2026-07-02T06:11:33.473Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\kheIsutra\web\src\lib\query.ts
[/external_agent_tool_call]
```

67. assistant (2026-07-02T06:11:33.473Z)

```
[external_agent_tool_result]
1      // Deterministic, dependency-free query compiler for the rally picker
```

```

(PER_RALLY_SELECTION.md
2      // D2/D5). NO LLM, NO network ? a tiny regex grammar over the ONLY honest, queryable
axes the
3      // engine returns today: rally LENGTH (duration), ORDER (start_time / duration /
shot_count),
4      // COUNT (top/first N), and SHOT COUNT (metadata.shot_count).
5      //
6      // Everything else a user might type ? shot TYPE (smash/clear/drop?), PLAYER, SCORE ?
has no
7      // detector and no column, so we FLAG it as "not available yet" and NEVER fabricate a
result.
8      // Parsing is pure and unit-tested; the QueryBar React component is a thin shell over
it.
9      import type { RallySegment } from "../api/types";
10
11     export type CompareOp = ">" | ">=" | "<" | "<=" | "=";
12     export type FilterField = "duration" | "shot_count";
13     /** One ANDed predicate over a rally field (e.g. duration > 10s). */
14     export interface Filter {
15         field: FilterField;
16         op: CompareOp;
17         value: number; // seconds for duration; a raw count for shot_count
18     }
19
20     export type SortKey = "duration" | "start_time" | "shot_count";
21     export interface Sort {
22         key: SortKey;
23         dir: "asc" | "desc";
24     }
25
26     /** A recognized-but-unsupported concept ? surfaced as a greyed "not available yet"
hint. */
27     export type UnavailableKind = "shot-type" | "player" | "score";
28     export interface Unavailable {
29         token: string;
30         kind: UnavailableKind;
31     }
32
33     export interface ParsedQuery {
34         raw: string;
35         /** ANDed together. */
36         filters: Filter[];
37         sort?: Sort;
38         /** "top N" / "first N" / "N longest". */
39         limit?: number;
40         /** Concepts we understood but cannot honour (shot-type / player / score). */
41         unavailable: Unavailable[];
42         /** True when at least one actionable clause (filter | sort | limit) was understood.
*/
43         recognized: boolean;
44     }
45
46     export interface QueryResult {
47         /** ALL rallies in display order (sorted if a sort was given, else input order). */
48         ordered: RallySegment[];
49         /** Ids matching the filters (then limited), in display order. */
50         selectedIds: number[];
51     }
52
53     // ?? grammar fragments ?????????????????????????????????????????????????????????
54     // `(!\d)` stops a greedy digit-run from backtracking and leaving stray trailing digits
55     // (so "over 10 smashes" can't degrade to "over 1" + "0 smashes").
56     const NUM = "(\\d+(?:\\.\\d+)?)(?!\\d)";
57     // Time / "shots" unit tokens. "shots?" is FIRST so it isn't eaten by the bare "s";
"mins?"

```

```

58 // already covers "min" (no redundant bare "min"); \b stops "m" matching "minutes" etc.
59 const UNIT_INNER = "shots?|seconds?|secs?|minutes?|mins?|m|s";
60 const UNIT = `(?:\\s*(${UNIT_INNER})\\b)?`;
61
62 // Shot-TYPE / player / score nouns ? concepts with NO detector and NO column. A number
trailing
63 // one of these must NEVER become a real filter (honesty: never reroute an unsupported
concept
64 // onto the duration axis ? PER_RALLY_SELECTION.md §3/§4.3). These stems are shared with
the
65 // "not available yet" flagging below so the two can never drift apart.
66 const SHOT_TYPE = "smash(?:es)?|clears?|drops?|dropshots?|net
?shots?|drives?|lifts?|serves?|kills?|taps?|pushes?|blocks?|slices?";
67 const PLAYER = "players?|opponents?|servers?|receivers?";
68 const SCORE = "points?|score|deuce|game point|set point|match point";
69 const UNAVAIL_WORD = `(?:${SHOT_TYPE}|${PLAYER}|${SCORE})`;
70 // Voids a comparator clause whose number is immediately followed by an unsupported
noun.
71 const NOT_UNAVAIL = `(!\\s*${UNAVAIL_WORD}\\b)`;
72
73 interface OpPattern {
74   op: CompareOp;
75   re: RegExp;
76   /** Which capture group holds the number / the unit. */
77   numAt: number;
78   unitAt: number;
79 }
80
81 // Leading comparator ("over 10s"). The "no more than" / "at least" guards come BEFORE
their
82 // bare counterparts ("more than" / "less than") so the negated phrase is consumed
first.
83 function lead(alt: string, op: CompareOp): OpPattern {...
[/external_agent_tool_result]

```

68. assistant (2026-07-02T06:11:33.473Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\i18n.test.tsx
[/external_agent_tool_call]

```

69. assistant (2026-07-02T06:11:33.474Z)

```

[external_agent_tool_result]
1 // @vitest-environment jsdom
2 //
3 // The SPA had NO render tests before this (verified) ? this file adds the jsdom safety
net for the
4 // i18n scaffold (tracker B1): real copy renders from the catalog, the switcher persists
the choice,
5 // missing-locale falls back to English, and ICU plurals format. It mounts the real <App
/>.
6 import { describe, it, expect, beforeEach, afterEach, vi } from "vitest";
7 import { render, screen, cleanup, fireEvent } from "@testing-library/react";
8 import i18n, { LANG_STORAGE_KEY } from "../index";
9 import App from "../App";
10
11 beforeEach(async () => {
12   // No engine in the test env: stub fetch so App's getConfig()/listRallies effects
reject quietly
13   // (App handles this gracefully ? cfg stays null) instead of hitting the network.
14   vi.stubGlobal("fetch", vi.fn(() => Promise.reject(new Error("no engine in test"))));
15   window.localStorage.clear();
16   await i18n.changeLanguage("en"); // also guarantees init completed before the first
render

```

```

17     });
18
19     afterEach(() => {
20         cleanup();
21         vi.unstubAllGlobals();
22     });
23
24     describe("i18n scaffold (B1)", () => {
25         it("renders user-facing copy from the en catalog (not raw keys)", () => {
26             render(<App />);
27             expect(screen.getByRole("heading", { level: 1 }).textContent).toBe("Pick the rallies
for your reel");
28             expect(screen.getByRole("button", { name: "Select all" })).toBeTruthy();
29         });
30
31         it("B2: the extracted component strings render (no leftover raw keys across the
page)", () => {
32             const { container } = render(<App />);
33             expect(screen.getByRole("button", { name: "Apply" })).toBeTruthy(); // QueryBar
34             expect(screen.getByRole("button", { name: /Build reel/ })).toBeTruthy(); //
SelectionFooter
35             expect(container.textContent).toContain("rally length"); // <Trans> hero intro +
query support
36             expect(container.textContent).toMatch(/d+ rallies selected · d+:d+ total/); //
ICU footer summary
37             expect(container.textContent).toMatch(/Rally d+/); // RallyGrid card
38             // No untranslated catalog keys leaked into the DOM.
39             expect(container.textContent).not.toMatch(/b(app|grid|footer|query|mode|result|buil
d)\.[a-zA-Z]+/);
40         });
41
42         it("offers a language switcher and persists the choice (persisted pref wins on next
visit)", () => {
43             render(<App />);
44             const knBtn = screen.getByRole("button", { name: "?????" });
45             fireEvent.click(knBtn);
46             expect(i18n.language).toBe("kn");
47             expect(window.localStorage.getItem(LANG_STORAGE_KEY)).toBe("kn");
48             expect(knBtn.getAttribute("aria-pressed")).toBe("true"); // highlights the chosen
locale
49             // Switching actually re-renders the page into Kannada (the kn catalog, B5).
50             expect(screen.getByRole("heading", { level: 1 }).textContent).toBe("????? ????????
????????????? ?????");
51         });
52
53         it("B5: switching to Kannada renders the kn catalog (Kannada copy, not English)",
async () => {
54             await i18n.changeLanguage("kn");
55             render(<App />);
56             expect(screen.getByRole("heading", { level: 1 }).textContent).toBe("????? ????????
????????????? ?????");
57             expect(screen.getByRole("button", { name: "?????????" })).toBeTruthy(); // Apply ?
Kannada
58             expect(screen.getByRole("button", { name: "????????? ??????????" })).toBeTruthy(); //
Select all ? Kannada
59             expect(document.documentElement.lang).toBe("kn"); // <html lang> tracks the active
locale (a11y)
60         });
61
62         it("B6: switching to Hindi renders the hi catalog (extensibility proof ? just a
catalog)", async () => {
63             await i18n.changeLanguage("hi");
64             render(<App />);
65             expect(screen.getByRole("heading", { level: 1 }).textContent).toBe("???? ???? ? ?
???????? ?????");

```

```
66         expect(screen.getByRole("button", { name: "???? ???? " })).toBeTruthy(); // Apply ?
Hindi
67         expect(document.documentElement.lang).toBe("hi");
68     });
69
70     it("a locale with no catalog falls back to ...
[/external_agent_tool_result]
```

70. assistant (2026-07-02T06:11:33.474Z)

I have the full picture now. The `unavailable` kinds are `shot-type`/`player`/`score`. Let me check the kn/hi catalogs so any new key I add matches their tone.

71. assistant (2026-07-02T06:11:33.474Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khehsutra\web\src\i18n\locales\kn\common.json
[/external_agent_tool_call]
```

72. assistant (2026-07-02T06:11:33.474Z)

```
[external_agent_tool_result]
1      {
2        "app": {
3          "badge": "?????-????? ???? . ????",
4          "title": "????? ?????????? ?????????????? ??????",
5          "intro": "<b>????? ????</b>? ?????????? ?????? ???? ? ???. <i>????? ?????? ???
15?</i> ? ??? ? ?????????????? ?????? ???? ?????????????? ??????. ?????? <b>?????????</b>
?????????????????; ?????????????? ???? ??????????."
6        },
7        "lang": {
8          "label": "????",
9          "en": "English",
10         "kn": "?????",
11         "hi": "?????"
12       },
13       "bulk": {
14         "selectAll": "????????? ??????????",
15         "clear": "?????????????",
16         "invert": "?????????"
17       },
18       "mode": {
19         "demo": "???? ???? ? ?????? ?????? ???? ?????? ?????????????? ??????????????
<code>?video=?video_id?</code> ???? ??????????",
20         "loading": "<code>{videoId}</code> ???? ?????????????? ???? ??????????????????",
21         "error": "<code>{videoId}</code> ???? ?????????????? ???? ??????????????: {error}",
22         "loaded": "? ? ?????????? <b>{count}</b> ?????????????? ???? ??????????".
23       },
24       "query": {
25         "placeholder": "???. over 10s · top 15 longest · shortest first",
26         "ariaLabel": "????, ?????? ???? ???? ? ?????????? ?????????????? ??????? ????",
27         "apply": "?????????",
28         "notAvailable": "????? ??????????",
29         "notAvailableTitle": "?{token}? ?? {kind} ?????? ???? ? ?????? ??????????",
30         "unreadable": "?????? ?????????? ? ?over 10s? ???? ?top 15 longest? ??????????",
31         "hint": "10 ?????????????? ?????? ?????????????? over 10s ???? ???? ???? ? ???? ?????
????????? ?????? ????.",
32         "try": "????? ??????????:",
33         "support": "<b>????? ????</b>? ?????????? ?????? ?????????? ??????????
????????????? ????-????? <x>?????????</x> (?????????????????)."
34       },
35       "grid": {
36         "empty": "?????? ???? ?????? ??????????????",
37         "rally": "?????? {n}",
```

```
38     "inReel": "????? {n}, ?????????",
39     "notInReel": "????? {n}, ??????????",
40     "start": "?????",
41     "long": "????",
42     "shots": "~{count} ?????? . ??????",
43     "shotsTitle": "???? ???? ?????????????? (?????????????) ????"
44     "shotsConfidence": " ? ??????: {conf}"
45 },
46 "footer": {
47     "count": "{count, plural, one {# ?????} other {# ?????????}}",
48     "summaryRest": "????????? . ???? {total}",
49     "reelName": "???? ?????",
50     "reelNamePlaceholder": "my-highlights",
51     "build": "???? ???? ({count})",
52     "building": "?????????????",
53     "floorOne": "<b>???? ???? ?????????? ??????</b> {min} ????? ???? ??????????"
54     "floorAll": "<b>????????? ???? {count} ??????????</b> {min} ????? ??"
55     "floorSomeLead": "{count, plural, one {# ?????} other {# ?????????}}",
56     "floorSomeRest": "{min} ????? ???? ?????????????? ?????? ??????????????"
57 },
58 "result": {
59     "ready": "???? ???? ??????????",
60     "failed": "???? ??????????",
61     "dismiss": "?????????",
62     "download": "{filename} ????????? ??"
63 },
64 "build": {
65     "noVideo": "???? ?????? ?????????????? ?????????????? ?video=<video_id> ??? ??"
66 },
67 "ingest": {
68     "title": "????? ??????????????",
69     "intro": "????????????? ???? ?????? ???? ? ???? ???? ???? ??????????????"
70     "uriPlaceholder": "gs://your-bucket/match.mp4",
71     "uriAriaLabel": "????? ???? ? gs:// ???? URL ???? ?????? ?????????????? ??",
72     "hint": "gs:// ???? ???? , ???? ?????? ?????????????? ?????? ?????? ??????"
73     "submit": "????? ??????????????",
74     "submitting": "?????????????????",
75     "cancel": "?????????????",
76     "working": "????? ??..."
77 }
78 [/external_agent_tool_result]
```

73. assistant (2026-07-02T06:11:33.474Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\hi\common.json
[/external_agent_tool_call]
```

74. assistant (2026-07-02T06:11:33.474Z)

```
[external_agent_tool_result]
1  {
2    "app": {
3      "badge": "?????-??? ???? . ??????",
4      "title": "???? ???? ???? ??????? ??????",
5      "intro": "<b>???? ????</b> ???? ???? ???? ???? ???? ???? ???? ???? ??"
15?</i> ? ??? ????? ???? ???? ?????? ???? ???? ???? ???? ???? ???? ???? ??"
16 }
```

```
6      },
7      "lang": {
8          "label": "?????",
9          "en": "English",
10         "kn": "??????",
11         "hi": "??????"
12     },
13     "bulk": {
14         "selectAll": "??? ?????",
15         "clear": "???? ?????",
16         "invert": "?????"
17     },
18     "mode": {
19         "demo": "???? ???? ? ???? ???? ??? ???? ?? ??? ???? ???? ?? ???
<code>?video=?video_id</code> ?? ??????",
20         "loading": "<code>{videoId}</code> ?? ??? ???????? ??? ?? ??? ?????",
21         "error": "<code>{videoId}</code> ?? ??? ???????? ??? ????? ?? ?????: {error}",
22         "loaded": "? ???? ? <b>{count}</b> ?????? ??? ??????"
23     },
24     "query": {
25         "placeholder": "??? . over 10s · top 15 longest · shortest first",
26         "ariaLabel": "?????, ?????? ?? ???? ?? ???? ?? ??????? ??????? ?????",
27         "apply": "???? ?????",
28         "notAvailable": "??? ?????? ?????",
29         "notAvailableTitle": "{token}? ?? ??? {kind} ????? ?????? ? ??? ?????? ?????",
30         "unreadable": "? ???? ???? ? ? ?over 10s? ?? ?top 15 longest? ????????",
31         "hint": "10 ?????? ?? ???? ???????? ?? ??? over 10s ???? ???? ? ? ???? ? ? ????
??? ??????",
32         "try": "????? ??????:",
33         "support": "<b>???? ? ????</b> ?? ???? ?? ??????? ???? ???? ?? ??????? ???
???-????? <x>????????????</x> ?? (?????????)?"
34     },
35     "grid": {
36         "empty": "?????? ?? ??? ?? ???? ???? ???? ????",
37         "rally": "???? {n}",
38         "inReel": "???? {n}, ??? ???",
39         "notInReel": "???? {n}, ??? ??? ?????",
40         "start": "?????",
41         "long": "?????",
42         "shots": "~{count} ????? · ??????????",
43         "shotsTitle": "??? ?????? ???????????? ?? (?????????) ?? ??? ?? ???? ??{conf}? ???
???? ???? ?? ?????? ?? ??????? ??????",
44         "shotsConfidence": " ? ??????: {conf}"
45     },
46     "footer": {
47         "count": "{count, plural, one {#} other {#}}",
48         "summaryRest": "????? · ??? {total}",
49         "reelName": "??? ?? ???",
50         "reelNamePlaceholder": "my-highlights",
51         "build": "??? ????? ({count})",
52         "building": "? ? ? ?",
53         "floorOne": "<b>???? ?????? ?????? ????</b> ?? {min} ?? ? ???? ? ? ? ? ? ? ? ?
???? ???? , ????? ?????? ???? ???? ? ???? ???? ??????",
54         "floorAll": "<b>??? {count} ????? ??????</b> ?? {min} ?? ? ???? ? ? ? ? ? ? ? ?
?? ???? ???? ? ? , ????? ?????? ???? ???? ???? ????????? ??????",
55         "floorSomeLead": "{count, plural, one {#} other {#}}",
56         "floorSomeRest": "????? {min} ?? ? ???? ? ? , ????? ???? ? ? ???? ? ? ?????? (???
?? <code>min_shot_count</code> ????)"
57     },
58     "result": {
59         "ready": "???? ? ? ? ? ? ",
60         "failed": "???? ? ? ? ? ",
61         "dismiss": "???? ? ? ? ? ",
62         "download": "{filename} ?????? ? ? ? ? "
63     },
```


file: C:\Users\avidu\Projects\khelsutra\web\src\components\QueryBar.tsx
[/external_agent_tool_call]

82. assistant (2026-07-02T06:11:33.474Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\QueryBar.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

83. assistant (2026-07-02T06:11:33.474Z)

Now add the visible unavailable-help line after the pill row, and enlarge the sample chips to ?44px:

84. assistant (2026-07-02T06:11:33.475Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\QueryBar.tsx
[/external_agent_tool_call]

85. assistant (2026-07-02T06:11:33.475Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\QueryBar.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

86. assistant (2026-07-02T06:11:33.475Z)

Now App.tsx bulk buttons (?44px):

87. assistant (2026-07-02T06:11:33.475Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\App.tsx
[/external_agent_tool_call]

88. assistant (2026-07-02T06:11:33.475Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\App.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

89. assistant (2026-07-02T06:11:33.475Z)

Now SelectionFooter (reel-name input + Build button ?44px):

90. assistant (2026-07-02T06:11:33.475Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\SelectionFooter.tsx
[/external_agent_tool_call]

91. assistant (2026-07-02T06:11:33.475Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\SelectionFooter.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

92. assistant (2026-07-02T06:11:33.475Z)

Now IngestPanel (input, submit, cancel, retry ?44px):

93. assistant (2026-07-02T06:11:33.475Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx
[/external_agent_tool_call]
```

94. assistant (2026-07-02T06:11:33.475Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

95. assistant (2026-07-02T06:11:33.475Z)

Now the IngestPanel retry button:

96. assistant (2026-07-02T06:11:33.475Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx
[/external_agent_tool_call]
```

97. assistant (2026-07-02T06:11:33.475Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

98. assistant (2026-07-02T06:11:33.475Z)

Now LanguageSwitcher (?44px tap targets ? important for this audience):

99. assistant (2026-07-02T06:11:33.475Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\LanguageSwitcher.tsx
[/external_agent_tool_call]
```

100. assistant (2026-07-02T06:11:33.475Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\LanguageSwitcher.tsx has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

101. assistant (2026-07-02T06:11:33.475Z)

Now CompileResult (dismiss ? ? 44px square; download link ? ?44px):

102. assistant (2026-07-02T06:11:33.476Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\CompileResult.tsx
[/external_agent_tool_call]
```

103. assistant (2026-07-02T06:11:33.476Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\CompileResult.tsx has been updated
```

successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

104. assistant (2026-07-02T06:11:33.476Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\CompileResult.tsx
[/external_agent_tool_call]

105. assistant (2026-07-02T06:11:33.476Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\CompileResult.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

106. assistant (2026-07-02T06:11:33.476Z)

Now add the `query.notAvailableHelp` key to all three catalogs (key-parity CI). en first:

107. assistant (2026-07-02T06:11:33.476Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json
[/external_agent_tool_call]

108. assistant (2026-07-02T06:11:33.476Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

109. assistant (2026-07-02T06:11:33.476Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json
[/external_agent_tool_call]

110. assistant (2026-07-02T06:11:33.476Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

111. assistant (2026-07-02T06:11:33.476Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\hi\common.json
[/external_agent_tool_call]

112. assistant (2026-07-02T06:11:33.476Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\hi\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

113. assistant (2026-07-02T06:11:33.476Z)

Now a focused A4 regression test. Let me check the test setup first.

114. assistant (2026-07-02T06:11:33.476Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\kheIsutra\web\src\test\setup.ts
[/external_agent_tool_call]
```

115. assistant (2026-07-02T06:11:33.476Z)

```
[external_agent_tool_result]
1      // Vitest setup (runs before each test file's imports). DOM-env only: jsdom under Node
26 does not
2      // expose a Web Storage `localStorage`, which the i18n language detector and the
persistence test
3      // both need. Node-env test files (typeof window === "undefined") skip this entirely.
4      if (typeof window !== "undefined" && typeof globalThis.localStorage === "undefined") {
5          let store: Record<string, string> = {};
6          const mem: Storage = {
7              get length() {
8                  return Object.keys(store).length;
9              },
10             clear() {
11                 store = {};
12             },
13             getItem(key: string) {
14                 return Object.prototype.hasOwnProperty.call(store, key) ? store[key] : null;
15             },
16             key(index: number) {
17                 return Object.keys(store)[index] ?? null;
18             },
19             removeItem(key: string) {
20                 delete store[key];
21             },
22             setItem(key: string, value: string) {
23                 store[key] = String(value);
24             },
25         };
26         Object.defineProperty(window, "localStorage", { value: mem, configurable: true,
writable: true });
27         Object.defineProperty(globalThis, "localStorage", { value: mem, configurable: true,
writable: true });
28     }
29
[/external_agent_tool_result]
```

116. assistant (2026-07-02T06:11:33.476Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\kheIsutra\web\src\components\IngestPanel.test.tsx
[/external_agent_tool_call]
```

117. assistant (2026-07-02T06:11:33.477Z)

```
[external_agent_tool_result]
1      // @vitest-environment jsdom
2      //
3      // The ingest screen end to end: type a URI ? submit ? (mocked engine) ? onLoaded gets
the video_id;
4      // and the two failure surfaces a non-technical user actually hits ? the engine's own
400 detail,
5      // and a friendly "is the engine running?" when it's unreachable. Mounts the real i18n
catalog.
6      import { describe, it, expect, vi, beforeEach, afterEach } from "vitest";
7      import { render, screen, fireEvent, cleanup, act } from "@testing-library/react";
8      import i18n from "../i18n/index";
```

```

9      import { IngestPanel } from "../IngestPanel";
10     import type { JobStatus } from "../api/types";
11
12     type FetchFn = typeof fetch;
13
14     function httpOk(status: number, json: unknown) {
15         return { ok: status >= 200 && status < 300, status, json: async () => json, text:
async () => JSON.stringify(json) };
16     }
17
18     function fetchSeq(process: unknown, jobs: JobStatus[]): FetchFn {
19         let i = 0;
20         return vi.fn(async (url: string | URL, init?: RequestInit) => {
21             const u = String(url);
22             if (u === "/api/process" && (init?.method ?? "GET") === "POST") return process;
23             if (u.startsWith("/api/jobs/")) {
24                 const body = jobs[Math.min(i, jobs.length - 1)];
25                 i += 1;
26                 return httpOk(200, body);
27             }
28             throw new Error(`unexpected fetch ${u}`);
29         }) as unknown as FetchFn;
30     }
31
32     beforeEach(async () => {
33         window.localStorage.clear();
34         await i18n.changeLanguage("en"); // guarantees init completed + a known locale for the
assertions
35     });
36
37     afterEach(() => {
38         cleanup();
39         vi.restoreAllMocks();
40         vi.unstubAllGlobals();
[/external_agent_tool_result]

```

118. assistant (2026-07-02T06:11:33.477Z)

Now the A4 regression test:

119. assistant (2026-07-02T06:11:33.477Z)

```

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\kheIsutra\web\src\components\A4.touch.test.tsx
[/external_agent_tool_call]

```

120. assistant (2026-07-02T06:11:33.477Z)

```

[external_agent_tool_result]
File created successfully at:
C:\Users\avidu\Projects\kheIsutra\web\src\components\A4.touch.test.tsx (file state is current in
your context ? no need to Read it back)
[/external_agent_tool_result]

```

121. assistant (2026-07-02T06:11:33.477Z)

Now run the web test suite + typecheck + build to verify A4:

122. assistant (2026-07-02T06:11:33.477Z)

```

[external_agent_tool_call: Bash]
description: Run web vitest suite
command: cd /c/Users/avidu/Projects/kheIsutra/web && echo "=== test ===" && npm test --silent
2>&1 | tail -40

```

[/external_agent_tool_call]

123. assistant (2026-07-02T06:11:33.477Z)

[external_agent_tool_result]

=== test ===

```
{"ts":"2026-06-21T14:56:04.701Z","level":"error","event":"api.network_error","cid":"9fde149b-122a-43ba-9c4d-ee095d25c008","method":"GET","url":"/api/config","error":"Error: no engine in test"}
```

stderr | src/i18n/i18n.test.tsx > i18n scaffold (B1) > B5:

switching to Kannada renders the kn catalog (Kannada copy, not English)

```
{"ts":"2026-06-21T14:56:04.701Z","level":"warn","event":"config.unavailable","error":"ApiError: Network error calling GET /api/config"}
```

stdout | src/i18n/i18n.test.tsx > i18n scaffold (B1) > B6:

switching to Hindi renders the hi catalog (extensibility proof ? just a catalog)

```
{"ts":"2026-06-21T14:56:04.717Z","level":"info","event":"api.request","cid":"2ae0b903-ef26-4b3c-a67b-b01f07d11ce5","method":"GET","url":"/api/config"}
```

stderr | src/i18n/i18n.test.tsx > i18n scaffold (B1) > B6:

switching to Hindi renders the hi catalog (extensibility proof ? just a catalog)

```
{"ts":"2026-06-21T14:56:04.743Z","level":"error","event":"api.network_error","cid":"2ae0b903-ef26-4b3c-a67b-b01f07d11ce5","method":"GET","url":"/api/config","error":"Error: no engine in test"}
```

stderr | src/i18n/i18n.test.tsx > i18n scaffold (B1) > B6:

switching to Hindi renders the hi catalog (extensibility proof ? just a catalog)

```
{"ts":"2026-06-21T14:56:04.743Z","level":"warn","event":"config.unavailable","error":"ApiError: Network error calling GET /api/config"}
```

stdout | src/i18n/i18n.test.tsx > i18n scaffold (B1) > a

locale with no catalog falls back to English (never a raw key)

```
{"ts":"2026-06-21T14:56:04.757Z","level":"info","event":"api.request","cid":"d98643cf-6242-4fc3-b0e3-b24b9e67154c","method":"GET","url":"/api/config"}
```

stderr | src/i18n/i18n.test.tsx > i18n scaffold (B1) > a

locale with no catalog falls back to English (never a raw key)

```
{"ts":"2026-06-21T14:56:04.761Z","level":"error","event":"api.network_error","cid":"d98643cf-6242-4fc3-b0e3-b24b9e67154c","method":"GET","url":"/api/config","error":"Error: no engine in test"}
```

stderr | src/i18n/i18n.test.tsx > i18n scaffold (B1) > a

locale with no catalog falls back to English (never a raw key)

```
{"ts":"2026-06-21T14:56:04.761Z","level":"warn","event":"config.unavailable","error":"ApiError: Network error calling GET /api/config"}
```

stdout | src/i18n/i18n.test.tsx > browser-language negotiation (B1:

N1/N3?N7) > N6: an unsupported browser language (Tamil) serves English content ?

never a locale we can't render

```
{"ts":"2026-06-21T14:56:04.778Z","level":"info","event":"api.request","cid":"ab7eed16-f5a4-403b-bde5-1fc272fa064b","method":"GET","url":"/api/config"}
```

stderr | src/i18n/i18n.test.tsx > browser-language negotiation (B1:

N1/N3?N7) > N6: an unsupported browser language (Tamil) serves English content ?

never a locale we can't render

```
{"ts":"2026-06-21T14:56:04.780Z","level":"error","event":"api.network_error","cid":"ab7eed16-f5a4-403b-bde5-1fc272fa064b","method":"GET","url":"/api/config","error":"Error: no engine in test"}
```

stderr | src/i18n/i18n.test.tsx > browser-language negotiation (B1:

N1/N3?N7) > N6: an unsupported browser language (Tamil) serves English content ?

```
never a locale we can't render
{"ts":"2026-06-
21T14:56:04.781Z","level":"warn","event":"config.unavailable","error":"ApiError: Network error
calling GET /api/config"}

? src/i18n/i18n.test.tsx (13 tests)
586ms
? i18n scaffold (B1) > renders user-facing copy from the en
catalog (not raw keys) 309ms

Test Files 12 passed■[9...
[/external_agent_tool_result]
```

124. assistant (2026-07-02T06:11:33.477Z)

All 127 pass (123 + my 4 A4 tests). Now typecheck, lint, and build:

125. assistant (2026-07-02T06:11:33.477Z)

```
[external_agent_tool_call: Bash]
description: Typecheck, lint, build the web app
command: echo "=== tsc ===" && npx tsc --noEmit 2>&1 | tail -15 && echo "EXIT tsc: $?" && echo
"=== lint ===" && (npm run lint --silent 2>&1 | tail -20 || echo "(no lint script or lint
failed)") && echo "=== build ===" && npm run build --silent 2>&1 | tail -15
[/external_agent_tool_call]
```

126. assistant (2026-07-02T06:11:33.477Z)

```
[external_agent_tool_result]
=== tsc ===
EXIT tsc: 0
=== lint ===
=== build ===
vite v6.4.3 building for production...
transforming...
? 88 modules transformed.
rendering chunks...
computing gzip size...
dist/index.html
  0.40 kB ? gzip: 0.27 kB
dist/assets/noto-sans-devanagari-latin-ext-wght-normal-CLKupaxV.woff2
 13.85 kB
dist/assets/noto-sans-kannada-latin-ext-wght-normal-MdBBF41x.woff2
 13.99 kB
dist/assets/noto-sans-devanagari-latin-wght-normal-Bxfr4UJH.woff2
 25.38 kB
dist/assets/noto-sans-kannada-latin-wght-normal-DeLZCgFT.woff2
 27.58 kB
dist/assets/noto-sans-kannada-kannada-wght-normal-DuwB-P8d.woff2
 90.80 kB
dist/assets/noto-sans-devanagari-devanagari-wght-normal-DWI_L5Cu.woff2
121.19 kB
dist/assets/index-CUYZ8r6F.css
 20.83 kB ? gzip: 4.84 kB
dist/assets/index-BQA7GRLW.js
347.66 kB ? gzip: 109.59 kB
? built in 1.21s
[/external_agent_tool_result]
```

127. assistant (2026-07-02T06:11:33.477Z)

tsc/lint/build all clean. A4 is an observable mobile-UX change, so let me verify in the browser. Loading the preview tools.

128. assistant (2026-07-02T06:11:33.477Z)

```
[external_agent_tool_call: ToolSearch]
input: {"query":"select:mcp__Claude_Preview__preview_start,mcp__Claude_Preview__preview_resize,mcp__Claude_Preview__preview_screenshot,mcp__Claude_Preview__preview_fill,mcp__Claude_Preview__preview_snapshot,mcp__Claude_Preview__preview_console_logs","max_results":8}
[/external_agent_tool_call]
```

129. assistant (2026-07-02T06:11:33.477Z)

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

130. assistant (2026-07-02T06:11:33.477Z)

```
[external_agent_tool_call: Bash]
description: Check for existing launch.json configs
command: cd /c/Users/avidu/Projects/khelsutra && echo "=== khelsutra/.claude ===" && find .claude -type f 2>/dev/null | head && echo "" && echo "=== primary .claude/launch.json? ===" && cat /c/Users/avidu/Projects/badminton-highlight-indexer/.claude/launch.json 2>/dev/null || echo "(none in primary)" && echo "" && cat .claude/launch.json 2>/dev/null || echo "(none in khelsutra)"
[/external_agent_tool_call]
```

131. assistant (2026-07-02T06:11:33.477Z)

```
[external_agent_tool_result]
=== khelsutra/.claude ===
.claude/launch.json

=== primary .claude/launch.json? ===
{
  "version": "0.0.1",
  "configurations": [
    {
      "name": "landing",
      "runtimeExecutable": "python",
      "runtimeArgs": ["-m", "http.server", "8099", "--directory", "artifacts/landing"],
      "port": 8099
    },
    {
      "name": "khelsutra-web",
      "runtimeExecutable": "npm",
      "runtimeArgs": ["--prefix", "C:/Users/avidu/Projects/khelsutra/web", "run", "dev", "--", "--port", "5174", "--strictPort"],
      "port": 5174
    },
    {
      "name": "khelsutra-site",
      "runtimeExecutable": "python",
      "runtimeArgs": ["-m", "http.server", "8123", "--directory", "C:/Users/avidu/Projects/khelsutra/site/public"],
      "port": 8123
    },
    {
      "name": "khelsutra-node",
      "runtimeExecutable": "node",
      "runtimeArgs": ["C:/Users/avidu/Projects/khelsutra/site/server.js"],
      "port": 8080
    }
  ]
}
{
```

```
"version": "0.0.1",
"configurations": [
  {
    "name": "khelsutra-web",
    "runtimeExecutable": "npm",
    "runtimeArgs": ["--prefix", "web", "run", "dev", "--", "--port", "5173", "--strictPort"],
    "port": 5173
  },
  {
    "name": "khelsutra-site",
    "runtimeExecutable": "node",
    "runtimeArgs": ["site/server.js"],
    "port": 8080
  }
]
}
[/external_agent_tool_result]
```

132. assistant (2026-07-02T06:11:33.477Z)

The primary `claude/launch.json` has a `khelsutra-web` config (absolute path, port 5174). Let me start it and verify A4 at mobile width.

133. assistant (2026-07-02T06:11:33.477Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_start]
input: {"name":"khelsutra-web"}
[/external_agent_tool_call]
```

134. assistant (2026-07-02T06:11:33.477Z)

```
[external_agent_tool_result]
{
  "serverId": "d4a4a2a4-7c2e-46e2-889e-f619477293b4",
  "port": 5174,
  "name": "khelsutra-web",
  "reused": false
}
```

Server started successfully on port 5174.

```
[/external_agent_tool_result]
```

135. assistant (2026-07-02T06:11:33.477Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_resize]
input: {"serverId":"d4a4a2a4-7c2e-46e2-889e-f619477293b4","preset":"mobile"}
[/external_agent_tool_call]
```

136. assistant (2026-07-02T06:11:33.478Z)

```
[external_agent_tool_result]
Viewport set to 375x812 (mobile).
[/external_agent_tool_result]
```

137. assistant (2026-07-02T06:11:33.478Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_screenshot]
input: {"serverId":"d4a4a2a4-7c2e-46e2-889e-f619477293b4"}
[/external_agent_tool_call]
```

138. assistant (2026-07-02T06:11:33.478Z)

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

139. assistant (2026-07-02T06:11:33.478Z)

Mobile layout looks clean ? taller language switcher, Process video, inputs, Build reel all comfortably tappable. Now let me verify the visible "not available" help line by typing an unsupported concept, then screenshot the query area.

140. assistant (2026-07-02T06:11:33.478Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_fill]
input: {"serverId":"d4a4a2a4-7c2e-46e2-889e-f619477293b4","selector":"input[aria-label=\"Filter rallies by length, count or order\"]","value":"over 10s smashes by player"}
[/external_agent_tool_call]
```

141. assistant (2026-07-02T06:11:33.478Z)

```
[external_agent_tool_result]
Successfully filled: input[aria-label="Filter rallies by length, count or order"]
[/external_agent_tool_result]
```

142. assistant (2026-07-02T06:11:33.478Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_snapshot]
input: {"serverId":"d4a4a2a4-7c2e-46e2-889e-f619477293b4"}
[/external_agent_tool_call]
```

143. assistant (2026-07-02T06:11:33.478Z)

```
[external_agent_tool_result]
[1] RootWebArea: "KhelSutra Studio"
  [5] main
    [6] sectionheader
      [7] none
      [9] StaticText: "Khel"
      [10] InlineTextBox: "Khel"
      [11] StaticText: "?"
      [12] InlineTextBox: "?"
      [13] StaticText: "Sutra "
      [14] InlineTextBox: "Sutra "
      [15] StaticText: "Studio"
      [16] InlineTextBox: "Studio"
      [17] navigation: "Language"
      [18] button: "English"
      [19] StaticText: "English"
      [20] InlineTextBox: "English"
      [21] button: "?????"
      [22] StaticText: "?????"
      [23] InlineTextBox: "?????"
      [24] button: "?????"
      [25] StaticText: "?????"
      [26] InlineTextBox: "?????"
    [27] generic
      [29] StaticText: "pick your rallies . alpha"
      [30] InlineTextBox: "pick your rallies . alpha"
      [31] heading: "Pick the rallies for your reel"
      [32] StaticText: "Pick the rallies for your reel"
      [33] InlineTextBox: "Pick the rallies for your "
      [34] InlineTextBox: "reel"
      [35] paragraph
        [36] StaticText: "Filter by "
        [37] InlineTextBox: "Filter by "
        [38] strong
          [39] StaticText: "rally length"
          [40] InlineTextBox: "rally length"
        [41] StaticText: " ? e.g. "
```

```

[42] InlineTextBox: " ? e.g. "
[43] emphasis
[44] StaticText: "?top 15 longest?"
[45] InlineTextBox: "?top 15 longest?"
[46] StaticText: " ? then fine-tune by tapping cards. Everything starts "
[47] InlineTextBox: "? then fine-tune by tapping cards. Everything "
[48] InlineTextBox: "starts "
[49] strong
[50] StaticText: "selected"
[51] InlineTextBox: "selected"
[52] StaticText: "; just drop the ones you don't want."
[53] InlineTextBox: "; just drop the ones you don't "
[54] InlineTextBox: "want."
[55] generic
[56] heading: "Process a video"
[57] StaticText: "Process a video"
[58] InlineTextBox: "Process a video"
[59] paragraph
[60] StaticText: "Point KhelSutra at a video and it finds the rallies for you ? this
can take a few minutes."
[61] InlineTextBox: "Point KhelSutra at a video and it finds the "
[62] InlineTextBox: "rallies for you ? this can take a few minutes."
[63] generic
[64] textbox: "Video location ? a gs:// bucket URL or a path on the engine host"
[67] button: "Process video"
[68] StaticText: "Process video"
[69] InlineTextBox: "Process video"
[70] paragraph
[71] StaticText: "Paste a cloud-storage link (gs://?), or a path to a video on the
machine running KhelSutra."
[72] InlineTextBox: "Paste a cloud-storage link (gs://?), or a path to a "
[73] InlineTextBox: "video on the machine running KhelSutra."
[75] paragraph
[76] StaticText: "These are sample rallies ? process a video above to work on a real
match."
[77] InlineTextBox: "These are sample rallies ? process a video above to "
[78] InlineTextBox: "work on a real match."
[80] none
[81] generic
[82] textbox: "Filter rallies by length, count or order" (value: "over 10s smashes by
player")
[85] StaticText: "over 10s smashes by player"
[86] InlineTextBox: "over 10s smashes by player"
[87] button: "Apply"
[88] StaticText: "Apply"
[89] InlineTextBox: "Apply"
[92] StaticText: "length > 10s"
[93] InlineTextBox: "length > 10s"
[95] StaticText: "smashes"
[96] InlineTextBox: "smashes"
[97] StaticText: " . "
[98] InlineTextBox: " . "
[99] StaticText: "not available yet"
[100] InlineTextBox: "not available yet"
[102] StaticText: "player"
[103] InlineTextBox: "player"
[104] S...
[/external_agent_tool_result]

```

144. assistant (2026-07-02T06:11:33.478Z)

The visible help line renders ([108?111]: "Shot type, players and score aren't available yet ? filter by rally length instead.") right after the "not available yet" pills. Let me confirm no console errors, then I'm done verifying.

145. assistant (2026-07-02T06:11:33.478Z)

```
[external_agent_tool_call: mcp_Claude_Preview_preview_console_logs]
input: {"serverId":"d4a4a2a4-7c2e-46e2-889e-f619477293b4","level":"error"}
[/external_agent_tool_call]
```

146. assistant (2026-07-02T06:11:33.478Z)

```
[external_agent_tool_result]
[error] {"ts":"2026-06-21T14:56:50.505Z","level":"error","event":"api.http_error","cid":"2b69255a-938f-4471-9224-8ffb2ee652a","method":"GET","url":"/api/config","status":500,"body":""}
[error] {"ts":"2026-06-21T14:56:50.505Z","level":"error","event":"api.http_error","cid":"2b69255a-938f-4471-9224-8ffb2ee652a","method":"GET","url":"/api/config","status":500,"body":""}
[error] {"ts":"2026-06-21T14:56:50.505Z","level":"error","event":"api.http_error","cid":"2b69255a-938f-4471-9224-8ffb2ee652a","method":"GET","url":"/api/config","status":500,"body":""}
[error] {"ts":"2026-06-21T14:56:50.505Z","level":"error","event":"api.http_error","cid":"2b69255a-938f-4471-9224-8ffb2ee652a","method":"GET","url":"/api/config","status":500,"body":""}
[error] {"ts":"2026-06-21T14:56:50.505Z","level":"error","event":"api.http_error","cid":"2b69255a-938f-4471-9224-8ffb2ee652a","method":"GET","url":"/api/config","status":500,"body":""}
[error] {"ts":"2026-06-21T14:56:50.505Z","level":"error","event":"api.http_error","cid":"2b69255a-938f-4471-9224-8ffb2ee652a","method":"GET","url":"/api/config","status":500,"body":""}
[error] {"ts":"2026-06-21T14:56:50.509Z","level":"error","event":"api.http_error","cid":"2c4a37fb-6d1d-41f6-bc6c-e39e3c9640d1","method":"GET","url":"/api/config","status":500,"body":""}
[error] {"ts":"2026-06-21T14:56:50.509Z","level":"error","event":"api.http_error","cid":"2c4a37fb-6d1d-41f6-bc6c-e39e3c9640d1","method":"GET","url":"/api/config","status":500,"body":""}
[error] {"ts":"2026-06-21T14:56:50.509Z","level":"error","event":"api.http_error","cid":"2c4a37fb-6d1d-41f6-bc6c-e39e3c9640d1","method":"GET","url":"/api/config","status":500,"body":""}
[error] {"ts":"2026-06-21T14:56:50.509Z","level":"error","event":"api.http_error","cid":"2c4a37fb-6d1d-41f6-bc6c-e39e3c9640d1","method":"GET","url":"/api/config","status":500,"body":""}
[error] {"ts":"2026-06-21T14:56:50.509Z","level":"error","event":"api.http_error","cid":"2c4a37fb-6d1d-41f6-bc6c-e39e3c9640d1","method":"GET","url":"/api/config","status":500,"body":""}
[error] {"ts":"2026-06-21T14:56:50.509Z","level":"error","event":"api.http_error","cid":"2c4a37fb-6d1d-41f6-bc6c-e39e3c9640d1","method":"GET","url":"/api/config","status":500,"body":""}
[error] {"ts":"2026-06-21T14:56:50.679Z","level":"error","event":"api.http_error","cid":"a6d12cab-9428-430f-92be-b1b7f106e7ac","method":"GET","url":"/api/config","status":500,"body":""}
[error] {"ts":"2026-06-21T14:56:50.679Z","level":"error","event":"api.http_error","cid":"a6d12cab-9428-430f-92be-b1b7f106e7ac","method":"GET","url":"/api/config","status":500,"body":""}
[error] {"ts":"2026-06-21T14:56:50.679Z","level":"error","event":"api.http_error","cid":"a6d12cab-9428-430f-92be-b1b7f106e7ac","method":"GET","url":"/api/config","status":500,"body":""}
[error] {"ts":"2026-06-21T14:56:50.679Z","level":"error","event":"api.http_error","cid":"a6d12cab-9428-430f-92be-b1b7f106e7ac","method":"GET","url":"/api/config","status":500,"body":""}
[error] {"ts":"2026-06-21T14:56:50.679Z","level":"error","event":"api.http_error","cid":"a6d12cab-9428-430f-92be-b1b7f106e7ac","method":"GET","url":"/api/config","status":500,"body":""}
[error] {"ts":"2026-06-21T14:56:50.683Z","level":"error","event":"api.http_error","cid":"040e10ec-56c9-48a9-b22f-
```

```
e04ab393b6a1", "method": "GET", "url": "/api/config", "status": 500, "body": ""}
[error] {"ts": "2026-06-21T14:56:50.683Z", "level": "error", "event": "api.http_error", "cid": "040e10ec-56c9-48a9-b22f-e04ab393b6a1", "method": "GET", "url": "/api/config", "status": 500, "body": ""}
[error] {"ts": "2026-06-21T14:56:50.683Z", "level": "error", "event": "api.http_error", "cid": "040e10ec-56c9-48a9-b22f-e04ab393b6a1", "method": "GET", "url": "/api/config", "status": 500, "body": ""}
[error] {"ts": "2026-06-21T14:56:50.683Z", "level": "error", "event": "api.http_error", "cid": "04...
[/external_agent_tool_result]
```

147. assistant (2026-07-02T06:11:33.478Z)

A4 verified: 244px targets visible in the screenshot, the help line renders for unsupported concepts, 127 tests + tsc + lint + build all green. The only console errors are expected `^/api/config` 500s (no engine on :8000 ? handled gracefully, unrelated to A4).

Per ultracode, let me run an adversarial review workflow over the A4 diff before merging ? to catch any missed interactive control or regression. First, capture the diff.

148. assistant (2026-07-02T06:11:33.478Z)

```
[external_agent_tool_call: Bash]
description: Capture A4 diff for review
command: git diff --stat && echo "====FULL DIFF====" && git diff
[/external_agent_tool_call]
```

149. assistant (2026-07-02T06:11:33.478Z)

```
[external_agent_tool_result]
web/src/App.tsx | 6 +++--
web/src/components/CompileResult.tsx | 4 ++--
web/src/components/IngestPanel.tsx | 8 +++----
web/src/components/LanguageSwitcher.tsx | 2 +-
web/src/components/QueryBar.tsx | 17 ++++++++-----
web/src/components/SelectionFooter.tsx | 4 ++--
web/src/i18n/locales/en/common.json | 1 +
web/src/i18n/locales/hi/common.json | 1 +
web/src/i18n/locales/kn/common.json | 1 +
9 files changed, 27 insertions(+), 17 deletions(-)
====FULL DIFF====
diff --git a/web/src/App.tsx b/web/src/App.tsx
index b43debd..85361d6 100644
--- a/web/src/App.tsx
+++ b/web/src/App.tsx
@@ -244,9 +244,9 @@ export default function App() {
  </div>

  <div className="mt-6 flex flex-wrap gap-2 text-sm">
    - <button type="button" onClick={() => onBulk("select_all")} className="rounded-full
border border-black/15 px-3 py-1 hover:border-green">{t("bulk.selectAll")}</button>
    - <button type="button" onClick={() => onBulk("clear")} className="rounded-full border
border-black/15 px-3 py-1 hover:border-green">{t("bulk.clear")}</button>
    - <button type="button" onClick={() => onBulk("invert")} className="rounded-full border
border-black/15 px-3 py-1 hover:border-green">{t("bulk.invert")}</button>
    + <button type="button" onClick={() => onBulk("select_all")} className="inline-flex
min-h-11 items-center rounded-full border border-black/15 px-4 hover:border-
green">{t("bulk.selectAll")}</button>
    + <button type="button" onClick={() => onBulk("clear")} className="inline-flex min-h-11
items-center rounded-full border border-black/15 px-4 hover:border-
green">{t("bulk.clear")}</button>
    + <button type="button" onClick={() => onBulk("invert")} className="inline-flex
min-h-11 items-center rounded-full border border-black/15 px-4 hover:border-
green">{t("bulk.invert")}</button>
  </div>
```

```

    <div className="mt-5">
diff --git a/web/src/components/CompileResult.tsx b/web/src/components/CompileResult.tsx
index 441c7d7..619d22b 100644
--- a/web/src/components/CompileResult.tsx
+++ b/web/src/components/CompileResult.tsx
@@ -36,7 +36,7 @@ export function CompileResult({ build, onDismiss }: Props) {
    type="button"
    onClick={onDismiss}
    aria-label={t("result.dismiss")}
-    className="rounded-full px-2 text-ink/50 hover:text-ink"
+    className="flex h-11 w-11 shrink-0 items-center justify-center rounded-full text-ink/50 hover:text-ink"
  >
    ?
  </button>
@@ -55,7 +55,7 @@ export function CompileResult({ build, onDismiss }: Props) {
    <a
      href={build.url}
      download={build.filename}
-      className="mt-3 inline-block rounded-full bg-green px-4 py-2 text-sm font-semibold text-[#04140D]"
+      className="mt-3 inline-flex min-h-11 items-center rounded-full bg-green px-4 py-2 text-sm font-semibold text-[#04140D]"
    >
      {t("result.download", { filename: build.filename })}
    </a>
diff --git a/web/src/components/IngestPanel.tsx b/web/src/components/IngestPanel.tsx
index 8c6ffec..322ffb8 100644
--- a/web/src/components/IngestPanel.tsx
+++ b/web/src/components/IngestPanel.tsx
@@ -47,12 +47,12 @@ export function IngestPanel({ onLoaded }: Props) {
    placeholder={t("ingest.uriPlaceholder")}
    aria-label={t("ingest.uriAriaLabel")}
    disabled={busy}
-    className="min-w-[18rem] flex-1 rounded-xl border border-black/15 bg-white px-4 py-2 text-sm text-ink outline-none focus:border-green disabled:opacity-50"
+    className="min-h-11 min-w-[18rem] flex-1 rounded-xl border border-black/15 bg-white px-4 py-2 text-sm text-ink outline-none focus:border-green disabled:opacity-50"
  />
  <button
    type="submit"
    disabled={busy}
-    className="rounded-xl bg-ink px-4 py-2 text-sm font-semibold text-white disabled:opacity-40"...
[/external_agent_tool_result]

```

150. assistant (2026-07-02T06:11:33.478Z)

The diff is complete and tidy. Now an adversarial review fan-out over it ? three independent lenses (touch-target completeness, correctness/regression, A4-faithfulness), then I'll verify and fold anything real.

151. assistant (2026-07-02T06:11:33.478Z)

```

[external_agent_tool_call: Workflow]
input: {"script":"export const meta = {\n  name: 'a4-review',\n  description: 'Adversarial 3-lens review of the A4 mobile-touch-targets diff',\n  phases: [{ title: 'Review' }],\n}\n\nconst BASE = 'C:/Users/avidu/Projects/khelsutra/web/src'\nconst DIFF = args.diff\n\nconst SCHEMA = {\n  type: 'object',\n  additionalProperties: false,\n  properties: {\n    summary: { type: 'string' },\n    findings: {\n      type: 'array',\n      items: {\n        type: 'object',\n        additionalProperties: false,\n        properties: {\n          severity: { type: 'string', enum: ['blocker', 'major', 'minor', 'nit'] },\n          file: { type: 'string' },\n          location: { type: 'string' },\n          title: { type: 'string' },\n          detail: { type: 'string' },\n          suggestion: { type: 'string' },\n          required: ['severity', 'file', 'title', 'detail'],\n        },\n      },\n    },\n  },\n}

```

```
required: ['summary', 'findings'],\n}\n\nconst LENSES = [\n  {\n    key: 'touch-completeness',\n    prompt: `You are auditing a UI diff for COMPLETENESS of mobile touch targets (WCAG 2.5.5: interactive controls should be >=44x44 CSS px). The Tailwind class min-h-11 = 2.75rem = 44px; h-11/w-11 = 44px.\n\nRead EVERY component under ${BASE}/components and ${BASE}/App.tsx. List any INTERACTIVE control (button, link <a>, <input>, <select>, role=button, onClick element) that a user taps but that does NOT get a >=44px height after this diff. Do NOT flag display-only spans/labels. The rally CARD buttons in RallyGrid.tsx are large (p-3) ? only flag them if they're actually < 44px tall. Be precise with file:line. If everything interactive is covered, return an empty findings array and say so.`,\n  },\n  {\n    key: 'correctness',\n    prompt: `You are reviewing a UI diff for CORRECTNESS and REGRESSION. Read the changed files under ${BASE}. Check:\n- Did adding min-h-11 / inline-flex / items-center break vertical alignment, wrapping, or the sticky footer layout in SelectionFooter.tsx?\n- T...\n  }\n]\n\n[/external_agent_tool_call]
```

152. assistant (2026-07-02T06:11:33.479Z)

```
[external_agent_tool_result]
Workflow launched in background. Task ID: wqh6khdbw
Summary: Adversarial 3-lens review of the A4 mobile-touch-targets diff
Transcript dir: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\2ba49afa-99f4-41f2-86a2-27a289d432f3\subagents\workflows\wf_cd064e9b-e5c
Script file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-khelsutra\2ba49afa-99f4-41f2-86a2-27a289d432f3\workflows\scripts\a4-review-wf_cd064e9b-e5c.js
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath: "C:\Users\avidu\claude\projects\C--Users-avidu-Projects-khelsutra\2ba49afa-99f4-41f2-86a2-27a289d432f3\workflows\scripts\a4-review-wf_cd064e9b-e5c.js"} to iterate without resending the script.)
Run ID: wf_cd064e9b-e5c
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\claude\projects\C--Users-avidu-Projects-khelsutra\2ba49afa-99f4-41f2-86a2-27a289d432f3\workflows\scripts\a4-review-wf_cd064e9b-e5c.js", resumeFromRunId: "wf_cd064e9b-e5c"}) ? completed agents return cached results.
```

You will be notified when it completes. Use /workflows to watch live progress.

```
[/external_agent_tool_result]
```

153. assistant (2026-07-02T06:11:33.479Z)

The 3-lens review runs in the background. While it works, let me read the A5-relevant files (the build path already uses `parseDetail`; A5 will reuse `classifyError`). Reading the client, the ingest hook, and the errors test to prep.

154. assistant (2026-07-02T06:11:33.479Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\web\src\api\client.ts
[/external_agent_tool_call]
```

155. assistant (2026-07-02T06:11:33.479Z)

```
[external_agent_tool_result]
1      // Typed client for the engine FastAPI. SAME-ORIGIN relative `/api/*` (the appliance serves
2      // UI + engine behind one origin; in dev Vite proxies `/api` ? :8000). Every call carries a
3      // correlation id (header + logs) so a journey is traceable end to end.
4      import { log, newCorrelationId } from "../lib/log";
5      import type { RallySegment, QueryResponse, CompileResponse, EngineConfig, ProcessResponse, JobStatus } from "../types";
6
7      const BASE = "/api";
8
9      export class ApiError extends Error {
10         readonly cid?: string;
```



```

11     readonly status?: number;
12     readonly body?: string;
13     constructor(message: string, opts: { cid?: string; status?: number; body?: string;
cause?: unknown } = {}) {
14         super(message);
15         this.name = "ApiError";
16         this.cid = opts.cid;
17         this.status = opts.status;
18         this.body = opts.body;
19         if (opts.cause !== undefined) (this as { cause?: unknown }).cause = opts.cause;
20     }
21 }
22
23 interface RequestOptions {
24     method?: string;
25     body?: string;
26     headers?: Record<string, string>;
27     /** Correlation id to thread through this call; auto-generated if absent. */
28     cid?: string;
29 }
30
31 async function request<T>(path: string, opts: RequestOptions = {}): Promise<T> {
32     const cid = opts.cid ?? newCorrelationId();
33     const url = `${BASE}${path}`;
34     const method = opts.method ?? "GET";
35     log("info", "api.request", { cid, method, url });
36
37     let res: Response;
38     try {
39         res = await fetch(url, {
40             method,
41             body: opts.body,
42             headers: { "content-type": "application/json", "x-correlation-id": cid,
...(opts.headers ?? {}) },
43         });
44     } catch (err) {
45         // Loud failure, never silent (LOCAL_APPLIANCE_ARCHITECTURE.md §5).
46         log("error", "api.network_error", { cid, method, url, error: String(err) });
47         throw new ApiError(`Network error calling ${method} ${url}`, { cid, cause: err });
48     }
49
50     if (!res.ok) {
51         const body = await res.text().catch(() => "");
52         log("error", "api.http_error", { cid, method, url, status: res.status, body:
body.slice(0, 500) });
53         throw new ApiError(`${method} ${url} failed: ${res.status}`, { cid, status:
res.status, body });
54     }
55
56     const data = (await res.json()) as T;
57     log("info", "api.response", { cid, method, url, status: res.status });
58     return data;
59 }
60
61 /** Submit a video (a `gs://`/`gcs://`/`s3://` bucket URI, or a path on the engine host)
for
62 * processing. POST /api/process {video_path} ? 202 {job_id, ...} [main.py:260]. Fast-
fails 400
63 * (unsupported scheme / out-of-allowed-root) or 404 (missing LOCAL file) ? surfaced as
ApiError. */
64 export async function processVideo(videoPath: string, cid?: string):
Promise<ProcessResponse> {
65     return request<ProcessResponse>("/process", {
66         method: "POST",
67         body: JSON.stringify({ video_path: videoPath }),

```

```

68         cid,
69     });
70 }
71
72 /** Poll a submitted job. GET /api/jobs/{job_id} [main.py:303]. 404 ? unknown job_id. */
73 export async function getJob(jobId: string, cid?: string): Promise<JobStatus> {
74     return request<JobStatus>(`/jobs/${encodeURIComponent(jobId)}`, { method: "GET", cid
75 });
76 }
77 /** List a video's detected rallies. POST /api/query {video_id} ? {play_segments}
[main.py:331]. */
78 export async function listRallies(videoId: string, cid?: string):
Promise<RallySegment[]> {
79     const data = await request<QueryResponse>("/query", {
80         method: "POST",
81         body: JSON.stringify({ video_id: videoId }),
82         cid,
83     });
84     return data.play_segments ?? [];
85 }
86
87 /** Stitch selected rallies into one reel. POST /api/compile {video_id, segment_ids,
output_filename} [main.py:342]. */
88 export async function compileReel(
89     videoId: string,
90     segmentIds: number[],
91     outputFilename: string,
92     cid?: string,
93 ): Promise<CompileResponse> {
94     return request<CompileResponse>("/compile", {
95         method: "POST",
96     });
97 }
98
99 [/_external_agent_tool_result]

```

156. assistant (2026-07-02T06:11:33.479Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\web\src\hooks\useIngestJob.ts
[/external_agent_tool_call]

```

157. assistant (2026-07-02T06:11:33.479Z)

```

[external_agent_tool_result]
1 // Drives the front-half ingest: submit a video to POST /api/process, then poll GET
/api/jobs/{id}
2 // until it finishes, and hand the resulting video_id to the existing rally-loading
back-half.
3 //
4 // Kept i18n-FREE (returns stable IngestErrorCodes, not prose) so it unit-tests without
a
5 // translation context ? IngestPanel maps codes to localized strings. The poll loop is a
RECURSIVE
6 // setTimeout (never setInterval): each tick waits for the prior getJob to resolve, so a
slow engine
7 // can't pile up overlapping requests. It backs off, gives up after a ceiling, and is
fully
8 // cancellable on unmount / user-cancel / a superseding submit.
9 import { useCallback, useEffect, useRef, useState } from "react";
10 import { getJob, processVideo } from "../api/client";
11 import { classifyError, type IngestError } from "../lib/errors";
12 import { log, newCorrelationId } from "../lib/log";
13 import type { JobStatus } from "../api/types";
14
15 const POLL_MIN_MS = 1000;

```

```

16     const POLL_MAX_MS = 5000;
17     const POLL_BACKOFF = 1.5;
18     const POLL_CEILING_MS = 10 * 60 * 1000; // a wedged 'running' job won't poll forever
19
20     export type IngestPhase =
21       | { status: "idle" }
22       | { status: "submitting" }
23       | { status: "polling"; progress: string | null }
24       | { status: "done"; videoId: string }
25       | { status: "error"; error: IngestError };
26
27     interface Run {
28       cancelled: boolean;
29       timer?: ReturnType<typeof setTimeout>;
30     }
31
32     export function useIngestJob(onLoaded?: (videoId: string) => void) {
33       const [phase, setPhase] = useState<IngestPhase>({ status: "idle" });
34       const runRef = useRef<Run | null>(null);
35       // Keep the latest onLoaded without making submit depend on it (a parent re-render
won't restart polls).
36       const onLoadedRef = useRef(onLoaded);
37       onLoadedRef.current = onLoaded;
38
39       const stop = useCallback(() => {
40         const run = runRef.current;
41         if (run) {
42           run.cancelled = true;
43           if (run.timer) clearTimeout(run.timer);
44         }
45         runRef.current = null;
46       }, []);
47
48       // Cancel any in-flight poll loop when the component unmounts (App.tsx:80-82 alive-
flag idiom).
49       useEffect(() => stop, [stop]);
50
51       const submit = useCallback(
52         (rawUri: string) => {
53           const uri = rawUri.trim();
54           if (!uri) {
55             setPhase({ status: "error", error: { code: "invalidUri" } });
56             return;
57           }
58
59           stop(); // supersede any prior run
60           const run: Run = { cancelled: false };
61           runRef.current = run;
62
63           const cid = newCorrelationId();
64           const startedAt = Date.now();
65           setPhase({ status: "submitting" });
66           log("info", "ingest.submit_start", { cid, uri });
67
68           const fail = (error: IngestError, event: string) => {
69             if (run.cancelled) return;
70             setPhase({ status: "error", error });
71             const level = error.code === "timeout" || error.code === "emptyResult" ? "warn"
: "error";
72             log(level, event, { cid, uri, code: error.code, detail: error.detail });
73           };
74
75           const poll = (jobId: string, delay: number) => {
76             run.timer = setTimeout(() => {
77               void (async () => {

```

```

78         if (run.cancelled) return;
79         let job: JobStatus;
80         try {
81             job = await getJob(jobId, cid);
82         } catch (err) {
83             fail(classifyError(err), "ingest.poll_error");
84             return;
85         }
86         if (run.cancelled) return;
87
88         if (job.status === "done") {
89             const videoId = job.video_id ?? job.result?.video_id ?? null;
90             // Two status layers: a `done` job with a `failed` pipeline verdict (or no
video_id) is
91             // a success-shaped FAILURE ? never load it, never hang.
92             if (job.result?.status === "failed" || !videoId) {
93                 fail({ code: "emptyResult..."
[/external_agent_tool_result]

```

158. assistant (2026-07-02T06:11:33.479Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\web\src\lib\errors.test.ts
[/external_agent_tool_call]

```

159. assistant (2026-07-02T06:11:33.479Z)

```

[external_agent_tool_result]
1     import { describe, it, expect } from "vitest";
2     import { parseDetail, classifyError } from "../errors";
3     import { ApiError } from "../api/client";
4
5     describe("parseDetail", () => {
6         it("extracts FastAPI's {detail}", () => {
7             expect(parseDetail(JSON.stringify({ detail: "unsupported storage URI scheme"
}))).toBe(
8                 "unsupported storage URI scheme",
9             );
10        });
11        it("returns the raw body when it isn't JSON or has no string detail", () => {
12            expect(parseDetail("plain text")).toBe("plain text");
13            expect(parseDetail(JSON.stringify({ other: 1 }))).toBe(JSON.stringify({ other: 1
})));
14            expect(parseDetail(JSON.stringify({ detail: { nested: true }
}))).toBe(JSON.stringify({ detail: { nested: true } }));
15        });
16    });
17
18    describe("classifyError", () => {
19        it("a network throw (ApiError with no status) ? networkError", () => {
20            expect(classifyError(new ApiError("net", { cid: "c" }))).toEqual({ code:
"networkError" });
21        });
22        it("an HTTP error with a body ? engine + the FastAPI detail", () => {
23            const err = new ApiError("x", { status: 400, body: JSON.stringify({ detail:
"unsupported storage URI scheme" }) });
24            expect(classifyError(err)).toEqual({ code: "engine", detail: "unsupported storage
URI scheme" });
25        });
26        it("a bare 5xx with no body ? networkError (friendly 'is the engine running?', not
'HTTP 500')", () => {
27            expect(classifyError(new ApiError("x", { status: 500 }))).toEqual({ code:
"networkError" });
28            expect(classifyError(new ApiError("x", { status: 502 }))).toEqual({ code:
"networkError" });

```

```

29     });
30     it("a 5xx WITH a detail still surfaces the engine's message", () => {
31         const err = new ApiError("x", { status: 500, body: JSON.stringify({ detail: "failed
to persist job record" }) });
32         expect(classifyError(err)).toEqual({ code: "engine", detail: "failed to persist job
record" });
33     });
34     it("a 4xx with no body keeps the status (a client-side mistake, not unreachable)", ()
=> {
35         expect(classifyError(new ApiError("x", { status: 404 }))).toEqual({ code: "engine",
detail: "HTTP 404" });
36     });
37     it("a non-ApiError ? engine + the stringified error", () => {
38         expect(classifyError(new Error("boom"))).toEqual({ code: "engine", detail: "Error:
boom" });
39     });
40     });
41
[/external_agent_tool_result]

```

160. assistant (2026-07-02T06:11:33.479Z)

A5 is now well-scoped (build catch reuses `classifyError`; shared `errors.\*` namespace via a small `errorText(t, e)` helper; unify IngestPanel onto it). Let me read the rest of the IngestPanel test (so the A5 migration keeps it green) while the A4 review runs.

161. assistant (2026-07-02T06:11:33.479Z)

```

[/external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.test.tsx
[/external_agent_tool_call]

```

162. assistant (2026-07-02T06:11:33.479Z)

```

[/external_agent_tool_result]
40     vi.unstubAllGlobals();
41     vi.useRealTimers();
42     });
43
44     const typeUri = (value: string) =>
45         fireEvent.change(screen.getByLabelText(/video location/i), { target: { value } });
46     const clickProcess = () => fireEvent.click(screen.getByRole("button", { name: /process
video/i }));
47
48     describe("IngestPanel", () => {
49         it("submits a URI, polls, and hands the video_id to onLoaded", async () => {
50             vi.useFakeTimers();
51             vi.stubGlobal(
52                 "fetch",
53                 fetchSeq(httpOk(202, { job_id: "j", status: "queued", poll: "/api/jobs/j" }), [
54                     { job_id: "j", status: "running", progress: "segmenting" },
55                     { job_id: "j", status: "done", video_id: "vidZ", result: { status: "success",
video_id: "vidZ" } } ],
56                 ),
57             );
58             const onLoaded = vi.fn();
59             render(<IngestPanel onLoaded={onLoaded} />);
60             typeUri("gs://bucket/match.mp4");
61             clickProcess();
62
63             await act(async () => {
64                 await vi.advanceTimersByTimeAsync(0); // flush submit ? start polling
65             });
66             await act(async () => {

```

```

67         await vi.advanceTimersByTimeAsync(1000); // running tick
68     });
69     await act(async () => {
70         await vi.advanceTimersByTimeAsync(1500); // done tick
71     });
72     expect(onLoaded).toHaveBeenCalled("vidZ");
73 });
74
75     it("shows the engine's own 400 detail (e.g. an unsupported scheme)", async () => {
76         vi.stubGlobal("fetch", fetchSeq(httpOk(400, { detail: "unsupported storage URI
scheme: ftp://" }), []));
77         render(<IngestPanel onLoaded={vi.fn()} />);
78         typeUri("ftp://nope");
79         clickProcess();
80         expect(await screen.findByText(/unsupported storage URI scheme/i)).toBeTruthy();
81     });
82
83     it("shows a friendly, localized message when the engine is unreachable", async () => {
84         vi.stubGlobal("fetch", vi.fn(async () => {
85             throw new Error("offline");
86         }) as unknown as FetchFn);
87         render(<IngestPanel onLoaded={vi.fn()} />);
88         typeUri("gs://bucket/match.mp4");
89         clickProcess();
90         expect(await screen.findByText(/can't reach khelsutra/i)).toBeTruthy();
91     });
92
93     it("rejects an empty URI with a hint and never calls the engine", () => {
94         const f = vi.fn();
95         vi.stubGlobal("fetch", f);
96         render(<IngestPanel onLoaded={vi.fn()} />);
97         clickProcess();
98         expect(screen.getByTestId("ingest-error")).toBeTruthy();
99         expect(f).not.toHaveBeenCalled();
100     });
101 });
102
[/external_agent_tool_result]

```

163. assistant (2026-07-02T06:11:33.479Z)

A5 migration is safe (tests assert rendered text, which I'll preserve). Now let me prep T4 (the keystone) by reading the engine's chunked-upload endpoints ? understanding the contract the SPA must call. This is read-only (I won't disturb the engine checkout, which a concurrent session owns).

164. assistant (2026-07-02T06:11:33.479Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\api\uploads.py
[/external_agent_tool_call]

```

165. assistant (2026-07-02T06:11:33.479Z)

```

[external_agent_tool_result]
1     """"Chunked upload (B2, PR-2) ? extracted from backend.main into its own router.
2
3     Browser uploads arrive in sequential parts ? security.max_part_mb (free-tier edge
4     proxies cap request bodies at ~100 MB ? HOSTING_PLAN $2), are assembled server-side
5     under security.upload_dir, then submitted to /api/process by the returned path.
6     Upload state is in-process by design (this is the single-box alpha): a server restart
7     aborts partial uploads and the client re-inits. Per-user partitioning of upload_dir
8     arrives with PR-3 identity scoping.
9

```

```

10 The startup config lives on ``backend.main.global_config`` (set ONLY in ``main()``).
11 ``_upload_cfg`` resolves it through the ``backend.main`` module object so it always
12 reads the live value the server set at startup (and so tests that monkeypatch
13 ``backend.main.global_config`` flow through unchanged).
14 """
15 import os
16 import threading
17 import uuid
18 from typing import Any, Dict, Optional
19
20 from fastapi import APIRouter, HTTPException, Request
21 from pydantic import BaseModel
22
23 from backend.utils.hashing import compute_video_id # canonical file-identity hashing
24 from backend.utils.paths import safe_output_filename
25
26 router = APIRouter()
27
28 _uploads: Dict[str, Dict[str, Any]] = {}
29 _uploads_lock = threading.Lock()
30
31
32 class UploadInitRequest(BaseModel):
33     filename: str
34     total_size_bytes: Optional[int] = None
35
36
37 class UploadCompleteRequest(BaseModel):
38     total_parts: int
39
40
41 def _upload_cfg():
42     # Resolved lazily (import inside the function) to read the LIVE startup config that
43     # main() sets on the module global, and to avoid a circular import at module load
44     # (main.py imports this router; this module must not import main at import time).
45     import backend.main as _main
46     sec = getattr(_main.global_config, "security", None)
47     upload_dir = getattr(sec, "upload_dir", None) or "output/uploads"
48     max_bytes = int(getattr(sec, "max_upload_mb", 4096) or 4096) * 1024 * 1024
49     part_bytes = int(getattr(sec, "max_part_mb", 95) or 95) * 1024 * 1024
50     exts = [e.lower().lstrip(".") for e in (getattr(sec, "allowed_upload_extensions",
51 None) or ["mp4", "mov"])]
52     return upload_dir, max_bytes, part_bytes, exts
53
54
55 def _abort_upload(upload_id: str) -> None:
56     with _uploads_lock:
57         st = _uploads.pop(upload_id, None)
58         if st:
59             try:
60                 os.remove(st["tmp_path"])
61             except OSError:
62                 pass
63
64
65 @router.post("/api/upload/init")
66 def upload_init(req: UploadInitRequest):
67     """Start a chunked upload. Returns upload_id; PUT sequential parts next."""
68     upload_dir, max_bytes, part_bytes, exts = _upload_cfg()
69     try:
70         name = safe_output_filename(req.filename)
71     except ValueError as e:
72         raise HTTPException(status_code=400, detail=str(e)) from e
73     ext = os.path.splitext(name)[1].lower().lstrip(".")
74     if ext not in exts:

```

```

74         raise HTTPException(status_code=400,
75                               detail=f"Extension '{ext}' not allowed. Allowed:
{sorted(exts)}")
76         if req.total_size_bytes is not None and req.total_size_bytes > max_bytes:
77             raise HTTPException(status_code=413,
78                                   detail=f"File exceeds the {max_bytes // (1024 * 1024)} MB
upload limit.")
79         os.makedirs(upload_dir, exist_ok=True)
80         upload_id = uuid.uuid4().hex
81         tmp_path = os.path.join(upload_dir, f"partial-{upload_id}")
82         open(tmp_path, "wb").close()
83         with _uploads_lock:
84             _uploads[upload_id] = {"filename": name, "tmp_path": tmp_path, "parts": 0,
"bytes": 0}
85         return {"upload_id": upload_id, "max_part_mb": part_bytes // (1024 * 1024),
86               "next_part": f"/api/upload/{upload_id}/part/0"}
87
88
89     @router.put("/api/upload/{upload_id}/part/{index}")
90     async def upload_part(upload_id: str, index: int, request: ...
[/external_agent_tool_result]

```

166. assistant (2026-07-02T06:11:33.479Z)

T4 upload contract is now clear (init ? PUT sequential parts ? complete returns a server `path` ? feed to `/api/process`). Solid for later.

Let me read the `site/` negotiation + test harness for B3 prep while the A4 review finishes.

167. assistant (2026-07-02T06:11:33.479Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\site\server.js
[/external_agent_tool_call]

```

168. assistant (2026-07-02T06:11:33.480Z)

```

[external_agent_tool_result]
1      // Always-on VM host for the KhelSutra site ? dependency-free (Node 22+ only).
2      // Serves public/ and POST /api/request-access into SQLite (node:sqlite), using the
3      // SAME handler logic as the Cloudflare Worker (src/index.js). This is the portability
4      // fallback "if Cloudflare acts out" ? see docs/PORTABILITY.md.
5      //
6      //   PORT=8080 DB_PATH=/var/lib/khelsutra/access.db node server.js
7      //
8      import { createServer } from "node:http";
9      import { readFile } from "node:fs/promises";
10     import { join, extname, normalize } from "node:path";
11     import { DatabaseSync } from "node:sqlite";
12     import { handleRequest } from "./src/index.js";
13     import { sqliteStore } from "./src/store.js";
14     import { noopNotify } from "./src/notify.js";
15
16     const PORT = Number(process.env.PORT) || 8080;
17     const ROOT = import.meta.dirname;
18     const PUBLIC = join(ROOT, "public");
19     const DB_PATH = process.env.DB_PATH || join(ROOT, "access.db");
20
21     const db = new DatabaseSync(DB_PATH);
22     db.exec(await readFile(join(ROOT, "schema.sql"), "utf8")); // CREATE TABLE IF NOT EXISTS
? idempotent
23     const store = sqliteStore(db);
24     const notify = noopNotify(); // VM host: no email yet (add SMTP later ? see
docs/PORTABILITY.md)

```



```

25
26     const TYPES = {
27         ".html": "text/html; charset=utf-8", ".png": "image/png", ".svg": "image/svg+xml",
28         ".css": "text/css", ".js": "text/javascript", ".ico": "image/x-icon", ".txt":
"text/plain",
29         ".mp4": "video/mp4", ".webm": "video/webm", ".gif": "image/gif", ".jpg": "image/jpeg",
".jpeg": "image/jpeg",
30     };
31
32     async function toRequest(req) {
33         const url = `http://${req.headers.host || "localhost"}${req.url}`;
34         let body;
35         if (req.method !== "GET" && req.method !== "HEAD") {
36             const chunks = [];
37             for await (const c of req) chunks.push(c);
38             body = Buffer.concat(chunks);
39         }
40         return new Request(url, { method: req.method, headers: req.headers, body });
41     }
42
43     async function send(res, response) {
44         res.statusCode = response.status;
45         for (const [k, v] of response.headers) res.setHeader(k, v);
46         res.end(Buffer.from(await response.arrayBuffer()));
47     }
48
49     async function serveStatic(pathname, res) {
50         const rel = pathname === "/" ? "index.html"
51             : pathname === "/capture" ? "capture.html"
52             : pathname === "/own-model" ? "own-model.html"
53             : pathname === "/flyer" ? "flyer.html"
54             : pathname === "/flyer-kn" ? "flyer-kn.html"
55             : pathname === "/flyer-hi" ? "flyer-hi.html"
56             : pathname.replace(/^\/+/, "");
57         const file = normalize(join(PUBLIC, rel));
58         if (!file.startsWith(PUBLIC)) { res.statusCode = 403; return res.end("forbidden"); }
59         try {
60             const data = await readFile(file);
61             res.setHeader("content-type", TYPES[extname(file)] || "application/octet-stream");
62             res.end(data);
63         } catch {
64             res.statusCode = 404;
65             res.end("not found");
66         }
67     }
68
69     createServer(async (req, res) => {
70         try {
71             const pathname = decodeURIComponent(new URL(`http://x${req.url}`).pathname);
72             if (pathname === "/api/request-access") {
73                 if (req.method !== "POST") { res.statusCode = 405; return
res.end('{"ok":false,"error":"method not allowed"}'); }
74                 return await send(res, await handleRequest(await toRequest(req), store, notify));
75             }
76             return await serveStatic(pathname, res);
77         } catch (err) {
78             res.statusCode = 500;
79             res.end('{"ok":false,"error":"server error"}');
80             console.error(err);
81         }
82     }).listen(PORT, () => console.log(`khelsutra site listening on :${PORT} (sqlite:
${DB_PATH})`));
83
[/external_agent_tool_result]

```

169. assistant (2026-07-02T06:11:33.480Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\site\src\assets.test.js
[/external_agent_tool_call]
```

170. assistant (2026-07-02T06:11:33.480Z)

```
[external_agent_tool_result]
1      // Static-site integrity ? the landing page + capture checklist + OG image ARE the
public
2      // product. server.test.js proves the VM host *serves* "/", "/capture", "/og.png"; this
3      // guards the CONTENT contract so a broken edit (deleted asset, broken form wiring, dead
4      // capture link, missing/!absolute OG image) fails CI instead of shipping to the live
site.
5      import { test } from "node:test";
6      import assert from "node:assert/strict";
7      import { readFileSync, statSync } from "node:fs";
8      import { join } from "node:path";
9
10     const PUBLIC = join(import.meta.dirname, "..", "public");
11     const read = (f) => readFileSync(join(PUBLIC, f), "utf8");
12     const size = (f) => statSync(join(PUBLIC, f)).size;
13
14     test("core public assets exist and are non-empty", () => {
15       for (const f of ["index.html", "capture.html", "og.png"]) {
16         assert.ok(size(f) > 0, `public/${f} is missing or empty`);
17       }
18     });
19
20     test("landing page wires the Request-Access form to the API", () => {
21       const html = read("index.html");
22       assert.match(html, /id="af"/, "the #af form is gone");
23       assert.match(html, /\api\/request-access/, "form no longer posts to /api/request-
access");
24       assert.match(html, /name="company"[^>]*tabindex="-1"/, "honeypot field removed (bots
will get through)");
25       for (const field of ["segment", "has_gpu", "storage"]) {
26         assert.match(html, new RegExp(`name="${field}"`), `form lost the '${field}' field (a
demand signal)`);
27       }
28     });
29
30     test("submitting the form shows an obvious in-flight state (no silent freeze)", () => {
31       const html = read("index.html");
32       assert.match(html, /Sending?/, "submit should swap to a 'Sending?' state while the
request is in flight");
33       assert.match(html, /aria-busy/, "submit should set aria-busy for assistive tech during
the request");
34       assert.match(html, /class="spin"/, "the in-flight spinner is gone");
35       assert.match(html, /\spin\{/, "the spinner style is gone");
36       // the button must be RE-ENABLED on failure so the user isn't stuck on a frozen
'Sending?'
37       assert.match(html, /btn\.disabled\s*=\s*false/, "submit button isn't restored on the
failure path");
38     });
39
40     test("OG + canonical meta point at the live https origin", () => {
41       const html = read("index.html");
42       assert.match(html,
/property="og:image"\s+content="https:\/\/khelsutra.guru\/og.png"/, "og:image must be the
absolute live URL");
43       assert.match(html, /property="og:url"\s+content="https:\/\/khelsutra.guru"/, "og:url
drift");
44     });
```

```

45
46     test("the sample reel is GATED behind sign-up + starts at a RANDOM point (not shown
openly)", () => {
47         const html = read("index.html");
48         assert.match(html, /ajTCZ7pJW_o/, "the sample-reel video id is gone");
49         assert.match(html, /youtube-nocookie\.com/, "reel should use the privacy nocookie
host");
50         assert.match(html, /id="reel-reveal"/, "the on-signup reveal host is gone");
51         assert.match(html, /see it in action/i, "the 'see it in action' sign-up teaser is
gone");
52         assert.match(html, /reel-count/, "the auto-close countdown is gone");
53         assert.match(html, /REEL_SECS = 30/, "the reel preview should be ~30s");
54         assert.match(html, /getDuration\(\)/, "random start should derive from the real video
duration");
55         assert.match(html, /Math\.random\(\)/, "the start point should be randomized");
56         assert.match(html, /seekTo\(/, "the player should seek to the random start");
57         // Must NOT be an openly-rendered public section ? it only appears after a successful
submit.
58         assert.doesNotMatch(html, /<section id="reel"/, "the reel must be sign-up-gated, not
an open public section");
59         assert.doesNotMatch(html, /id="reel-play"/, "the open click-to-play facade should be
removed");
60     });
61
62     test("the TOOL screencast is self-hosted, footage-free, and revealed only after sign-
up", () => {
63         const html = read("index.html");
64         assert.match(html, /\video\tool-cuj\.mp4/, "the tool-...
[/external_agent_tool_result]

```

171. assistant (2026-07-02T06:11:33.480Z)

I now understand the `site/` surface for B3 (vanilla, dependency-free, clean routes). Good prep banked.

The A4 review should be near done. Let me check its status.

172. assistant (2026-07-02T06:11:33.480Z)

```

[external_agent_tool_call: ToolSearch]
input: {"query":"select:TaskOutput,TaskGet","max_results":3}
[/external_agent_tool_call]

```

173. assistant (2026-07-02T06:11:33.480Z)

```

[external_agent_tool_result]
[/external_agent_tool_result]

```

174. assistant (2026-07-02T06:11:33.480Z)

```

[external_agent_tool_call: TaskOutput]
input: {"task_id":"wqh6khdbw","block":true,"timeout":180000}
[/external_agent_tool_call]

```

175. assistant (2026-07-02T06:11:33.480Z)

```

[external_agent_tool_result]
<retrieval_status>success</retrieval_status>

<task_id>wqh6khdbw</task_id>

<task_type>local_workflow</task_type>

<status>completed</status>

```

```

<output>
{
  "summary": "Adversarial 3-lens review of the A4 mobile-touch-targets diff",
  "agentCount": 3,
  "logs": [],
  "result": [
    {
      "lens": "touch-completeness",
      "summary": "All interactive controls meet the WCAG 2.5.5 44px floor; no violations
found.",
      "findings": [
        {
          "severity": "nit",
          "file": "web/src/components/RallyGrid.tsx",
          "title": "All-clear",
          "detail": "No touch-target violations. Every control uses min-h-11 or h-11/w-11;
RallyGrid card is p-3 plus multi-line content over 44px."
        }
      ]
    },
    {
      "lens": "correctness",
      "summary": "Reviewed the feat/a4-mobile-touch-targets diff (8 changed files under web/src
+ a new test). The change is a clean WCAG 2.5.5 touch-target pass: it bumps primary controls to
a 44px floor via Tailwind min-h-11/h-11, makes the QueryBar \"not available yet\" reason visible
as DOM text (not hover-only title), and adds the query.notAvailableHelp key. I verified all five
requested concerns and ran the gate locally.\n\nVERIFIED LOCALLY (web/ package):\n- typecheck:
`npm run typecheck` (tsc --noEmit) ? clean, no errors.\n- tests: `npm run test` (vitest) ? 12
files, 127 tests, all green, including the new components/a4.touch.test.tsx (4 tests), the i18n
key-parity hygiene test, and the no-bare-strings (BARE_ATTR) test. The error log lines in output
are expected (the suite stubs fetch to reject and asserts the config-unavailable fallback).\n-
No lint step exists in this package (no eslint/biome config, no lint script); CI relies on
typecheck + vitest, both green.\n\nConcern-by-concern:\n1. SelectionFooter (lib 53 row is flex
flex-wrap items-center gap-3): min-h-11 on the reel-name input (line 66) and Build button (line
74) only raises min height; items-center keeps them centered. No alignment/wrap/sticky-footer
regression ? the fixed footer container is untouched.\n2. CompileResult header (line 31 parent
is flex items-start justify-between gap-3): the dismiss button is now h-11 w-11 with internal
items-center justify-center. With items-start the button's TOP still aligns to the h2 top; the
larger 44px tap target extends downward and the glyph centers within it. This is a minor,
intentional cosmetic shift (standard top-aligned close-button-as-tap-target pattern), not a
break ? header does not wrap or overflow. Same applies to the download anchor at line 58
(inline-flex min-h-11 items-center is correct for a link turned into a tap target).\n3.
query.notAvailableHelp is present in ALL THREE catalogs (en/kn/hi) at the same path under
\"query\", so key-parity stays green (confirmed by the passing hygiene test). The string
contains no ICU placeholders ({token}/{kind}), so no i18next-icu interpolation risk.\n4. no-
bare-strings: ran the test's exact BARE_ATTR regex over all 7 watched files ? zero matches.
notAvailableHelp is rendered as {t(...)} in a <p> body, not an attribute. No new bare attribute
introduced.\n5. TypeScript/React: clean. The notAvailableHelp <p> is correctly gated on
parsed.unavailable.length > 0, mutually exclusive with the unreadable line (gated on length ===
0), so no double-messaging.\n\nNo correctness or regression issues found.",
      "findings": []
    },
    {
      "lens": "faithfulness",
      "findings": [
        {
          "severity": "minor",
          "file": "C:/Users/avidu/Projects/khelsutra/web/src/components/RallyGrid.tsx",
          "location": "lines 60-69 shot-count chip; grid.shotsConfidence in common.json line
45",
          "title": "Per-card shot-count confidence is hover-only",
          "detail": "The shot chip title at line 62 is the only place per-card
shot_count_confidence renders, and title= never surfaces on touch devices. Everything else in
the title is already visible elsewhere so is not lost on phones: experim...
[/external_agent_tool_result]

```

176. assistant (2026-07-02T06:11:33.480Z)

The 3-lens review came back essentially clean:

- **touch-completeness**: no violations ? all controls meet 44px.
- **correctness**: no issues; independently re-ran typecheck + 127 tests + the key-parity/no-bare-strings guards, all green.
- **faithfulness**: one **minor** (not blocker) ? the per-card shot-count **confidence** is the one datum still hover-only (in `grid.shotsTitle`), invisible on phones. The reviewer's suggested fix: surface it in the visible chip.

That's a legitimate, cheap win that fully honors A4's "tooltip-text ? visible." Let me apply it ? append the confidence to the visible chip when present (keeping the title as redundant desktop detail).

177. assistant (2026-07-02T06:11:33.480Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\RallyGrid.tsx
[/external_agent_tool_call]
```

178. assistant (2026-07-02T06:11:33.480Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\RallyGrid.tsx has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

179. assistant (2026-07-02T06:11:33.480Z)

Now add a coverage assertion for the visible confidence to the A4 test:

180. assistant (2026-07-02T06:11:33.480Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\A4.touch.test.tsx
[/external_agent_tool_call]
```

181. assistant (2026-07-02T06:11:33.480Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\A4.touch.test.tsx has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

182. assistant (2026-07-02T06:11:33.480Z)

Now re-run the suite + typecheck + build, and reload the preview to confirm the visible confidence.

183. assistant (2026-07-02T06:11:33.481Z)

```
[external_agent_tool_call: Bash]
description: Re-run tests, typecheck, build after RallyGrid change
command: cd /c/Users/avidu/Projects/khelsutra/web && npm test --silent 2>&1 | grep -E "Test
Files|Tests|FAIL|?" | head -20 && echo "=== tsc ===" && npx tsc --noEmit && echo "tsc OK" &&
echo "=== build ===" && npm run build --silent 2>&1 | tail -3
[/external_agent_tool_call]
```

184. assistant (2026-07-02T06:11:33.481Z)

```
[external_agent_tool_result]
Test Files 12 passed (12)
Tests 128 passed (128)
```

```
=== tsc ===
tsc OK
=== build ===
dist/assets/index-CUYZ8r6F.css
  20.83 kB ? gzip:  4.84 kB
dist/assets/index-BBxB3my1.js
347.72 kB ? gzip: 109.60 kB
? built in 1.21s
[/external_agent_tool_result]
```

185. assistant (2026-07-02T06:11:33.481Z)

128 tests pass (+1 confidence test), tsc clean, build green. The A4 test pins the visible `confidence: High` text. Let me grab a quick desktop screenshot as visual proof of the confidence on the chip, then commit.

186. assistant (2026-07-02T06:11:33.481Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_resize]
input: {"serverId":"d4a4a2a4-7c2e-46e2-889e-f619477293b4","preset":"desktop"}
[/external_agent_tool_call]
```

187. assistant (2026-07-02T06:11:33.481Z)

```
[external_agent_tool_result]
Viewport reset to native size (desktop).
[/external_agent_tool_result]
```

188. assistant (2026-07-02T06:11:33.481Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_screenshot]
input: {"serverId":"d4a4a2a4-7c2e-46e2-889e-f619477293b4"}
[/external_agent_tool_call]
```

189. assistant (2026-07-02T06:11:33.481Z)

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

190. assistant (2026-07-02T06:11:33.481Z)

Confirmed: Rally cards now show `~9 shots · experimental ? confidence: High` (and Medium on Rally 3) visibly ? no longer hover-only. The full A4 change is verified (touch targets, visible help line, visible confidence).

Let me commit A4 (re-verifying branch/HEAD and staging explicit paths per shared-checkout discipline).

191. assistant (2026-07-02T06:11:33.481Z)

```
[external_agent_tool_call: Bash]
description: Verify branch/HEAD and stage A4 paths
command: cd /c/Users/avidu/Projects/khelsutra && echo "branch: $(git branch --show-current)" &&
echo "HEAD (mine): $(git log --oneline -1)" && echo "--- staging A4 paths ---" && git add
web/src/App.tsx web/src/components/CompileResult.tsx web/src/components/IngestPanel.tsx
web/src/components/LanguageSwitcher.tsx web/src/components/QueryBar.tsx
web/src/components/RallyGrid.tsx web/src/components/SelectionFooter.tsx
web/src/components/a4.touch.test.tsx web/src/i18n/locales/en/common.json
web/src/i18n/locales/hi/common.json web/src/i18n/locales/kn/common.json && echo "--- staged ---"
&& git diff --cached --name-only
[/external_agent_tool_call]
```

192. assistant (2026-07-02T06:11:33.481Z)

```
[external_agent_tool_result]
branch: feat/a4-mobile-touch-targets
HEAD (mine): cdcfcfd feat(web): A1 ? plain-language copy pass on the en catalog (de-jargon)
(#60)
--- staging A4 paths ---
--- staged ---
web/src/App.tsx
web/src/components/CompileResult.tsx
web/src/components/IngestPanel.tsx
web/src/components/LanguageSwitcher.tsx
web/src/components/QueryBar.tsx
web/src/components/RallyGrid.tsx
web/src/components/SelectionFooter.tsx
web/src/components/a4.touch.test.tsx
web/src/i18n/locales/en/common.json
web/src/i18n/locales/hi/common.json
web/src/i18n/locales/kn/common.json
[/external_agent_tool_result]
```

193. assistant (2026-07-02T06:11:33.481Z)

Staged exactly the 11 A4 paths (not the untracked `site/access.db` or `.claude/`). HEAD is the clean A1 base. Committing:

194. assistant (2026-07-02T06:11:33.481Z)

```
[external_agent_tool_call: Bash]
description: Commit A4
command: git commit -q -F - <<'EOF'
feat(web): A4 ? mobile touch targets (?44px) + visible tooltip text
```

Non-tech-savvy / phone-first UX pass on the Studio SPA:

- ?44px tap targets (Tailwind min-h-11 / h-11) on every primary control: query input+Apply, bulk buttons, sample-query chips, reel-name input, Build, the language switcher, the ingest input+submit+cancel+retry, and the compile dismiss (44px square) + download link.
- Enlarge the tiny parsed-pill + sample chips (text-xs ? text-sm) for legibility.
- Surface hover-only title= info as VISIBLE text (a phone never shows title=): a "not available yet" reason line in QueryBar (new query.notAvailableHelp, en/kn/hi) and per-card shot-count confidence inline on the RallyGrid chip.

Adds a4.touch.test.tsx pinning the 44px floor + the visible help/confidence. 128 web tests green; tsc + build clean; key-parity + no-bare-strings hold. Reviewed via a 3-lens adversarial pass (touch-completeness / correctness / faithfulness); the confidence-visibility fix folds the one minor finding.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>

EOF

```
echo "--- committed ---" && git log --oneline -1 && echo "--- vs origin/master (must be ONLY my
commit) ---" && git log --oneline origin/master..HEAD
[/external_agent_tool_call]
```

195. assistant (2026-07-02T06:11:33.481Z)

```
[external_agent_tool_result]
--- committed ---
981e254 feat(web): A4 ? mobile touch targets (?44px) + visible tooltip text
--- vs origin/master (must be ONLY my commit) ---
981e254 feat(web): A4 ? mobile touch targets (?44px) + visible tooltip text
[/external_agent_tool_result]
```

196. assistant (2026-07-02T06:11:33.481Z)

Clean ? only my commit on the branch. Push and open the PR:

197. assistant (2026-07-02T06:11:33.481Z)

```
[external_agent_tool_call: Bash]
description: Push A4 branch
command: git push -u origin feat/a4-mobile-touch-targets 2>&1 | tail -5
[/external_agent_tool_call]
```

198. assistant (2026-07-02T06:11:33.481Z)

```
[external_agent_tool_result]
remote:      https://github.com/avidullu/khelsutra/pull/new/feat/a4-mobile-touch-targets
remote:
branch 'feat/a4-mobile-touch-targets' set up to track 'origin/feat/a4-mobile-touch-targets'.
To https://github.com/avidullu/khelsutra.git
 * [new branch]      feat/a4-mobile-touch-targets -> feat/a4-mobile-touch-targets
[/external_agent_tool_result]
```

199. assistant (2026-07-02T06:11:33.481Z)

```
[external_agent_tool_call: Bash]
description: Create A4 PR
command: gh pr create --title "feat(web): A4 ? mobile touch targets (?44px) + visible tooltip
text" --base master --head feat/a4-mobile-touch-targets --body "$(cat <<'EOF'
```

A4 ? non-tech-savvy / phone-first UX (tracker `FRIENDLY\_AND\_MULTILINGUAL.md` §7)

The Studio SPA was desktop-shaped: tiny tap targets and explanations hidden behind hover-only `title=` (a phone never shows `title=`). This makes it usable on a phone, for the Bangalore coach/parent audience.

What changed

- **44px tap targets** (WCAG 2.5.5 / platform HIG) via Tailwind `min-h-11` / `h-11` on every primary control: the query input + Apply, the bulk buttons, the **sample-query chips**, the reel-name input + Build, the **language switcher**, the ingest input + submit + cancel + retry, and the compile **dismiss** (44px square) + **download** link.
- **Legibility**: the tiny parsed-pill + sample chips go `text-xs ? text-sm`.
- **Tooltip-text ? visible** (the A4 promise): the **"not available yet"** reason now renders as a visible line in the QueryBar (new `query.notAvailableHelp`, in en/kn/hi), and per-card **shot-count confidence** is shown inline on the RallyGrid chip instead of only in the hover title.

Verification

- **128 web tests** green (new `a4.touch.test.tsx` pins the 44px floor + the visible help line + visible confidence); `tsc --noEmit` clean; `vite build` clean; **key-parity** + **no-bare-strings** guards hold.
- **Browser-verified** at mobile (375px) and desktop: enlarged controls, the visible **"Shot type, players and score aren't available yet?"** line, and `~N shots · experimental ? confidence: High` on the cards.
- **Reviewed by a 3-lens adversarial workflow** (touch-completeness / correctness / faithfulness): touch-completeness clean, correctness clean (re-ran the gate independently), and the one minor finding (per-card confidence was hover-only) is folded into this PR.

Honesty notes

- kn/hi `notAvailableHelp` is AI-drafted (native review pending, consistent with the existing kn/hi catalogs).
- RallyGrid's `title=` is kept as redundant desktop hover detail; all essential info is now visible.

```
? Generated with [Claude Code](https://claude.com/claude-code)
EOF
)" 2>&1 | tail -3
[/external_agent_tool_call]
```

200. assistant (2026-07-02T06:11:33.481Z)


```
[external_agent_tool_result]
https://github.com/avidullu/khelsutra/pull/61
[/external_agent_tool_result]
```

201. assistant (2026-07-02T06:11:33.481Z)

PR [#61](https://github.com/avidullu/khelsutra/pull/61) created. Now update the \$7 tracker to mark A4 ? (and fix A1's stale ? ? it merged in #60), plus a changelog note, on the same branch.

202. assistant (2026-07-02T06:11:33.482Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\docs\FRIENDLY_AND_MULTILINGUAL.md
[/external_agent_tool_call]
```

203. assistant (2026-07-02T06:11:33.482Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\docs\FRIENDLY_AND_MULTILINGUAL.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

204. assistant (2026-07-02T06:11:33.482Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\docs\FRIENDLY_AND_MULTILINGUAL.md
[/external_agent_tool_call]
```

205. assistant (2026-07-02T06:11:33.482Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\docs\FRIENDLY_AND_MULTILINGUAL.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

206. assistant (2026-07-02T06:11:33.482Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\docs\FRIENDLY_AND_MULTILINGUAL.md
[/external_agent_tool_call]
```

207. assistant (2026-07-02T06:11:33.482Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\docs\FRIENDLY_AND_MULTILINGUAL.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

208. assistant (2026-07-02T06:11:33.482Z)

```
[external_agent_tool_call: Bash]
description: Commit + push tracker update
command: git add docs/FRIENDLY_AND_MULTILINGUAL.md && git commit -q -F - <<'EOF'
docs(a4): mark A1 (#60) + A4 (#61) done in the $7 tracker + changelog
```

A1 was merged in #60 but the tracker still showed it ? ? corrected to ?.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>

EOF

```
git push -q 2>&1 | tail -2 && echo "pushed" && git log --oneline origin/master..HEAD
[/external_agent_tool_call]
```

209. assistant (2026-07-02T06:11:33.482Z)

```
[external_agent_tool_result]
pushed
5dfb91f docs(a4): mark A1 (#60) + A4 (#61) done in the S7 tracker + changelog
981e254 feat(web): A4 ? mobile touch targets (?44px) + visible tooltip text
[/external_agent_tool_result]
```

210. assistant (2026-07-02T06:11:33.482Z)

Both commits on the branch are mine. Now let me check CI on PR #61 before merging.

211. assistant (2026-07-02T06:11:33.482Z)

```
[external_agent_tool_call: Bash]
description: Check PR 61 CI status
command: sleep 20 && gh pr checks 61 2>&1 | head -20
[/external_agent_tool_call]
```

212. assistant (2026-07-02T06:11:33.482Z)

```
[external_agent_tool_result]
Workers Builds: hidden-king-khelsutra pass 0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/8e9fd510-94c9-4340-8f6d-629b0a654473
site pass 11s
https://github.com/avidullu/khelsutra/actions/runs/27908339306/job/82580947234
web pending 0
https://github.com/avidullu/khelsutra/actions/runs/27908339306/job/82580947225
[/external_agent_tool_result]
```

213. assistant (2026-07-02T06:11:33.482Z)

site + CF build pass; web is still running. Let me wait for it.

214. assistant (2026-07-02T06:11:33.482Z)

```
[external_agent_tool_call: Bash]
description: Poll web CI until done
command: for i in 1 2 3 4 5 6; do s=$(gh pr checks 61 2>&1 | grep -E "^web\b" | awk '{print $2}'); echo "poll $i: web=$s"; if [ "$s" = "pass" ] || [ "$s" = "fail" ]; then break; fi; sleep 15; done && echo "=== final ===" && gh pr checks 61 2>&1 | head
[/external_agent_tool_call]
```

215. assistant (2026-07-02T06:11:33.482Z)

```
[external_agent_tool_result]
poll 1: web=pass
=== final ===
Workers Builds: hidden-king-khelsutra pass 0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/8e9fd510-94c9-4340-8f6d-629b0a654473
site pass 11s
https://github.com/avidullu/khelsutra/actions/runs/27908339306/job/82580947234
web pass 25s
https://github.com/avidullu/khelsutra/actions/runs/27908339306/job/82580947225
[/external_agent_tool_result]
```

216. assistant (2026-07-02T06:11:33.482Z)

All green (web, site, Cloudflare build). Merging A4 per my merge authorization for green UX PRs.

217. assistant (2026-07-02T06:11:33.482Z)

```
[external_agent_tool_call: Bash]
```

```
description: Squash-merge A4 and sync master
command: gh pr merge 61 --squash --delete-branch 2>&1 | tail -5 && echo "--- sync local master
---" && git checkout master -q && git pull --ff-only -q 2>&1 | tail -3 && git log --oneline -3
[/external_agent_tool_call]
```

218. assistant (2026-07-02T06:11:33.482Z)

```
[external_agent_tool_result]
web/src/i18n/locales/en/common.json      | 1 +
web/src/i18n/locales/hi/common.json      | 1 +
web/src/i18n/locales/kn/common.json      | 1 +
12 files changed, 116 insertions(+), 19 deletions(-)
create mode 100644 web/src/components/a4.touch.test.tsx
--- sync local master ---
95195a5 feat(web): A4 ? mobile touch targets (?44px) + visible tooltip text (#61)
ccdcfcd feat(web): A1 ? plain-language copy pass on the en catalog (de-jargon) (#60)
a20002f docs: propose Multilingual Expansion ? 7 new locales (th/id/da/es/zh-Hans/ja/ko) (#58)
[/external_agent_tool_result]
```

219. assistant (2026-07-02T06:11:33.482Z)

A4 merged (#61 ? `95195a5`) and local master synced. Marking A4 done and starting **A5 ? friendly engine-error mapping on the build path**.

220. assistant (2026-07-02T06:11:33.483Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"1","status":"completed"}
[/external_agent_tool_call]
```

221. assistant (2026-07-02T06:11:33.483Z)

```
[external_agent_tool_result]
Updated task #1 status
[/external_agent_tool_result]
```

222. assistant (2026-07-02T06:11:33.483Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"2","status":"in_progress"}
[/external_agent_tool_call]
```

223. assistant (2026-07-02T06:11:33.483Z)

```
[external_agent_tool_result]
Updated task #2 status
[/external_agent_tool_result]
```

224. assistant (2026-07-02T06:11:33.483Z)

```
[external_agent_tool_call: mcp_ccd_session__mark_chapter]
input: {"title":"A5 ? friendly build errors","summary":"A4 merged (#61). Now reusing
classifyError on the build path + a shared errors.* namespace."}
[/external_agent_tool_call]
```

225. assistant (2026-07-02T06:11:33.483Z)

```
[external_agent_tool_result]
Chapter marked: "A5 ? friendly build errors". Continue your current work.
[/external_agent_tool_result]
```

226. assistant (2026-07-02T06:11:33.483Z)

```
[external_agent_tool_call: Bash]
description: Cut A5 branch from origin/master
command: git fetch origin -q && git checkout -b feat/a5-friendly-build-errors origin/master 2>&1
| tail -2 && git log --oneline -1
[/external_agent_tool_call]
```

227. assistant (2026-07-02T06:11:33.483Z)

```
[external_agent_tool_result]
branch 'feat/a5-friendly-build-errors' set up to track 'origin/master'.
Switched to a new branch 'feat/a5-friendly-build-errors'
95195a5 feat(web): A4 ? mobile touch targets (?44px) + visible tooltip text (#61)
[/external_agent_tool_result]
```

228. assistant (2026-07-02T06:11:33.483Z)

A5 branch cut (includes A4). Now implementing A5: a shared `errors.\*` namespace + an `errorText(t, e)` helper, wiring `classifyError` into the build path and unifying IngestPanel onto it.

First, the catalogs ? move the ingest code-messages into a shared `errors` namespace (+ a new `engineGeneric` fallback). en:

229. assistant (2026-07-02T06:11:33.483Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json
[/external_agent_tool_call]
```

230. assistant (2026-07-02T06:11:33.483Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

231. assistant (2026-07-02T06:11:33.483Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json
[/external_agent_tool_call]
```

232. assistant (2026-07-02T06:11:33.483Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

233. assistant (2026-07-02T06:11:33.483Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\hi\common.json
[/external_agent_tool_call]
```

234. assistant (2026-07-02T06:11:33.483Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\hi\common.json has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

235. assistant (2026-07-02T06:11:33.483Z)

Now the `errorText` helper:

236. assistant (2026-07-02T06:11:33.483Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\khelsutra\web\src\lib\errorText.ts
[/external_agent_tool_call]
```

237. assistant (2026-07-02T06:11:33.483Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\khelsutra\web\src\lib\errorText.ts (file
state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

238. assistant (2026-07-02T06:11:33.483Z)

Now wire the build path in App.tsx ? swap `parseDetail` for `classifyError` + `errorText`:

239. assistant (2026-07-02T06:11:33.483Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\App.tsx
[/external_agent_tool_call]
```

240. assistant (2026-07-02T06:11:33.484Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\App.tsx has been updated successfully. (file
state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

241. assistant (2026-07-02T06:11:33.484Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\App.tsx
[/external_agent_tool_call]
```

242. assistant (2026-07-02T06:11:33.484Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\App.tsx has been updated successfully. (file
state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

243. assistant (2026-07-02T06:11:33.484Z)

Now unify IngestPanel onto `errorText`:

244. assistant (2026-07-02T06:11:33.484Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx
[/external_agent_tool_call]
```

245. assistant (2026-07-02T06:11:33.484Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

246. assistant (2026-07-02T06:11:33.484Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx
[/external_agent_tool_call]
```

247. assistant (2026-07-02T06:11:33.484Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

248. assistant (2026-07-02T06:11:33.484Z)

Now the tests ? a unit test for `errorText` and an integration test proving the build path shows the friendly message:

249. assistant (2026-07-02T06:11:33.484Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\khelsutra\web\src\lib\errorText.test.ts
[/external_agent_tool_call]
```

250. assistant (2026-07-02T06:11:33.484Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\khelsutra\web\src\lib\errorText.test.ts
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

251. assistant (2026-07-02T06:11:33.484Z)

Now the build-path integration test (proves `App.onBuild` shows the friendly mapping):

252. assistant (2026-07-02T06:11:33.484Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\build-error.test.tsx
[/external_agent_tool_call]
```

253. assistant (2026-07-02T06:11:33.484Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\khelsutra\web\src\components\build-
error.test.tsx (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

254. assistant (2026-07-02T06:11:33.484Z)

Now run the suite + typecheck + build for A5:

255. assistant (2026-07-02T06:11:33.484Z)

```
[external_agent_tool_call: Bash]
description: Run A5 tests, typecheck, build
command: cd /c/Users/avidu/Projects/khelsutra/web && npm test --silent 2>&1 | grep -E "Test
Files|Tests|FAIL|?|×|Error:" | head -30 && echo "=== tsc ===" && npx tsc --noEmit && echo "tsc
OK" && echo "=== build ===" && npm run build --silent 2>&1 | tail -2
[/external_agent_tool_call]
```

256. assistant (2026-07-02T06:11:33.484Z)

```
[external_agent_tool_result]
{"ts":"2026-06-
```

21T15:10:58.491Z", "level": "error", "event": "api.network_error", "cid": "c209481f-6e05-4ea4-b2b4-9ff55e51391a", "method": "GET", "url": "/api/config", "error": "Error: offline"}
{ "ts": "2026-06-
21T15:11:01.857Z", "level": "error", "event": "ingest.job_failed", "cid": "5266ab2c-3ec4-4863-a3ee-2615b7ebddc2", "uri": "gs://b/x.mp4", "code": "engine", "detail": "RuntimeError: boom"}
{ "ts": "2026-06-
21T15:11:00.864Z", "level": "error", "event": "api.network_error", "cid": "ee7eb32b-47c1-4d22-a904-299da0fc1579", "method": "POST", "url": "/api/process", "error": "Error: offline"}
{ "ts": "2026-06-
21T15:11:01.204Z", "level": "error", "event": "api.network_error", "cid": "28cc7b77-6e61-40aa-abac-49c68c3ace00", "method": "POST", "url": "/api/process", "error": "Error: offline"}
{ "ts": "2026-06-
21T15:11:01.271Z", "level": "warn", "event": "config.unavailable", "error": "ApiError: GET /api/config failed: 500"}
{ "ts": "2026-06-
21T15:11:01.338Z", "level": "error", "event": "api.network_error", "cid": "6d7cc958-5f94-45ed-89f5-2637d9e11fb1", "method": "GET", "url": "/api/config", "error": "Error: no engine in test"}
{ "ts": "2026-06-
21T15:11:01.339Z", "level": "warn", "event": "config.unavailable", "error": "ApiError: Network error calling GET /api/config"}
{ "ts": "2026-06-
21T15:11:01.352Z", "level": "error", "event": "api.network_error", "cid": "dc9658af-4520-428a-a7dc-6eba7d7078a4", "method": "POST", "url": "/api/compile", "error": "Error: offline"}
{ "ts": "2026-06-
21T15:11:01.358Z", "level": "error", "event": "build.failed", "cid": "dc9658af-4520-428a-a7dc-6eba7d7078a4", "video_id": "vidx", "filename": "highlights.mp4", "code": "networkError", "error": "ApiError: Network error calling POST /api/compile"}
{ "ts": "2026-06-
21T15:11:01.384Z", "level": "error", "event": "api.network_error", "cid": "f2a01c01-eaf5-4a5f-bc92-64438cb62a2a", "method": "GET", "url": "/api/config", "error": "Error: no engine in test"}
{ "ts": "2026-06-
21T15:11:01.384Z", "level": "warn", "event": "config.unavailable", "error": "ApiError: Network error calling GET /api/config"}
{ "ts": "2026-06-
21T15:11:01.415Z", "level": "error", "event": "api.network_error", "cid": "ea56608a-f652-4bd7-b568-b48f45f0f365", "method": "GET", "url": "/api/config", "error": "Error: no engine in test"}
{ "ts": "2026-06-
21T15:11:01.415Z", "level": "warn", "event": "config.unavailable", "error": "ApiError: Network error calling GET /api/config"}
{ "ts": "2026-06-
21T15:11:01.417Z", "level": "warn", "event": "config.unavailable", "error": "ApiError: GET /api/config failed: 500"}
{ "ts": "2026-06-
21T15:11:01.430Z", "level": "error", "event": "api.network_error", "cid": "c9bf0794-14a3-42ae-b2f1-e8d4b2bd6041", "method": "GET", "url": "/api/config", "error": "Error: no engine in test"}
{ "ts": "2026-06-
21T15:11:01.430Z", "level": "warn", "event": "config.unavailable", "error": "ApiError: Network error calling GET /api/config"}
{ "ts": "2026-06-
21T15:11:01.488Z", "level": "error", "event": "build.failed", "cid": "4e3bfb21-b465-48f0-9804-3cfd45f8c34f", "video_id": "vidx", "filename": "highlights.mp4", "status": 400, "code": "engine", "detail": "rally below min_shot_count", "error": "ApiError: POST /api/compile failed: 400"}
{ "ts": "2026-06-
21T15:11:01.493Z", "level": "error", "event": "api.network_error", "cid": "b1c967ee-5dac-4813-bdf9-8727cddbc64a", "method": "GET", "url": "/api/config", "error": "Error: no engine in test"}
{ "ts": "2026-06-
21T15:11:01.494Z", "level": "warn", "event": "config.unavailable", "error": "ApiError: Network error calling GET /api/config"}
{ "ts": "2026-06-
21T15:11:01.512Z", "level": "error", "event": "api.network_error", "cid": "ffd4a5df-ba65-45ae-a588-821e075346f1", "method": "GET", "url": "/api/config", "error": "Error: no engine in test"}
{ "ts": "2026-06-
21T15:11:01.512Z", "level": "warn", "event": "config.unavailable", "error": "ApiError: Network error calling GET /api/config"}
{ "ts": "2026-06-

```
{"ts":"2026-06-21T15:11:01.518Z","level":"error","event":"api.network_error","cid":"22100c51-77a...[/external_agent_tool_result]}
```

257. assistant (2026-07-02T06:11:33.484Z)

The build logs confirm A5 working (`build.failed code:networkError` and `code:engine detail:"rally below min\_shot\_count"`). tsc + build clean. Let me get the clean test summary:

258. assistant (2026-07-02T06:11:33.484Z)

```
[external_agent_tool_call: Bash]
description: Clean A5 test summary
command: npm test --silent 2>/dev/null | grep -E "Test Files|Tests |passed|failed" | tail -5
[/external_agent_tool_call]
```

259. assistant (2026-07-02T06:11:33.484Z)

```
[external_agent_tool_result]
stdout | src/hooks/useIngestJob.test.ts > useIngestJob >
treats a done-but-failed pipeline verdict as emptyResult ? never a load
stdout | src/hooks/useIngestJob.test.ts > useIngestJob >
treats a done-but-failed pipeline verdict as emptyResult ? never a load
stdout | src/hooks/useIngestJob.test.ts > useIngestJob >
treats a done-but-failed pipeline verdict as emptyResult ? never a load
Test Files 14 passed (14)
Tests 135 passed (135)
[/external_agent_tool_result]
```

260. assistant (2026-07-02T06:11:33.485Z)

135 tests pass (+5 errorText, +2 build-error), tsc + build clean. Let me browser-verify the shared `errorText` wiring via the ingest path (same helper the build path now uses) ? with no engine on :8000, a submit should show the friendly localized message.

261. assistant (2026-07-02T06:11:33.485Z)

```
[external_agent_tool_call: ToolSearch]
input: {"query":"select:mcp__Claude_Preview__preview_click,mcp__Claude_Preview__preview_eval","max_results":3}
[/external_agent_tool_call]
```

262. assistant (2026-07-02T06:11:33.485Z)

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

263. assistant (2026-07-02T06:11:33.485Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_eval]
input: {"serverId":"d4a4a2a4-7c2e-46e2-889e-f619477293b4","expression":"window.location.href = window.location.origin + \"\\\"/\\\"; \\\"navigated to clean root\\\"\"}
[/external_agent_tool_call]
```

264. assistant (2026-07-02T06:11:33.485Z)

```
[external_agent_tool_result]
"navigated to clean root"
[/external_agent_tool_result]
```

265. assistant (2026-07-02T06:11:33.485Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_fill]
```



```
input: {"serverId":"d4a4a2a4-7c2e-46e2-889e-f619477293b4","selector":"input[aria-label^=\"Video location\"]","value":"gs://demo-bucket/match.mp4"}
[/external_agent_tool_call]
```

266. assistant (2026-07-02T06:11:33.485Z)

```
[external_agent_tool_result]
Successfully filled: input[aria-label^="Video location"]
[/external_agent_tool_result]
```

267. assistant (2026-07-02T06:11:33.485Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_click]
input: {"serverId":"d4a4a2a4-7c2e-46e2-889e-f619477293b4","selector":"button[type=\"submit\"]"}
[/external_agent_tool_call]
```

268. assistant (2026-07-02T06:11:33.485Z)

```
[external_agent_tool_result]
Successfully clicked: button[type="submit"]
[/external_agent_tool_result]
```

269. assistant (2026-07-02T06:11:33.485Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_eval]
input: {"serverId":"d4a4a2a4-7c2e-46e2-889e-f619477293b4","expression":"(async () => { await new Promise(r => setTimeout(r, 800)); const el = document.querySelector('[data-testid=\"ingest-error\"]'); return el ? el.textContent : \"(no error yet)\"; })()"}
[/external_agent_tool_call]
```

270. assistant (2026-07-02T06:11:33.485Z)

```
[external_agent_tool_result]
"Can't reach KhelSutra. Is it running?Try again"
[/external_agent_tool_result]
```

271. assistant (2026-07-02T06:11:33.485Z)

The shared `errorText` renders the friendly localized message in the browser (`"Can't reach KhelSutra. Is it running?" + Try again) ? same helper the build path now uses. A5 verified: 135 tests, tsc, build all green; key-parity holds.

Let me capture the A5 diff and run an adversarial review before merging.

272. assistant (2026-07-02T06:11:33.485Z)

```
[external_agent_tool_call: Bash]
description: Capture A5 diff
command: git --no-pager diff --stat && echo "====DIFF====" && git --no-pager diff -- web/src/App.tsx web/src/components/IngestPanel.tsx web/src/lib/errorText.ts && echo "====CATALOG (en only, kn/hi parallel)====" && git --no-pager diff -- web/src/i18n/locales/en/common.json
[/external_agent_tool_call]
```

273. assistant (2026-07-02T06:11:33.485Z)

```
[external_agent_tool_result]
web/src/App.tsx | 11 ++++++----
web/src/components/IngestPanel.tsx | 15 ++++++-----
web/src/i18n/locales/en/common.json | 10 ++++++----
web/src/i18n/locales/hi/common.json | 10 ++++++----
web/src/i18n/locales/kn/common.json | 10 ++++++----
5 files changed, 31 insertions(+), 25 deletions(-)
```

```
=====DIFF=====
=====CATALOG (en only, kn/hi parallel)=====
[/external_agent_tool_result]
```

274. assistant (2026-07-02T06:11:33.485Z)

Let me get a clean diff (the new files are untracked, so `git diff` skipped them) and run a focused 2-lens review:

275. assistant (2026-07-02T06:11:33.485Z)

```
[external_agent_tool_call: Bash]
description: Show A5 code diff (intent-add new files)
command: git add -N web/src/lib/errorText.ts web/src/lib/errorText.test.ts
web/src/components/build-error.test.tsx && git --no-pager diff -- web/src/App.tsx
web/src/components/IngestPanel.tsx | cat
[/external_agent_tool_call]
```

276. assistant (2026-07-02T06:11:33.485Z)

```
[external_agent_tool_result: error]
Exit code 128
warning: could not open directory 'web/web/src/': No such file or directory
fatal: pathspec 'web/src/lib/errorText.ts' did not match any files
[/external_agent_tool_result]
```

277. assistant (2026-07-02T06:11:33.485Z)

Working directory was still `khelsutra/web`. Let me run from the repo root:

278. assistant (2026-07-02T06:11:33.485Z)

```
[external_agent_tool_call: Bash]
description: Show A5 code diff from repo root
command: cd /c/Users/avidu/Projects/khelsutra && git add -N web/src/lib/errorText.ts
web/src/lib/errorText.test.ts web/src/components/build-error.test.tsx && git --no-pager diff --
web/src/App.tsx web/src/components/IngestPanel.tsx | cat
[/external_agent_tool_call]
```

279. assistant (2026-07-02T06:11:33.486Z)

```
[external_agent_tool_result]
diff --git a/web/src/App.tsx b/web/src/App.tsx
index 85361d6..4231ba3 100644
--- a/web/src/App.tsx
+++ b/web/src/App.tsx
@@ -10,7 +10,8 @@ import { IngestPanel } from "../components/IngestPanel";
   import { runQuery, type ParsedQuery } from "../lib/query";
   import { floorRisk } from "../lib/floor";
   import { reelFilename } from "../lib/format";
-  import { parseDetail } from "../lib/errors";
+  import { classifyError } from "../lib/errors";
+  import { errorText } from "../lib/errorText";
   import { log, newCorrelationId } from "../lib/log";
   import { ApiError, compileReel, getConfig, highlightUrl, listRallies, minShotCount } from
"../api/client";
   import type { EngineConfig, RallySegment } from "../api/types";
@@ -143,10 +144,12 @@ export default function App() {
   setBuild({ status: "done", url, filename });
   log("info", "build.done", { cid, video_id: videoId, filename, filepath: res.filepath, url
});
} catch (err) {
+  // Same friendly mapping the ingest flow uses: a bare 5xx / network throw reads "is the
engine
```

```

+ // running?", a real engine 4xx surfaces its own detail ? never a raw "HTTP 500" (A5).
const status = err instanceof ApiError ? err.status : undefined;
- const detail = err instanceof ApiError && err.body ? parseDetail(err.body) : String(err);
- setBuild({ status: "error", error: detail });
- log("error", "build.failed", { cid, video_id: videoId, filename, status, error:
String(err) });
+ const e = classifyError(err);
+ setBuild({ status: "error", error: errorText(t, e) });
+ log("error", "build.failed", { cid, video_id: videoId, filename, status, code: e.code,
detail: e.detail, error: String(err) });
}
};

```

```
diff --git a/web/src/components/IngestPanel.tsx b/web/src/components/IngestPanel.tsx
```

```
index 322ffb8..87607f8 100644
```

```
--- a/web/src/components/IngestPanel.tsx
```

```
+++ b/web/src/components/IngestPanel.tsx
```

```
@@ -8,6 +8,7 @@ import { useState } from "react";
```

```
import type { FormEvent } from "react";
```

```
import { useTranslation } from "react-i18next";
```

```
import { useIngestJob } from "../hooks/useIngestJob";
```

```
+import { errorText } from "../lib/errorText";
```

```
interface Props {
```

```
  /** Called with the engine's video_id once a submitted job finishes successfully. */
```

```
@@ -24,15 +25,11 @@ export function IngestPanel({ onLoad: Props } {
```

```
  submit(uri);
```

```
};
```

```
- // `engine` failures carry the engine's own message (a 400 scheme/path detail, a worker
error);
```

```
- // every other code maps to a friendly localized line ? the network case especially, since
this
```

```
- // is the first screen a fresh user hits and "is the engine running?" must read plainly.
```

```
- const errorMsg =
```

```
-   phase.status === "error"
```

```
-     ? phase.error.code === "engine"
```

```
-       ? (phase.error.detail ?? t("ingest.failed", { error: "" })))
```

```
-       : t(`ingest.${phase.error.code}`)
```

```
-       : null;
```

```
+ // Shared mapping with the build flow (lib/errorText): `engine` failures carry the engine's
own
```

```
+ // message (a 400 scheme/path detail, a worker error); every other code maps to a friendly
```

```
+ // localized line ? the network case especially, since this is the first screen a fresh user
hits
```

```
+ // and "is the engine running?" must read plainly.
```

```
+ const errorMsg = phase.status === "error" ? errorText(t, phase.error) : null;
```

```
  return (
```

```
    <section className="mt-6 rounded-2xl border border-black/10 bg-white p-5" data-
```

```
    testid="ingest-panel">
```

```
[/external_agent_tool_result]
```

280. assistant (2026-07-02T06:11:33.486Z)

Clean A5 diff. Now a focused 2-lens adversarial review (regression/orphan-keys + faithfulness/observability):

281. assistant (2026-07-02T06:11:33.486Z)

```
[external_agent_tool_call: Workflow]
```

```
input: {"script": "export const meta = {\n  name: 'a5-review',\n  description: 'Adversarial
review of the A5 shared-error-mapping diff',\n  phases: [{ title: 'Review' }],\n}\n\nconst BASE
= 'C:/Users/avidu/Projects/khelsutra/web/src'\n\nconst SCHEMA = {\n  type: 'object',\n
additionalProperties: false,\n  properties: {\n    summary: { type: 'string' },\n    findings:
```

```
{\n      type: 'array',\n      items: {\n        type: 'object',\n        additionalProperties: false,\n        properties: {\n          severity: { type: 'string', enum: ['blocker', 'major', 'minor', 'nit'] },\n          file: { type: 'string' },\n          location: { type: 'string' },\n          title: { type: 'string' },\n          detail: { type: 'string' },\n          suggestion: { type: 'string' },\n          required: ['severity', 'file', 'title', 'detail'],\n          },\n        },\n        required: ['summary', 'findings'],\n      },\n    }\n\nconst CONTEXT = 'A5 refactor under ${BASE}: a shared error->text mapping.\n- NEW lib/errorText.ts: errorText(t, e) ? engine code -> e.detail ?? t(`errors.engineGeneric`); else t(`errors.\\${e.code}\\`).\n- App.tsx onBuild catch now: classifyError(err) -> errorText(t, e) (was: parseDetail(err.body) ?? String(err)). parseDetail import removed.\n- IngestPanel.tsx: errorMsg now = errorText(t, phase.error) (was: inline ingest.\\${code} map).\n- Catalogs (en/kn/hi): MOVED ingest.{networkError,invalidUri,emptyResult,timeout} -> new errors.* namespace; ADDED errors.engineGeneric; REMOVED ingest.failed. ingest.* kept: title/intro/uriPlaceholder/uriAriaLabel/hint/submit/submitting/cancel/working/progress/retry.\nTests: 135 green incl. errorText.test.ts, build-error.test.tsx, the existing IngestPanel.test.tsx, i18n key-parity.\n\nconst LENSES = [\n  {\n    key: 'regression-orphans',\n    prompt: `You are reviewing for REGRESSIONS and ORPHANED i18n keys. ${CONTEXT}\n\nDo these checks against the files on disk:\n1. ORPHANED KEYS: grep the ENTIRE ${BASE} tree for any lingering ...`,\n  },\n];\n\n[external_agent_tool_call]
```

282. assistant (2026-07-02T06:11:33.486Z)

```
[external_agent_tool_result]
Workflow launched in background. Task ID: w8jaidc9f
Summary: Adversarial review of the A5 shared-error-mapping diff
Transcript dir: C:\Users\avidu\claudu\projects\C--Users-avidu-Projects-badminton-highlight-indexer\2ba49afa-99f4-41f2-86a2-27a289d432f3\subagents\workflows\wf_d954e9ee-fd0
Script file: C:\Users\avidu\claudu\projects\C--Users-avidu-Projects-khelsutra\2ba49afa-99f4-41f2-86a2-27a289d432f3\workflows\scripts\a5-review-wf_d954e9ee-fd0.js
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath: "C:\Users\avidu\claudu\projects\C--Users-avidu-Projects-khelsutra\2ba49afa-99f4-41f2-86a2-27a289d432f3\workflows\scripts\a5-review-wf_d954e9ee-fd0.js"})
to iterate without resending the script.)
Run ID: wf_d954e9ee-fd0
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\claudu\projects\C--Users-avidu-Projects-khelsutra\2ba49afa-99f4-41f2-86a2-27a289d432f3\workflows\scripts\a5-review-wf_d954e9ee-fd0.js", resumeFromRunId: "wf_d954e9ee-fd0"}) ? completed agents return cached results.
```

You will be notified when it completes. Use /workflows to watch live progress.
[external_agent_tool_result]

283. assistant (2026-07-02T06:11:33.486Z)

While the A5 review runs, I've scoped A8 (reassurance + icon+text status chips in IngestPanel progress and the build/CompileResult states). Let me wait for the A5 review to gate the merge.

284. assistant (2026-07-02T06:11:33.486Z)

```
[external_agent_tool_call: TaskOutput]
input: {"task_id":"w8jaidc9f","block":true,"timeout":240000}
[/external_agent_tool_call]
```

285. assistant (2026-07-02T06:11:33.486Z)

```
[external_agent_tool_result]
<retrieval_status>timeout</retrieval_status>

<task_id>w8jaidc9f</task_id>

<task_type>local_workflow</task_type>

<status>running</status>
[/external_agent_tool_result]
```

286. assistant (2026-07-02T06:11:33.486Z)

```
[external_agent_tool_call: TaskOutput]
input: {"task_id":"w8jaidc9f","block":true,"timeout":180000}
[/external_agent_tool_call]
```

287. assistant (2026-07-02T06:11:33.486Z)

```
[external_agent_tool_result]
<retrieval_status>success</retrieval_status>
```

```
<task_id>w8jaidc9f</task_id>
```

```
<task_type>local_workflow</task_type>
```

```
<status>completed</status>
```

```
<output>
```

```
{
  "summary": "Adversarial review of the A5 shared-error-mapping diff",
  "agentCount": 2,
  "logs": [],
  "result": [
    {
      "lens": "regression-orphans",
      "summary": "clean",
      "findings": [
        {
          "severity": "nit",
          "file": "web/src/lib/errors.ts",
          "title": "parseDetail now internal-only",
          "detail": "After A5, parseDetail is only used inside errors.ts and its test; still
exported. Harmless."
        }
      ]
    },
    {
      "lens": "faithfulness-observability",
      "findings": [
        {
          "severity": "nit",
          "file": "web/src/App.tsx",
          "title": "Rally-load failure banner still shows raw String(err) (out of A5 scope)",
          "detail": "rallies.load_failed catch sets loadState.error=String(err), rendered by
ModeBanner mode.error (App.tsx 85-89, 290-296); a 500/network throw shows raw ApiError text. Out
of A5 scope (build path + ingest unification both done), not a spec violation; natural next
consolidation.",
          "location": "App.tsx 85-89, 290-296",
          "suggestion": "Map through errorText(t, classifyError(err)) and store in
loadState.error."
        }
      ],
      "summary": "A5 faithful; one out-of-scope nit. Build path uses classifyError->errorText
(friendly+localized, not raw String(err)/HTTP500), proven by build-error.test.tsx (network->'is
it running', 4xx->engine detail verbatim). errors.* namespace shared by IngestPanel+App via the
single errorText helper; the engine/non-engine branch lives ONLY in lib/errorText.ts (no
duplication; old inline ingest.${code} map and App parseDetail removed). errors.ts stays
i18n-free/testable. Observability: build.failed retains cid+status+raw error and ADDS
code+detail (nothing dropped). engineGeneric is a safe last-resort (only when code==engine AND
detail nullish; every producer sets a non-null detail). Catalogs:
ingest.{networkError,invalidUri,emptyResult,timeout} moved to errors.*, engineGeneric added,
ingest.failed removed, 11 ingest.* keys retained; identical en/kn/hi key sets with real
translations. Verified locally: web suite 135/135 green, tsc --noEmit clean (no eslint config
for web/, lint N/A there)."
    }
  ]
}
```

```
]
}
</output>
[/external_agent_tool_result]
```

288. assistant (2026-07-02T06:11:33.486Z)

A5 review is clean ? two ****nits****, no blockers:

1. `parseDetail` is now internal-only but still exported (harmless ? it's used by `classifyError` and tested).
2. The rally-****load**** failure banner still shows raw `String(err)` ? "out of A5 scope but a natural next consolidation."

Nit #2 is the same raw-error class on the one remaining SPA surface (the load path), and it's a one-line application of the helper I just built. Leaving load showing `ApiError: Network error...` while build is friendly would be inconsistent and violates the DoD's "no raw engine jargon." I'll fold it in to fully complete "friendly errors everywhere."

289. assistant (2026-07-02T06:11:33.486Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\App.tsx
[/external_agent_tool_call]
```

290. assistant (2026-07-02T06:11:33.486Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\App.tsx has been updated successfully. (file
state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

291. assistant (2026-07-02T06:11:33.486Z)

Now extend the test to cover the load path too:

292. assistant (2026-07-02T06:11:33.486Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\build-error.test.tsx
[/external_agent_tool_call]
```

293. assistant (2026-07-02T06:11:33.487Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\build-error.test.tsx has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

294. assistant (2026-07-02T06:11:33.487Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\build-error.test.tsx
[/external_agent_tool_call]
```

295. assistant (2026-07-02T06:11:33.487Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\build-error.test.tsx has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

296. assistant (2026-07-02T06:11:33.487Z)

```
[external_agent_tool_call: Bash]
description: Re-run A5 tests after load-path fold-in
command: cd /c/Users/avidu/Projects/khelsutra/web && npm test --silent 2>/dev/null | grep -E
"Test Files|Tests |FAIL" | tail -5 && echo "=== tsc ===" && npx tsc --noEmit && echo "tsc OK" &&
echo "=== build ===" && npm run build --silent 2>&1 | tail -1
[/external_agent_tool_call]
```

297. assistant (2026-07-02T06:11:33.487Z)

```
[external_agent_tool_result]
  Test Files   14 passed (14)
    Tests     136 passed (136)
=== tsc ===
tsc OK
=== build ===
? built in 1.41s
[/external_agent_tool_result]
```

298. assistant (2026-07-02T06:11:33.487Z)

136 tests pass (+1 load-failure test), tsc + build clean. A5 is complete (build + load paths + ingest unified). Committing ? verify branch/HEAD, stage explicit paths:

299. assistant (2026-07-02T06:11:33.487Z)

```
[external_agent_tool_call: Bash]
description: Stage A5 paths, verify clean base
command: cd /c/Users/avidu/Projects/khelsutra && echo "branch: $(git branch --show-current)" &&
git add web/src/App.tsx web/src/components/IngestPanel.tsx web/src/components/build-
error.test.tsx web/src/lib/errorText.ts web/src/lib/errorText.test.ts
web/src/i18n/locales/en/common.json web/src/i18n/locales/kn/common.json
web/src/i18n/locales/hi/common.json && echo "--- staged ---" && git diff --cached --name-only &&
echo "--- HEAD must be A4-merged master ---" && git log --oneline origin/master..HEAD
[/external_agent_tool_call]
```

300. assistant (2026-07-02T06:11:33.487Z)

```
[external_agent_tool_result]
branch: feat/a5-friendly-build-errors
--- staged ---
web/src/App.tsx
web/src/components/IngestPanel.tsx
web/src/components/build-error.test.tsx
web/src/i18n/locales/en/common.json
web/src/i18n/locales/hi/common.json
web/src/i18n/locales/kn/common.json
web/src/lib/errorText.test.ts
web/src/lib/errorText.ts
--- HEAD must be A4-merged master ---
[/external_agent_tool_result]
```

301. assistant (2026-07-02T06:11:33.487Z)

```
[external_agent_tool_call: Bash]
description: Commit + push A5
command: git commit -q -F - <<'EOF'
feat(web): A5 ? friendly engine-error mapping across build + load (shared errors.*)
```

The build and load paths showed raw client text on failure (e.g. "ApiError: Network error calling POST /api/compile", "HTTP 500"). Now every SPA surface maps errors the same friendly way the ingest flow already did:

- New lib/errorText.ts: errorText(t, e) ? `engine` shows the engine's OWN detail verbatim (most specific); every other code maps to a localized line in a new

- shared `errors.\*` namespace (engineGeneric is the detail-less last resort).
- App.onBuild + the rally-load catch now go classifyError ? errorText (localized), not String(err)/parseDetail. build.failed / rallies.load_failed logs gain code+detail (observability) and keep cid+status.
- IngestPanel unified onto errorText (drops its inline ingest.\${code} map).
- Catalogs: moved ingest.{networkError,invalidUri,emptyResult,timeout} ? errors.*, added errors.engineGeneric, removed the now-unused ingest.failed (en/kn/hi).

Tests: errorText.test.ts (mapping + kn localization) + build-error.test.tsx (build network?"is it running?", 4xx?engine detail verbatim, load?friendly). 136 web tests green; tsc + build clean; key-parity + no-bare-strings hold. Reviewed via a 2-lens adversarial pass; the load-path fold-in addresses its "natural next consolidation" nit.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>

EOF

```
git log --oneline -1 && git push -u origin feat/a5-friendly-build-errors 2>&1 | tail -2
[/external_agent_tool_call]
```

302. assistant (2026-07-02T06:11:33.487Z)

```
[external_agent_tool_result]
1f28484 feat(web): A5 ? friendly engine-error mapping across build + load (shared errors.*)
To https://github.com/avidullu/khelsutra.git
* [new branch]      feat/a5-friendly-build-errors -> feat/a5-friendly-build-errors
[/external_agent_tool_result]
```

303. assistant (2026-07-02T06:11:33.487Z)

```
[external_agent_tool_call: Bash]
description: Create A5 PR
command: gh pr create --title "feat(web): A5 ? friendly engine-error mapping across build + load (shared errors.*)" --base master --head feat/a5-friendly-build-errors --body "$(cat <<'EOF'
```

A5 ? friendly engine-error mapping (tracker `FRIENDLY\_AND\_MULTILINGUAL.md` §7)

The ingest flow already mapped failures to friendly, localized lines ? but the **build** path and the **rally-load** path still surfaced raw client text (ApiError: Network error calling POST /api/compile, HTTP 500). For a non-technical coach/parent that's a dead end. This unifies all three SPA surfaces on one mapping.

What changed

- **New lib/errorText.ts** ? errorText(t, e): an engine error shows the engine's **own** detail verbatim (the most specific, actionable message); every other code maps to a localized line in a **new shared errors.*** namespace (engineGeneric is the detail-less last resort). Kept out of the i18n-free errors.ts so that stays unit-testable.
- **App.onBuild** and the **rally-load** catch now go classifyError ? errorText (localized), not String(err)/parseDetail. The build.failed / rallies.load_failed logs gain code+detail (observability) and keep cid+status.
- **IngestPanel** unified onto errorText (drops its inline ingest.\${code} map).
- **Catalogs (en/kn/hi)**: moved ingest.{networkError,invalidUri,emptyResult,timeout} ? errors.*, added errors.engineGeneric, removed the now-unused ingest.failed.

Verification

- **136 web tests** green: new errorText.test.ts (mapping + kn localization) and build-error.test.tsx (build network?"is it running?", 4xx?engine detail verbatim, **load**?friendly). tsc + vite build clean; **key-parity** + **no-bare-strings** hold.
- **Browser-verified**: with no engine on :8000, a submit shows "Can't reach KhelSutra. Is it running?" via the shared helper.
- **Reviewed by a 2-lens adversarial workflow** (regression/orphaned-keys + faithfulness/observability): no remaining ingest.\${code} references, key sets identical across en/kn/hi, nothing dropped from the logs. The load-path fix in this PR folds the review's one "natural next consolidation" nit.

Honesty notes

- kn/hi `errors.\*` reuse the existing (AI-drafted) ingest translations + a new `engineGeneric` draft ? native review still pending.
- A stored error string doesn't re-localize on a mid-error language switch (consistent with the existing build/mode handling); errors are transient.

? Generated with [Claude Code](https://claude.com/claude-code)

EOF

)" 2>&1 | tail -2

[/external_agent_tool_call]

304. assistant (2026-07-02T06:11:33.487Z)

[external_agent_tool_result]

<https://github.com/avidullu/khelsutra/pull/62>

[/external_agent_tool_result]

305. assistant (2026-07-02T06:11:33.487Z)

PR [#62](https://github.com/avidullu/khelsutra/pull/62) created. Now mark A5 ? in the tracker:

306. assistant (2026-07-02T06:11:33.487Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\khelsutra\docs\FRIENDLY_AND_MULTILINGUAL.md

[/external_agent_tool_call]

307. assistant (2026-07-02T06:11:33.487Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\khelsutra\docs\FRIENDLY_AND_MULTILINGUAL.md has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

308. assistant (2026-07-02T06:11:33.487Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\khelsutra\docs\FRIENDLY_AND_MULTILINGUAL.md

[/external_agent_tool_call]

309. assistant (2026-07-02T06:11:33.487Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\khelsutra\docs\FRIENDLY_AND_MULTILINGUAL.md has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

310. assistant (2026-07-02T06:11:33.488Z)

[external_agent_tool_call: Bash]

description: Commit tracker + push + check CI

command: git add docs/FRIENDLY_AND_MULTILINGUAL.md && git commit -q -m "docs(a5): mark A5 (#62)

done in the \$7 tracker + changelog

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>" && git push -q 2>&1 | tail -1 && echo "pushed; CI poll..." && sleep 25 && gh pr checks 62 2>&1 | head

[/external_agent_tool_call]

311. assistant (2026-07-02T06:11:33.488Z)

[external_agent_tool_result]

pushed; CI poll...

site pass 16s

<https://github.com/avidullu/khelsutra/actions/runs/27908758417/job/82582082654>

```
Workers Builds: hidden-king-khelsutra    pending 0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/78ca8f09-a900-4337-819b-5d57a21e3501
web      pending 0
https://github.com/avidullu/khelsutra/actions/runs/27908758417/job/82582082635
[/external_agent_tool_result]
```

312. assistant (2026-07-02T06:11:33.488Z)

```
[external_agent_tool_call: Bash]
description: Poll A5 CI until done
command: for i in 1 2 3 4 5 6 7 8; do s=$(gh pr checks 62 2>&1 | grep -E "^web\b" | awk '{print $2}'); c=$(gh pr checks 62 2>&1 | grep -iE "Workers Builds" | awk '{print $4}'); echo "poll $i: web=$s cf=$c"; if { [ "$s" = "pass" ] || [ "$s" = "fail" ]; } && { [ "$c" = "pass" ] || [ "$c" = "fail" ]; }; then break; fi; sleep 15; done && echo "=== final ===" && gh pr checks 62 2>&1 | head
[/external_agent_tool_call]
```

313. assistant (2026-07-02T06:11:33.488Z)

```
[external_agent_tool_result]
poll 1: web=pending cf=pass
poll 2: web=pass cf=pass
=== final ===
Workers Builds: hidden-king-khelsutra    pass    0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/78ca8f09-a900-4337-819b-5d57a21e3501
site      pass    16s
https://github.com/avidullu/khelsutra/actions/runs/27908758417/job/82582082654
web       pass    30s
https://github.com/avidullu/khelsutra/actions/runs/27908758417/job/82582082635
[/external_agent_tool_result]
```

314. assistant (2026-07-02T06:11:33.488Z)

All three checks green (web, site, Cloudflare). Merging A5:

315. assistant (2026-07-02T06:11:33.488Z)

```
[external_agent_tool_call: Bash]
description: Squash-merge A5 and sync master
command: gh pr merge 62 --squash --delete-branch 2>&1 | tail -3 && git checkout master -q && git pull --ff-only -q && git log --oneline -2
[/external_agent_tool_call]
```

316. assistant (2026-07-02T06:11:33.488Z)

```
[external_agent_tool_result]
create mode 100644 web/src/components/build-error.test.tsx
create mode 100644 web/src/lib/errorText.test.ts
create mode 100644 web/src/lib/errorText.ts
501c014 feat(web): A5 ? friendly engine-error mapping across build + load (shared errors.*)
(#62)
95195a5 feat(web): A4 ? mobile touch targets (?44px) + visible tooltip text (#61)
[/external_agent_tool_result]
```

317. assistant (2026-07-02T06:11:33.488Z)

A5 merged (#62 ? `501c014`). Now **A8 ? background-processing reassurance + icon+text status chips**.

318. assistant (2026-07-02T06:11:33.488Z)

```
[external_agent_tool_call: TaskUpdate]
```

```
input: {"taskId":"2","status":"completed"}
[/external_agent_tool_call]
```

319. assistant (2026-07-02T06:11:33.488Z)

```
[external_agent_tool_result]
Updated task #2 status
[/external_agent_tool_result]
```

320. assistant (2026-07-02T06:11:33.488Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"3","status":"in_progress"}
[/external_agent_tool_call]
```

321. assistant (2026-07-02T06:11:33.488Z)

```
[external_agent_tool_result]
Updated task #3 status
[/external_agent_tool_result]
```

322. assistant (2026-07-02T06:11:33.488Z)

```
[external_agent_tool_call: mcp_ccd_session__mark_chapter]
input: {"title":"A8 ? status chips + reassurance","summary":"A4+A5 merged (#61,#62). Now
icon+text status chips + background-processing reassurance in IngestPanel + build state."}
[/external_agent_tool_call]
```

323. assistant (2026-07-02T06:11:33.488Z)

```
[external_agent_tool_result]
Chapter marked: "A8 ? status chips + reassurance". Continue your current work.
[/external_agent_tool_result]
```

324. assistant (2026-07-02T06:11:33.488Z)

```
[external_agent_tool_call: Bash]
description: Cut A8 branch from origin/master
command: git fetch origin -q && git checkout -b feat/a8-status-chips origin/master 2>&1 | tail
-2 && git log --oneline -1
[/external_agent_tool_call]
```

325. assistant (2026-07-02T06:11:33.488Z)

```
[external_agent_tool_result]
branch 'feat/a8-status-chips' set up to track 'origin/master'.
Switched to a new branch 'feat/a8-status-chips'
501c014 feat(web): A5 ? friendly engine-error mapping across build + load (shared errors.*)
(#62)
[/external_agent_tool_result]
```

326. assistant (2026-07-02T06:11:33.488Z)

A8 branch cut. First, a small reusable `StatusChip` (always icon+text) + a shared `Spinner`:

327. assistant (2026-07-02T06:11:33.488Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\StatusChip.tsx
[/external_agent_tool_call]
```

328. assistant (2026-07-02T06:11:33.488Z)

```
[external_agent_tool_result]
File created successfully at:
C:\Users\avidu\Projects\khelsutra\web\src\components\StatusChip.tsx (file state is current in
your context ? no need to Read it back)
[/external_agent_tool_result]
```

329. assistant (2026-07-02T06:11:33.488Z)

Let me use the proven spinner classes (matching the original) for robustness:

330. assistant (2026-07-02T06:11:33.488Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\StatusChip.tsx
[/external_agent_tool_call]
```

331. assistant (2026-07-02T06:11:33.488Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\StatusChip.tsx has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

332. assistant (2026-07-02T06:11:33.489Z)

Now wire the IngestPanel ? import StatusChip/Spinner, use chips + reassurance, drop the local Spinner:

333. assistant (2026-07-02T06:11:33.489Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx
[/external_agent_tool_call]
```

334. assistant (2026-07-02T06:11:33.489Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

335. assistant (2026-07-02T06:11:33.489Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx
[/external_agent_tool_call]
```

336. assistant (2026-07-02T06:11:33.489Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

337. assistant (2026-07-02T06:11:33.489Z)

Now wire `CompileResult` ? add a building reassurance card + status icons on the done/failed headers:

338. assistant (2026-07-02T06:11:33.489Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\CompileResult.tsx
```

[/external_agent_tool_call]

339. assistant (2026-07-02T06:11:33.489Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\khelsutra\web\src\components\CompileResult.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

340. assistant (2026-07-02T06:11:33.489Z)

Now the catalog keys (en/kn/hi) for `result.building`, `result.buildingReassure`, `ingest.reassure`, `ingest.statusFailed`:

341. assistant (2026-07-02T06:11:33.489Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json

[/external_agent_tool_call]

342. assistant (2026-07-02T06:11:33.489Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

343. assistant (2026-07-02T06:11:33.489Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json

[/external_agent_tool_call]

344. assistant (2026-07-02T06:11:33.489Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

345. assistant (2026-07-02T06:11:33.489Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json

[/external_agent_tool_call]

346. assistant (2026-07-02T06:11:33.489Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

347. assistant (2026-07-02T06:11:33.489Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json

[/external_agent_tool_call]

348. assistant (2026-07-02T06:11:33.489Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json has been updated

successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

349. assistant (2026-07-02T06:11:33.489Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khehsutra\web\src\i18n\locales\hi\common.json
[/external_agent_tool_call]

350. assistant (2026-07-02T06:11:33.489Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khehsutra\web\src\i18n\locales\hi\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

351. assistant (2026-07-02T06:11:33.490Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khehsutra\web\src\i18n\locales\hi\common.json
[/external_agent_tool_call]

352. assistant (2026-07-02T06:11:33.490Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khehsutra\web\src\i18n\locales\hi\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

353. assistant (2026-07-02T06:11:33.490Z)

Now an A8 test (CompileResult building/done/error chips + IngestPanel reassurance/error chip):

354. assistant (2026-07-02T06:11:33.490Z)

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\khehsutra\web\src\components\a8.status.test.tsx
[/external_agent_tool_call]

355. assistant (2026-07-02T06:11:33.490Z)

[external_agent_tool_result]
File created successfully at:
C:\Users\avidu\Projects\khehsutra\web\src\components\a8.status.test.tsx (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

356. assistant (2026-07-02T06:11:33.490Z)

[external_agent_tool_call: Bash]
description: Run A8 tests, typecheck, build
command: cd /c/Users/avidu/Projects/khehsutra/web && npm test --silent 2>/dev/null | grep -E "Test Files|Tests |FAIL|?" | tail -8 && echo "=== tsc ===" && npx tsc --noEmit && echo "tsc OK" && echo "=== build ===" && npm run build --silent 2>&1 | tail -1
[/external_agent_tool_call]

357. assistant (2026-07-02T06:11:33.490Z)

[external_agent_tool_result]
Test Files 15 passed (15)
Tests 141 passed (141)
=== tsc ===
tsc OK

```
=== build ===
? built in 1.29s
[/external_agent_tool_result]
```

358. assistant (2026-07-02T06:11:33.490Z)

141 tests pass (+5 A8 tests), tsc + build clean. Let me browser-verify the reachable A8 states (the ingest error chip is reachable with no engine; working/building are covered by tests). Reload and submit:

359. assistant (2026-07-02T06:11:33.490Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_eval]
input: {"serverId":"d4a4a2a4-7c2e-46e2-889e-f619477293b4","expression":"window.location.href =
window.location.origin + \"/\"; \"ok\""}
[/external_agent_tool_call]
```

360. assistant (2026-07-02T06:11:33.490Z)

```
[external_agent_tool_result]
"ok"
[/external_agent_tool_result]
```

361. assistant (2026-07-02T06:11:33.490Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_fill]
input: {"serverId":"d4a4a2a4-7c2e-46e2-889e-f619477293b4","selector":"input[aria-label=\"Video
location\"]", "value":"gs://demo-bucket/match.mp4"}
[/external_agent_tool_call]
```

362. assistant (2026-07-02T06:11:33.490Z)

```
[external_agent_tool_result]
Successfully filled: input[aria-label="Video location"]
[/external_agent_tool_result]
```

363. assistant (2026-07-02T06:11:33.490Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_click]
input: {"serverId":"d4a4a2a4-7c2e-46e2-889e-f619477293b4","selector":"button[type=\"submit\"]"}
[/external_agent_tool_call]
```

364. assistant (2026-07-02T06:11:33.490Z)

```
[external_agent_tool_result]
Successfully clicked: button[type="submit"]
[/external_agent_tool_result]
```

365. assistant (2026-07-02T06:11:33.490Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_eval]
input: {"serverId":"d4a4a2a4-7c2e-46e2-889e-f619477293b4","expression":"(async () => { await new
Promise(r => setTimeout(r, 900)); const el = document.querySelector('[data-testid=\"ingest-
error\"]'); return el ? el.textContent : \"(no error)\"; })()"}
[/external_agent_tool_call]
```

366. assistant (2026-07-02T06:11:33.490Z)

```
[external_agent_tool_result]
"?Couldn't processCan't reach KhelSutra. Is it running?Try again"
[/external_agent_tool_result]
```

367. assistant (2026-07-02T06:11:33.490Z)

The A8 error chip renders live: `***? Couldn't process**` + the friendly message + Try again. Let me grab a screenshot for proof, then review.

368. assistant (2026-07-02T06:11:33.491Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_screenshot]
input: {"serverId":"d4a4a2a4-7c2e-46e2-889e-f619477293b4"}
[/external_agent_tool_call]
```

369. assistant (2026-07-02T06:11:33.491Z)

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

370. assistant (2026-07-02T06:11:33.491Z)

A8 error chip renders cleanly (`***? Couldn't process**` + message + Try again). Working/building/done states are covered by the 5 new tests. Let me capture the diff and run a lean adversarial review before committing.

371. assistant (2026-07-02T06:11:33.491Z)

```
[external_agent_tool_call: Bash]
description: Capture A8 component diff
command: cd /c/Users/avidu/Projects/khelsutra && git add -N web/src/components/StatusChip.tsx
web/src/components/a8.status.test.tsx && git --no-pager diff --
web/src/components/IngestPanel.tsx web/src/components/CompileResult.tsx
web/src/components/StatusChip.tsx | cat
[/external_agent_tool_call]
```

372. assistant (2026-07-02T06:11:33.491Z)

```
[external_agent_tool_result]
diff --git a/web/src/components/CompileResult.tsx b/web/src/components/CompileResult.tsx
index 619d22b..04c40a5 100644
--- a/web/src/components/CompileResult.tsx
+++ b/web/src/components/CompileResult.tsx
@@ -4,6 +4,7 @@
 // failure (e.g. the min_shot_count 400) we show the engine's reason ? loud, never swallowed.

 import { useTranslation } from "react-i18next";
+import { StatusChip, Spinner } from "../StatusChip";

 export type BuildState =
   | { status: "idle" }

@@ -18,18 +19,29 @@ interface Props {

 export function CompileResult({ build, onDismiss }: Props) {
   const { t } = useTranslation();
-  if (build.status === "idle" || build.status === "building") return null;
+  if (build.status === "idle") return null;
+  if (build.status === "building") return null;

+  // Build state (A8): a visible "stitching?" card with an icon+text chip + reassurance, so the
+  // user
+  // isn't staring at a frozen footer wondering whether anything is happening.
+  if (build.status === "building") {
+    return (
+      <section data-testid="compile-result" className="mt-6 rounded-xl border border-black/10
bg-paper p-4" aria-live="polite">
+        <StatusChip tone="info" icon={<Spinner />} label={t("result.building")} />
+        <p className="mt-2 text-sm text-ink/60">{t("result.buildingReassure")}</p>
+      </section>
+    );
+  }
 }
```



```

+
+ const isError = build.status === "error";
+ return (
+   <section
+     data-testid="compile-result"
+     className={
+       "mt-6 rounded-xl border p-4 " +
+       (build.status === "error" ? "border-red-200 bg-red-50" : "border-green/30 bg-green/5")
+     }
+     className={"mt-6 rounded-xl border p-4 " + (isError ? "border-red-200 bg-red-50" :
"border-green/30 bg-green/5")}
+   >
+     <div className="flex items-start justify-between gap-3">
+       <h2 className="font-semibold text-ink">
+         {/* Status as icon+text (never icon-only): ? ready / ? failed (A8). */}
+       <h2 className="flex items-center gap-2 font-semibold text-ink">
+         <span aria-hidden="true" className={isError ? "text-red-700" : "text-green"}>{isError
? "?" : "?"}</span>
+         {build.status === "done" ? t("result.ready") : t("result.failed")}
+       </h2>
+       <button
diff --git a/web/src/components/IngestPanel.tsx b/web/src/components/IngestPanel.tsx
index 87607f8..6f0c37f 100644
--- a/web/src/components/IngestPanel.tsx
+++ b/web/src/components/IngestPanel.tsx
@@ -9,6 +9,7 @@ import type { FormEvent } from "react";
import { useTranslation } from "react-i18next";
import { useIngestJob } from "../hooks/useIngestJob";
import { errorText } from "../lib/errorText";
+import { StatusChip, Spinner } from "../StatusChip";

interface Props {
  /** Called with the engine's video_id once a submitted job finishes successfully. */
@@ -66,23 +67,29 @@ export function IngestPanel({ onLoad }: Props) {

  <p className="mt-2 text-xs text-ink/40">{t("ingest.hint")}</p>

-  {/* Live status (aria-live so screen readers hear progress). Indeterminate: the engine
reports
-    phase words ("hashing", "segmenting"), not a percentage ? so a spinner + the phase
word. */}
+  {/* Live status (aria-live so screen readers hear progress) as icon+text chips ? never
icon-only
+    (A8). Indeterminate: the engine reports phase words ("hashing", "segmenting"), not a
%. */}
  <div aria-live="polite" className="mt-3 text-sm" data-testid="ingest-status">
    {busy && (
-      <p className="flex items-center gap-2 text-ink/70">
-        <Spinner />
-        <span>
-          {t("ingest.working")}
-          {phase.status === "polling" && phase.progress
-            ? ` ` $${t("ingest.progress", { progress: phase.progress })}`
-            : ""}
-        </span>
-      </p>
+      <div>
+        <StatusChip
+          tone="info"
+          icon={<Spinner />}
+          label={
+            t("ingest.working") +
+            (phase.status === "polling" && phase.progress
+
[/external_agent_tool_result]

```

373. assistant (2026-07-02T06:11:33.491Z)

Clean diff. One tidy-up before review: the `success` tone in `StatusChip` is currently unused (the done header uses an inline ?). Let me drop it to avoid dead code (trivial to re-add):

374. assistant (2026-07-02T06:11:33.491Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\StatusChip.tsx
[/external_agent_tool_call]
```

375. assistant (2026-07-02T06:11:33.491Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\StatusChip.tsx has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

376. assistant (2026-07-02T06:11:33.491Z)

Now the A8 adversarial review (a11y/correctness + faithfulness):

377. assistant (2026-07-02T06:11:33.491Z)

```
[external_agent_tool_call: Workflow]
input: {"script":"export const meta = {\n  name: 'a8-review',\n  description: 'Adversarial
review of the A8 status-chips + reassurance diff',\n  phases: [{ title: 'Review' }],\n}\n\nconst
BASE = 'C:/Users/avidu/Projects/khelsutra/web/src'\n\nconst SCHEMA = {\n  type: 'object',\n
additionalProperties: false,\n  properties: {\n    summary: { type: 'string' },\n    findings:
{\n      type: 'array',\n      items: {\n        type: 'object',\n        additionalProperties:
false,\n        properties: {\n          severity: { type: 'string', enum: ['blocker', 'major',
'minor', 'nit'] },\n          file: { type: 'string' },\n          location: { type: 'string'
},\n          title: { type: 'string' },\n          detail: { type: 'string' },\n
suggestion: { type: 'string' },\n          },\n          required: ['severity', 'file', 'title',
'detail'],\n        },\n      },\n    },\n    required: ['summary', 'findings'],\n}\n\nconst CONTEXT =
`A8 under ${BASE}: background-processing reassurance + status as icon+text chips (never icon-
only).\n- NEW components/StatusChip.tsx: <StatusChip icon label tone={info|error}/> (icon aria-
hidden, label is the text) + exported <Spinner/>.\n- IngestPanel.tsx: working state ->
StatusChip(info, Spinner, \"Processing your video?[· phase]\") + a reassurance
<p>ingest.reassure</p>; error state -> StatusChip(error, \"?\", ingest.statusFailed) above the
message; removed its local Spinner (now imported).\n- CompileResult.tsx: NEW building branch ->
a card with StatusChip(info, Spinner, result.building) + <p>result.buildingReassure</p>;
done/error headers gain an inline ?? (aria-hidden) paired with result.ready/result.failed.\n-
Catalogs (en/kn/hi): +ingest.reassure, +ingest.statusFailed, +result.building,
+result.buildingReassure.\n141 web tests green; tsc + build clean.`\n\nconst LENSES = [\n {
key: 'a11y-correctness',\n  prompt: `Review for ACCESSIBILITY + CORRECTNESS.
${CONTEXT}`\n}\n\nRead components/{StatusChip,IngestPanel,CompileResult}.tsx + the...
[/external_agent_tool_call]
```

378. assistant (2026-07-02T06:11:33.491Z)

```
[external_agent_tool_result]
Workflow launched in background. Task ID: whphp65hw
Summary: Adversarial review of the A8 status-chips + reassurance diff
Transcript dir: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\2ba49afa-99f4-41f2-86a2-27a289d432f3\subagents\workflows\wf_268ea95a-a09
Script file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-
khelsutra\2ba49afa-99f4-41f2-86a2-27a289d432f3\workflows\scripts\a8-review-wf_268ea95a-a09.js
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath:
"C:\Users\avidu\claude\projects\C--Users-avidu-Projects-
khelsutra\2ba49afa-99f4-41f2-86a2-27a289d432f3\workflows\scripts\a8-review-wf_268ea95a-a09.js"})
to iterate without resending the script.)
Run ID: wf_268ea95a-a09
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\claude\projects\C--
```

```
Users-avidu-Projects-khelsutra\2ba49afa-99f4-41f2-86a2-27a289d432f3\workflows\scripts\a8-review-wf_268ea95a-a09.js", resumeFromRunId: "wf_268ea95a-a09"}) ? completed agents return cached results.
```

You will be notified when it completes. Use /workflows to watch live progress.
[/external_agent_tool_result]

379. assistant (2026-07-02T06:11:33.491Z)

While the A8 review runs, let me re-verify tests after the `success`-tone removal, and prep B3 by reading the `site/index.html` head/structure.

380. assistant (2026-07-02T06:11:33.491Z)

```
[external_agent_tool_call: Bash]
description: Re-verify after success-tone removal
command: cd /c/Users/avidu/Projects/khelsutra/web && npm test --silent 2>/dev/null | grep -E
"Test Files|Tests " | tail -3 && npx tsc --noEmit && echo "tsc OK"
[/external_agent_tool_call]
```

381. assistant (2026-07-02T06:11:33.491Z)

```
[external_agent_tool_result]
Test Files 15 passed (15)
Tests 141 passed (141)
tsc OK
[/external_agent_tool_result]
```

382. assistant (2026-07-02T06:11:33.491Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\site\public\index.html
[/external_agent_tool_call]
```

383. assistant (2026-07-02T06:11:33.491Z)

```
[external_agent_tool_result]
1      <!doctype html>
2      <html lang="en">
3      <head>
4      <meta charset="utf-8" />
5      <meta name="viewport" content="width=device-width, initial-scale=1" />
6      <title>KhelSutra ? every rally indexed, the dead time gone</title>
7      <meta name="description" content="KhelSutra turns a full badminton match into a clean
index of rallies with the dead time stripped out ? a highlight reel you can watch and download.
Invite-only alpha, Bengaluru." />
8      <meta property="og:title" content="KhelSutra ? every rally indexed, the dead time gone"
/>
9      <meta property="og:description" content="A full match becomes a clean highlight reel ?
only the real play, no dead time. Invite-only alpha for academies and players in Bengaluru." />
10     <meta property="og:type" content="website" />
11     <meta property="og:url" content="https://khelsutra.guru" />
12     <meta property="og:image" content="https://khelsutra.guru/og.png" />
13     <meta property="og:image:width" content="1200" />
14     <meta property="og:image:height" content="630" />
15     <meta name="twitter:card" content="summary_large_image" />
16     <meta name="twitter:title" content="KhelSutra ? every rally indexed, the dead time gone"
/>
17     <meta name="twitter:description" content="A full match becomes a clean highlight reel ?
only the real play, no dead time. Invite-only alpha, Bengaluru." />
18     <meta name="twitter:image" content="https://khelsutra.guru/og.png" />
19     <meta name="theme-color" content="#0C261F" />
20     <link rel="icon" href="data:image/svg+xml,%3Csvg xmlns='http://www.w3.org/2000/svg'
viewBox='0 0 32 32'%3E%3Crect width='32' height='32' rx='8' fill='%230C261F'/%3E%3Cpath d='M13
```

```

10 L23 16 L13 22 Z' fill='%2319B36B'/%3E%3C/svg%3E" />
21 <link rel="preconnect" href="https://fonts.googleapis.com" />
22 <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />
23 <link href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@400;500;600;
700;800&display=swap" rel="stylesheet" />
24 <style>
25 :root{
26 --ink:#0C261F; --ink2:#123A2E; --green:#19B36B; --green-d:#149A5C;
27 --lime:#C7F2B0; --paper:#F6F8F4; --card:#fff;
28 --tx:#14271F; --mut:#566B61; --line:#E4E9E2;
29 --on-dark:#EAF3EE; --on-dark-mut:#A7C3B7;
30 --r:14px; --rs:10px; --maxw:1080px;
31 --font:'Plus Jakarta Sans',system-ui,-apple-system,Segoe
UI,Roboto,Helvetica,Arial,sans-serif;
32 }
33 *{box-sizing:border-box}
34 html{scroll-behavior:smooth}
35 body{margin:0;font-family:var(--font);color:var(--tx);background:var(--paper);line-
height:1.65;-webkit-font-smoothing:antialiased}
36 a{color:inherit}
37 .wrap{max-width:var(--maxw);margin:0 auto;padding:0 22px}
38 h1,h2,h3{line-height:1.15;margin:0;letter-spacing:-.02em}
39 h1{font-size:clamp(2rem,5vw,3.4rem);font-weight:800}
40 h2{font-size:clamp(1.5rem,3.2vw,2.2rem);font-weight:700}
41 h3{font-size:1.15rem;font-weight:700}
42 p{margin:.6rem 0}
43 .btn{display:inline-flex;align-items:center;gap:.5rem;font-weight:600;font-
size:1rem;border-radius:999px;padding:.78rem 1.4rem;text-decoration:none;border:1px solid
transparent;cursor:pointer;transition:.15s}
44 .btn-primary{background:var(--green);color:#04140D}
45 .btn-primary:hover{background:var(--green-d)}
46 .btn-ghost{background:transparent;border-color:rgba(255,255,255,.28);color:var(--on-
dark)}
47 .btn-ghost:hover{border-color:rgba(255,255,255,.6)}
48 .btn-dark{background:var(--ink);color:#fff}
49 .btn-dark:hover{background:var(--ink2)}
50 .btn[disabled]{opacity:.75;cursor:progress}
51 /* in-flight spinner for the request-access submit */
52 .spin{display:inline-block;width:1em;height:1em;border:2px solid currentColor;border-
right-color:transparent;border-radius:50%;animation:spin .6s linear infinite}
53 @keyframes spin{to{transform:rotate(360deg)}}
54 .eyebrow{display:inline-flex;align-items:center;gap:.5rem;font-size:.82rem;font-
weight:600;letter-spacing:.02em;
55 background:rgba(25,179,107,.14);color:#0a6c45;border:1px solid
rgba(25,179,107,.3);padding:...
[/external_agent_tool_result]

```

384. assistant (2026-07-02T06:11:33.492Z)

B3 is the next item but it carries an explicit open owner decision (the catalog-hoist approach) and is architecturally significant ? I'll surface that after A8 merges. Let me wait for the A8 review.

385. assistant (2026-07-02T06:11:33.492Z)

```

[external_agent_tool_call: TaskOutput]
input: {"task_id":"whphp65hw","block":true,"timeout":240000}
[/external_agent_tool_call]

```

386. assistant (2026-07-02T06:11:33.492Z)

```

[external_agent_tool_result]
<retrieval_status>success</retrieval_status>

```

```

<task_id>whphp65hw</task_id>

<task_type>local_workflow</task_type>

<status>completed</status>

<output>
{
  "summary": "Adversarial review of the A8 status-chips + reassurance diff",
  "agentCount": 2,
  "logs": [],
  "result": [
    {
      "lens": "a11y-correctness",
      "summary": "A8 reviewed and verified clean.",
      "findings": [
        {
          "severity": "minor",
          "file": "web/src/components/CompileResult.tsx",
          "title": "aria-live on conditionally-mounted building section; done/error has none",
          "detail": "IngestPanel uses a stable always-mounted aria-live region. CompileResult puts aria-live on the building section only, and done/error has none, so SRs may not announce the terminal status. Tests assert text not announcement. Minor.",
          "location": "line 28 and lines 36-40",
          "suggestion": "Use one stable always-mounted aria-live wrapper, or add aria-live to the done/error section."
        }
      ]
    },
    {
      "lens": "faithfulness",
      "summary": "A8 faithful and verified locally.",
      "findings": [
        {
          "severity": "nit",
          "file": "web/src/i18n/locales/en/common.json",
          "location": "result.buildingReassure line 63",
          "title": "under a minute is the one timed promise",
          "detail": "Only line naming a time. Hedge usually is honest and stitch is fast concat. A slow box or long reel can exceed a minute. kn and hi share it.",
          "suggestion": "Optional. Drop the hard number to match ingest.reassure. As is is fine."
        }
      ]
    }
  ]
}
</output>
[/external_agent_tool_result]

```

387. assistant (2026-07-02T06:11:33.492Z)

A8 review: faithful + clean, two small items worth folding (both improve a11y/honesty, which the owner's launch bar values):

1. **minor** ? the done/error CompileResult section lacks `aria-live` (only the building card has it), so a screen reader may not announce the terminal status.
2. **nit** ? `result.buildingReassure` is the one hard timed promise ("under a minute"); soften to a hedge.

Applying both:

388. assistant (2026-07-02T06:11:33.492Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\khelsutra\web\src\components\CompileResult.tsx

[/external_agent_tool_call]

389. assistant (2026-07-02T06:11:33.492Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\khelsutra\web\src\components\CompileResult.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

390. assistant (2026-07-02T06:11:33.492Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json

[/external_agent_tool_call]

391. assistant (2026-07-02T06:11:33.492Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

392. assistant (2026-07-02T06:11:33.492Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json

[/external_agent_tool_call]

393. assistant (2026-07-02T06:11:33.492Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

394. assistant (2026-07-02T06:11:33.492Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\hi\common.json

[/external_agent_tool_call]

395. assistant (2026-07-02T06:11:33.492Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\hi\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

396. assistant (2026-07-02T06:11:33.492Z)

The A8 test asserts the old reassurance copy ? update it to match:

397. assistant (2026-07-02T06:11:33.492Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\khelsutra\web\src\components\a8.status.test.tsx

[/external_agent_tool_call]

398. assistant (2026-07-02T06:11:33.492Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\khelsutra\web\src\components\a8.status.test.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)

```
[/external_agent_tool_result]
```

399. assistant (2026-07-02T06:11:33.492Z)

```
[external_agent_tool_call: Bash]
description: Re-run A8 tests after folding review findings
command: npm test --silent 2>/dev/null | grep -E "Test Files|Tests |FAIL" | tail -3 && npx tsc
--noEmit && echo "tsc OK" && npm run build --silent 2>&1 | tail -1
[/external_agent_tool_call]
```

400. assistant (2026-07-02T06:11:33.492Z)

```
[external_agent_tool_result]
  Test Files   15 passed (15)
     Tests    141 passed (141)
tsc OK
? built in 1.28s
[/external_agent_tool_result]
```

401. assistant (2026-07-02T06:11:33.492Z)

141 tests, tsc, build all green after folding the A8 findings. Committing A8:

402. assistant (2026-07-02T06:11:33.493Z)

```
[external_agent_tool_call: Bash]
description: Stage A8 paths, verify clean base
command: cd /c/Users/avidu/Projects/khelsutra && echo "branch: $(git branch --show-current)" &&
git add web/src/components/StatusChip.tsx web/src/components/IngestPanel.tsx
web/src/components/CompileResult.tsx web/src/components/a8.status.test.tsx
web/src/i18n/locales/en/common.json web/src/i18n/locales/kn/common.json
web/src/i18n/locales/hi/common.json && git diff --cached --name-only && echo "--- base check
(only my work) ---" && git log --oneline origin/master..HEAD
[/external_agent_tool_call]
```

403. assistant (2026-07-02T06:11:33.493Z)

```
[external_agent_tool_result]
branch: feat/a8-status-chips
web/src/components/CompileResult.tsx
web/src/components/IngestPanel.tsx
web/src/components/StatusChip.tsx
web/src/components/a8.status.test.tsx
web/src/i18n/locales/en/common.json
web/src/i18n/locales/hi/common.json
web/src/i18n/locales/kn/common.json
--- base check (only my work) ---
[/external_agent_tool_result]
```

404. assistant (2026-07-02T06:11:33.493Z)

```
[external_agent_tool_call: Bash]
description: Commit + push A8
command: git commit -q -F - <<'EOF'
feat(web): A8 ? background-processing reassurance + icon+text status chips
```

Honest expectations for non-tech users on slow uplinks, status never icon-only:

- New StatusChip (always icon + text, color is never the only signal) + a shared Spinner.
- IngestPanel: the processing state is now a working chip + a "safe to wait ? runs on the server" reassurance line; failures show a ? "Couldn't process" chip above the friendly message.
- CompileResult: a new building state shows a "Stitching your reel?" chip + reassurance (so the user isn't staring at a frozen footer); the done/failed headers gain a ?? icon paired with the text, and the terminal card is aria-live for screen readers.

- Catalogs (en/kn/hi): +ingest.reassure, +ingest.statusFailed, +result.building, +result.buildingReassure (hedged copy ? no hard time promise).

Adds a8.status.test.tsx (building/done/error chips + ingest reassurance/error chip).
141 web tests green; tsc + build clean; key-parity + no-bare-strings hold. Reviewed
via a 2-lens adversarial pass; folded its a11y (aria-live) + honest-copy findings.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>
EOF
git push -u origin feat/a8-status-chips 2>&1 | tail -2
[/external_agent_tool_call]

405. assistant (2026-07-02T06:11:33.493Z)

[external_agent_tool_result]
To https://github.com/avidullu/khelsutra.git
* [new branch] feat/a8-status-chips -> feat/a8-status-chips
[/external_agent_tool_result]

406. assistant (2026-07-02T06:11:33.493Z)

[external_agent_tool_call: Bash]
description: Create A8 PR
command: gh pr create --title "feat(web): A8 ? background-processing reassurance + icon+text status chips" --base master --head feat/a8-status-chips --body "\$(cat <<'EOF'

A8 ? honest expectations + icon+text status (tracker `FRIENDLY\_AND\_MULTILINGUAL.md` §7)

For a non-tech user on a slow uplink, a long silent wait reads as "broken". This adds
background-processing reassurance and shows status as **icon + text chips (never icon-
only)** across both the ingest progress and the build state.

What changed

- **New `StatusChip`** ? always icon **** text (color is never the only signal, per the HCI4D guidance in §4) + a shared `Spinner`.
- **`IngestPanel`**: processing is now a *working* chip + a "safe to wait ? KhelSutra keeps working on the server" reassurance line; a failure shows a **? "Couldn't process" chip above the friendly message.
- **`CompileResult`**: a new **building** state ? a "Stitching your reel?" chip + reassurance, so the user isn't staring at a frozen footer; the **done / failed** headers gain a **? / ?** icon paired with the text, and the terminal card is `aria-live` so a screen reader announces the outcome.
- **Catalogs (en/kn/hi)**: `+ingest.reassure`, `+ingest.statusFailed`, `+result.building`, `+result.buildingReassure` (hedged ? no hard time promise).

Verification

- **141 web tests** green (new `a8.status.test.tsx` covers building/done/error chips + ingest reassurance + the failure chip); `tsc` + `vite build` clean; **key-parity** + **no-bare-strings** hold.
- **Browser-verified**: the ingest failure chip renders live as **? Couldn't process** + "Can't reach KhelSutra. Is it running?" + Try again.
- **Reviewed by a 2-lens adversarial workflow** (a11y/correctness + faithfulness); folded both findings ? added `aria-live` to the terminal CompileResult card, and softened the build reassurance to a hedge (no hard "under a minute" claim).

Honesty notes

- kn/hi strings are AI-drafted (native review pending).
- Screen-reader announcement of a freshly-mounted live region varies by SR; this is strictly better than the prior no-`aria-live` state. A fully stable always-mounted region is a possible later refinement.

? Generated with [Claude Code](https://claude.com/claude-code)
EOF
)" 2>&1 | tail -2
[/external_agent_tool_call]

407. assistant (2026-07-02T06:11:33.493Z)

```
[external_agent_tool_result]
https://github.com/avidullu/khelsutra/pull/63
[/external_agent_tool_result]
```

408. assistant (2026-07-02T06:11:33.493Z)

PR [#63](https://github.com/avidullu/khelsutra/pull/63) created. Marking A8 ? in the tracker:

409. assistant (2026-07-02T06:11:33.493Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\docs\FRIENDLY_AND_MULTILINGUAL.md
[/external_agent_tool_call]
```

410. assistant (2026-07-02T06:11:33.493Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\docs\FRIENDLY_AND_MULTILINGUAL.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

411. assistant (2026-07-02T06:11:33.493Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\docs\FRIENDLY_AND_MULTILINGUAL.md
[/external_agent_tool_call]
```

412. assistant (2026-07-02T06:11:33.493Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\docs\FRIENDLY_AND_MULTILINGUAL.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

413. assistant (2026-07-02T06:11:33.493Z)

```
[external_agent_tool_call: Bash]
description: Commit tracker, push, check A8 CI
command: git add docs/FRIENDLY_AND_MULTILINGUAL.md && git commit -q -m "docs(a8): mark A8 (#63)
done in the $? tracker + changelog
```

```
Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>" && git push -q 2>&1 | tail -1 && echo
"pushed" && sleep 25 && gh pr checks 63 2>&1 | head
[/external_agent_tool_call]
```

414. assistant (2026-07-02T06:11:33.493Z)

```
[external_agent_tool_result]
pushed
web      pending 0
https://github.com/avidullu/khelsutra/actions/runs/27909148385/job/82583116449
site     pass    15s
https://github.com/avidullu/khelsutra/actions/runs/27909148385/job/82583116448
Workers Builds: hidden-king-khelsutra pending 0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/33bb9960-d839-41d5-958c-57e77871035c
[/external_agent_tool_result]
```

415. assistant (2026-07-02T06:11:33.494Z)

```
[external_agent_tool_call: Bash]
```

```
description: Poll A8 CI
command: for i in $(seq 1 8); do s=$(gh pr checks 63 2>&1 | grep -E "^web\b" | awk '{print $2}'); c=$(gh pr checks 63 2>&1 | grep -iE "Workers Builds" | awk '{print $4}'); echo "poll $i: web=$s cf=$c"; if { [ "$s" = "pass" ] || [ "$s" = "fail" ]; } && { [ "$c" = "pass" ] || [ "$c" = "fail" ]; }; then break; fi; sleep 15; done && echo "=== final ===" && gh pr checks 63 2>&1 | head
[/external_agent_tool_call]
```

416. assistant (2026-07-02T06:11:33.494Z)

```
[external_agent_tool_result]
poll 1: web=pending cf=pass
poll 2: web=pass cf=pass
=== final ===
Workers Builds: hidden-king-khelsutra pass 0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-khelsutra/production/builds/33bb9960-d839-41d5-958c-57e77871035c
site pass 15s
https://github.com/avidullu/khelsutra/actions/runs/27909148385/job/82583116448
web pass 32s
https://github.com/avidullu/khelsutra/actions/runs/27909148385/job/82583116449
[/external_agent_tool_result]
```

417. assistant (2026-07-02T06:11:33.494Z)

All green. Merging A8:

418. assistant (2026-07-02T06:11:33.494Z)

```
[external_agent_tool_call: Bash]
description: Squash-merge A8 and sync master
command: gh pr merge 63 --squash --delete-branch 2>&1 | tail -2 && git checkout master -q && git pull --ff-only -q && git log --oneline -3
[/external_agent_tool_call]
```

419. assistant (2026-07-02T06:11:33.494Z)

```
[external_agent_tool_result]
create mode 100644 web/src/components/StatusChip.tsx
create mode 100644 web/src/components/a8.status.test.tsx
92a6c45 feat(web): A8 ? background-processing reassurance + icon+text status chips (#63)
501c014 feat(web): A5 ? friendly engine-error mapping across build + load (shared errors.*) (#62)
95195a5 feat(web): A4 ? mobile touch targets (?44px) + visible tooltip text (#61)
[/external_agent_tool_result]
```

420. assistant (2026-07-02T06:11:33.494Z)

A8 merged (#63 ? `92a6c45`). **All three SPA UX A-items (A4, A5, A8) are done and on master.**

421. assistant (2026-07-02T06:11:33.494Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"3","status":"completed"}
[/external_agent_tool_call]
```

422. assistant (2026-07-02T06:11:33.494Z)

```
[external_agent_tool_result]
Updated task #3 status
[/external_agent_tool_result]
```

423. assistant (2026-07-02T06:11:33.494Z)

```
[external_agent_tool_call: mcp_ccd_session__mark_chapter]
input: {"title": "B3 / North-Star decision point", "summary": "A4/A5/A8 merged (#61/#62/#63).
Surfacing the B3 catalog-hoist decision + next-priority before the bigger items."}
[/external_agent_tool_call]
```

424. assistant (2026-07-02T06:11:33.494Z)

```
[external_agent_tool_result]
Chapter marked: "B3 / North-Star decision point". Continue your current work.
[/external_agent_tool_result]
```

425. assistant (2026-07-02T06:11:33.494Z)

Three clean PRs merged this session (A4 [#61](https://github.com/avidullu/khelsutra/pull/61), A5 [#62](https://github.com/avidullu/khelsutra/pull/62), A8 [#63](https://github.com/avidullu/khelsutra/pull/63)) ? each browser-verified, adversarially reviewed, CI-green, tracker updated.

The next items diverge, and two carry genuine calls that are yours. **B3** is flagged in your handoff §3 as an open decision ("confirm the catalog-hoist"), it's architecturally significant (adds a build step to the deliberately build-free `site/`), and it's a large translation-infra lift. **T4** (in-UI upload) is your stated North-Star #1 and is fully autonomous. So before I commit to the bigger work, let me confirm direction.

426. assistant (2026-07-02T06:11:33.494Z)

```
[external_agent_tool_call: AskUserQuestion]
input: {"questions":[{"question": "A4/A5/A8 are merged. Which should I build
next?", "header": "Next up", "multiSelect": false, "options": [{"label": "T4 ? in-UI upload (North
Star)", "description": "Wire the SPA to the engine's existing chunked-upload endpoints so alpha
users can upload a FILE (not just a gs:// URI) ? process ? reel. Your stated #1 priority, and
fully autonomous (no owner gate). Big but high-value."}, {"label": "B3 ? site/ trilingual
plumbing", "description": "The shared t() shim + data-i18n + switcher + Accept-Language
negotiation over the marketing site, en-only first (kn/hi content follows). Per the mission
order, but needs the catalog-home decision below."}, {"label": "MULTILINGUAL Phase A
(es/da/id)", "description": "Add Spanish/Danish/Indonesian catalogs to the SPA (catalog-only; es
needs the 3-form ICU plural). Owner-approved (#58), smaller, mechanical."}, {"label": "T11 ?
retention sweep", "description": "Small: an output/ + uploads/ retention sweep before non-owner
exposure. You'd still set the retention window."}], {"question": "For B3, where should the single
source of truth for locale catalogs live? (Applies whenever B3 runs.)", "header": "B3
catalogs", "multiSelect": false, "options": [{"label": "Hoist to repo-root
/locales/", "description": "Recommended in the handoff. SPA + site both consume one source; a tiny
site prebuild copies them to site/public/locales/ + a drift-guard test. Cost: one new build step
on the currently build-free site, and re-pointing the SPA's 3 imports."}, {"label": "Keep in
web/src/i18n/locales/", "description": "No repo-root move; the site prebuild copies from web/'s
path. Less churn now, but couples the marketing site to the SPA's internal directory
layout."}, {"label": "Decide later", "description": "I'll fully scope B3 (shim, switcher,
negotiation, tests) and hold the catalog-home choice until you weigh in."}]}]}
[/external_agent_tool_call]
```

427. assistant (2026-07-02T06:11:33.494Z)

```
[external_agent_tool_result]
Your questions have been answered: "A4/A5/A8 are merged. Which should I build next?"="1 now but
do record 4, 2 and 3 as immediate next or feel free to spin off a parallel session to complete
them if thats reasonable", "For B3, where should the single source of truth for locale catalogs
live? (Applies whenever B3 runs.)"="Hoist to repo-root /locales/". You can now continue with
these answers in mind.
[/external_agent_tool_result]
```

428. assistant (2026-07-02T06:11:33.494Z)

The owner confirmed: **build T4 (North Star) next**; keep B3, MULTILINGUAL Phase A, and T11 recorded as immediate-next (and I can spin them off as parallel sessions once T4 is in, to avoid

catalog-file races); and **B3 will hoist catalogs to repo-root `~/locales/`** (decision locked).

Let me start T4. First, read the API types to slot in the upload helpers.

429. assistant (2026-07-02T06:11:33.494Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"5","status":"in_progress"}
[/external_agent_tool_call]
```

430. assistant (2026-07-02T06:11:33.494Z)

```
[external_agent_tool_result]
Updated task #5 status
[/external_agent_tool_result]
```

431. assistant (2026-07-02T06:11:33.494Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\web\src\api\types.ts
[/external_agent_tool_call]
```

432. assistant (2026-07-02T06:11:33.494Z)

```
[external_agent_tool_result]
1      // Types mirror the engine's verified contract (badminton-highlight-indexer).
2      // Anchors: backend/infrastructure/database.py:124-135 (play_segments),
3      // backend/main.py:331-396 (query/compile), :307-328 (highlights), :121 (config).
4
5      export interface RallyMetadata {
6          /** Gemini's shuttle_exchange_count (fallback max(3, dur/0.8)); absent on some
detectors. */
7          shot_count?: number;
8          /** Gemini confidence STRING (often "Unknown") ? not a numeric meter. */
9          shot_count_confidence?: string;
10         detector?: string;
11         [key: string]: unknown;
12     }
13
14     export interface RallySegment {
15         /** The integer the UI selects and passes to /api/compile as segment_ids. */
16         id: number;
17         start_time: number;
18         end_time: number;
19         duration: number;
20         /** motion-segmenter FABRICATION (random) ? DO NOT surface (see
PER_RALLY_SELECTION.md). */
21         server: string | null;
22         /** motion-segmenter FABRICATION (random) ? DO NOT surface. */
23         receiver: string | null;
24         metadata: RallyMetadata;
25         /** motion-only fabricated sub-events ? DO NOT surface. */
26         events?: unknown[];
27     }
28
29     export interface QueryResponse {
30         play_segments: RallySegment[];
31     }
32
33     export interface CompileResponse {
34         status: string;
35         /** server-local path (NOT a URL) ? build the download URL with highlightUrl(). */
36         filepath: string;
37     }
```

```

38
39     export interface EngineConfig {
40         stitching?: {
41             /** The shot-count floor /api/compile silently enforces ? surface it, don't hide it.
42             */
43             min_shot_count?: number;
44             [key: string]: unknown;
45         };
46         [key: string]: unknown;
47     }
48     /** POST /api/process ? 202 [backend/main.py:260]. The job is queued; poll GET
49     /api/jobs/{job_id}. */
50     export interface ProcessResponse {
51         job_id: string;
52         /** Always "queued" at submit. */
53         status: string;
54         /** Relative poll path, e.g. "/api/jobs/<id>". */
55         poll: string;
56     }
57     /** GET /api/jobs/{job_id} [backend/main.py:303]. Two status layers:
58     * `status` = the EXECUTION state; `result.status` = the pipeline VERDICT
59     (success|failed).
60     * A `done` job whose `result.status` === "failed" (or with no video_id) is a success-
61     shaped
62     * FAILURE ? the UI must not treat it as a load. */
63     export interface JobStatus {
64         job_id: string;
65         status: "queued" | "running" | "done" | "failed";
66         /** Free-text coarse phase ("hashing", "segmenting (gemini)", "finished") ? NOT a
67         percentage. */
68         progress?: string | null;
69         /** The canonical video_id, surfaced top-level on success (also in result.video_id).
70         */
71         video_id?: string | null;
72         /** Execution error (worker crash / unexpected) when status === "failed". */
73         error?: string | null;
74         result?: { status?: string; video_id?: string; message?: string } | null;
75     }
76     [/external_agent_tool_result]

```

433. assistant (2026-07-02T06:11:33.494Z)

```

[external_agent_tool_call: mcp_ccd_session__mark_chapter]
input: {"title":"T4 ? in-UI file upload (North Star)","summary":"Owner picked T4 next. Building
chunked in-UI upload wired to the existing engine endpoints."}
[/external_agent_tool_call]

```

434. assistant (2026-07-02T06:11:33.495Z)

```

[external_agent_tool_result]
Chapter marked: "T4 ? in-UI file upload (North Star)". Continue your current work.
[/external_agent_tool_result]

```

435. assistant (2026-07-02T06:11:33.495Z)

Now I have the full T4 contract. Let me add the upload types, then build the upload client, hook, and wire IngestPanel. First, types:

436. assistant (2026-07-02T06:11:33.495Z)

```

[external_agent_tool_call: Edit]

```

file: C:\Users\avidu\Projects\khelsutra\web\src\api\types.ts
[/external_agent_tool_call]

437. assistant (2026-07-02T06:11:33.495Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\api\types.ts has been updated successfully.
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

438. assistant (2026-07-02T06:11:33.495Z)

Now build the chunked-upload client. Add the upload helpers + orchestrator to `client.ts`:

439. assistant (2026-07-02T06:11:33.495Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\api\client.ts
[/external_agent_tool_call]

440. assistant (2026-07-02T06:11:33.495Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\api\client.ts has been updated successfully.
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

441. assistant (2026-07-02T06:11:33.495Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\api\client.ts
[/external_agent_tool_call]

442. assistant (2026-07-02T06:11:33.495Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\api\client.ts has been updated successfully.
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

443. assistant (2026-07-02T06:11:33.495Z)

Now the `useUpload` hook:

444. assistant (2026-07-02T06:11:33.495Z)

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\khelsutra\web\src\hooks\useUpload.ts
[/external_agent_tool_call]

445. assistant (2026-07-02T06:11:33.495Z)

[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\khelsutra\web\src\hooks\useUpload.ts (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

446. assistant (2026-07-02T06:11:33.495Z)

Now rewrite IngestPanel to add the upload mode. Let me read its current state after A8:

447. assistant (2026-07-02T06:11:33.495Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\kheIsutra\web\src\components\IngestPanel.tsx
[/external_agent_tool_call]
```

448. assistant (2026-07-02T06:11:33.495Z)

```
[external_agent_tool_result]
1      // The front-half ingest screen (appliance S1/S2, tracker B-P2). Point the tool at a
video ? a
2      // `gs://` bucket URI or a path on the engine host ? and it submits to POST
/api/process, shows
3      // live progress while polling GET /api/jobs/{id}, and hands the resulting video_id to
the existing
4      // rally-selection back-half (App lifts ?video=<id>). Honest by construction: the engine
reports
5      // coarse phase WORDS (not a %), and a success-shaped failure (done but no rallies) is
shown, never
6      // silently treated as a load.
7      import { useState } from "react";
8      import type { FormEvent } from "react";
9      import { useTranslation } from "react-i18next";
10     import { useIngestJob } from "../hooks/useIngestJob";
11     import { errorText } from "../lib/errorText";
12     import { StatusChip, Spinner } from "../StatusChip";
13
14     interface Props {
15       /** Called with the engine's video_id once a submitted job finishes successfully. */
16       onLoaded: (videoId: string) => void;
17     }
18
19     export function IngestPanel({ onLoaded }: Props) {
20       const { t } = useTranslation();
21       const [uri, setUri] = useState("");
22       const { phase, submit, cancel, reset, busy } = useIngestJob(onLoaded);
23
24       const onSubmit = (e: FormEvent) => {
25         e.preventDefault();
26         submit(uri);
27       };
28
29       // Shared mapping with the build flow (lib/errorText): `engine` failures carry the
engine's own
30       // message (a 400 scheme/path detail, a worker error); every other code maps to a
friendly
31       // localized line ? the network case especially, since this is the first screen a
fresh user hits
32       // and "is the engine running?" must read plainly.
33       const errorMsg = phase.status === "error" ? errorText(t, phase.error) : null;
34
35       return (
36         <section className="mt-6 rounded-2xl border border-black/10 bg-white p-5" data-
testid="ingest-panel">
37           <h2 className="text-lg font-bold">{t("ingest.title")}</h2>
38           <p className="mt-1 max-w-2xl text-sm text-ink/60">{t("ingest.intro")}</p>
39
40           <form onSubmit={onSubmit} className="mt-3 flex flex-wrap items-center gap-2">
41             <input
42               type="text"
43               value={uri}
44               onChange={(e) => setUri(e.target.value)}
45               placeholder={t("ingest.uriPlaceholder")}
46               aria-label={t("ingest.uriAriaLabel")}
47               disabled={busy}
48               className="min-h-11 min-w-[18rem] flex-1 rounded-xl border border-black/15 bg-
white px-4 py-2 text-sm text-ink outline-none focus:border-green disabled:opacity-50"

```

```

49         />
50         <button
51             type="submit"
52             disabled={busy}
53             className="min-h-11 rounded-xl bg-ink px-4 py-2 text-sm font-semibold text-
white disabled:opacity-40"
54         >
55             {phase.status === "submitting" ? t("ingest.submitting") : t("ingest.submit")}
56         </button>
57         {busy && (
58             <button
59                 type="button"
60                 onClick={cancel}
61                 className="min-h-11 rounded-xl border border-black/15 px-4 py-2 text-sm
hover:border-green"
62             >
63                 {t("ingest.cancel")}
64             </button>
65         )}
66     </form>
67
68     <p className="mt-2 text-xs text-ink/40">{t("ingest.hint")}</p>
69
70     {/* Live status (aria-live so screen readers hear progress) as icon+text chips ?
never icon-only
71         (A8). Indeterminate: the engine reports phase words ("hashing", "segmenting"),
not a %. */}
72     <div aria-live="polite" className="mt-3 text-sm" data-testid="ingest-status">
73         {busy && (
74             <div>
75                 <StatusChip
76                     tone="info"
77                     icon={<Spinner />}
78                     label={
79                         t("ingest.working") +
80                         (phase.status === "polling" && phase.progress
81                             ? ` · ${t("ingest.progress", { progress: phase.progress })}`
82                             : "")
83                     }
84                 />
85             </div>
86         )}
87     </div>
88     {/* Reass...
[/external_agent_tool_result]

```

449. assistant (2026-07-02T06:11:33.495Z)

Now I'll rewrite IngestPanel with a **both-visible** design (file upload primary, cloud-link secondary) ? this adds the upload path while keeping the existing URI input present, so all existing IngestPanel/A8 tests still pass. Wiring `useUpload` ? on complete ? `submit(path)`:

450. assistant (2026-07-02T06:11:33.495Z)

```

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx
[/external_agent_tool_call]

```

451. assistant (2026-07-02T06:11:33.495Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

452. assistant (2026-07-02T06:11:33.496Z)

Now the 5 upload keys (en/kn/hi):

453. assistant (2026-07-02T06:11:33.496Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json
[/external_agent_tool_call]
```

454. assistant (2026-07-02T06:11:33.496Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

455. assistant (2026-07-02T06:11:33.496Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json
[/external_agent_tool_call]
```

456. assistant (2026-07-02T06:11:33.496Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

457. assistant (2026-07-02T06:11:33.496Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\hi\common.json
[/external_agent_tool_call]
```

458. assistant (2026-07-02T06:11:33.496Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\hi\common.json has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

459. assistant (2026-07-02T06:11:33.496Z)

Now T4 tests. Let me check the existing client test conventions first:

460. assistant (2026-07-02T06:11:33.496Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\web\src\api\client.test.ts
[/external_agent_tool_call]
```

461. assistant (2026-07-02T06:11:33.496Z)

```
[external_agent_tool_result]
1      import { describe, it, expect, vi, afterEach } from "vitest";
2      import { listRallies, compileReel, highlightUrl, getConfig, minShotCount, ApiError }
from "./client";
3
4      type FetchFn = typeof fetch;
5
6      function mockFetch(status: number, json: unknown, text?: string): FetchFn {
7          return vi.fn(async () => ({
8              ok: status >= 200 && status < 300,
```

```

9      status,
10      json: async () => json,
11      text: async () => text ?? JSON.stringify(json),
12  ))) as unknown as FetchFn;
13  }
14
15  function lastCall(f: FetchFn): [string, RequestInit] {
16      const calls = (f as unknown as { mock: { calls: [string, RequestInit][] }
17  }).mock.calls;
18      return calls[calls.length - 1];
19  }
20
21  afterEach(() => {
22      vi.restoreAllMocks();
23      vi.unstubAllGlobals();
24  });
25
26  describe("highlightUrl", () => {
27      it("builds an encoded same-origin path", () => {
28          expect(highlightUrl("abc123", "my
29  reel.mp4")).toBe("/api/highlights/abc123/my%20reel.mp4");
30      });
31  });
32
33  describe("listRallies", () => {
34      it("POSTs video_id to /api/query and returns play_segments", async () => {
35          const seg = { id: 1, start_time: 0, end_time: 5, duration: 5, server: null,
36  receiver: null, metadata: { shot_count: 7 } };
37          const f = mockFetch(200, { play_segments: [seg] });
38          vi.stubGlobal("fetch", f);
39
40          const rallies = await listRallies("vid");
41          expect(rallies).toHaveLength(1);
42          expect(rallies[0].id).toBe(1);
43
44          const [url, init] = lastCall(f);
45          expect(url).toBe("/api/query");
46          expect(init.method).toBe("POST");
47          expect(JSON.parse(init.body as string)).toEqual({ video_id: "vid" });
48          expect((init.headers as Record<string, string>)["x-correlation-id"]).toBeTruthy();
49      });
50
51      it("returns [] when play_segments is missing", async () => {
52          vi.stubGlobal("fetch", mockFetch(200, {}));
53          expect(await listRallies("vid")).toEqual([]);
54      });
55  });
56
57  describe("compileReel", () => {
58      it("POSTs the selection to /api/compile", async () => {
59          const f = mockFetch(200, { status: "success", filepath: "/out/reel.mp4" });
60          vi.stubGlobal("fetch", f);
61
62          const res = await compileReel("vid", [1, 2, 3], "reel.mp4");
63          expect(res.status).toBe("success");
64
65          const [url, init] = lastCall(f);
66          expect(url).toBe("/api/compile");
67          expect(JSON.parse(init.body as string)).toEqual({
68              video_id: "vid",
69              segment_ids: [1, 2, 3],
70              output_filename: "reel.mp4",
71          });
72      });
73  });

```

```

71
72 describe("getConfig + minShotCount", () => {
73   it("reads the stitching floor", async () => {
74     vi.stubGlobal("fetch", mockFetch(200, { stitching: { min_shot_count: 4 } }));
75     const cfg = await getConfig();
76     expect(minShotCount(cfg)).toBe(4);
77   });
78
79   it("returns undefined when no floor is configured", async () => {
80     vi.stubGlobal("fetch", mockFetch(200, {}));
81     expect(minShotCount(await getConfig())).toBeUndefined();
82   });
83 });
84
85 describe("error handling (loud, never silent)", () => {
86   it("throws ApiError with status + correlation id on a non-ok response", async () => {
87     vi.stubGlobal("fetch", mockFetch(400, { detail: "bad" }, "bad request"));
88     await expect(listRallies("vid")).rejects.toMatchObject({ name: "ApiError", status:
400 });
89   });
90
91   it("throws ApiError on a network failure", async () => {
92     vi.stubGlobal("fetch", vi.fn(async () => { throw new Error("offline"); }) as unknown
as FetchFn);
93     await expect(getConfig()).rejects.toBeInstanceOf(ApiError);
94   });
95 });
96
[/external_agent_tool_result]

```

462. assistant (2026-07-02T06:11:33.496Z)

Let me check how the existing hook test drives a hook (renderHook vs wrapper):

463. assistant (2026-07-02T06:11:33.496Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\web\src\hooks\useIngestJob.test.ts
[/external_agent_tool_call]

```

464. assistant (2026-07-02T06:11:33.496Z)

```

[external_agent_tool_result]
1      // @vitest-environment jsdom
2      //
3      // The ingest state machine: submit ? poll (queued/running) ? done | error. Drives the
recursive
4      // setTimeout poll loop with fake timers so each tick is deterministic, and pins the
load-bearing
5      // behaviours: the success-shaped-failure (done + no rallies ? emptyResult, never a
load), the
6      // two error surfaces (engine detail vs friendly network), and cancellation stopping the
loop.
7      import { describe, it, expect, vi, afterEach } from "vitest";
8      import { renderHook, act } from "@testing-library/react";
9      import { useIngestJob } from "../useIngestJob";
10     import type { JobStatus } from "../api/types";
11
12     type FetchFn = typeof fetch;
13
14     function httpOk(status: number, json: unknown) {
15       return { ok: status >= 200 && status < 300, status, json: async () => json, text:
async () => JSON.stringify(json) };
16     }

```

```

17
18     /** A fetch that answers POST /api/process with `process`, then walks `jobs` on each GET
/api/jobs/*. */
19     function fetchSeq(process: unknown, jobs: JobStatus[]): FetchFn {
20         let i = 0;
21         return vi.fn(async (url: string | URL, init?: RequestInit) => {
22             const u = String(url);
23             const method = init?.method ?? "GET";
24             if (u === "/api/process" && method === "POST") return process;
25             if (u.startsWith("/api/jobs/")) {
26                 const body = jobs[Math.min(i, jobs.length - 1)];
27                 i += 1;
28                 return httpOk(200, body);
29             }
30             throw new Error(`unexpected fetch ${method} ${u}`);
31         }) as unknown as FetchFn;
32     }
33
34     /** Flush the submit microtasks, then walk the backoff ladder (1000 ? 1500 ? 2250 ? ?)
for n ticks. */
35     async function drain(ticks: number) {
36         await act(async () => {
37             await vi.advanceTimersByTimeAsync(0); // flush processVideo ? schedule the first
poll
38         });
39         let delay = 1000;
40         for (let k = 0; k < ticks; k += 1) {
41             await act(async () => {
42                 await vi.advanceTimersByTimeAsync(delay);
43             });
44             delay = Math.min(delay * 1.5, 5000);
45         }
46     }
47
48     afterEach(() => {
49         vi.restoreAllMocks();
50         vi.unstubAllGlobals();
51         vi.useRealTimers();
52     });
53
54     describe("useIngestJob", () => {
55         it("rejects an empty URI before any network call", () => {
[/external_agent_tool_result]

```

465. assistant (2026-07-02T06:11:33.496Z)

Now the T4 tests. First, the upload client (chunking + abort-cleanup + error):

466. assistant (2026-07-02T06:11:33.496Z)

```

[/external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\khelsutra\web\src\api\upload.test.ts
[/external_agent_tool_call]

```

467. assistant (2026-07-02T06:11:33.496Z)

```

[/external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\khelsutra\web\src\api\upload.test.ts (file
state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

468. assistant (2026-07-02T06:11:33.497Z)

Now the `useUpload` hook test:

469. assistant (2026-07-02T06:11:33.497Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\khelsutra\web\src\hooks\useUpload.test.ts
[/external_agent_tool_call]
```

470. assistant (2026-07-02T06:11:33.497Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\khelsutra\web\src\hooks\useUpload.test.ts
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

471. assistant (2026-07-02T06:11:33.497Z)

Now the IngestPanel upload?process?onLoaded integration test:

472. assistant (2026-07-02T06:11:33.497Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\upload.test.tsx
[/external_agent_tool_call]
```

473. assistant (2026-07-02T06:11:33.497Z)

```
[external_agent_tool_result]
File created successfully at:
C:\Users\avidu\Projects\khelsutra\web\src\components\upload.test.tsx (file state is current in
your context ? no need to Read it back)
[/external_agent_tool_result]
```

474. assistant (2026-07-02T06:11:33.497Z)

```
[external_agent_tool_call: Bash]
description: Run full suite + tsc + build for T4
command: cd /c/Users/avidu/Projects/khelsutra/web && npm test --silent 2>/dev/null | grep -E
"Test Files|Tests |FAIL|?|x|?" | tail -20 && echo "=== tsc ===" && npx tsc --noEmit && echo "tsc
OK" && echo "=== build ===" && npm run build --silent 2>&1 | tail -1
[/external_agent_tool_call]
```

475. assistant (2026-07-02T06:11:33.497Z)

```
[external_agent_tool_result]
? src/components/upload.test.tsx (2 tests | 1
failed) 5101ms
  × T4: in-UI file upload > uploads the picked file, feeds the server
path to /api/process, and loads the resulting video_id 5014ms
  Test Files  1 failed | 17 passed
(18)
    Tests  1 failed | 149 passed
(150)
=== tsc ===
tsc OK
=== build ===
? built in 1.14s
[/external_agent_tool_result]
```

476. assistant (2026-07-02T06:11:33.497Z)

One integration test failed ? `waitFor` doesn't advance fake timers (a known gotcha). Let me switch it to the manual drain pattern (like `useIngestJob.test`):

477. assistant (2026-07-02T06:11:33.497Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\upload.test.tsx
[/external_agent_tool_call]
```

478. assistant (2026-07-02T06:11:33.497Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\upload.test.tsx has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

479. assistant (2026-07-02T06:11:33.497Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\upload.test.tsx
[/external_agent_tool_call]
```

480. assistant (2026-07-02T06:11:33.497Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\upload.test.tsx has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

481. assistant (2026-07-02T06:11:33.497Z)

```
[external_agent_tool_call: Bash]
description: Re-run suite after fixing upload integration test
command: npm test --silent 2>/dev/null | grep -E "Test Files|Tests |FAIL|?|×|? src" | tail -20
[/external_agent_tool_call]
```

482. assistant (2026-07-02T06:11:33.497Z)

```
[external_agent_tool_result]
Test Files  18 passed (18)
Tests      150 passed (150)
[/external_agent_tool_result]
```

483. assistant (2026-07-02T06:11:33.497Z)

All 150 tests pass (+8 T4 tests), tsc + build green. Let me browser-verify the new upload UI renders (the actual upload?process flow is test-covered; a live upload needs a running engine).

484. assistant (2026-07-02T06:11:33.497Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_eval]
input: {"serverId":"d4a4a2a4-7c2e-46e2-889e-f619477293b4","expression":"window.location.href =
window.location.origin + \"/\"; \"ok\""}
[/external_agent_tool_call]
```

485. assistant (2026-07-02T06:11:33.497Z)

```
[external_agent_tool_result: error]
Server not found. No running servers for this workspace.
[/external_agent_tool_result]
```

486. assistant (2026-07-02T06:11:33.497Z)

The preview server stopped. Let me restart it and verify the T4 UI:

487. assistant (2026-07-02T06:11:33.498Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_start]
```

```
input: {"name":"khelsutra-web"}
[/external_agent_tool_call]
```

488. assistant (2026-07-02T06:11:33.498Z)

```
[external_agent_tool_result]
{
  "serverId": "001260a2-3d4b-4722-8222-5e253669c429",
  "port": 5174,
  "name": "khelsutra-web",
  "reused": false
}
Server started successfully on port 5174.
[/external_agent_tool_result]
```

489. assistant (2026-07-02T06:11:33.498Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_snapshot]
input: {"serverId":"001260a2-3d4b-4722-8222-5e253669c429"}
[/external_agent_tool_call]
```

490. assistant (2026-07-02T06:11:33.498Z)

```
[external_agent_tool_result]
[1] RootWebArea: "KhelSutra Studio"
[5] main
[6] sectionheader
[7] none
[9] StaticText: "Khel"
[10] InlineTextBox: "Khel"
[11] StaticText: "?"
[12] InlineTextBox: "?"
[13] StaticText: "Sutra "
[14] InlineTextBox: "Sutra "
[15] StaticText: "Studio"
[16] InlineTextBox: "Studio"
[17] navigation: "Language"
[18] button: "English"
[19] StaticText: "English"
[20] InlineTextBox: "English"
[21] button: "?????"
[22] StaticText: "?????"
[23] InlineTextBox: "?????"
[24] button: "??????"
[25] StaticText: "??????"
[26] InlineTextBox: "??????"
[27] generic
[29] StaticText: "pick your rallies · alpha"
[30] InlineTextBox: "pick your rallies · alpha"
[31] heading: "Pick the rallies for your reel"
[32] StaticText: "Pick the rallies for your reel"
[33] InlineTextBox: "Pick the rallies for your reel"
[34] paragraph
[35] StaticText: "Filter by "
[36] InlineTextBox: "Filter by "
[37] strong
[38] StaticText: "rally length"
[39] InlineTextBox: "rally length"
[40] StaticText: " ? e.g. "
[41] InlineTextBox: " ? e.g. "
[42] emphasis
[43] StaticText: "?top 15 longest?"
[44] InlineTextBox: "?top 15 longest?"
[45] StaticText: " ? then fine-tune by tapping cards. Everything starts "
```

```

[46] InlineTextBox: " ? then fine-tune by tapping cards. Everything "
[47] InlineTextBox: "starts "
[48] strong
[49] StaticText: "selected"
[50] InlineTextBox: "selected"
[51] StaticText: "; just drop the ones you don't want."
[52] InlineTextBox: "; just drop the ones you don't want."
[53] generic
[54] heading: "Process a video"
[55] StaticText: "Process a video"
[56] InlineTextBox: "Process a video"
[57] paragraph
[58] StaticText: "Point KhelSutra at a video and it finds the rallies for you ? this
can take a few minutes."
[59] InlineTextBox: "Point KhelSutra at a video and it finds the rallies for you ?
this can take a few minutes."
[60] none
[61] LabelText
[62] StaticText: "Choose a video file"
[63] InlineTextBox: "Choose a video file"
[64] button: "Choose a video file to upload" (value: "No file chosen")
[65] paragraph
[66] StaticText: "Pick an .mp4 or .mov from this device ? it uploads, then KhelSutra
finds the rallies."
[67] InlineTextBox: "Pick an .mp4 or .mov from this device ? it uploads, then
KhelSutra finds the rallies."
[68] none
[69] paragraph
[70] StaticText: "OR USE A CLOUD LINK"
[71] InlineTextBox: "OR USE A CLOUD LINK"
[72] generic
[73] textbox: "Video location ? a gs:// bucket URL or a path on the engine host"
[76] button: "Process video"
[77] StaticText: "Process video"
[78] InlineTextBox: "Process video"
[79] paragraph
[80] StaticText: "Paste a cloud-storage link (gs://?), or a path to a video on the
machine running KhelSutra."
[81] InlineTextBox: "Paste a cloud-storage link (gs://?), or a path to a video on
the machine running KhelSutra."
[83] paragraph
[84] StaticText: "These are sample rallies ? process a video above to work on a real
match."
[85] InlineTextBox: "These are sample rallies ? process a video above to work on a
real match."
[87] none
[88] generic
[89] textbox: "Filter rallies by length, count or order"
[92] button: "Apply"
[93] StaticText: "Apply"
[94] InlineTextBox: "Apply"
[97] StaticText: "Type over 10s for rallies longer than 10 seconds ? ..."
[/external_agent_tool_result]

```

491. assistant (2026-07-02T06:11:33.498Z)

The T4 UI renders correctly: a labelled *****Choose a video file***** picker ([60-67]) with hint, plus the *****OR USE A CLOUD LINK***** section ([68-81]). Let me trigger a real file pick via eval to verify the live upload wiring (engine down on :8000 ? expect the friendly error chip):

492. assistant (2026-07-02T06:11:33.498Z)

```

[external_agent_tool_call: mcp__Claude_Preview__preview_eval]
input: {"serverId":"001260a2-3d4b-4722-8222-5e253669c429","expression":"(async () => {\n  const
input = document.querySelector('input[type=file]');\n  const dt = new DataTransfer();\n

```



```
dt.items.add(new File([new Uint8Array(500000)], 'demo-match.mp4', { type: 'video/mp4' }));\ninput.files = dt.files;\n input.dispatchEvent(new Event('change', { bubbles: true }));\n await\n new Promise(r => setTimeout(r, 1200));\n const status = document.querySelector('[data-\ntestid=\"ingest-status\"]');\n return status ? status.textContent : \"(no status)\";\n})();\n[/external_agent_tool_call]
```

493. assistant (2026-07-02T06:11:33.498Z)

```
[external_agent_tool_result]\n"?Couldn't processCan't reach KhelSutra. Is it running?Try again"\n[/external_agent_tool_result]
```

494. assistant (2026-07-02T06:11:33.498Z)

The live upload path works end-to-end: picking a file triggers the real upload (`init` call ? friendly **? Couldn't process / "Can't reach KhelSutra. Is it running?"** chip, since no engine is on :8000). T4 verified: 150 tests, tsc, build green; file picker labelled + both paths visible.

This is the North-Star keystone ? let me run a thorough 3-lens adversarial review. Capturing the diff:

495. assistant (2026-07-02T06:11:33.498Z)

```
[external_agent_tool_call: Bash]\ndescription: Capture T4 diff overview\ncommand: cd /c/Users/avidu/Projects/khelsutra && git add -N web/src/hooks/useUpload.ts\nweb/src/hooks/useUpload.test.ts web/src/api/upload.test.ts web/src/components/upload.test.tsx &&\necho "=== files ===" && git status --short web/ && echo "=== client.ts upload additions ===" &&\ngit --no-pager diff -- web/src/api/client.ts web/src/api/types.ts | head -160\n[/external_agent_tool_call]
```

496. assistant (2026-07-02T06:11:33.498Z)

```
[external_agent_tool_result]\n=== files ===\n M web/src/api/client.ts\n M web/src/api/types.ts\n A web/src/api/upload.test.ts\n M web/src/components/IngestPanel.tsx\n A web/src/components/upload.test.tsx\n A web/src/hooks/useUpload.test.ts\n A web/src/hooks/useUpload.ts\n M web/src/i18n/locales/en/common.json\n M web/src/i18n/locales/hi/common.json\n M web/src/i18n/locales/kn/common.json\n=== client.ts upload additions ===\ndiff --git a/web/src/api/client.ts b/web/src/api/client.ts\nindex f59884e..92cb9fb 100644\n--- a/web/src/api/client.ts\n+++ b/web/src/api/client.ts\n@@ -2,7 +2,17 @@\n // UI + engine behind one origin; in dev Vite proxies `/api` ? :8000). Every call carries a\n // correlation id (header + logs) so a journey is traceable end to end.\n import { log, newCorrelationId } from "../lib/log";\n -import type { RallySegment, QueryResponse, CompileResponse, EngineConfig, ProcessResponse,\n JobStatus } from "../types";\n +import type {\n + RallySegment,\n + QueryResponse,\n + CompileResponse,\n + EngineConfig,\n + ProcessResponse,\n + JobStatus,
```

```
+ UploadInitResponse,  
+ UploadPartResponse,  
+ UploadCompleteResponse,  
+} from "../types";  
  
const BASE = "/api";  
  
@@ -112,3 +122,98 @@ export async function getConfig(cid?: string): Promise<EngineConfig> {  
export function minShotCount(cfg: EngineConfig): number | undefined {  
    return cfg.stitching?.min_shot_count;  
}  
+  
+// ?? chunked upload (backend/api/uploads.py) ??????????????????????????????  
+// A browser uploads a LOCAL file in sequential parts ? max_part_mb (free-tier edge proxies cap  
+// bodies ~100 MB), then `complete` returns a server-host `path` to feed straight into  
+/api/process.  
+  
+/** Start a chunked upload. 400 = bad extension; 413 = over the size limit [uploads.py:64]. */  
+export async function uploadInit(filename: string, totalSizeBytes: number, cid?: string):  
Promise<UploadInitResponse> {  
+    return request<UploadInitResponse>("/upload/init", {  
+        method: "POST",  
+        body: JSON.stringify({ filename, total_size_bytes: totalSizeBytes }),  
+        cid,  
+    });  
+}  
+  
+/** Append ONE part (raw bytes ? not JSON). Parts are strictly sequential 0,1,2,?  
+[uploads.py:89]. */  
+export async function uploadPart(  
+    uploadId: string,  
+    index: number,  
+    chunk: Blob,  
+    cid?: string,  
+    signal?: AbortSignal,  
+): Promise<UploadPartResponse> {  
+    const c = cid ?? newCorrelationId();  
+    const url = `${BASE}/upload/${encodeURIComponent(uploadId)}/part/${index}`;  
+    log("info", "api.request", { cid: c, method: "PUT", url });  
+    let res: Response;  
+    try {  
+        res = await fetch(url, {  
+            method: "PUT",  
+            body: chunk,  
+            headers: { "content-type": "application/octet-stream", "x-correlation-id": c },  
+            signal,  
+        });  
+    } catch (err) {  
+        log("error", "api.network_error", { cid: c, method: "PUT", url, error: String(err) });  
+        throw new ApiError(`Network error calling PUT ${url}`, { cid: c, cause: err });  
+    }  
+    if (!res.ok) {  
+        const body = await res.text().catch(() => "");  
+        log("error", "api.http_error", { cid: c, method: "PUT", url, status: res.status, body:  
body.slice(0, 500) });  
+        throw new ApiError(`PUT ${url} failed: ${res.status}`, { cid: c, status: res.status, body  
});  
+    }  
+    log("info", "api.response", { cid: c, method: "PUT", url, status: res.status });  
+    return (await res.json()) as UploadPartResponse;  
+}  
+  
+/** Finalize: returns {path, video_id} ? feed `path` to processVideo() [uploads.py:123]. */  
+export async function uploadComplete(uploadId: string, totalParts: number, cid?: string):  
Promise<UploadCompleteResponse> {
```

```
+ return request<UploadCompleteResponse>(`/upload/${encodeURIComponent(uploadId)}/complete`, {
+   method: "POST",
+   body: JSON.stringify({ total_parts: totalParts }),
+   cid,
+ });
+}
+
+/** Abort an in-flight upload (best-effort cleanup of the server-side partial)
[uploads.py:147]. */
+export async function uploadAbort(uploadId: string, cid?: string): Promise<void> {
+  await request<unknown>(`/upload/${encodeURIComponent(uploadId)}`, { method: "DELETE", cid
+}...
[/external_agent_tool_result]
```

497. assistant (2026-07-02T06:11:33.498Z)

T4 diff is clean. Now the 3-lens adversarial review (correctness/chunking, a11y/UX faithfulness, engine-contract fidelity) ? agents read the SPA + the engine's `uploads.py`:

498. assistant (2026-07-02T06:11:33.498Z)

```
[external_agent_tool_call: Workflow]
input: {"script":"export const meta = {\n  name: 't4-review',\n  description: 'Adversarial
3-lens review of the T4 in-UI upload',\n  phases: [{ title: 'Review' }],\n}\n\nconst WEB =
'C:/Users/avidu/Projects/khelsutra/web/src'\n\nconst ENGINE = 'C:/Users/avidu/Projects/badminton-
highlight-indexer/backend/api/uploads.py'\n\nconst SCHEMA = {\n  type: 'object',\n
additionalProperties: false,\n  properties: {\n    summary: { type: 'string' },\n    findings:
{\n      type: 'array',\n      items: {\n        type: 'object',\n        additionalProperties:
false,\n        properties: {\n          severity: { type: 'string', enum: ['blocker', 'major',
'minor', 'nit' ] },\n          file: { type: 'string' },\n          location: { type: 'string'
},\n          title: { type: 'string' },\n          detail: { type: 'string' },\n
suggestion: { type: 'string' },\n          },\n          required: ['severity', 'file', 'title',
'detail'],\n          },\n          },\n          required: ['summary', 'findings'],\n}\n\nconst CONTEXT =
`T4 (North Star keystone): in-UI local file UPLOAD wired to the engine's chunked-upload
endpoints.\nFiles under ${WEB}: \n- api/client.ts: uploadInit/uploadPart(raw octet-stream
PUT)/uploadComplete/uploadAbort + uploadFile(file,{cid,signal,onProgress}) orchestrator (init ->
slice into max_part_mb parts -> sequential PUT 0,1,2 -> complete; best-effort DELETE on
failure/abort).\n- api/types.ts: UploadInit/Part/CompleteResponse.\n- hooks/useUpload.ts:
start(file) -> uploading{pct} -> onComplete(serverPath) | error(classifyError); rejects 0-byte;
AbortController cancel/unmount.\n- components/IngestPanel.tsx: file picker (primary) + the
existing gs:// URI form (secondary); on upload complete -> useIngestJob.submit(path) (the SAME
process->poll->onLoaded back-half); upload % progressbar; A8 chips reused.\n- catalogs en/kn/hi:
+ingest.chooseFile/uploadAria/fileHint/orLink/uploading.\nEngine contract at ${ENGINE}.\n150 web
tests green; tsc + build clean.`\n\nconst LENSES = [\n ...
[/external_agent_tool_call]
```

499. assistant (2026-07-02T06:11:33.498Z)

```
[external_agent_tool_result]
Workflow launched in background. Task ID: wrmvkuioo
Summary: Adversarial 3-lens review of the T4 in-UI upload
Transcript dir: C:\Users\avidu\claud\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\2ba49afa-99f4-41f2-86a2-27a289d432f3\subagents\workflows\wf_a48e6aa3-748
Script file: C:\Users\avidu\claud\projects\C--Users-avidu-Projects-
khelsutra\2ba49afa-99f4-41f2-86a2-27a289d432f3\workflows\scripts\t4-review-wf_a48e6aa3-748.js
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath:
"C:\Users\avidu\claud\projects\C--Users-avidu-Projects-
khelsutra\2ba49afa-99f4-41f2-86a2-27a289d432f3\workflows\scripts\t4-review-wf_a48e6aa3-748.js"})
to iterate without resending the script.)
Run ID: wf_a48e6aa3-748
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\claud\projects\C--
Users-avidu-Projects-khelsutra\2ba49afa-99f4-41f2-86a2-27a289d432f3\workflows\scripts\t4-review-
wf_a48e6aa3-748.js", resumeFromRunId: "wf_a48e6aa3-748"}) ? completed agents return cached
results.
```

You will be notified when it completes. Use /workflows to watch live progress.
[/external_agent_tool_result]

500. assistant (2026-07-02T06:11:33.498Z)

While the T4 review runs, let me wait for it (it gates the merge). I'll do the memory checkpoint + spawn-offs after T4 lands to minimize edit churn.

501. assistant (2026-07-02T06:11:33.498Z)

```
[external_agent_tool_call: TaskOutput]
input: {"task_id":"wrmvkuioo","block":true,"timeout":300000}
[/external_agent_tool_call]
```

502. assistant (2026-07-02T06:11:33.499Z)

```
[external_agent_tool_result]
<retrieval_status>timeout</retrieval_status>

<task_id>wrmvkuioo</task_id>

<task_type>local_workflow</task_type>

<status>running</status>
[/external_agent_tool_result]
```

503. assistant (2026-07-02T06:11:33.499Z)

```
[external_agent_tool_call: TaskOutput]
input: {"task_id":"wrmvkuioo","block":true,"timeout":180000}
[/external_agent_tool_call]
```

504. assistant (2026-07-02T06:11:33.499Z)

```
[external_agent_tool_result]
<retrieval_status>success</retrieval_status>

<task_id>wrmvkuioo</task_id>

<task_type>local_workflow</task_type>

<status>completed</status>

<output>
{
  "summary": "Adversarial 3-lens review of the T4 in-UI upload",
  "agentCount": 3,
  "logs": [],
  "result": [
    {
      "lens": "correctness",
      "summary": "T4 upload review",
      "findings": [
        {
          "severity": "nit",
          "file": "C:/Users/avidu/Projects/khelsutra/web/src/api/client.ts",
          "location": "199-200",
          "title": "max_part_mb guard catches 0 but not NaN/missing",
          "detail": "Math.max(1, init.max_part_mb) covers 0 but not undefined/NaN since Math.max(1, NaN) is NaN; partBytes and totalParts become NaN, the for-loop uploads zero parts, and uploadComplete(NaN) serializes to null and 422s as a silent zero-byte upload. Engine returns an int so defensive only, but it is the 0-or-missing gap the prompt asked about.",
          "suggestion": "Coerce with Number.isFinite (fallback 95) before scaling to bytes."
        }
      ],
    }
  ],
}
```

```

{
  "severity": "nit",
  "file": "C:/Users/avidu/Projects/khelsutra/web/src/api/client.ts",
  "location": "211",
  "title": "Per-part 413 is terminal; no retry without re-init",
  "detail": "Engine per-part overflow (uploads.py:113-115) aborts the whole upload, so a
413 part error is terminal for the upload_id; client part size is byte-exact vs engine so
steady-state never hits it. Fine for the alpha; a documented edge.",
  "suggestion": "Note in the uploadFile docstring that a 413 part error needs re-init."
}
],
{
  "lens": "a11y-ux",
  "summary": "T4 in-UI file upload is solid; nothing blocking. See findings.",
  "findings": [
    {
      "severity": "minor",
      "file": "C:/Users/avidu/Projects/khelsutra/web/src/components/IngestPanel.tsx",
      "location": "lines 59-74 (file-picker label/input)",
      "title": "Keyboard focus on the file picker is invisible (WCAG 2.4.7 Focus Visible)",
      "detail": "The file input is sr-only (correct for staying focusable+labelled) inside a
styled label, but the label has no focus-within style. A keyboard user who tabs to the picker
gets no visible focus ring on the Choose-a-video-file button because focus lands on the hidden
input. By contrast the gs:// input and the Process button both get a visible focus:border-green.
The picker is reachable and SR-labelled but not focus-visible. Size is fine (min-h-11=44px);
this is the focus indicator only.",
      "suggestion": "Add a focus-within ring to the label so the visible button reflects the
hidden input focus. Worth a one-line a4-style test asserting the label carries a focus-within
class."
    },
    {
      "severity": "nit",
      "file": "C:/Users/avidu/Projects/khelsutra/web/src/components/IngestPanel.tsx",
      "location": "lines 108-114 (progressbar div)",
      "title": "Progressbar has no accessible name",
      "detail": "role=progressbar has aria-valuenow/min/max but no aria-label or aria-
labelledby, so a screen reader announces a bare percent progressbar with no indication of what
is progressing. The adjacent StatusChip under aria-live carries the meaning, so not a functional
gap; a named bar is just more robust. The determinate-vs-indeterminate honesty itself is
correct.",
      "suggestion": "Add aria-label via t() to the progressbar div, or wire aria-labelledby
to the chip text."
    },
    {
      "severity": "nit",
      "file": "C:/Users/avidu/Projects/khelsutra/web/src/components/IngestPanel.tsx",
      "location": "lines 104-149 (aria-live region with Cancel + the percent chip)",
      "title": "Interactive Cancel and a per-tick percent chip sit inside the aria-live
region",
      "detail": "Both busy blocks (each with a Cancel button) live inside the aria-
live=polite wrapper. Nesting an interactive control plu...
[/external_agent_tool_result]

```

505. assistant (2026-07-02T06:11:33.499Z)

T4 review: **contract verified clean, no blockers.** Worth folding: one **a11y MINOR** (file-picker keyboard focus invisible ? WCAG 2.4.7), plus cheap robustness nits (NaN-safe ``max_part_mb``, named progressbar, avoid chatty per-tick re-announcement). The hosted-mode ``allowed_video_root`` note is a T2 deployment item (I'll carry it into the T2 spawn-off). Folding the real ones:

506. assistant (2026-07-02T06:11:33.499Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\khelsutra\web\src\api\client.ts
[/external_agent_tool_call]

507. assistant (2026-07-02T06:11:33.499Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\api\client.ts has been updated successfully.
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

508. assistant (2026-07-02T06:11:33.499Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx
[/external_agent_tool_call]

509. assistant (2026-07-02T06:11:33.499Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

510. assistant (2026-07-02T06:11:33.499Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx
[/external_agent_tool_call]

511. assistant (2026-07-02T06:11:33.499Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

512. assistant (2026-07-02T06:11:33.499Z)

Now update the catalogs ? replace `uploading` with `uploadingStatus` + `uploadProgress` (all three):

513. assistant (2026-07-02T06:11:33.499Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json
[/external_agent_tool_call]

514. assistant (2026-07-02T06:11:33.499Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

515. assistant (2026-07-02T06:11:33.499Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json
[/external_agent_tool_call]

516. assistant (2026-07-02T06:11:33.499Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

517. assistant (2026-07-02T06:11:33.500Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\hi\common.json
[/external_agent_tool_call]

518. assistant (2026-07-02T06:11:33.500Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\hi\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

519. assistant (2026-07-02T06:11:33.500Z)

Now add a focus-visible regression assertion to the upload test, then re-run:

520. assistant (2026-07-02T06:11:33.500Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\upload.test.tsx
[/external_agent_tool_call]

521. assistant (2026-07-02T06:11:33.500Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\upload.test.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

522. assistant (2026-07-02T06:11:33.500Z)

[external_agent_tool_call: Bash]
description: Re-run suite after folding T4 review findings
command: cd /c/Users/avidu/Projects/khelsutra/web && npm test --silent 2>/dev/null | grep -E "Test Files|Tests |FAIL|?|x|? src" | tail -10 && echo "=== tsc ===" && npx tsc --noEmit && echo "tsc OK" && echo "=== build ===" && npm run build --silent 2>&1 | tail -1
[/external_agent_tool_call]

523. assistant (2026-07-02T06:11:33.500Z)

[external_agent_tool_result]
Test Files 18 passed (18)
Tests 150 passed (150)
=== tsc ===
tsc OK
=== build ===
? built in 1.14s
[/external_agent_tool_result]

524. assistant (2026-07-02T06:11:33.500Z)

150 tests, tsc, build all green after folding the T4 findings. Let me confirm no orphaned `ingest.uploading` reference, then commit:

525. assistant (2026-07-02T06:11:33.500Z)

[external_agent_tool_call: Grep]
input: {"pattern":"ingest\\.uploading\\b|\"uploading\"","path":"C:\\Users\\avidu\\Projects\\khel

```
sutra\\web\\src","output_mode":"content"}
[/external_agent_tool_call]
```

526. assistant (2026-07-02T06:11:33.500Z)

```
[external_agent_tool_result]
src\\hooks\\useUpload.ts:17:  | { status: "uploading"; pct: number; filename: string }
src\\hooks\\useUpload.ts:47:      setPhase({ status: "uploading", pct: 0, filename: file.name });
src\\hooks\\useUpload.ts:58:          setPhase({ status: "uploading", pct, filename: file.name
});
src\\hooks\\useUpload.ts:86:  const busy = phase.status === "uploading";
src\\components\\IngestPanel.tsx:107:      {upload.phase.status === "uploading" && (
[/external_agent_tool_result]
```

527. assistant (2026-07-02T06:11:33.500Z)

All matches are the `"uploading"` phase-status string (state machine), not the removed `i18n` key
? clean. Committing T4:

528. assistant (2026-07-02T06:11:33.500Z)

```
[external_agent_tool_call: Bash]
description: Stage T4 paths, verify clean base
command: cd /c/Users/avidu/Projects/khelsutra && echo "branch: $(git branch --show-current)" &&
git add web/src/api/types.ts web/src/api/client.ts web/src/api/upload.test.ts
web/src/hooks/useUpload.ts web/src/hooks/useUpload.test.ts web/src/components/IngestPanel.tsx
web/src/components/upload.test.tsx web/src/i18n/locales/en/common.json
web/src/i18n/locales/kn/common.json web/src/i18n/locales/hi/common.json && git diff --cached
--name-only && echo "--- base (only my work) ---" && git log --oneline origin/master..HEAD
[/external_agent_tool_call]
```

529. assistant (2026-07-02T06:11:33.500Z)

```
[external_agent_tool_result]
branch: master
web/src/api/client.ts
web/src/api/types.ts
web/src/api/upload.test.ts
web/src/components/IngestPanel.tsx
web/src/components/upload.test.tsx
web/src/hooks/useUpload.test.ts
web/src/hooks/useUpload.ts
web/src/i18n/locales/en/common.json
web/src/i18n/locales/hi/common.json
web/src/i18n/locales/kn/common.json
--- base (only my work) ---
[/external_agent_tool_result]
```

530. assistant (2026-07-02T06:11:33.500Z)

I'm on `master` ? I skipped cutting the T4 branch. Caught before any commit. Let me move these
staged changes onto a feature branch (`checkout -b` preserves the working tree):

531. assistant (2026-07-02T06:11:33.500Z)

```
[external_agent_tool_call: Bash]
description: Move T4 changes to a feature branch
command: git fetch origin -q && echo "origin/master: $(git rev-parse --short origin/master)" &&
echo "local master: $(git rev-parse --short master)" && git checkout -b feat/t4-in-ui-upload
2>&1 | tail -2 && echo "--- now on ---" && git branch --show-current && echo "--- staged still
present ---" && git diff --cached --name-only | head && echo "--- base vs origin/master (should
be empty: nothing committed yet) ---" && git log --oneline origin/master..HEAD
[/external_agent_tool_call]
```


532. assistant (2026-07-02T06:11:33.500Z)

```
[external_agent_tool_result]
origin/master: 4627830
local master: 92a6c45
Switched to a new branch 'feat/t4-in-ui-upload'
--- now on ---
feat/t4-in-ui-upload
--- staged still present ---
web/src/api/client.ts
web/src/api/types.ts
web/src/api/upload.test.ts
web/src/components/IngestPanel.tsx
web/src/components/upload.test.tsx
web/src/hooks/useUpload.test.ts
web/src/hooks/useUpload.ts
web/src/i18n/locales/en/common.json
web/src/i18n/locales/hi/common.json
web/src/i18n/locales/kn/common.json
--- base vs origin/master (should be empty: nothing committed yet) ---
[/external_agent_tool_result]
```

533. assistant (2026-07-02T06:11:33.500Z)

`origin/master` moved to `4627830` (a sibling merged after my A8) ? my branch is based on the older `92a6c45`. Let me check what landed and rebase onto `origin/master` per branching discipline. First, commit T4, then rebase:

534. assistant (2026-07-02T06:11:33.500Z)

```
[external_agent_tool_call: Bash]
description: Inspect the sibling commit on origin/master
command: echo "=== what is 4627830 ===" && git log --oneline 92a6c45..origin/master && echo "===
files it touched ===" && git diff --name-only 92a6c45 origin/master
[/external_agent_tool_call]
```

535. assistant (2026-07-02T06:11:33.501Z)

```
[external_agent_tool_result]
=== what is 4627830 ===
4627830 docs(deploy): alpha deploy kit ? make Studio a shareable page (split: Pages + api.*
tunnel + CF Access) (#65)
8bbc557 feat(web): VITE_API_BASE ? let the SPA target a split api.* origin (default same-origin
/api) (#64)
=== files it touched ===
README.md
deploy/DEPLOY_ALPHA.md
deploy/cloudflared.config.example.yml
docs/LOCAL_APPLIANCE_ARCHITECTURE.md
web/src/api/client.test.ts
web/src/api/client.ts
web/src/vite-env.d.ts
[/external_agent_tool_result]
```

536. assistant (2026-07-02T06:11:33.501Z)

#64 modified `web/src/api/client.ts` (VITE_API_BASE) ? the same file I touched. I'll commit T4, then rebase onto `origin/master` and resolve. Committing:

537. assistant (2026-07-02T06:11:33.501Z)

```
[external_agent_tool_call: Bash]
description: Commit T4 and rebase onto origin/master
command: git commit -q -F - <<'EOF'
```

feat(web): T4 ? in-UI file upload (the runnable-tool keystone)

Alpha users have a FILE, not a gs:// URI. The ingest screen now lets them upload a local video straight from the browser, wired to the engine's existing chunked-upload endpoints (backend/api/uploads.py) and converging on the SAME process?poll?load back-half:

- api/client.ts: uploadInit / uploadPart (raw octet-stream PUT) / uploadComplete / uploadAbort + a uploadFile() orchestrator ? slices the file into sequential parts ? the engine's max_part_mb, reports byte progress, is AbortController-cancellable, and best-effort DELETES the server-side partial on any failure/abort. The returned server PATH feeds straight into POST /api/process.
- hooks/useUpload.ts: idle ? uploading{pct} ? hand the path to onComplete | error; rejects a 0-byte file before any request; classifyError on failures.
- IngestPanel: a file picker (primary ? keyboard-focusable, focus-visible, ?44px) + the existing gs:// URI form (secondary); a determinate % progressbar (named, with a sighted-only %); reuses the A8 chips + the errorText mapping; one Cancel/Retry.
- catalogs (en/kn/hi): +ingest.chooseFile/uploadAria/fileHint/orLink/uploadingStatus/uploadProgress.

Tests: api/upload.test.ts (chunk math, sequential order, abort cleanup, init reject), hooks/useUpload.test.ts (empty-file guard, happy path ? onComplete, classified error), components/upload.test.tsx (pick ? upload ? /api/process gets the server path ? onLoaded with the job's canonical video_id; focus-visible). 150 web tests green; tsc + build clean. Reviewed via a 3-lens adversarial pass (correctness/chunking · a11y/UX · engine-contract); contract verified clean, folded its focus-visible + NaN-guard + named-progressbar findings.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>

EOF

```
echo "committed: $(git log --oneline -1)" && echo "=== rebase onto origin/master ===" && git
rebase origin/master 2>&1 | tail -15
[/external_agent_tool_call]
```

538. assistant (2026-07-02T06:11:33.501Z)

```
[external_agent_tool_result]
committed: c06ee54 feat(web): T4 ? in-UI file upload (the runnable-tool keystone)
=== rebase onto origin/master ===
Rebasing (1/1)
```

Successfully rebased and updated refs/heads/feat/t4-in-ui-upload.
[/external_agent_tool_result]

539. assistant (2026-07-02T06:11:33.501Z)

The rebase merged cleanly ? #64's `apiBase()`/`VITE\_API\_BASE` and my upload helpers coexist (my code uses `\${BASE}`, now resolved via `apiBase()`). Let me re-verify the full suite on the rebased base (which now includes #64's client changes):

540. assistant (2026-07-02T06:11:33.501Z)

```
[external_agent_tool_call: Bash]
description: Re-verify after rebase onto origin/master
command: cd /c/Users/avidu/Projects/khelsutra/web && npm test --silent 2>/dev/null | grep -E
"Test Files|Tests |FAIL|?|x|? src" | tail -10 && echo "=== tsc ===" && npx tsc --noEmit && echo
"tsc OK" && echo "=== build ===" && npm run build --silent 2>&1 | tail -1 && echo "=== branch
base ===" && git log --oneline origin/master..HEAD
[/external_agent_tool_call]
```

541. assistant (2026-07-02T06:11:33.501Z)

```
[external_agent_tool_result]
Test Files 18 passed (18)
Tests 153 passed (153)
```

```
=== tsc ===
tsc OK
=== build ===
? built in 1.12s
=== branch base ===
a154121 feat(web): T4 ? in-UI file upload (the runnable-tool keystone)
[/external_agent_tool_result]
```

542. assistant (2026-07-02T06:11:33.501Z)

Rebased cleanly ? **153 tests** pass (my 150 + #64's 3 new client tests), tsc + build green, and my T4 commit is the only one atop `origin/master`. Pushing + PR:

543. assistant (2026-07-02T06:11:33.501Z)

```
[external_agent_tool_call: Bash]
description: Push T4 branch
command: cd /c/Users/avidu/Projects/khelsutra && git push -u origin feat/t4-in-ui-upload 2>&1 |
tail -3
[/external_agent_tool_call]
```

544. assistant (2026-07-02T06:11:33.501Z)

```
[external_agent_tool_result]
branch 'feat/t4-in-ui-upload' set up to track 'origin/feat/t4-in-ui-upload'.
To https://github.com/avidullu/khelsutra.git
 * [new branch]      feat/t4-in-ui-upload -> feat/t4-in-ui-upload
[/external_agent_tool_result]
```

545. assistant (2026-07-02T06:11:33.501Z)

```
[external_agent_tool_call: Bash]
description: Create T4 PR
command: gh pr create --title "feat(web): T4 ? in-UI file upload (the runnable-tool keystone)"
--base master --head feat/t4-in-ui-upload --body "$(cat <<'EOF'
```

T4 ? in-UI local file upload (North Star keystone · `LOCAL\_APPLIANCE\_ARCHITECTURE.md` §9)

Alpha users have a **file**, not a `gs://` URI. The ingest screen now lets them upload a local video straight from the browser, wired to the engine's **existing** chunked-upload endpoints (`backend/api/uploads.py`) and converging on the **same** process ? poll ? load ? compile ? download back-half. This is the piece that makes the tool point-and-go for a non-technical user.

What changed (100% `web/`, zero engine code)

- **api/client.ts** ? `uploadInit` / `uploadPart` (raw octet-stream PUT) / `uploadComplete` / `uploadAbort` + a **uploadFile()** orchestrator: slices the file into sequential parts ? the engine's `max\_part\_mb`, reports byte progress, is `AbortController`-cancellable, and best-effort **DELETE**s the server-side partial on any failure/abort. The returned server **path** feeds straight into `POST /api/process`.
- **hooks/useUpload.ts** ? `idle` ? uploading{pct} ? hand the path to onComplete | error; rejects a 0-byte file before any request; `classifyError` on failures.
- **IngestPanel** ? a **file picker** (primary; keyboard-focusable, **focus-visible**, ?44px) + the existing `gs://` URI form (secondary); a determinate **progressbar** (named, with a sighted-only %); reuses the A8 status chips + the A5 `errorText` mapping; one Cancel / Retry for either half.
- **Catalogs (en/kn/hi)** ? `+ingest.chooseFile` / `uploadAria` / `fileHint` / `orLink` / `uploadingStatus` / `uploadProgress`.

Verification

- **153 web tests** green: `api/upload.test.ts` (chunk math, strict 0,1,2 order, abort cleanup, init-reject), `hooks/useUpload.test.ts` (empty-file guard, happy path ? `onComplete`, classified error), `components/upload.test.tsx` (pick ? upload ? `/api/process` gets the **server path** ? `onLoaded` with the **job's canonical** video_id; focus-visible). `tsc` + `vite build` clean; key-parity + no-bare-strings hold.

- **Browser-verified**: the labelled file picker + "OR USE A CLOUD LINK" both render; picking a file triggers the real upload (`/api/upload/init`), surfacing the friendly **?** Couldn't process chip when no engine is up ? exactly as expected.

- **Reviewed by a 3-lens adversarial workflow** (correctness/chunking · a11y/UX · engine-contract). Contract **verified clean** against `uploads.py`; folded the focus-visible (WCAG 2.4.7), NaN-safe `max\_part\_mb`, and named-progressbar findings.

- Rebased onto `origin/master` (after #64/#65); `apiBase()`/`VITE\_API\_BASE` and the upload helpers coexist (the upload calls use `\${BASE}`).

Deployment note (for T2, not this PR)

In **hosted mode** with edge-auth ON, the engine's `allowed\_video\_root` must contain `upload\_dir` or `/api/process` will reject uploaded paths. Default single-user mode (the alpha target) needs no change.

? Generated with [Claude Code](https://claude.com/claude-code)

EOF

)" 2>&1 | tail -2

[/external_agent_tool_call]

546. assistant (2026-07-02T06:11:33.501Z)

[external_agent_tool_result]

<https://github.com/avidullu/khelsutra/pull/66>

[/external_agent_tool_result]

547. assistant (2026-07-02T06:11:33.501Z)

PR [#66](https://github.com/avidullu/khelsutra/pull/66) created. Let me update T4's status in its tracker doc (`LOCAL\_APPLIANCE\_ARCHITECTURE.md` §9). First, find the T4 row:

548. assistant (2026-07-02T06:11:33.501Z)

[external_agent_tool_call: Grep]

input: {"pattern":"T4|in-UI.*upload|chunked.upload|uploads\\.py","path":"C:\\Users\\avidu\\Projects\\khelsutra\\docs\\LOCAL_APPLIANCE_ARCHITECTURE.md","output_mode":"content","-n":true,"-C":1}

[/external_agent_tool_call]

549. assistant (2026-07-02T06:11:33.501Z)

[external_agent_tool_result]

41-## 3. Foundation ? what the engine gives us today `[verified this session unless noted]`

42:[Omitted long matching line]

43-- **In-process worker**: one module-global `ThreadPoolExecutor(max\_workers=1)` `[verified job\_worker.py:37]` ? strictly **one job at a time**; the DB `jobs` table is the clock; `fail\_stale\_jobs` sweeps queued/running on boot. Gemini chunk concurrency is a **separate** knob `indexing.ai\_max\_workers` (**default 5**) `[verified models.py:183]`; set to 1 to serialize) ? it is **not** already 1.

--

68-? control: jobs table + ThreadPoolExecutor(max_workers=1) drain ?

69:? data: uploads.py chunked upload · /api/compile · /api/highlights ?

70-? ? ?

--

85-|---|---|---|

86:| **Local upload** | SPA chunks the file ? `max\_part\_mb` via the existing `/api/upload/init|part|complete` `[verified uploads.py]` ? server path ? `POST /api/process`. (Scale: also `StorageBackend.put` to `s3:///gcs:///`) | **wire client only** (endpoints exist)

87-[Omitted long context line]

--

112-### 4.6 Reuse map

113:[Omitted long matching line]

114-

--

160-| T3 | L3 live posture test (raw `:8000`/`:900x` refused from outside) | T2 | no | ? | ? |

```

161:| T4 | S1 ingest (local upload wired) + segmenter picker | T1 | no | ? | ? |
162:| T5 | S2 progress (poll + one-job + restart-failure) | T4 | no | ? | ? |
163:| T6 | S3 embed **RallyPicker** + compile + download | T4 | no | ? | ? |
164:| T7 | *(engine)* `GET /api/capabilities` + `GET /api/health` + banner | T1 | coord | ? | ? |
|
--
166:| T9 | S5 Settings (config+capabilities; disabled compute/storage target) | T7,T8 | no | ? | ? |
? |
167:| T10 | Observability pass: correlation id end-to-end + structured logs + metrics ($5) | T4 | no | ? | ? |
|
168:| T11 | Retention sweep for `output/`+`uploads/` | ? | owner | ? | ? |
169:| T12 | Ingest from **bucket URI** (`s3://`/`gs://`) wiring ? engine accepts URIs
(M2/[#280]) *** SPA ingest screen shipped (B-P2, [#56])**; in-UI `gs://`?reel CUJ proven live
e2e (`DRY_RUN_CUJ.md` $9) | T4 | ? | ? | engine #280 + SPA #56 |
170:| T13 | **GPU host stand-up** (provider-agnostic) + wire `detector_impl=native`; owner byte-
parity (P4) | branch | ? owner | ? | ? |
--
191-## 12. References
192:[Omitted long matching line]
193-
[/external_agent_tool_result]

```

550. assistant (2026-07-02T06:11:33.501Z)

T4's row bundles "local upload wired *** segmenter picker***" ? my PR delivers the upload keystone but not the picker. I'll mark it ? honestly (upload shipped, picker pending):

551. assistant (2026-07-02T06:11:33.501Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\kheIsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md
[/external_agent_tool_call]

```

552. assistant (2026-07-02T06:11:33.501Z)

```

[external_agent_tool_result: error]
<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>
[/external_agent_tool_result]

```

553. assistant (2026-07-02T06:11:33.501Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\kheIsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md
[/external_agent_tool_call]

```

554. assistant (2026-07-02T06:11:33.501Z)

```

[external_agent_tool_result]
158      | T1 | Mock engine + capability fixtures + browser-replay CUJ scripts (L1) | T0 | no | ? | ? |
159      | T2 | Appliance shell: Caddyfile + systemd units; firewall; **edge auth**; rotate key (P1) | ? | owner | ? | ? |
160      | T3 | L3 live posture test (raw `:8000`/`:900x` refused from outside) | T2 | no | ? | ? |
161      | T4 | S1 ingest (local upload wired) + segmenter picker | T1 | no | ? | ? |
162      | T5 | S2 progress (poll + one-job + restart-failure) | T4 | no | ? | ? |
163      | T6 | S3 embed **RallyPicker** + compile + download | T4 | no | ? | ? |
164      | T7 | *(engine)* `GET /api/capabilities` + `GET /api/health` + banner | T1 | coord | ? | ? |
165      | T8 | *(engine)* `GET /api/videos` + `GET /api/reels/{video_id}` (S4 reload-survival) | ? | coord | ? | ? |
166      | T9 | S5 Settings (config+capabilities; disabled compute/storage target) | T7,T8 | no | ? | ? |
167      | T10 | Observability pass: correlation id end-to-end + structured logs + metrics ($5) |

```

```
T4 | no | ? | ? |  
168 | T11 | Retention sweep for `output/`+`uploads/` | ? | owner | ? | ? |  
169 | T12 | Ingest from **bucket URI** (`s3://`/`gs://`) wiring ? engine accepts URIs  
(M2/[#280]) *** SPA ingest screen shipped (B-P2, [#56])**; in-UI `gs://`?reel CUJ proven live  
e2e (`DRY_RUN_CUJ.md` §9) | T4 | ? | ? | engine #280 + SPA #56 |  
170 | T13 | **GPU host stand-up** (provider-agnostic) + wire `detector_impl=native`; owner  
byte-parity (P4) | branch | ? owner | ? | ? |  
171 | T14 | *(reserved)* Bucket-worker bridge ? remote-GPU "compute as a setting" (P5) |  
T8,T12 | ? owner | ? | ? |  
[/external_agent_tool_result]
```

555. assistant (2026-07-02T06:11:33.501Z)

```
[external_agent_tool_call: Edit]  
file: C:\Users\avidu\Projects\khelsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md  
[/external_agent_tool_call]
```

556. assistant (2026-07-02T06:11:33.502Z)

```
[external_agent_tool_result]  
The file C:\Users\avidu\Projects\khelsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md has been updated  
successfully. (file state is current in your context ? no need to Read it back)  
[/external_agent_tool_result]
```

557. assistant (2026-07-02T06:11:33.502Z)

```
[external_agent_tool_call: Bash]  
description: Commit T4 tracker + push + check CI  
command: git add docs/LOCAL_APPLIANCE_ARCHITECTURE.md && git commit -q -m "docs(t4): mark local-  
upload-wired ? (#66) in the §9 tracker (T4 ?; picker pending)
```

```
Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>" && git push -q 2>&1 | tail -1 && echo  
"pushed" && sleep 25 && gh pr checks 66 2>&1 | head  
[/external_agent_tool_call]
```

558. assistant (2026-07-02T06:11:33.502Z)

```
[external_agent_tool_result]  
pushed  
site pass 13s  
https://github.com/avidullu/khelsutra/actions/runs/27910960369/job/82587898644  
Workers Builds: hidden-king-khelsutra pending 0  
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-  
khelsutra/production/builds/b93606ec-542f-41ca-b341-cc54516bf214  
web pending 0  
https://github.com/avidullu/khelsutra/actions/runs/27910960369/job/82587898656  
[/external_agent_tool_result]
```

559. assistant (2026-07-02T06:11:33.502Z)

```
[external_agent_tool_call: Bash]  
description: Poll T4 CI  
command: for i in $(seq 1 8); do s=$(gh pr checks 66 2>&1 | grep -E "^web\b" | awk '{print  
$2}'); c=$(gh pr checks 66 2>&1 | grep -iE "Workers Builds" | awk '{print $4}'); echo "poll $i:  
web=$s cf=$c"; if { [ "$s" = "pass" ] || [ "$s" = "fail" ]; } && { [ "$c" = "pass" ] || [ "$c" =  
"fail" ]; }; then break; fi; sleep 15; done && echo "=== final ===" && gh pr checks 66 2>&1 |  
head  
[/external_agent_tool_call]
```

560. assistant (2026-07-02T06:11:33.502Z)

```
[external_agent_tool_result]  
poll 1: web=pending cf=pass  
poll 2: web=pass cf=pass
```

```
=== final ===
Workers Builds: hidden-king-khelsutra pass 0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/b93606ec-542f-41ca-b341-cc54516bf214
site pass 13s
https://github.com/avidullu/khelsutra/actions/runs/27910960369/job/82587898644
web pass 30s
https://github.com/avidullu/khelsutra/actions/runs/27910960369/job/82587898656
[/external_agent_tool_result]
```

561. assistant (2026-07-02T06:11:33.502Z)

All green. Merging T4 (the North-Star keystone):

562. assistant (2026-07-02T06:11:33.502Z)

```
[external_agent_tool_call: Bash]
description: Squash-merge T4, sync master
command: gh pr merge 66 --squash --delete-branch 2>&1 | tail -2 && git checkout master -q && git
pull --ff-only -q && git log --oneline -3
[/external_agent_tool_call]
```

563. assistant (2026-07-02T06:11:33.502Z)

```
[external_agent_tool_result]
create mode 100644 web/src/hooks/useUpload.ts
create mode 100644 web/src/vite-env.d.ts
ca619fb feat(web): T4 ? in-UI file upload (the runnable-tool keystone) (#66)
4627830 docs(deploy): alpha deploy kit ? make Studio a shareable page (split: Pages + api.*
tunnel + CF Access) (#65)
8bbc557 feat(web): VITE_API_BASE ? let the SPA target a split api.* origin (default same-origin
/api) (#64)
[/external_agent_tool_result]
```

564. assistant (2026-07-02T06:11:33.502Z)

****T4 merged (#66 ? `ca619fb`) ? the North-Star keystone is in.**** Note: the sibling's #64/#65 already shipped a `VITE\_API\_BASE` split-origin mode + an ****alpha deploy kit**** (Cloudflare Pages + `api.\*` tunnel + CF Access) ? that's most of T2's plumbing.

That's ****4 merged PRs this session**** (A4/A5/A8/T4). Per your direction, let me record the remaining immediate-next items (B3, MULTILINGUAL Phase A, T11) as spawn-off chips you can launch as parallel sessions, each self-contained with the locked decisions + sequencing.

565. assistant (2026-07-02T06:11:33.502Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"5","status":"completed"}
[/external_agent_tool_call]
```

566. assistant (2026-07-02T06:11:33.502Z)

```
[external_agent_tool_result]
Updated task #5 status
[/external_agent_tool_result]
```

567. assistant (2026-07-02T06:11:33.502Z)

```
[external_agent_tool_call: mcp_ccd_session_spawn_task]
input: {"title":"B3 ? make site/ marketing trilingual (shared t()
shim)","cwd":"C:\\Users\\avidu\\Projects\\khelsutra","tldr":"Make the marketing site
(homepage/capture/own-model) trilingual using a shared vanilla t() shim, hoisting the locale
catalogs to a repo-root /locales/ shared by the SPA and the site.","prompt":"Implement B3 in the
khelsutra repo (C:\\Users\\avidu\\Projects\\khelsutra): make the marketing `site/` trilingual
```

via a shared, dependency-free `t()`/`data-i18n` shim, en-only content first (this proves the shared-catalog architecture across both surfaces; kn/hi marketing translation is a later effort).\n\nLOCKED OWNER DECISION (D-CATALOG-HOME): hoist the locale catalogs to repo-root `/locales/{en,kn,hi}/common.json` as the single source of truth.\n- Re-point the SPA's catalog imports (`web/src/i18n/index.ts` currently imports `./locales/{en,kn,hi}/common.json`) to the hoisted location, or add a Vite `@locales` alias. Keep `web/` tests green.\n- Add a tiny `site` prebuild step that copies `/locales` ? `site/public/locales/` + a drift-guard test (the site is dependency-free/no-build today ? this copy is the only new build step).\n\nBuild:\n- `site/public/i18n.js` ? a ~70-LOC vanilla `t(key, vars)` + a `data-i18n` attribute walker, with ICU-lite interpolation/plurals for the 2-form languages (mirror i18next behaviour for en/kn/hi).\n- A persistent language switcher on `site/public/{index,capture,own-model}.html`.\n- `site/server.js` Accept-Language negotiation.\n- Negotiation MUST match the SPA EXACTLY: persisted `localStorage` key `khelsutra.lang` ? `?lang=` ? `navigator/`Accept-Language` ? `en`; a missing key degrades to en, never a raw key.\n- Tests: extend `site/src/{server,assets}.test.js` + a shim unit test + the \$6.5 N8 negotiation test (`Accept-Language: kn` / `?lang=kn` ? kn-rendered).\n\nConventions (shared checkout ? multiple sessions): `git fetch origin` then `git checkout -b feat/b3-site-i18n origin/master`; re-verify HEAD before commit; stage...\n[/external_agent_tool_call]

568. assistant (2026-07-02T06:11:33.502Z)

[external_agent_tool_result]
Noted (position 1, task_id: task_b3060842). A chip is showing for the user ? they can start it in a fresh worktree with one click, or dismiss it. If this suggestion becomes stale or superseded, call dismiss_task with this task_id. Currently pending: task_b3060842 "B3 ? make site/ marketing trilingual (shared t() shim)". Continue your current work.\n[/external_agent_tool_result]

569. assistant (2026-07-02T06:11:33.502Z)

[external_agent_tool_call: mcp_ccd_session_spawn_task]
input: {"title":"MULTILINGUAL Phase A ? add es/da/id SPA catalogs","cwd":"C:\\Users\\avidu\\Projects\\khelsutra","tldr":"Add Spanish, Danish, and Indonesian to the Studio SPA (catalog-only, Latin script). Spanish needs a 3-form ICU plural. Owner-approved expansion #58.","prompt":"Implement MULTILINGUAL_EXPANSION Phase A in the khelsutra repo (C:\\Users\\avidu\\Projects\\khelsutra): add Spanish (es), Danish (da), and Indonesian (id) to the Studio SPA. Owner-approved (#58, `docs/MULTILINGUAL\_EXPANSION.md`). Phase A is Latin-script, catalog-only ? NO new fonts.\n\nCRITICAL ? Spanish plural: es is a 3-form CLDR language `{one, many, other}`; da and id are 2-form `{one, other}` like en. The catalog has ICU plural keys (`footer.count`, `footer.floorSomeLead`, `selection.count`). For es write `{count, plural, one {# ?} many {# ?} other {# ?}}`; for da/id use `{one, other}`.\n\nSteps:\n1. Add `web/src/i18n/locales/{es,da,id}/common.json`, each mirroring `en/common.json` EXACTLY (every key ? key-parity CI enforces it). AI-drafted ? native review pending (note it).\n2. In `web/src/i18n/index.ts`: add `es`,`da`,`id` to `SUPPORTED\_LOCALES`, import the 3 catalogs, add them to `resources`.\n3. Add the autonynms to the `lang.\*` block in ALL catalogs (key-parity needs `lang.es`/`lang.da`/`lang.id` in every locale): `lang.es`=\`Espa\u00f1ol\`, `lang.da`=\`Dansk\`, `lang.id`=\`Bahasa Indonesia\`.\n4. `LanguageSwitcher` renders from `SUPPORTED\_LOCALES`, so it auto-lists them ? but verify the header still fits (6 languages); if cramped, that's fine for alpha (note it).\n5. Tests: `hygiene.test.ts` key-parity auto-covers; add `i18n.test.tsx` render assertions for es/da/id + an es 3-form plural test (count 1 ? `one`, 2 ? `many`, 1000000 ? `other` per CLDR es) using `i18n.t(selection.count,{count})`.\n\nVerify: `cd web && npm test`, `npx tsc --noEmit`, `npm run build` all green.\n\nConventions (shared checkout): `git fetch origin` then `git checkout -b feat/mlx-phaseA-es-da-id origin/mas`...\n[/external_agent_tool_call]

570. assistant (2026-07-02T06:11:33.502Z)

[external_agent_tool_result]
Noted (position 2, task_id: task_e15e9643). A chip is showing for the user ? they can start it in a fresh worktree with one click, or dismiss it. If this suggestion becomes stale or superseded, call dismiss_task with this task_id. Currently pending: task_b3060842 "B3 ? make site/ marketing trilingual (shared t() shim)", task_e15e9643 "MULTILINGUAL Phase A ? add

es/da/id SPA catalogs". Continue your current work.
[/external_agent_tool_result]

571. assistant (2026-07-02T06:11:33.502Z)

```
[external_agent_tool_call: mcp_ccd_session_spawn_task]
input: {"title": "T11 ? retention sweep for output/ + uploads/", "tldr": "Add a retention sweep that purges old uploaded videos and generated reels/frames past a window, before non-owner exposure. The owner sets the retention window.", "prompt": "Implement T11 in the engine repo (badminton-highlight-indexer): a retention sweep for the engine's `output/` and `uploads/` directories, needed before non-owner (alpha) exposure of the hosted tool. Tracked in `docs/LOCAL_APPLIANCE_ARCHITECTURE.md` §9 (T11) and the `ui-driven-cloud-serving-gap` thread.\n\nGoal: a sweep (tool/job) that purges, past a retention window, the uploaded source videos (`uploads/`, written by `backend/api/uploads.py` ? default `output/uploads`) and generated artifacts (`output/` reels/frames). \n\nIMPORTANT: do NOT hand-roll this ? there is already a cache-hygiene policy + a dry-run-first cleanup tool (`tools/cleanup_caches.py`, plus a retention table in the cache-hygiene notes). EXTEND that tooling (or add a sibling sweep consistent with it: keep DB/CSVs/reels per the retention policy, purge stale frames/staged/uploads). Dry-run-first; loud structured logging.\n\nOWNER-GATED: the retention WINDOW is the owner's call ? surface a recommendation (e.g. uploads 7 days, output reels per existing policy) and ask before hardcoding; default to a conservative dry-run.\n\nConventions (? this engine checkout is SHARED with a concurrent serving session): `git fetch origin` then `git checkout -b fix/t11-retention-sweep origin/master`; re-verify HEAD before commit; stage EXPLICIT paths (never `git add -A`); ONE PR. Full local gate before sign-off: `python run_all_tests.py`, `ruff check .`, `mypy` per CI, coverage ? the repo floor.\n\nRead first: `docs/LOCAL_APPLIANCE_ARCHITECTURE.md` §9, `tools/cleanup_caches.py`, the cache-hygiene policy notes, `backend/api/uploads.py` (where uploads land), and the engine config for `security.upload_dir`."}
[/external_agent_tool_call]
```

572. assistant (2026-07-02T06:11:33.503Z)

```
[external_agent_tool_result]
Noted (position 3, task_id: task_2480b9e2). A chip is showing for the user ? they can start it in a fresh worktree with one click, or dismiss it. If this suggestion becomes stale or superseded, call dismiss_task with this task_id. Currently pending: task_b3060842 "B3 ? make site/ marketing trilingual (shared t() shim)", task_e15e9643 "MULTILINGUAL Phase A ? add es/da/id SPA catalogs", task_2480b9e2 "T11 ? retention sweep for output/ + uploads/". Continue your current work.
[/external_agent_tool_result]
```

573. assistant (2026-07-02T06:11:33.503Z)

All three immediate-next items are now spawn-off chips you can launch as parallel sessions. Let me refresh this track's handoff memory to capture the session's progress.

574. assistant (2026-07-02T06:11:33.503Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\next-session-ux-localization.md
[/external_agent_tool_call]
```

575. assistant (2026-07-02T06:11:33.503Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\next-session-ux-localization.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

576. assistant (2026-07-02T06:11:33.503Z)

```
[external_agent_tool_call: Edit]
```

file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\next-session-ux-localization.md
[/external_agent_tool_call]

577. assistant (2026-07-02T06:11:33.503Z)

[external_agent_tool_result]
The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\next-session-ux-localization.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

578. assistant (2026-07-02T06:11:33.503Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\next-session-ux-localization.md
[/external_agent_tool_call]

579. assistant (2026-07-02T06:11:33.503Z)

[external_agent_tool_result]
The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\next-session-ux-localization.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

580. assistant (2026-07-02T06:11:33.503Z)

Let me add a concise checkpoint to the shared `current-state.md`. Reading its current top first (it may have shifted via the concurrent session):

581. assistant (2026-07-02T06:11:33.503Z)

[external_agent_tool_call: Read]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md
[/external_agent_tool_call]

582. assistant (2026-07-02T06:11:33.503Z)

[external_agent_tool_result]
10 # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12 > ****Scope of this file = LIVE only:**** the current handoff + ***today's*** checkpoints + open PRs/decisions. Keep it read-first-short (?1 page). Older checkpoints are archived in [[current-state-archive]] ? when today's checkpoints age out (next clean-slate day), move them there, don't let them pile up here.
13
14 ****Checkpoint:**** 2026-06-21 (****ALPHA-SHAREABLE ? Claude's half DONE; 3 PRs merged****). Owner's #1 = "highlights from a page I can share, no SSH." ****KEY CORRECTION** (anti-spiral [[lesson-verify-engine-surface-before-scoping]]):****** the icebreaker said do "B-P2 ingest screen + M7 Cloud Run ***service***" ? but a recon ****workflow**** (`wf\_5de56b64`, 5 readers across both repos) proved ****B-P2 was ALREADY shipped**** (khelsutra #56, live e2e `DRY\_RUN\_CUJ.md` §9) + CORS is already env-driven ? ****alpha-shareable is a DEPLOY/AUTH task, not an SPA build****. ****Owner decisions (2x AskUserQuestion):**** topology ****i)** Tunnel-to-a-box******, sub-topology ****SPLIT**** (SPA on CF Pages + engine on `api.` cloudflared tunnel + CF Access; single-origin Caddy kept as the alt). ****3 PRs MERGED:**** ? ****engine** [#294](https://github.com/Khelsutra/badminton-highlight-indexer/pull/294)****** (`0c0e064`) ****exposure-safety boot posture**** ? `startup.py` fails fast at boot (server-only `include\_exposure=True`) when `edge\_auth\_enabled` is ON but half-configured (`EXPOSED\_NO\_VIDEO\_ROOT`/`EXPOSED\_EDGE\_AUTH\_INCOMPLETE` fatal, `EXPOSED\_AUTH\_EXTRA\_MISSING` warn); ****default-OFF****; aligned `auth.resolve\_tenant` whitespace; adversarial 3-lens review (`wf\_35b735a9`) ? no bypass; +16 tests; 1259 green cross-OS. ? ****khelsutra** [#64](https://github.com/avidullu/khelsutra/pull/64)****** (`8bbc557`) ****SPA `VITE\_API\_BASE`**** ?

```

`apiBase()` env-driven (default same-origin `/api`) + `vite-env.d.ts` + 3 tests; build-smoke
proved the override bakes in. ? **khelsutra
[#65](https://github.com/avidullu/khelsutra/pull/65)** (`4627830`) **deploy kit** ?
`deploy/DEPLOY_ALPHA.md` (split runbook + CORS-preflight gotcha + posture test) +
`cloudflared.config.example.yml` + tracker pointers. Each isolated/default-OFF/adversarial-
reviewed/CI-green/merged-on-green; khelsutra built in a dedicated **worktree** (main tree had
sibling A8 WIP, since merged #63; ? `rmdir` the `node_modules` junction BEFORE `git worktree
remove`). **? REMAINING for alpha = OWNER stand-up only** (cloudflared tunnel + CF Pages + CF
Access + key rotation per `khelsutra/deploy/DEPLOY_ALPHA.md`). **? NEXT engine code (Claude-
doable, batch-authorized): M12** (gdrive-api, clean, no DB migration) then **M11** (flywheel
skeleton, privacy-gated on v7 consent). [[ui-driven-cloud-serving-gap]] [[lesson-verify-engine-
surface-before-scoping]]
15
16      ***? NEXT SESSION FOCUS (2026-06-21, after the big localization+dry-run session) ? ?
SUPERSEDED by the checkpoint above (B-P2 confirmed DONE):** SPA is fully **en+kn+hi (DONE)**; a
real **UI?engine?reel?download CUJ ran (en+hi)**; a **critical engine bug fixed** (#281). **?
NEXT BUILD (owner-picked): (1) Path B ? point at a bucket video IN THE UI.** ? **B-P1 VERIFIED
ALREADY DONE ? NO ENGINE CODE (do NOT re-scope it as a feature):** `POST /api/process` has
accepted `gs://`/`gcs://` URIs since **M2/#280** (submit resolves the URI + skips the local-
exists-404 for non-local backends `main.py:283`; the worker fetches to scratch
`acquire_local_input main.py:159` ? same hash?validate?segment?DB pipeline ? queryable API-DB
`video_id`); **TESTED**
(`tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches` + hosted-
mode-reject / unknown-scheme-400 / CLI variants). Confirmed by a **3-skeptic verification
workflow (0 refuted, gaps:[])**. **Scope pin:** holds in DEFAULT single-user mode (hosted-mode
`allowed_video_root` correctly **400s** a bucket URI ? by design). Stale "local-path-only / net-
new" docs **CORRECTED ? khelsutra [#55]** (DRY_RUN_CUJ $1/2/4/6/8 + LOCAL_APPLIANCE + README).
**? So Path B = ONL...
[/external_agent_tool_result]

```

583. assistant (2026-07-02T06:11:33.503Z)

The concurrent session added the ALPHA-SHAREABLE checkpoint (#294/#64/#65) ? confirming #64/#65 are its T2 deploy-kit work. Let me insert my session checkpoint above it:

584. assistant (2026-07-02T06:11:33.503Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md
[/external_agent_tool_call]

```

585. assistant (2026-07-02T06:11:33.503Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\current-state.md has been updated successfully. (file state is current in your
context ? no need to Read it back)
[/external_agent_tool_result]

```

586. assistant (2026-07-02T06:11:33.504Z)

Let me verify the final clean git state and stop the dev server:

587. assistant (2026-07-02T06:11:33.504Z)

```

[external_agent_tool_call: ToolSearch]
input: {"query":"select:mcp__Claude_Preview__preview_stop","max_results":2}
[/external_agent_tool_call]

```

588. assistant (2026-07-02T06:11:33.504Z)

```

[external_agent_tool_result]
[/external_agent_tool_result]

```

589. assistant (2026-07-02T06:11:33.504Z)

```
[external_agent_tool_call: Bash]
description: Final git state check on khelsutra
command: echo "=== branch + sync ===" && git branch --show-current && git fetch origin -q &&
echo "local master: $(git rev-parse --short master) origin/master: $(git rev-parse --short
origin/master)" && echo "=== uncommitted (should be only untracked .claude/ + site/access.db)
=== " && git status --short && echo "=== my merged PRs this session ===" && git log --oneline -6
origin/master
[/external_agent_tool_call]
```

590. assistant (2026-07-02T06:11:33.504Z)

```
[external_agent_tool_result]
=== branch + sync ===
master
local master: ca619fb origin/master: ca619fb
=== uncommitted (should be only untracked .claude/ + site/access.db) ===
?? .claude/
?? site/access.db
=== my merged PRs this session ===
ca619fb feat(web): T4 ? in-UI file upload (the runnable-tool keystone) (#66)
4627830 docs(deploy): alpha deploy kit ? make Studio a shareable page (split: Pages + api.*
tunnel + CF Access) (#65)
8bbc557 feat(web): VITE_API_BASE ? let the SPA target a split api.* origin (default same-origin
/api) (#64)
92a6c45 feat(web): A8 ? background-processing reassurance + icon+text status chips (#63)
501c014 feat(web): A5 ? friendly engine-error mapping across build + load (shared errors.*)
(#62)
95195a5 feat(web): A4 ? mobile touch targets (?44px) + visible tooltip text (#61)
[/external_agent_tool_result]
```

591. assistant (2026-07-02T06:11:33.504Z)

Clean wrap: on `master`, local ? `origin/master` (`ca619fb`); only the pre-existing untracked
`.claude/` + `site/access.db` remain (not mine); all feature branches deleted on merge. Stopping
the dev server:

592. assistant (2026-07-02T06:11:33.504Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_stop]
input: {"serverId":"001260a2-3d4b-4722-8222-5e253669c429"}
[/external_agent_tool_call]
```

593. assistant (2026-07-02T06:11:33.504Z)

```
[external_agent_tool_result]
Server 001260a2-3d4b-4722-8222-5e253669c429 stopped
[/external_agent_tool_result]
```

594. assistant (2026-07-02T06:11:33.504Z)

Done. Here's the session summary.

Shipped this session ? 4 PRs merged (all green CI, browser-verified, each adversarially reviewed via a workflow)

```
| PR | What |
|----|-----|
| [#61](https://github.com/avidullu/khelsutra/pull/61) **A4** | ?44px mobile tap targets across
the SPA; enlarged parsed-pill + sample chips; **hover-only `title=` text made visible** (the
"not available yet" reason + per-card shot confidence ? phones never show `title=`). |
| [#62](https://github.com/avidullu/khelsutra/pull/62) **A5** | One shared
[ `lib/errorText.ts` ](web/src/lib/errorText.ts) + `errors.*` namespace maps **build + load +
ingest** failures the same friendly, localized way ? no more raw `ApiError: ?`/`HTTP 500`. Logs
```

```
gained `code`+`detail`. |
| [#63](https://github.com/avidullu/khelsutra/pull/63) **A8** | Background-processing
reassurance + icon+text [StatusChip](web/src/components/StatusChip.tsx) (never icon-only); a
"Stitching your reel?" building card + ?? status icons + `aria-live`. |
| [#66](https://github.com/avidullu/khelsutra/pull/66) **T4 ? North-Star keystone** | **In-UI
file upload**: a file picker ? chunked `/api/upload/*` ([`api/client.ts`](web/src/api/client.ts)
+ [`hooks/useUpload.ts`](web/src/hooks/useUpload.ts)) ? server path ? `/api/process` ? the
existing poll?load?compile?download. Alpha users can now point the tool at a file, not just a
`gs://` URI. **153 web tests.** |
```

Every item folded its review findings (e.g. A8's `aria-live` + honest copy; T4's WCAG focus-visible, NaN-safe chunking, named progressbar). Trackers in `FRIENDLY\_AND\_MULTILINGUAL.md` §7 and `LOCAL\_APPLIANCE\_ARCHITECTURE.md` §9 updated; memory handoff + current-state refreshed.

Decisions you made

- ****Build T4 first**** ?; ****B3 catalogs hoist to repo-root** ``/locales/`` (locked, baked into the chip).

Recorded as spawn-off chips (one-click ? parallel session/worktree)

- ****B3**** ? ``site/`` trilingual via the shared shim (hoist approach baked in)
- ****MULTILINGUAL Phase A**** ? es/da/id catalogs (es 3-form plural noted)
- ****T11**** ? retention sweep (engine repo; you set the window)

? B3 + MULTILINGUAL both edit the catalogs ? don't run them at the same time.

Two things worth your attention

1. ****T2** is nearly done by the concurrent session ? their #64 (``VITE_API_BASE``) + #65 (alpha deploy kit: CF Pages + ``api.*`` tunnel + CF Access, see ``deploy/DEPLOY_ALPHA.md``) cover the plumbing. ****Remaining is owner stand-up only:**** set up Cloudflare Access, ****rotate** ``GEMINI_API_KEY``, run the deploy. One caveat surfaced by the T4 review: in hosted mode set the engine's ``allowed_video_root`` to a parent of ``upload_dir``, or ``/api/process`` will reject uploaded files.
2. ****Process honesty:**** I started the T4 edits on ``master`` by mistake and caught it ****before any commit**** ? moved them to a feature branch (``checkout -b`` preserves the tree), then rebased onto ``origin/master`` after your sibling session's #64/#65 landed (clean merge). No master pollution.

Repo is clean: on ``master`` ? ``origin/master`` (``ca619fb``), no open PRs of mine, all branches deleted, dev server stopped. The two untracked files (``.`` `claude/``, ``site/access.db``) pre-existed this session.

Next natural picks: launch the B3/MULTILINGUAL/T11 chips, do the T2 owner stand-up, and tackle A9/A3 + native review.

595. user (2026-07-02T06:11:33.504Z)

please keep merging the green CI PRs and ensure to keep improving the test coverage

596. assistant (2026-07-02T06:11:33.504Z)

On it ? I'll keep building the queued items myself, one PR each, merging on green and adding tests each time. Since I'm doing these here, I'll dismiss the corresponding spawn-off chips so they don't get double-launched. Starting with ****MULTILINGUAL Phase A**** (mechanical, high coverage value, owner-approved #58).

Let me read the expansion plan's specifics first, and dismiss its chip.

597. assistant (2026-07-02T06:11:33.504Z)

```
[external_agent_tool_call: mcp_ccd_session_dismiss_task]
input: {"task_id":"task_e15e9643","reason":"doing it now in this session per owner \"keep
merging green CI PRs\""}
[/external_agent_tool_call]
```

598. assistant (2026-07-02T06:11:33.504Z)

[external_agent_tool_result]

Task task_e15e9643 withdrawn ? the chip is no longer shown to the user. Continue your current work.

[/external_agent_tool_result]

599. assistant (2026-07-02T06:11:33.505Z)

[external_agent_tool_call: Read]

file: C:\Users\avidu\Projects\khelsutra\docs\MULTILINGUAL_EXPANSION.md

[/external_agent_tool_call]

600. assistant (2026-07-02T06:11:33.505Z)

[external_agent_tool_result]

```
1      # Multilingual Expansion ? 7 new locales (th · id · da · es · zh-Hans · ja · ko)
2
3      > **Status:** `DRAFT` ? **owner-gated (approval needed to START this line)** .
**Owner:** Avi Dullu . **Created:** 2026-06-21 · **Last updated:** 2026-06-21
4      > **Lifecycle:** `DRAFT` ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
5      > **Tracking anchors:** $7 progress tracker is the source of truth; index in
`docs/README.md`; pointer in memory `current-state`; thread in `NEXT_STEPS.md` once work starts.
6      > **Relation to existing docs:** **PEER-OF / extends**
[ `FRIENDLY_AND_MULTILINGUAL.md` ] (FRIENDLY_AND_MULTILINGUAL.md) (the `en`/`kn`/`hi` foundation)
and the engine design `I18N_PLAN.md`. **Reuses that foundation verbatim** (react-i18next +
i18next-icu + browser detector + key-parity/no-bare-strings CI + self-hosted OFL Noto via
`@fontsource` + Gemini-draft?native-review). This is a **parallel** line: it adds locales, it
does **not** re-architect i18n.
7      > **Honesty note:** each claim tagged `[verified]` (checked vs the working tree this
session), `[researched]` (sourced, needs sign-off), or `[design]` (proposed). Not legal advice ?
the OFL-font carve-out is the owner's standing call (D7 in FRIENDLY).
8
9      ---
10
11     ## 0. TL;DR
12     Add **Thai, Indonesian, Danish, Spanish, Chinese (Simplified), Japanese, Korean** to the
Studio SPA with the **same rigor as en/kn/hi**. The good news `[verified]`: the i18n foundation
already scales ? **zero changes to the negotiation contract or the catalog format** ; adding a
locale is "drop a reviewed `common.json` + (if non-Latin) bundle one OFL Noto font + keep CI
key-parity green." The work splits cleanly into **Phase A** (es/da/id ? *catalog-only, no
fonts*), **Phase B** (Thai ? *one small OFL font*), and **Phase C** (zh-Hans/ja/ko ? *large OFL
CJK fonts + lazy-load + an offline-bundle subsetting decision + the zh script-variant call*).
**The single most important caveats:** (1) **native-speaker review per language** is the real
cost and the public-launch gate (Gemini drafts are *inference-only*, never the source of truth ?
the standing guardrail); (2) **CJK fonts add tens of MB** to the otherwise-tiny air-gapped
bundle; (3) **Spanish now has a 3-form plural** (`{one, many, other}`) ? the one authoring
wrinkle.
13
14     ## 1. Problem & goal
15     **Why now.** en/kn/hi shipped and proved the foundation is locale-pluggable (a Kannada-
default browser auto-renders Kannada; the switcher walks kn?hi?en). The owner wants to broaden
reach beyond the India beachhead ? **SE Asia (th, id), Europe (da, es), and East Asia (zh, ja,
ko)** .
16     **Concrete outcome ("good"):** a user whose browser is in any of the 7 languages lands
on the Studio (and, once `site/` is localized via FRIENDLY B3, the marketing pages too), and ?
without help ? understands the product, runs the tool, and gets a reel, **entirely in their
language** , with the fonts rendering crisply offline.
17     **Read to get current:** this doc $3? $4;
[ `FRIENDLY_AND_MULTILINGUAL.md` ] (FRIENDLY_AND_MULTILINGUAL.md) (the foundation this rides on,
esp. $4 design + $6.5 negotiation tests); `web/src/i18n/index.ts` (the detector/ICU/fallback
config) + `web/src/main.tsx` (the `@fontsource` font-load pattern) +
`web/src/i18n/hygiene.test.ts` (key-parity + no-bare-strings + offline guards).
18
```

```

19      ## 2. Decisions locked (inherited from FRIENDLY unless noted)
20      | # | Decision | Source / date | Implication |
21      |---|-----|-----|-----|
22      | ED1 | **Same framework, no re-architecture** ? react-i18next + i18next-icu + `i18next-
browser-languagedetector`; one `en` source catalog + per-locale siblings; **always fall back to
`en`** | FRIENDLY D-* / `i18n/index.ts` `[verified]` | Each new locale = add to
`SUPPORTED_LOCALES` + a `common.json`; negotiation/format unchanged |
23      | ED2 | **Self-host OFL Noto via `@fontsource`** (the D7 OFL-for-media carve-out) ?
**extend it to the 4 new Noto fonts** (Thai, SC, JP, KR) | FRIENDLY D7 (owner, 2026...
[/external_agent_tool_result]

```

601. assistant (2026-07-02T06:11:33.505Z)

I have the Phase A specs: **es** = 3-form `{one, many, other}`, **da** = 2-form `{one, other}`, **id** = 1-form `{other}` only. Let me read the current complete en catalog to translate from:

602. assistant (2026-07-02T06:11:33.505Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\kheilsutra\web\src\i18n\locales\en\common.json
[/external_agent_tool_call]

```

603. assistant (2026-07-02T06:11:33.505Z)

```

[external_agent_tool_result]
1      {
2      "app": {
3          "badge": "pick your rallies · alpha",
4          "title": "Pick the rallies for your reel",
5          "intro": "Filter by <b>rally length</b> ? e.g. <i>?top 15 longest?</i> ? then fine-
tune by tapping cards. Everything starts <b>selected</b>; just drop the ones you don't want."
6      },
7      "lang": {
8          "label": "Language",
9          "en": "English",
10         "kn": "?????",
11         "hi": "?????"
12     },
13     "bulk": {
14         "selectAll": "Select all",
15         "clear": "Clear",
16         "invert": "Invert"
17     },
18     "mode": {
19         "demo": "These are sample rallies ? process a video above to work on a real match.",
20         "loading": "Finding the rallies in your video?",
21         "error": "Couldn't load this video's rallies: {error}",
22         "loaded": "Found <b>{count}</b> rallies in your video."
23     },
24     "query": {
25         "placeholder": "e.g. over 10s · top 15 longest · shortest first",
26         "ariaLabel": "Filter rallies by length, count or order",
27         "apply": "Apply",
28         "notAvailable": "not available yet",
29         "notAvailableTitle": "?{token}? needs {kind} detection ? not available yet",
30         "notAvailableHelp": "Shot type, players and score aren't available yet ? filter by
rally length instead.",
31         "unreadable": "Couldn't read that ? try ?over 10s? or ?top 15 longest?.",
32         "hint": "Type over 10s for rallies longer than 10 seconds ? or tap a sample below.",
33         "try": "Sample queries:",
34         "support": "Filtering by <b>rally length</b> is fully supported. Shot-count is
<x>experimental</x> (under development)."
35     },
36     "grid": {

```

```

37     "empty": "No rallies to show yet.",
38     "rally": "Rally {n}",
39     "inReel": "Rally {n}, in reel",
40     "notInReel": "Rally {n}, not in reel",
41     "start": "start",
42     "long": "long",
43     "shots": "~{count} shots · experimental",
44     "shotsTitle": "Shot count is experimental (under development) and may be
inaccurate{conf}. Filter by rally length instead.",
45     "shotsConfidence": " ? confidence: {conf}"
46 },
47     "footer": {
48         "count": "{count, plural, one {# rally} other {# rallies}}",
49         "summaryRest": "selected · {total} total",
50         "reelName": "Reel name",
51         "reelNamePlaceholder": "my-highlights",
52         "build": "Build reel ({count})",
53         "building": "Building?",
54         "floorOne": "<b>Your only selected rally</b> has fewer than {min} shots ? it would
be skipped, so the build would fail. Pick a longer rally.",
55         "floorAll": "<b>All {count} selected rallies</b> have fewer than {min} shots ?
they'd all be skipped, so the build would fail. Pick longer rallies.",
56         "floorSomeLead": "{count, plural, one {# rally} other {# rallies}}",
57         "floorSomeRest": "with fewer than {min} shots will be skipped when building (they're
too short).",
58     },
59     "result": {
60         "ready": "Your reel is ready",
61         "failed": "Build failed",
62         "building": "Stitching your reel?",
63         "buildingReassure": "Joining your selected rallies into one video ? this usually
only takes a moment.",
64         "dismiss": "Dismiss",
65         "download": "Download {filename}"
66     },
67     "build": {
68         "noVideo": "These are sample rallies ? process a real video first to build a reel."
69     },
70     "ingest": {
71         "title": "Process a video",
72         "intro": "Point KhelSutra at a video and it finds the rallies for you ? this can
take a few minutes.",
73         "uriPlaceholder": "gs://your-bucket/match.mp4",
74         "uriAriaLabel": "Video location ? a gs:// bucket URL or a path on the engine host",
75         "hint": "Paste a cloud-storage link (gs://?), or a path to a video on the machine
running KhelSutra.",
76         "submit": "Process video",
77         "submitting": "Submitting?",
78         "cancel": "Cancel",
79         "working": "Processing your video?",
80         "progress": "{progress}",
81         "retry": "Try again",
82         "reassure": "Safe to wait ? KhelSutra keeps working even if your connection is slow;
a long video can take sev...
[/external_agent_tool_result]

```

604. assistant (2026-07-02T06:11:33.505Z)

Now I have the complete en catalog. Let me set up the MULTILINGUAL Phase A branch and mark progress:

605. assistant (2026-07-02T06:11:33.505Z)

```

[external_agent_tool_call: TaskUpdate]
input: {"taskId":"8","status":"in_progress"}

```


[/external_agent_tool_call]

606. assistant (2026-07-02T06:11:33.505Z)

[external_agent_tool_result]
Updated task #8 status
[/external_agent_tool_result]

607. assistant (2026-07-02T06:11:33.505Z)

[external_agent_tool_call: mcp_ccd_session__mark_chapter]
input: {"title":"MULTILINGUAL Phase A ? es/da/id","summary":"Owner: keep merging green PRs. Adding Spanish/Danish/Indonesian SPA catalogs (es 3-form plural)."}
[/external_agent_tool_call]

608. assistant (2026-07-02T06:11:33.505Z)

[external_agent_tool_result]
Chapter marked: "MULTILINGUAL Phase A ? es/da/id". Continue your current work.
[/external_agent_tool_result]

609. assistant (2026-07-02T06:11:33.505Z)

[external_agent_tool_call: Bash]
description: Cut MULTILINGUAL Phase A branch
command: cd /c/Users/avidu/Projects/khelsutra && git fetch origin -q && git checkout -b feat/mlx-phaseA-es-da-id origin/master 2>&1 | tail -2 && git log --oneline -1
[/external_agent_tool_call]

610. assistant (2026-07-02T06:11:33.505Z)

[external_agent_tool_result]
branch 'feat/mlx-phaseA-es-da-id' set up to track 'origin/master'.
Switched to a new branch 'feat/mlx-phaseA-es-da-id'
ca619fb feat(web): T4 ? in-UI file upload (the runnable-tool keystone) (#66)
[/external_agent_tool_result]

611. assistant (2026-07-02T06:11:33.505Z)

Now add the `lang.es/da/id` autonyms to the existing en/kn/hi catalogs (key-parity needs them everywhere):

612. assistant (2026-07-02T06:11:33.505Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json
[/external_agent_tool_call]

613. assistant (2026-07-02T06:11:33.505Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

614. assistant (2026-07-02T06:11:33.505Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json
[/external_agent_tool_call]

615. assistant (2026-07-02T06:11:33.506Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khehsutra\web\src\i18n\locales\kn\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

616. assistant (2026-07-02T06:11:33.506Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khehsutra\web\src\i18n\locales\hi\common.json
[/external_agent_tool_call]

617. assistant (2026-07-02T06:11:33.506Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khehsutra\web\src\i18n\locales\hi\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

618. assistant (2026-07-02T06:11:33.506Z)

Now create the three Phase A catalogs. Spanish first (with the 3-form `{one, many, other}` plural):

619. assistant (2026-07-02T06:11:33.506Z)

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\khehsutra\web\src\i18n\locales\es\common.json
[/external_agent_tool_call]

620. assistant (2026-07-02T06:11:33.506Z)

[external_agent_tool_result]
File created successfully at:
C:\Users\avidu\Projects\khehsutra\web\src\i18n\locales\es\common.json (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

621. assistant (2026-07-02T06:11:33.506Z)

Now Danish (2-form `{one, other}`):

622. assistant (2026-07-02T06:11:33.506Z)

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\khehsutra\web\src\i18n\locales\da\common.json
[/external_agent_tool_call]

623. assistant (2026-07-02T06:11:33.506Z)

[external_agent_tool_result]
File created successfully at:
C:\Users\avidu\Projects\khehsutra\web\src\i18n\locales\da\common.json (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

624. assistant (2026-07-02T06:11:33.506Z)

Now Indonesian (1-form `{other}` only ? CLDR `id` has no plural distinction):

625. assistant (2026-07-02T06:11:33.506Z)

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\khehsutra\web\src\i18n\locales\id\common.json

[/external_agent_tool_call]

626. assistant (2026-07-02T06:11:33.506Z)

[external_agent_tool_result]

File created successfully at:

C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\id\common.json (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

627. assistant (2026-07-02T06:11:33.506Z)

Now wire the three locales into `i18n/index.ts`:

628. assistant (2026-07-02T06:11:33.506Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\index.ts

[/external_agent_tool_call]

629. assistant (2026-07-02T06:11:33.506Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\index.ts has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

630. assistant (2026-07-02T06:11:33.506Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\index.ts

[/external_agent_tool_call]

631. assistant (2026-07-02T06:11:33.507Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\index.ts has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

632. assistant (2026-07-02T06:11:33.507Z)

Now tests for es/da/id ? render, plural correctness, and negotiation. Appending to `i18n.test.tsx`:

633. assistant (2026-07-02T06:11:33.507Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\i18n.test.tsx

[/external_agent_tool_call]

634. assistant (2026-07-02T06:11:33.507Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\i18n.test.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

635. assistant (2026-07-02T06:11:33.507Z)

Now the structural 3-form check in `hygiene.test.ts` (the catalog-tooling test):

636. assistant (2026-07-02T06:11:33.507Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\hygiene.test.ts
[/external_agent_tool_call]
```

637. assistant (2026-07-02T06:11:33.507Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\hygiene.test.ts has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

638. assistant (2026-07-02T06:11:33.507Z)

```
[external_agent_tool_call: Bash]
description: Run suite + tsc + build for MULTILINGUAL Phase A
command: cd /c/Users/avidu/Projects/khelsutra/web && npm test --silent 2>/dev/null | grep -E
"Test Files|Tests |FAIL|?|x|? src" | tail -12 && echo "=== tsc ===" && npx tsc --noEmit && echo
"tsc OK" && echo "=== build ===" && npm run build --silent 2>&1 | tail -1
[/external_agent_tool_call]
```

639. assistant (2026-07-02T06:11:33.507Z)

```
[external_agent_tool_result: error]
Exit code 2
  Test Files   18 passed (18)
     Tests    163 passed (163)
=== tsc ===
src/components/LanguageSwitcher.tsx(6,7): error TS2739: Type '{ en: string; kn: string; hi:
string; }' is missing the following properties from type 'Record<"en" | "kn" | "hi" | "es" |
"da" | "id", string>': es, da, id
[/external_agent_tool_result]
```

640. assistant (2026-07-02T06:11:33.507Z)

163 tests pass (+10), but tsc caught `LanguageSwitcher`'s `LANG\_LABEL\_KEY` needs the new locales (the `Locale` type expanded). Fixing:

641. assistant (2026-07-02T06:11:33.507Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\LanguageSwitcher.tsx
[/external_agent_tool_call]
```

642. assistant (2026-07-02T06:11:33.507Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\LanguageSwitcher.tsx has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

643. assistant (2026-07-02T06:11:33.507Z)

```
[external_agent_tool_call: Bash]
description: Re-verify after LanguageSwitcher fix
command: cd /c/Users/avidu/Projects/khelsutra/web && npx tsc --noEmit && echo "tsc OK" && npm
test --silent 2>/dev/null | grep -E "Test Files|Tests " | tail -2 && npm run build --silent 2>&1
| tail -1
[/external_agent_tool_call]
```

644. assistant (2026-07-02T06:11:33.508Z)

```
[external_agent_tool_result]
tsc OK
```

```
Test Files 18 passed (18)
Tests 163 passed (163)
? built in 1.17s
[/external_agent_tool_result]
```

645. assistant (2026-07-02T06:11:33.508Z)

tsc, 163 tests, build all green. Let me browser-verify the new locales render and the switcher shows all six. Starting the preview:

646. assistant (2026-07-02T06:11:33.508Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_start]
input: {"name":"khelsutra-web"}
[/external_agent_tool_call]
```

647. assistant (2026-07-02T06:11:33.508Z)

```
[external_agent_tool_result]
{
  "serverId": "b116717c-076f-463c-b562-6b4ea2cbaab3",
  "port": 5174,
  "name": "khelsutra-web",
  "reused": false
}
Server started successfully on port 5174.
[/external_agent_tool_result]
```

648. assistant (2026-07-02T06:11:33.508Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_eval]
input: {"serverId":"b116717c-076f-463c-b562-6b4ea2cbaab3","expression":"(async () => {\n
localStorage.setItem('khelsutra.lang','es');\n location.reload();\n await new
Promise(r=>setTimeout(r,1200));\n return 'reloaded es';\n})()"}
[/external_agent_tool_call]
```

649. assistant (2026-07-02T06:11:33.508Z)

```
[external_agent_tool_result: error]
Eval failed: Inspected target navigated or closed
[/external_agent_tool_result]
```

650. assistant (2026-07-02T06:11:33.508Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_eval]
input: {"serverId":"b116717c-076f-463c-b562-6b4ea2cbaab3","expression":"(() => {\n const h1 =
document.querySelector('h1')?.textContent;\n const footer = document.querySelector('[data-
testid=\"selection-summary\"]')?.textContent;\n const langs =
[...document.querySelectorAll('nav[aria-label] button')].map(b=>b.textContent);\n const
htmlLang = document.documentElement.lang;\n return { h1, footer, langs, htmlLang }; \n})()"}
[/external_agent_tool_call]
```

651. assistant (2026-07-02T06:11:33.508Z)

```
[external_agent_tool_result]
{
  "footer": "6 jugadas seleccionadas · 1:22 en total",
  "h1": "Elige las jugadas para tu reel",
  "htmlLang": "es",
  "langs": [
    "English",
    "?????",
    "??????",
    "Español",
  ]
}
```

```
    "Dansk",
    "Bahasa Indonesia"
  ]
}
[/external_agent_tool_result]
```

652. assistant (2026-07-02T06:11:33.508Z)

Spanish renders perfectly ? h1 in Spanish, the **3-form plural works** ("6 jugadas seleccionadas"), `<html lang="es">`, and the switcher shows all **6 languages**. Let me confirm Danish + Indonesian too:

653. assistant (2026-07-02T06:11:33.508Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_eval]
input: {"serverId":"b116717c-076f-463c-b562-6b4ea2cbaab3","expression":"(async () => {\n  for
(const l of ['da','id']) localStorage.setItem('khelsutra.lang', l), 0;\n
localStorage.setItem('khelsutra.lang','da'); location.reload();\n})()"}
[/external_agent_tool_call]
```

654. assistant (2026-07-02T06:11:33.508Z)

```
[external_agent_tool_result]
undefined
[/external_agent_tool_result]
```

655. assistant (2026-07-02T06:11:33.509Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_eval]
input: {"serverId":"b116717c-076f-463c-b562-6b4ea2cbaab3","expression":"({ da_h1:
document.querySelector('h1')?.textContent, da_footer: document.querySelector('[data-
testid=\"selection-summary\"]')?.textContent, lang: document.documentElement.lang })"}
[/external_agent_tool_call]
```

656. assistant (2026-07-02T06:11:33.509Z)

```
[external_agent_tool_result]
{
  "da_footer": "6 dueller valgt · 1:22 i alt",
  "da_h1": "Vælg duellerne til din reel",
  "lang": "da"
}
[/external_agent_tool_result]
```

657. assistant (2026-07-02T06:11:33.509Z)

Danish renders too ("Vælg duellerne til din reel", 2-form plural "6 dueller valgt"). es + da confirmed in-browser; id covered by tests.

Let me add one more durable guard that raises coverage across **all** locales ? a **placeholder-parity** test (catches a dropped `{count}/{error}`/etc. in any translation, which key-parity doesn't):

658. assistant (2026-07-02T06:11:33.509Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\hygiene.test.ts
[/external_agent_tool_call]
```

659. assistant (2026-07-02T06:11:33.509Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\hygiene.test.ts has been updated
```

successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

660. assistant (2026-07-02T06:11:33.509Z)

```
[external_agent_tool_call: Bash]
description: Run suite with the new placeholder-parity guard
command: npm test --silent 2>/dev/null | grep -E "Test Files|Tests
|FAIL|?|×|placeholder|preserves" | tail -15
[/external_agent_tool_call]
```

661. assistant (2026-07-02T06:11:33.509Z)

```
[external_agent_tool_result]
  ? placeholder parity ? every locale preserves en's {placeholders} > da
preserves every {placeholder} that en uses 1ms
  ? placeholder parity ? every locale preserves en's {placeholders} > es
preserves every {placeholder} that en uses 1ms
    × placeholder parity ? every locale preserves en's {placeholders} >
hi preserves every {placeholder} that en uses 15ms
  ? placeholder parity ? every locale preserves en's {placeholders} > id
preserves every {placeholder} that en uses 1ms
    × placeholder parity ? every locale preserves en's {placeholders} >
kn preserves every {placeholder} that en uses 2ms
  ? no-bare-strings ? user-facing component attributes go through t() >
App.tsx: no hardcoded placeholder/aria-label/title 1ms
  ? no-bare-strings ? user-facing component attributes go through t() >
components/QueryBar.tsx: no hardcoded placeholder/aria-label/title 0ms
  ? no-bare-strings ? user-facing component attributes go through t() >
components/RallyGrid.tsx: no hardcoded placeholder/aria-label/title 0ms
  ? no-bare-strings ? user-facing component attributes go through t() >
components/SelectionFooter.tsx: no hardcoded placeholder/aria-label/title
0ms
  ? no-bare-strings ? user-facing component attributes go through t() >
components/CompileResult.tsx: no hardcoded placeholder/aria-label/title
0ms
  ? no-bare-strings ? user-facing component attributes go through t() >
components/LanguageSwitcher.tsx: no hardcoded placeholder/aria-label/title
0ms
  ? no-bare-strings ? user-facing component attributes go through t() >
components/IngestPanel.tsx: no hardcoded placeholder/aria-label/title
1ms
Test Files   1 failed | 17 passed
(18)
Tests       2 failed | 166 passed
(168)
[/external_agent_tool_result]
```

662. assistant (2026-07-02T06:11:33.509Z)

The new guard immediately caught **real** pre-existing placeholder bugs in the kn and hi catalogs** (es/da/id pass). Let me see exactly what drifted:

663. assistant (2026-07-02T06:11:33.509Z)

```
[external_agent_tool_call: Bash]
description: Show the kn/hi placeholder mismatches
command: npx vitest run src/i18n/hygiene.test.ts 2>&1 | grep -A 30 "preserves every" | grep -E
"key|en:|got:|kn|hi|AssertionError|Expected|Received|\\{ key" | head -60
[/external_agent_tool_call]
```

664. assistant (2026-07-02T06:11:33.509Z)

```
[external_agent_tool_result]
```

```

    × placeholder parity ? every locale preserves en's {placeholders} >
hi preserves every {placeholder} that en uses 10ms
    ? expected [ { key: 'mode.loading', ?(2) }, ?(1) ] to deeply equal []
    × placeholder parity ? every locale preserves en's {placeholders} >
kn preserves every {placeholder} that en uses 1ms
    ? expected [ { key: 'mode.loading', ?(2) }, ?(1) ] to deeply equal []
    ? offline / self-contained ? no external runtime assets > web/index.html
loads no absolute-URL <link>/<script src> (everything is bundled/local) 0ms
FAIL src/i18n/hygiene.test.ts > placeholder parity ? every locale
preserves en's {placeholders} > hi preserves every {placeholder} that en uses
FAIL src/i18n/hygiene.test.ts > placeholder parity ? every locale
preserves en's {placeholders} > kn preserves every {placeholder} that en uses
AssertionError: expected [ { key: 'mode.loading', ?(2) }, ?(1) ] to deeply equal
[]
- Expected
+ Received
+   "key": "mode.loading",
+   "key": "mode.error",
+   77|         if (a.join(",") !== b.join(",")) mismatches.push({ key, en: a,?
[/external_agent_tool_result]

```

665. assistant (2026-07-02T06:11:33.510Z)

Excellent ? the guard immediately earned its keep: it caught a **real pre-existing bug**. The drift is in `mode.loading` + `mode.error` for **kn** and **hi** ? they still carry a `{videoId}` placeholder that A1 (#60) removed from the en copy. This is exactly the A1 translation-drift I flagged at the start of the session; now it's caught by a test. Fixing it (re-draft the stale kn/hi `mode.\*` + `build.noVideo` to match A1's friendly en copy):

666. assistant (2026-07-02T06:11:33.510Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json
[/external_agent_tool_call]

```

667. assistant (2026-07-02T06:11:33.510Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

668. assistant (2026-07-02T06:11:33.510Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json
[/external_agent_tool_call]

```

669. assistant (2026-07-02T06:11:33.510Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

670. assistant (2026-07-02T06:11:33.510Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\hi\common.json
[/external_agent_tool_call]

```

671. assistant (2026-07-02T06:11:33.510Z)

```

[external_agent_tool_result]

```


The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\hi\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

672. assistant (2026-07-02T06:11:33.510Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\hi\common.json
[/external_agent_tool_call]

673. assistant (2026-07-02T06:11:33.510Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\hi\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

674. assistant (2026-07-02T06:11:33.510Z)

[external_agent_tool_call: Bash]
description: Re-run suite after fixing kn/hi A1 drift
command: npm test --silent 2>/dev/null | grep -E "Test Files|Tests |FAIL|?|×" | tail -5 && echo "=== tsc ===" && npx tsc --noEmit && echo "tsc OK" && echo "=== build ===" && npm run build --silent 2>&1 | tail -1
[/external_agent_tool_call]

675. assistant (2026-07-02T06:11:33.510Z)

[external_agent_tool_result]
Test Files 18 passed (18)
Tests 168 passed (168)
=== tsc ===
tsc OK
=== build ===
? built in 1.05s
[/external_agent_tool_result]

676. assistant (2026-07-02T06:11:33.510Z)

All **168 tests** green (placeholder guard now passes for every locale), tsc + build clean, and the kn/hi A1 drift is fixed. Committing MULTILINGUAL Phase A:

677. assistant (2026-07-02T06:11:33.511Z)

[external_agent_tool_call: Bash]
description: Stage MULTILINGUAL Phase A, verify clean base
command: cd /c/Users/avidu/Projects/khelsutra && echo "branch: \$(git branch --show-current)" && git add web/src/i18n/index.ts web/src/components/LanguageSwitcher.tsx web/src/i18n/locales/en/common.json web/src/i18n/locales/kn/common.json web/src/i18n/locales/hi/common.json web/src/i18n/locales/es/common.json web/src/i18n/locales/da/common.json web/src/i18n/locales/id/common.json web/src/i18n/i18n.test.tsx web/src/i18n/hygiene.test.ts && git diff --cached --name-only && echo "--- base (only my work) ---" && git log --oneline origin/master..HEAD
[/external_agent_tool_call]

678. assistant (2026-07-02T06:11:33.511Z)

[external_agent_tool_result]
branch: feat/mlx-phaseA-es-da-id
web/src/components/LanguageSwitcher.tsx
web/src/i18n/hygiene.test.ts
web/src/i18n/i18n.test.tsx
web/src/i18n/index.ts
web/src/i18n/locales/da/common.json

```
web/src/i18n/locales/en/common.json
web/src/i18n/locales/es/common.json
web/src/i18n/locales/hi/common.json
web/src/i18n/locales/id/common.json
web/src/i18n/locales/kn/common.json
--- base (only my work) ---
[/external_agent_tool_result]
```

679. assistant (2026-07-02T06:11:33.511Z)

```
[external_agent_tool_call: Bash]
description: Commit + push MULTILINGUAL Phase A
command: git commit -q -F - <<'EOF'
feat(web): MULTILINGUAL Phase A ? Spanish/Danish/Indonesian SPA catalogs (#58)
```

Adds es/da/id to the Studio SPA (catalog-only, Latin ? no new fonts), same rigor as en/kn/hi. The i18n foundation scales unchanged: add to SUPPORTED_LOCALES + a common.json.

- es/da/id `common.json` (AI-drafted ? native review pending) + `lang.{es,da,id}` autonoms in every catalog (key-parity). Wired into i18n/index.ts resources + LanguageSwitcher.
- Plurals per CLDR: `**es is 3-form {one, many, other}**` (the one authoring wrinkle), da is 2-form `{one, other}`, id is single-form `{other}`.
- Tests: es/da/id render + auto-negotiation (es-ES/da-DK/id-ID ? base) + es/id plural word-forms; a hygiene check that es plural messages carry one/many/other arms.

Also adds a `**placeholder-parity guard**` (every locale's value keeps en's `{placeholder}` set) ? which immediately caught a real pre-existing bug: kn/hi `mode.loading`/mode.error`` still carried the `{videoId}` placeholder A1 (#60) removed from en. Re-drafted the stale kn/hi `mode.*`` + `build.noVideo`` to match A1's friendly copy.

168 web tests green; tsc + build clean; key-parity + no-bare-strings + offline guards hold. Browser-verified: es ("Elige las jugadas?", "6 jugadas seleccionadas") + da render, switcher lists all 6 languages, `<html lang>` syncs. Phases B (Thai) / C (CJK) stay owner-gated.

```
Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>
EOF
git push -u origin feat/mlx-phaseA-es-da-id 2>&1 | tail -2
[/external_agent_tool_call]
```

680. assistant (2026-07-02T06:11:33.511Z)

```
[external_agent_tool_result]
To https://github.com/avidullu/khelsutra.git
* [new branch] feat/mlx-phaseA-es-da-id -> feat/mlx-phaseA-es-da-id
[/external_agent_tool_result]
```

681. assistant (2026-07-02T06:11:33.511Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_stop]
input: {"serverId":"b116717c-076f-463c-b562-6b4ea2cbaab3"}
[/external_agent_tool_call]
```

682. assistant (2026-07-02T06:11:33.511Z)

```
[external_agent_tool_result]
Server b116717c-076f-463c-b562-6b4ea2cbaab3 stopped
[/external_agent_tool_result]
```

683. assistant (2026-07-02T06:11:33.511Z)

```
[external_agent_tool_call: Bash]
description: Stop preview + create Phase A PR
command: gh pr create --title "feat(web): MULTILINGUAL Phase A ? Spanish/Danish/Indonesian SPA catalogs" --base master --head feat/mlx-phaseA-es-da-id --body "$(cat <<'EOF'
```

Adds ****Spanish, Danish, Indonesian**** to the Studio SPA ? Latin-script, ****catalog-only, no new fonts****, same rigor as en/kn/hi. The foundation scales unchanged: add to `SUPPORTED\_LOCALES` + a `common.json`.

```
- `web/src/i18n/locales/{es,da,id}/common.json` (AI-drafted ? **native review pending**) +
`lang.{es,da,id}` autonoms in **every** catalog (key-parity). Wired into `i18n/index.ts`
resources + `LanguageSwitcher`.
- **Plurals per CLDR:** **es` is 3-form `{one, many, other}` (the one authoring wrinkle),
`da` is 2-form `{one, other}`, `id` is single-form `{other}`.
- **Tests (coverage up):** es/da/id render + auto-negotiation (`es-ES`/`da-DK`/`id-ID` ? base) +
es/id plural word-forms; a hygiene check that the es plural messages carry `one`/`many`/`other`
arms.
```

Added a **placeholder-parity** guard (every locale's value must keep en's `{placeholder}` set to something key-parity doesn't check). It immediately flagged a **pre-existing bug**: `kn/hi`mode.loading` + `mode.error`` still carried the `{videoId}` placeholder that **A1 (#60)** removed from the en copy. Re-drafted the stale `kn/hi`mode.*` + `build.noVideo`` to match A1's friendly copy.

```
- **168 web tests** green; `tsc` + `vite build` clean; key-parity + no-bare-strings + offline
guards hold.
- **Browser-verified:** es renders ("Elige las jugadas para tu reel", footer "6 jugadas
seleccionadas" ? 3-form plural), da renders ("Vælg duellerne til din reel", "6 dueller valgt"),
the switcher lists all **6 languages**, `` syncs.
```

Phase ****B**** (Thai ? needs one OFL font) and ****C**** (zh-Hans/ja/ko ? large CJK fonts + offline-subset) stay ****owner-gated**** (``MULTILINGUAL_EXPANSION.md` §8 Q2/Q3/Q4`). All AI drafts await native-speaker review before wide public launch.

684. assistant (2026-07-02T06:11:33.511Z)

685. assistant (2026-07-02T06:11:33.511Z)

686. assistant (2026-07-02T06:11:33.511Z)

687. assistant (2026-07-02T06:11:33.511Z)

```
[external_agent_tool_result]
73      | ID | Deliverable | Phase | Depends on | Gated? | Status | PR |
74      |----|-----|-----|-----|-----|-----|----|
75      | E0 | This proposal doc + inbound pointers (README/NEXT STEPS/memory) | ? | ? | **Owner
```

```
approval to start** | ? | ? |
76      | EA1 | **Spanish (`es`)** catalog (Gemini-draft) + ICU **3-form `{one,many,other}`**
handling + extend key-parity/no-bare-strings tooling for the `many` arm + negotiation test | A |
E0 | No (after approval) | ? | ? |
77      | EA2 | **Danish (`da`)** catalog + negotiation test | A | E0 | No | ? | ? |
78      | EA3 | **Indonesian (`id`)** catalog + negotiation test | A | E0 | No | ? | ? |
79      | EB1 | **Noto Sans Thai** bundle (OFL, `@fontsource-variable`) + Thai no-space word-
wrap CSS safety net | B | E0 | No (ED2 carve-out applies) | ? | ? |
80      | EB2 | **Thai (`th`)** catalog ? render + negotiation test | B | EB1 | No | ? | ? |
[/external_agent_tool_result]
```

688. assistant (2026-07-02T06:11:33.511Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\docs\MULTILINGUAL_EXPANSION.md
[/external_agent_tool_call]
```

689. assistant (2026-07-02T06:11:33.512Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\docs\MULTILINGUAL_EXPANSION.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

690. assistant (2026-07-02T06:11:33.512Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\docs\MULTILINGUAL_EXPANSION.md
[/external_agent_tool_call]
```

691. assistant (2026-07-02T06:11:33.512Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\docs\MULTILINGUAL_EXPANSION.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

692. assistant (2026-07-02T06:11:33.512Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\docs\MULTILINGUAL_EXPANSION.md
[/external_agent_tool_call]
```

693. assistant (2026-07-02T06:11:33.512Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\docs\MULTILINGUAL_EXPANSION.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

694. assistant (2026-07-02T06:11:33.512Z)

```
[external_agent_tool_call: Bash]
description: Commit tracker, push, check Phase A CI
command: git add docs/MULTILINGUAL_EXPANSION.md && git commit -q -m "docs(mlx): mark Phase A
(es/da/id) shipped (#67) in $7 tracker + changelog

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>" && git push -q 2>&1 | tail -1 && echo
"pushed" && sleep 25 && gh pr checks 67 2>&1 | head
[/external_agent_tool_call]
```

695. assistant (2026-07-02T06:11:33.512Z)

```
[external_agent_tool_result]
pushed
site    pass    11s
https://github.com/avidullu/khelsutra/actions/runs/27911498681/job/82589300488
Workers Builds: hidden-king-khelsutra    pending 0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/8f1a264a-845e-4ea1-818d-718b4b67deb8
web     pending 0
https://github.com/avidullu/khelsutra/actions/runs/27911498681/job/82589300499
[/external_agent_tool_result]
```

696. assistant (2026-07-02T06:11:33.512Z)

```
[external_agent_tool_call: Bash]
description: Poll Phase A CI
command: for i in $(seq 1 8); do s=$(gh pr checks 67 2>&1 | grep -E "^web\b" | awk '{print
$2}'); c=$(gh pr checks 67 2>&1 | grep -iE "Workers Builds" | awk '{print $4}'); echo "poll $i:
web=$s cf=$c"; if { [ "$s" = "pass" ] || [ "$s" = "fail" ]; } && { [ "$c" = "pass" ] || [ "$c" =
"fail" ]; }; then break; fi; sleep 15; done && echo "=== final ===" && gh pr checks 67 2>&1 |
head
[/external_agent_tool_call]
```

697. assistant (2026-07-02T06:11:33.512Z)

```
[external_agent_tool_result]
poll 1: web=pending cf=pass
poll 2: web=pass cf=pass
=== final ===
Workers Builds: hidden-king-khelsutra    pass    0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/8f1a264a-845e-4ea1-818d-718b4b67deb8
site    pass    11s
https://github.com/avidullu/khelsutra/actions/runs/27911498681/job/82589300488
web     pass    29s
https://github.com/avidullu/khelsutra/actions/runs/27911498681/job/82589300499
[/external_agent_tool_result]
```

698. assistant (2026-07-02T06:11:33.512Z)

All green. Merging MULTILINGUAL Phase A:

699. assistant (2026-07-02T06:11:33.512Z)

```
[external_agent_tool_call: Bash]
description: Squash-merge Phase A, sync master
command: gh pr merge 67 --squash --delete-branch 2>&1 | tail -2 && git checkout master -q && git
pull --ff-only -q && git log --oneline -2
[/external_agent_tool_call]
```

700. assistant (2026-07-02T06:11:33.512Z)

```
[external_agent_tool_result]
create mode 100644 web/src/i18n/locales/es/common.json
create mode 100644 web/src/i18n/locales/id/common.json
a2292ef feat(web): MULTILINGUAL Phase A ? Spanish/Danish/Indonesian SPA catalogs (#67)
ca619fb feat(web): T4 ? in-UI file upload (the runnable-tool keystone) (#66)
[/external_agent_tool_result]
```

701. assistant (2026-07-02T06:11:33.513Z)

MULTILINGUAL Phase A merged (#67 ? `a2292ef`) ? that's **5 PRs this session** plus a coverage win: the new placeholder-parity guard caught and fixed a real kn/hi A1-drift bug.

Continuing the cadence with **A9 ? tap-only filter controls** (the \$6.5 bridge for non-English-

typing kn/hi users), a clean self-contained SPA feature with good coverage and no engine collision. Let me set it up.

702. assistant (2026-07-02T06:11:33.513Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"8","status":"completed"}
[/external_agent_tool_call]
```

703. assistant (2026-07-02T06:11:33.513Z)

```
[external_agent_tool_result]
Updated task #8 status
[/external_agent_tool_result]
```

704. assistant (2026-07-02T06:11:33.513Z)

```
[external_agent_tool_call: TaskCreate]
description: Localized tap controls (Sort/Length/Count) alongside the English query bar so non-
English-typing kn/hi users can filter without typing. Build ParsedQuery directly ? onApply. One
PR, tests, merge on green.
[/external_agent_tool_call]
```

705. assistant (2026-07-02T06:11:33.513Z)

```
[external_agent_tool_result]
Task #9 created successfully: A9 ? tap-only localized filter controls (SPA)
[/external_agent_tool_result]
```

706. assistant (2026-07-02T06:11:33.513Z)

```
[external_agent_tool_call: Bash]
description: Cut A9 branch from origin/master
command: git fetch origin -q && git checkout -b feat/a9-tap-filters origin/master 2>&1 | tail -2
&& git log --oneline -1
[/external_agent_tool_call]
```

707. assistant (2026-07-02T06:11:33.513Z)

```
[external_agent_tool_result]
branch 'feat/a9-tap-filters' set up to track 'origin/master'.
Switched to a new branch 'feat/a9-tap-filters'
a2292ef feat(web): MULTILINGUAL Phase A ? Spanish/Danish/Indonesian SPA catalogs (#67)
[/external_agent_tool_result]
```

708. assistant (2026-07-02T06:11:33.513Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\kheIsutra\web\src\components\QueryBar.examples.test.ts
[/external_agent_tool_call]
```

709. assistant (2026-07-02T06:11:33.513Z)

```
[external_agent_tool_result]
1      import { describe, it, expect } from "vitest";
2      import { EXAMPLES } from "../QueryBar";
3      import { parseQuery } from "../lib/query";
4
5      // The example chips apply on tap, so a broken example would silently do nothing (or
show a
6      // "couldn't read that"). Pin that EVERY shipped example is a recognized, fully-
available query
7      // (no unsupported tokens) ? a regression guard for the duration-led example set.
```

```

8      describe("QueryBar examples", () => {
9          it("are all recognized, actionable, and fully available (no experimental/unsupported
tokens)", () => {
10              for (const ex of EXAMPLES) {
11                  const q = parseQuery(ex);
12                  expect(q.recognized, ex).toBe(true);
13                  expect(q.unavailable, ex).toEqual([]);
14                  // every example must actually constrain or order the set
15                  expect(q.filters.length > 0 || q.sort != null || q.limit != null, ex).toBe(true);
16              }
17          });
18
19          it("lead with rally length, not the experimental shot-count axis", () => {
20              const shotExamples = EXAMPLES.filter((ex) => parseQuery(ex).filters.some((f) =>
f.field === "shot_count"));
21              expect(shotExamples).toEqual([]);
22          });
23      });
24
[/external_agent_tool_result]

```

710. assistant (2026-07-02T06:11:33.513Z)

```

[/external_agent_tool_call: mcp_ccd_session__mark_chapter]
input: {"title":"A9 ? tap-only localized filters","summary":"5 PRs merged. Building localized
tap filter controls so non-English-typing users can filter."}
[/external_agent_tool_call]

```

711. assistant (2026-07-02T06:11:33.514Z)

```

[/external_agent_tool_result]
Chapter marked: "A9 ? tap-only localized filters". Continue your current work.
[/external_agent_tool_result]

```

712. assistant (2026-07-02T06:11:33.514Z)

The `EXAMPLES` test pins the English chips. I'll keep those (English "type-this" hints) and add a complementary **localized** `TapFilters` control set ? and pin that each tap builds the exact same query as typing its English equivalent (tap == type). Building the component:

713. assistant (2026-07-02T06:11:33.514Z)

```

[/external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\kheIsutra\web\src\components\TapFilters.tsx
[/external_agent_tool_call]

```

714. assistant (2026-07-02T06:11:33.514Z)

```

[/external_agent_tool_result]
File created successfully at:
C:\Users\avidu\Projects\kheIsutra\web\src\components\TapFilters.tsx (file state is current in
your context ? no need to Read it back)
[/external_agent_tool_result]

```

715. assistant (2026-07-02T06:11:33.514Z)

Now the `tap.\*` keys in all six catalogs. en:

716. assistant (2026-07-02T06:11:33.515Z)

```

[/external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\kheIsutra\web\src\i18n\locales\en\common.json
[/external_agent_tool_call]

```

717. assistant (2026-07-02T06:11:33.515Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

718. assistant (2026-07-02T06:11:33.515Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json
[/external_agent_tool_call]
```

719. assistant (2026-07-02T06:11:33.515Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

720. assistant (2026-07-02T06:11:33.515Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\hi\common.json
[/external_agent_tool_call]
```

721. assistant (2026-07-02T06:11:33.515Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\hi\common.json has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

722. assistant (2026-07-02T06:11:33.515Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\es\common.json
[/external_agent_tool_call]
```

723. assistant (2026-07-02T06:11:33.515Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\es\common.json has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

724. assistant (2026-07-02T06:11:33.515Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\da\common.json
[/external_agent_tool_call]
```

725. assistant (2026-07-02T06:11:33.515Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\da\common.json has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

726. assistant (2026-07-02T06:11:33.515Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\id\common.json
```


[/external_agent_tool_call]

727. assistant (2026-07-02T06:11:33.516Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\id\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

728. assistant (2026-07-02T06:11:33.516Z)

Now wire `TapFilters` into App (below the QueryBar) and add it to the no-bare-strings file list:

729. assistant (2026-07-02T06:11:33.516Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\khelsutra\web\src\App.tsx

[/external_agent_tool_call]

730. assistant (2026-07-02T06:11:33.516Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\khelsutra\web\src\App.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

731. assistant (2026-07-02T06:11:33.516Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\khelsutra\web\src\App.tsx

[/external_agent_tool_call]

732. assistant (2026-07-02T06:11:33.516Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\khelsutra\web\src\App.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

733. assistant (2026-07-02T06:11:33.516Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\hygiene.test.ts

[/external_agent_tool_call]

734. assistant (2026-07-02T06:11:33.516Z)

[external_agent_tool_result]

The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\hygiene.test.ts has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

735. assistant (2026-07-02T06:11:33.516Z)

Now the TapFilters test (render, onApply, the tap==type invariant, localized labels, ?44px):

736. assistant (2026-07-02T06:11:33.517Z)

[external_agent_tool_call: Write]

file: C:\Users\avidu\Projects\khelsutra\web\src\components\TapFilters.test.tsx

[/external_agent_tool_call]

737. assistant (2026-07-02T06:11:33.517Z)

```
[external_agent_tool_result]
File created successfully at:
C:\Users\avidu\Projects\khelsutra\web\src\components\TapFilters.test.tsx (file state is current
in your context ? no need to Read it back)
[/external_agent_tool_result]
```

738. assistant (2026-07-02T06:11:33.517Z)

```
[external_agent_tool_call: Bash]
description: Run suite + tsc + build for A9
command: cd /c/Users/avidu/Projects/khelsutra/web && npm test --silent 2>/dev/null | grep -E
"Test Files|Tests |FAIL|?|x|? src" | tail -10 && echo "=== tsc ===" && npx tsc --noEmit && echo
"tsc OK" && echo "=== build ===" && npm run build --silent 2>&1 | tail -1
[/external_agent_tool_call]
```

739. assistant (2026-07-02T06:11:33.517Z)

```
[external_agent_tool_result]
Test Files 19 passed (19)
Tests 173 passed (173)
=== tsc ===
tsc OK
=== build ===
? built in 1.23s
[/external_agent_tool_result]
```

740. assistant (2026-07-02T06:11:33.517Z)

173 tests (+5), tsc + build clean. Let me browser-verify A9 ? the tap controls render and actually filter:

741. assistant (2026-07-02T06:11:33.517Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_start]
input: {"name":"khelsutra-web"}
[/external_agent_tool_call]
```

742. assistant (2026-07-02T06:11:33.517Z)

```
[external_agent_tool_result]
{
  "serverId": "db54b1fa-72ce-4237-be96-32791b41b21d",
  "port": 5174,
  "name": "khelsutra-web",
  "reused": false
}
```

Server started successfully on port 5174.

```
[/external_agent_tool_result]
```

743. assistant (2026-07-02T06:11:33.518Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_eval]
input: {"serverId":"db54b1fa-72ce-4237-be96-32791b41b21d","expression":"(() => {\n //
localStorage may be 'da' from the earlier session; reset to en for this check\n
localStorage.setItem('khelsutra.lang','en');\n const tap = document.querySelector('[data-
testid=\"tap-filters\"]');\n const labels = tap ?
[...tap.querySelectorAll('button')].map(b=>b.textContent) : null;\n const title =
tap?.querySelector('span')?.textContent;\n return { present: !!tap, title, labels, lang:
document.documentElement.lang };;\n}}()"}
[/external_agent_tool_call]
```

744. assistant (2026-07-02T06:11:33.518Z)

```
[external_agent_tool_result]
```

```
{
  "labels": [
    "Længste først",
    "Korteste først",
    "Over 10 sekunder",
    "Over 20 sekunder",
    "De 5 længste",
    "De 10 længste"
  ],
  "lang": "da",
  "present": true,
  "title": "Eller tryk for at filtrere:"
}
[/external_agent_tool_result]
```

745. assistant (2026-07-02T06:11:33.518Z)

TapFilters renders + localizes (showing Danish labels since the browser was still set to `da`).
Let me reload to en, tap "Over 10 seconds", and confirm it actually filters the selection:

746. assistant (2026-07-02T06:11:33.518Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_eval]
input: {"serverId":"db54b1fa-72ce-4237-be96-32791b41b21d","expression":"(()=>{localStorage.setItem('khelsutra.lang','en');location.reload();})();"}
[/external_agent_tool_call]
```

747. assistant (2026-07-02T06:11:33.518Z)

```
[external_agent_tool_result]
undefined
[/external_agent_tool_result]
```

748. assistant (2026-07-02T06:11:33.518Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_click]
input: {"serverId":"db54b1fa-72ce-4237-be96-32791b41b21d","selector":"[data-testid=\"tap-filters\"] button:nth-of-type(3)"}
[/external_agent_tool_call]
```

749. assistant (2026-07-02T06:11:33.518Z)

```
[external_agent_tool_result]
Successfully clicked: [data-testid="tap-filters"] button:nth-of-type(3)
[/external_agent_tool_result]
```

750. assistant (2026-07-02T06:11:33.518Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_eval]
input: {"serverId":"db54b1fa-72ce-4237-be96-32791b41b21d","expression":"(() => {\n  const\n  summary = document.querySelector('[data-testid=\"selection-summary\"]')?.textContent;\n  const\n  tapLabels = [...document.querySelectorAll('[data-testid=\"tap-filters\"]\n  button')].map(b=>b.textContent);\n  return { summary, clicked: 'Over 10 seconds (3rd)',\n  enLabels: tapLabels, lang: document.documentElement.lang };;\n})();"}
[/external_agent_tool_call]
```

751. assistant (2026-07-02T06:11:33.518Z)

```
[external_agent_tool_result]
{
  "clicked": "Over 10 seconds (3rd)",
  "enLabels": [
    "Longest first",
```

```

    "Shortest first",
    "Over 10 seconds",
    "Over 20 seconds",
    "Top 5 longest",
    "Top 10 longest"
  ],
  "lang": "en",
  "summary": "6 rallies selected · 1:22 total"
}
[/external_agent_tool_result]

```

752. assistant (2026-07-02T06:11:33.518Z)

The click raced the reload. Page is stable now (en). Let me click "Over 10 seconds" and re-check the selection:

753. assistant (2026-07-02T06:11:33.519Z)

```

[external_agent_tool_call: mcp__Claude_Preview__preview_click]
input: {"serverId":"db54b1fa-72ce-4237-be96-32791b41b21d","selector":"[data-testid=\\"tap-filters\\"] button:nth-of-type(3)"}
[/external_agent_tool_call]

```

754. assistant (2026-07-02T06:11:33.519Z)

```

[external_agent_tool_result]
Successfully clicked: [data-testid="tap-filters"] button:nth-of-type(3)
[/external_agent_tool_result]

```

755. assistant (2026-07-02T06:11:33.519Z)

```

[external_agent_tool_call: mcp__Claude_Preview__preview_eval]
input:
{"serverId":"db54b1fa-72ce-4237-be96-32791b41b21d","expression":"document.querySelector('[data-testid=\\"selection-summary\\"]')?.textContent"}
[/external_agent_tool_call]

```

756. assistant (2026-07-02T06:11:33.519Z)

```

[external_agent_tool_result]
"6 rallies selected · 1:22 total"
[/external_agent_tool_result]

```

757. assistant (2026-07-02T06:11:33.519Z)

Still 6 ? the tap isn't filtering. Let me check the console logs to see what `onApply` did:

758. assistant (2026-07-02T06:11:33.519Z)

```

[external_agent_tool_call: mcp__Claude_Preview__preview_console_logs]
input: {"serverId":"db54b1fa-72ce-4237-be96-32791b41b21d","level":"all","lines":15}
[/external_agent_tool_call]

```

759. assistant (2026-07-02T06:11:33.519Z)

```

[external_agent_tool_result]
[error] {"ts":"2026-06-21T17:15:44.663Z","level":"error","event":"api.http_error","cid":"5bf6d848-d685-4b01-83f6-8741b84b7430","method":"GET","url":"/api/config","status":500,"body":""}
[error] {"ts":"2026-06-21T17:15:44.663Z","level":"error","event":"api.http_error","cid":"5bf6d848-d685-4b01-83f6-8741b84b7430","method":"GET","url":"/api/config","status":500,"body":""}
[error] {"ts":"2026-06-

```

```

21T17:15:44.663Z", "level": "error", "event": "api.http_error", "cid": "5bf6d848-d685-4b01-83f6-8741b84b7430", "method": "GET", "url": "/api/config", "status": 500, "body": ""}
[warn]
{"ts": "2026-06-21T17:15:44.663Z", "level": "warn", "event": "config.unavailable", "error": "ApiError: GET /api/config failed: 500"}
[warn]
{"ts": "2026-06-21T17:15:44.663Z", "level": "warn", "event": "config.unavailable", "error": "ApiError: GET /api/config failed: 500"}
[warn]
{"ts": "2026-06-21T17:15:44.663Z", "level": "warn", "event": "config.unavailable", "error": "ApiError: GET /api/config failed: 500"}
[warn]
{"ts": "2026-06-21T17:15:44.663Z", "level": "warn", "event": "config.unavailable", "error": "ApiError: GET /api/config failed: 500"}
[error] {"ts": "2026-06-21T17:15:44.665Z", "level": "error", "event": "api.http_error", "cid": "7c83df6a-94c5-4ca0-a8e3-6f71725a6700", "method": "GET", "url": "/api/config", "status": 500, "body": ""}
[error] {"ts": "2026-06-21T17:15:44.665Z", "level": "error", "event": "api.http_error", "cid": "7c83df6a-94c5-4ca0-a8e3-6f71725a6700", "method": "GET", "url": "/api/config", "status": 500, "body": ""}
[error] {"ts": "2026-06-21T17:15:44.665Z", "level": "error", "event": "api.http_error", "cid": "7c83df6a-94c5-4ca0-a8e3-6f71725a6700", "method": "GET", "url": "/api/config", "status": 500, "body": ""}
[error] {"ts": "2026-06-21T17:15:44.665Z", "level": "error", "event": "api.http_error", "cid": "7c83df6a-94c5-4ca0-a8e3-6f71725a6700", "method": "GET", "url": "/api/config", "status": 500, "body": ""}
[warn]
{"ts": "2026-06-21T17:15:44.665Z", "level": "warn", "event": "config.unavailable", "error": "ApiError: GET /api/config failed: 500"}
[warn]
{"ts": "2026-06-21T17:15:44.665Z", "level": "warn", "event": "config.unavailable", "error": "ApiError: GET /api/config failed: 500"}
[warn]
{"ts": "2026-06-21T17:15:44.665Z", "level": "warn", "event": "config.unavailable", "error": "ApiError: GET /api/config failed: 500"}
[warn]
{"ts": "2026-06-21T17:15:44.665Z", "level": "warn", "event": "config.unavailable", "error": "ApiError: GET /api/config failed: 500"}

```

(Showing last 15 of 80 entries. Use 'lines' parameter (max 200) to see more.)
[/external_agent_tool_result]

760. assistant (2026-07-02T06:11:33.519Z)

Let me trigger the tap via a native click in-page (more reliable for React delegation) and read the result:

761. assistant (2026-07-02T06:11:33.519Z)

```

[external_agent_tool_call: mcp__Claude_Preview__preview_eval]
input: {"serverId": "db54b1fa-72ce-4237-be96-32791b41b21d", "expression": "(async () => {\n  const btns = document.querySelectorAll('[data-testid=\"tap-filters\"] button');\n  const label = btns[2].textContent;\n  btns[2].click();\n  await new Promise(r => setTimeout(r, 200));\n  const summary = document.querySelector('[data-testid=\"selection-summary\"]')?.textContent;\n  // also count selected cards (aria-pressed=true)\n  const selected = [...document.querySelectorAll('ul li button[aria-pressed=\"true\"]')].length;\n  return { clickedLabel: label, summary, selectedCards: selected }; \n})()"}
[/external_agent_tool_call]

```

762. assistant (2026-07-02T06:11:33.519Z)

```

[external_agent_tool_result]
{
  "clickedLabel": "Over 10 seconds",

```

```
"selectedCards": 3,
"summary": "3 rallies selected · 1:02 total"
}
[/external_agent_tool_result]
```

763. assistant (2026-07-02T06:11:33.519Z)

A9 works in-browser: tapping "Over 10 seconds" filtered to **3 rallies (1:02)** ? tap == type confirmed, and labels localize (Danish earlier, English now). 173 tests, tsc, build green.

Let me run a focused adversarial review, then commit. Capturing the diff:

764. assistant (2026-07-02T06:11:33.520Z)

```
[external_agent_tool_call: Bash]
description: Capture A9 component diff
command: cd /c/Users/avidu/Projects/khelsutra && git add -N web/src/components/TapFilters.tsx
web/src/components/TapFilters.test.tsx && git --no-pager diff -- web/src/App.tsx
web/src/components/TapFilters.tsx | cat
[/external_agent_tool_call]
```

765. assistant (2026-07-02T06:11:33.521Z)

```
[external_agent_tool_result]
diff --git a/web/src/App.tsx b/web/src/App.tsx
index 4ab5002..34f546e 100644
--- a/web/src/App.tsx
+++ b/web/src/App.tsx
@@ -3,6 +3,7 @@ import { useTranslation, Trans } from "react-i18next";
  import { useRallySelection } from "../hooks/useRallySelection";
  import { RallyGrid } from "../components/RallyGrid";
  import { QueryBar } from "../components/QueryBar";
+import { TapFilters } from "../components/TapFilters";
  import { SelectionFooter } from "../components/SelectionFooter";
  import { LanguageSwitcher } from "../components/LanguageSwitcher";
  import { CompileResult, type BuildState } from "../components/CompileResult";
@@ -247,6 +248,8 @@ export default function App() {

    <div className="mt-6">
      <QueryBar onApply={onApplyQuery} />
+    /* A9: localized tap controls for users who can't type the English query syntax. */
+    <TapFilters onApply={onApplyQuery} />
    </div>

    <div className="mt-6 flex flex-wrap gap-2 text-sm">
diff --git a/web/src/components/TapFilters.tsx b/web/src/components/TapFilters.tsx
new file mode 100644
index 0000000..c0ed250
--- /dev/null
+++ b/web/src/components/TapFilters.tsx
@@ -0,0 +1,58 @@
+// Localized tap-only filter controls (tracker A9). The query bar PARSES ENGLISH text (a
+// documented
+// limit) ? so a Kannada/Hindi/? user who can't type English needs to filter by TAPPING. Each
+// control
+// builds the SAME ParsedQuery the parser would for its English equivalent ? directly, no
+// English
+// round-trip ? and applies it through the shared onApply path. A test pins tap == type so a
+// control
+// can never silently drift from the parser. Length-led + fully-available only (never shot-
+// count).
+import { useTranslation } from "react-i18next";
+import type { ParsedQuery } from "../lib/query";
+
```

```

+interface TapFilter {
+  /** i18n label key (localized) ? the user sees this, not the English query. */
+  key: string;
+  /** The equivalent typed query; the tap MUST select identically to typing this (pinned by a
test). */
+  english: string;
+  parts: Pick<ParsedQuery, "filters" | "sort" | "limit">;
+}
+
+export const TAP_FILTERS: TapFilter[] = [
+  { key: "tap.longest", english: "longest first", parts: { filters: [], sort: { key:
"duration", dir: "desc" } } },
+  { key: "tap.shortest", english: "shortest first", parts: { filters: [], sort: { key:
"duration", dir: "asc" } } },
+  { key: "tap.over10", english: "over 10s", parts: { filters: [{ field: "duration", op: ">",
value: 10 } ] } },
+  { key: "tap.over20", english: "over 20s", parts: { filters: [{ field: "duration", op: ">",
value: 20 } ] } },
+  { key: "tap.top5", english: "top 5", parts: { filters: [], sort: { key: "duration", dir:
"desc" }, limit: 5 } },
+  { key: "tap.top10", english: "top 10", parts: { filters: [], sort: { key: "duration", dir:
"desc" }, limit: 10 } },
+];
+
+interface Props {
+  /** Same handler the QueryBar uses ? App runs runQuery + writes the shared selection. */
+  onApply: (q: ParsedQuery, source: "typed" | "example") => void;
+}
+
+export function TapFilters({ onApply }: Props) {
+  const { t } = useTranslation();
+  // 244px tap targets (A4) so the no-type path is comfortable on a phone.
+  const BTN = "inline-flex min-h-11 items-center rounded-full border border-black/15 px-4 text-
sm hover:border-green";
+
+  return (
+    <div className="mt-3" data-testid="tap-filters">
+      <span className="text-xs text-ink/40">{t("tap.title")}</span>
+      <div className="mt-1.5 flex flex-wrap gap-2">
+        {TAP_FILTERS.map((f) => (
+          <button
+            key={f.key}
+            type="button"
+            className={BTN}
+            onClick={() =>
+              onApply(
+                { raw: `tap:${f.key}`, filters: f.parts.filters, sort: f.parts.sort, limit:
f.parts.limit, unavailable: [], recognized: true },
+                "example",
+              )
+            }
+          >
+            {t(f.key)}
+          </button>
+        ))}
+      </div>
+    </div>
+  );
+}
[/external_agent_tool_result]

```

766. assistant (2026-07-02T06:11:33.521Z)

Clean diff. Running a 2-lens adversarial review (correctness/tap-invariant + UX/faithfulness):

767. assistant (2026-07-02T06:11:33.521Z)

```
[external_agent_tool_call: Workflow]
input: {"script": "export const meta = {\n  name: 'a9-review',\n  description: 'Adversarial review of the A9 tap-only localized filters',\n  phases: [{ title: 'Review' }],\n}\n\nconst WEB = 'C:/Users/avidu/Projects/khelsutra/web/src'\n\nconst SCHEMA = {\n  type: 'object',\n  additionalProperties: false,\n  properties: {\n    summary: { type: 'string' },\n    findings: { type: 'array', items: { type: 'object', additionalProperties: false,\n      properties: {\n        severity: { type: 'string', enum: ['blocker', 'major', 'minor', 'nit'] },\n        file: { type: 'string' },\n        location: { type: 'string' },\n        title: { type: 'string' },\n        detail: { type: 'string' },\n        suggestion: { type: 'string' },\n      },\n      required: ['severity', 'file', 'title', 'detail']\n    } },\n  },\n  required: ['summary', 'findings']\n}\n\nconst CONTEXT = `A9 (tracker FRIENDLY_AND_MULTILINGUAL $7): localized tap-only filter controls so a non-English-typing user can filter without typing the English query syntax.\n\nFiles under ${WEB}: NEW components/TapFilters.tsx (exports TAP_FILTERS data + the component); App.tsx renders <TapFilters onApply={onApplyQuery}/> below <QueryBar/>; catalogs en/kn/hi/es/da/id gained a tap.* block (title/longest/shortest/over10/over20/top5/top10). Each control builds a ParsedQuery DIRECTLY (no parsing) and calls the shared onApply(q,\"example\"). A test pins tap==type (parseQuery(f.english) deep-equals f.parts). 173 web tests green; tsc+build clean; browser-verified (tap \"Over 10 seconds\" ? 3 rallies; labels localize).\n\nconst LENSES = [\n  { key: 'correctness', prompt: `Review A9 for CORRECTNESS. ${CONTEXT}\n\nRead ${WEB}/components/TapFilters.tsx, ${WEB}/lib/query.ts (parseQuery + runQuery), ${WEB}/App.tsx (onApplyQuery).\n\n1. tap==type INVARIANT: for EACH TAP_FILTER, does parseQuery(english) truly produce the same {filters,sort,limit} as parts? Manually trace \"top 5\" (does parseQuery set sort duration desc + limit 5?), \"over 10s\", \"longest first\". Any drift the test would miss (e.g. pa...`
  }
]\n\n`"
  }
}

[/external_agent_tool_call]
```

768. assistant (2026-07-02T06:11:33.522Z)

```
[external_agent_tool_result]
Workflow launched in background. Task ID: wg7x6iy0g
Summary: Adversarial review of the A9 tap-only localized filters
Transcript dir: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\2ba49afa-99f4-41f2-86a2-27a289d432f3\subagents\workflows\wf_5d655769-ba4
Script file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-khelsutra\2ba49afa-99f4-41f2-86a2-27a289d432f3\workflows\scripts\a9-review-wf_5d655769-ba4.js
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath: "C:\Users\avidu\claude\projects\C--Users-avidu-Projects-khelsutra\2ba49afa-99f4-41f2-86a2-27a289d432f3\workflows\scripts\a9-review-wf_5d655769-ba4.js"})
to iterate without resending the script.)
Run ID: wf_5d655769-ba4
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\claude\projects\C--Users-avidu-Projects-khelsutra\2ba49afa-99f4-41f2-86a2-27a289d432f3\workflows\scripts\a9-review-wf_5d655769-ba4.js", resumeFromRunId: "wf_5d655769-ba4"}) ? completed agents return cached results.

You will be notified when it completes. Use /workflows to watch live progress.
[/external_agent_tool_result]
```

769. assistant (2026-07-02T06:11:33.522Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_stop]
input: {"serverId": "db54b1fa-72ce-4237-be96-32791b41b21d"}
[/external_agent_tool_call]
```

770. assistant (2026-07-02T06:11:33.522Z)

```
[external_agent_tool_result]
Server db54b1fa-72ce-4237-be96-32791b41b21d stopped
[/external_agent_tool_result]
```

771. assistant (2026-07-02T06:11:33.522Z)


```
[external_agent_tool_call: TaskOutput]
input: {"task_id":"wg7x6iy0g","block":true,"timeout":240000}
[/external_agent_tool_call]
```

772. assistant (2026-07-02T06:11:33.522Z)

```
[external_agent_tool_result]
<retrieval_status>success</retrieval_status>
```

```
<task_id>wg7x6iy0g</task_id>
```

```
<task_type>local_workflow</task_type>
```

```
<status>completed</status>
```

```
<output>
```

```
{
  "summary": "Adversarial review of the A9 tap-only localized filters",
  "agentCount": 2,
  "logs": [],
  "result": [
    {
      "lens": "correctness",
      "summary": "A9 is CORRECT. The tap==type invariant holds for all six controls ? I manually
traced each through parseQuery: \"longest first\" ? {sort:duration/desc}, \"shortest first\" ?
{sort:duration/asc}, \"over 10s\"/\"over 20s\" ? {filters:[duration>10]/[duration>20]} with no
sort and no limit, \"top 5\"/\"top 10\" ? {sort:duration/desc, limit:5/10} (the \"top N\" branch
sets sortFromLimit=true ? duration desc, validation passes for positive integers). All match the
hardcoded TAP_FILTERS.parts exactly. The pinning test deep-equals the full {filters,sort,limit}
triple plus asserts unavailable==[], so it would catch any drift including a sort the parts omit
(verified: over10/over20 produce sort:undefined on both sides, since no SORT_PATTERN matches and
no limit triggers a phantom sort). The constructed ParsedQuery (recognized:true, unavailable:[],
raw:\"tap:tap.longest\") flows through App.onApplyQuery identically to a typed query: it passes
the !recognized guard, runs runQuery + sel.apply + setOrder, and logs query.parse/query.apply.
The \"tap:...\" raw string is only ever echoed into log lines via parseTrace ? never re-parsed ?
so it's safe. All controls are length-led (sort key + filter field are duration only; no
shot_count, no unavailable tokens) per the doc. Full web suite green (173 tests, including
TapFilters.test.tsx 4, hygiene 28, query 46), tsc --noEmit exits 0. tap.* block present in all
six catalogs (en/kn/hi/es/da/id) with title + 6 control labels and key-parity enforced by the
hygiene test. The diff is scoped cleanly to A9. No findings.",
      "findings": []
    },
    {
      "lens": "ux-faithfulness",
      "summary": "A9 is solid, well-tested work that faithfully meets the tracker intent. The
TapFilters component builds a ParsedQuery directly (no English round-trip), routes through the
shared onApply path, is pinned tap==type by a test, uses real ?44px <button>s with localized
text labels, and ships localized tap.* blocks for all 6 locales. I reproduced the verification
locally: 173 web tests pass, tsc --noEmit is clean. The no-bare-strings hygiene test DOES cover
TapFilters (added to its FILES list, line 107). Findings below are all minor/nit ? the main one
is the genuine UX redundancy you flagged: two chip rows stacked directly on top of each other,
one English-only and one localized, which is confusing clutter for the exact non-tech, non-
English-typing user A9 targets. My concrete recommendation: differentiate the copy and, for non-
English locales, suppress the English-only QueryBar examples row (it is the only English-only
chip row left on the screen for a kn/hi user). FAITHFULNESS: A9 meets $6.4 (\"kn/hi users rely
on tap-only controls\") and $7 A9 (\"tap-only filter controls alongside the English query
bar\"); nothing essential is missing ? reset is covered by the existing Select-all bulk control.
No blockers.",
      "findings": [
        {
          "severity": "minor",
          "file": "web/src/components/QueryBar.tsx",
          "location": "QueryBar.tsx:90-102 (EXAMPLES row) + App.tsx:250-252 (TapFilters rendered
directly below)",
          "title": "Two stacked chip rows ? the English-only QueryBar examples row is dead
```

weight for the very user A9 targets",
"detail": "On screen the user now sees, in order: 'Sample queries:' + HARDCODED English chips ('over 10s', 'top 15 longest', 'shortest first', 'first 10') from QueryBar.EXAMPLES, then a support note, then 'Or tap to filter:' + the localized A9 chips. The EXAMPLES chips render literal English strings regardless of locale (QueryBar.tsx:99 renders `{ex}` directly, not a t() key) ? so a Kannada/Hindi user gets a row of untap...
[/external_agent_tool_result]

773. assistant (2026-07-02T06:11:33.522Z)

A9 review: ****correctness clean**** (tap==type traced for all 6 controls). One ****minor**** + 3 nits ? all genuinely good, so I'll fold them: (1) hide the English-only `EXAMPLES` row for non-English locales (it's clutter for the exact user A9 serves), (2) make `tap.title` self-contained ("Tap to filter:" ? needed once EXAMPLES is hidden), (3) log `raw` as the English equivalent (cleaner trace), (4) loop the applied-query test over all controls (closes the coverage gap).

Fix 1 ? QueryBar hides EXAMPLES for non-en:

774. assistant (2026-07-02T06:11:33.522Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\QueryBar.tsx
[/external_agent_tool_call]

775. assistant (2026-07-02T06:11:33.522Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\QueryBar.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

776. assistant (2026-07-02T06:11:33.522Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\QueryBar.tsx
[/external_agent_tool_call]

777. assistant (2026-07-02T06:11:33.522Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\QueryBar.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

778. assistant (2026-07-02T06:11:33.522Z)

Fix 3 ? log `raw` as the English equivalent:

779. assistant (2026-07-02T06:11:33.522Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\TapFilters.tsx
[/external_agent_tool_call]

780. assistant (2026-07-02T06:11:33.522Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\TapFilters.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

781. assistant (2026-07-02T06:11:33.523Z)

Fix 2 ? make `tap.title` self-contained across all six catalogs:

782. assistant (2026-07-02T06:11:33.523Z)

```
[external_agent_tool_call: Bash]
description: Locate tap.title in each catalog
command: for f in en kn hi es da id; do echo "=== $f tap.title ==="; grep -n '"title"'
web/src/i18n/locales/$f/common.json | tail -1; done
[/external_agent_tool_call]
```

783. assistant (2026-07-02T06:11:33.523Z)

```
[external_agent_tool_result]
=== en tap.title ===
102:   "title": "Or tap to filter:",
=== kn tap.title ===
102:   "title": "???? ?????? ????? ?????? ????:",
=== hi tap.title ===
102:   "title": "? ? ?????? ????? ?? ??? ??? ????:",
=== es tap.title ===
102:   "title": "0 toca para filtrar:",
=== da tap.title ===
102:   "title": "Eller tryk for at filtrere:",
=== id tap.title ===
102:   "title": "Atau ketuk untuk memfilter:",
[/external_agent_tool_result]
```

784. assistant (2026-07-02T06:11:33.523Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json
[/external_agent_tool_call]
```

785. assistant (2026-07-02T06:11:33.523Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\en\common.json has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

786. assistant (2026-07-02T06:11:33.523Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json
[/external_agent_tool_call]
```

787. assistant (2026-07-02T06:11:33.523Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\kn\common.json has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

788. assistant (2026-07-02T06:11:33.523Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\hi\common.json
[/external_agent_tool_call]
```

789. assistant (2026-07-02T06:11:33.523Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\hi\common.json has been updated
```

successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

790. assistant (2026-07-02T06:11:33.524Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\es\common.json
[/external_agent_tool_call]

791. assistant (2026-07-02T06:11:33.524Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\es\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

792. assistant (2026-07-02T06:11:33.524Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\da\common.json
[/external_agent_tool_call]

793. assistant (2026-07-02T06:11:33.524Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\da\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

794. assistant (2026-07-02T06:11:33.524Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\id\common.json
[/external_agent_tool_call]

795. assistant (2026-07-02T06:11:33.524Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\i18n\locales\id\common.json has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

796. assistant (2026-07-02T06:11:33.524Z)

Fix 4 ? strengthen the test to verify every control's emitted query equals the parser:

797. assistant (2026-07-02T06:11:33.524Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\TapFilters.test.tsx
[/external_agent_tool_call]

798. assistant (2026-07-02T06:11:33.524Z)

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\TapFilters.test.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

799. assistant (2026-07-02T06:11:33.524Z)

[external_agent_tool_call: Bash]
description: Re-run suite after folding A9 review findings

```
command: cd /c/Users/avidu/Projects/khelsutra/web && npm test --silent 2>/dev/null | grep -E
"Test Files|Tests |FAIL|?|x|? src" | tail -8 && echo "=== tsc ===" && npx tsc --noEmit && echo
"tsc OK" && echo "=== build ===" && npm run build --silent 2>&1 | tail -1
[/external_agent_tool_call]
```

800. assistant (2026-07-02T06:11:33.525Z)

```
[external_agent_tool_result]
  Test Files  19 passed (19)
    Tests     173 passed (173)
=== tsc ===
tsc OK
=== build ===
? built in 1.36s
[/external_agent_tool_result]
```

801. assistant (2026-07-02T06:11:33.525Z)

173 tests, tsc, build green. Let me quickly verify the key UX fix in-browser (kn hides the English EXAMPLES row, shows only localized TapFilters):

802. assistant (2026-07-02T06:11:33.525Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_start]
input: {"name":"khelsutra-web"}
[/external_agent_tool_call]
```

803. assistant (2026-07-02T06:11:33.525Z)

```
[external_agent_tool_result]
{
  "serverId": "98f833a4-2f35-49f7-9857-2efa97e11ae8",
  "port": 5174,
  "name": "khelsutra-web",
  "reused": false
}
Server started successfully on port 5174.
[/external_agent_tool_result]
```

804. assistant (2026-07-02T06:11:33.525Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_eval]
input: {"serverId":"98f833a4-2f35-49f7-9857-2efa97e11ae8","expression":"(()=>{localStorage.setItem('khelsutra.lang','kn');location.reload();
})();"}
[/external_agent_tool_call]
```

805. assistant (2026-07-02T06:11:33.525Z)

```
[external_agent_tool_result]
undefined
[/external_agent_tool_result]
```

806. assistant (2026-07-02T06:11:33.525Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_eval]
input: {"serverId":"98f833a4-2f35-49f7-9857-2efa97e11ae8","expression":"(() => {\n  const
bodyText = document.body.innerHTML;\n  const hasEnglishExamples = bodyText.includes(\"top 15
longest\") || bodyText.includes(\"Sample queries\");\n  const tap =
document.querySelector('[data-testid=\"tap-filters\"]');\n  const tapTitle =
tap?.querySelector('span')?.textContent;\n  const tapLabels = tap ?
[...tap.querySelectorAll('button')].map(b=>b.textContent) : null;\n  return { lang:
document.documentElement.lang, hasEnglishExamplesRow: hasEnglishExamples, tapTitle, tapLabels
};\n})();"}
[/external_agent_tool_call]
```

```
[/external_agent_tool_call]
```

807. assistant (2026-07-02T06:11:33.525Z)

```
[external_agent_tool_result]
{
  "hasEnglishExamplesRow": false,
  "lang": "kn",
  "tapLabels": [
    "??? ?????? ??????",
    "??? ????????? ??????",
    "10 ????????????? ??????",
    "20 ????????????? ??????",
    "??? ?????? 5",
    "??? ?????? 10"
  ],
  "tapTitle": "???????? ?????? ?????? ?????:"
}
[/external_agent_tool_result]
```

808. assistant (2026-07-02T06:11:33.526Z)

In Kannada: the English EXAMPLES row is ****hidden****, the self-contained "???????? ?????? ?????? ?????:" title shows, and all 6 controls are localized ? exactly the declutter the review asked for. Let me confirm en still shows both rows (no regression):

809. assistant (2026-07-02T06:11:33.526Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_eval]
input: {"serverId":"98f833a4-2f35-49f7-9857-2efa97e11ae8","expression":"(()=>{localStorage.setItem('kheIsutra.lang','en');location.reload();})();"}
[/external_agent_tool_call]
```

810. assistant (2026-07-02T06:11:33.526Z)

```
[external_agent_tool_result]
undefined
[/external_agent_tool_result]
```

811. assistant (2026-07-02T06:11:33.526Z)

```
[external_agent_tool_call: mcp__Claude_Preview__preview_eval]
input: {"serverId":"98f833a4-2f35-49f7-9857-2efa97e11ae8","expression":"(() => {\n  const t = document.body.innerText;\n  return { lang: document.documentElement.lang, hasSampleQueries: t.includes(\"Sample queries\"), hasEnglishChip: t.includes(\"top 15 longest\"), hasTapTitle: t.includes(\"Tap to filter:\") };\n})();"}
[/external_agent_tool_call]
```

812. assistant (2026-07-02T06:11:33.526Z)

```
[external_agent_tool_result]
{
  "hasEnglishChip": true,
  "hasSampleQueries": true,
  "hasTapTitle": true,
  "lang": "en"
}
[/external_agent_tool_result]
```

813. assistant (2026-07-02T06:11:33.526Z)

en shows both rows (English type-hints + localized taps); kn shows only the localized taps. No regression. A9 fully verified. Stopping preview and committing:

814. assistant (2026-07-02T06:11:33.526Z)

```
[external_agent_tool_call: mcp_Claude_Preview_preview_stop]
input: {"serverId":"98f833a4-2f35-49f7-9857-2efa97e11ae8"}
[/external_agent_tool_call]
```

815. assistant (2026-07-02T06:11:33.526Z)

```
[external_agent_tool_result]
Server 98f833a4-2f35-49f7-9857-2efa97e11ae8 stopped
[/external_agent_tool_result]
```

816. assistant (2026-07-02T06:11:33.526Z)

```
[external_agent_tool_call: Bash]
description: Stage A9, verify clean base
command: cd /c/Users/avidu/Projects/khelsutra && echo "branch: $(git branch --show-current)" &&
git add web/src/components/TapFilters.tsx web/src/components/TapFilters.test.tsx
web/src/components/QueryBar.tsx web/src/App.tsx web/src/i18n/hygiene.test.ts
web/src/i18n/locales/en/common.json web/src/i18n/locales/kn/common.json
web/src/i18n/locales/hi/common.json web/src/i18n/locales/es/common.json
web/src/i18n/locales/da/common.json web/src/i18n/locales/id/common.json && git diff --cached
--name-only && echo "--- base (only my work) ---" && git log --oneline origin/master..HEAD
[/external_agent_tool_call]
```

817. assistant (2026-07-02T06:11:33.526Z)

```
[external_agent_tool_result]
branch: feat/a9-tap-filters
web/src/App.tsx
web/src/components/QueryBar.tsx
web/src/components/TapFilters.test.tsx
web/src/components/TapFilters.tsx
web/src/i18n/hygiene.test.ts
web/src/i18n/locales/da/common.json
web/src/i18n/locales/en/common.json
web/src/i18n/locales/es/common.json
web/src/i18n/locales/hi/common.json
web/src/i18n/locales/id/common.json
web/src/i18n/locales/kn/common.json
--- base (only my work) ---
[/external_agent_tool_result]
```

818. assistant (2026-07-02T06:11:33.526Z)

```
[external_agent_tool_call: Bash]
description: Commit + push A9
command: git commit -q -F - <<'EOF'
feat(web): A9 ? localized tap-only filter controls
```

The query bar parses ENGLISH text (a documented limit), so a Kannada/Hindi/? user who can't type English had no way to filter. A9 adds localized TAP controls alongside it.

- New TapFilters: localized Sort (longest/shortest) + Length (over 10s/20s) + Count (top 5/10) controls (?44px, A4). Each builds the SAME ParsedQuery the parser would for its English equivalent ? directly, no English round-trip ? via the shared onApply path. A test pins tap == type (the emitted query equals parseQuery(english)) for EVERY control.
- QueryBar: the English example chips ("top 15 longest" ?) now show only in English ? for a non-English user they were an untappable wall above the localized tap row.
- catalogs en/kn/hi/es/da/id: a `tap.\*` block (self-contained "Tap to filter:" + 6 labels).

Tests: TapFilters.test.tsx (tap==type table + every-control emit + localized labels + ?44px). 173 web tests green; tsc + build clean; key-parity + no-bare-strings + placeholder parity hold. Browser-verified: en shows type-examples + tap controls; kn shows only the

localized tap controls; tapping "Over 10 seconds" ? 3 rallies. Reviewed via a 2-lens adversarial pass (correctness clean); folded its declutter + self-contained-copy + raw-trace + full-coverage findings.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>

EOF

```
git push -u origin feat/a9-tap-filters 2>&1 | tail -2
[/external_agent_tool_call]
```

819. assistant (2026-07-02T06:11:33.526Z)

```
[external_agent_tool_result]
To https://github.com/avidullu/khelsutra.git
* [new branch]      feat/a9-tap-filters -> feat/a9-tap-filters
[/external_agent_tool_result]
```

820. assistant (2026-07-02T06:11:33.527Z)

```
[external_agent_tool_call: Bash]
description: Create A9 PR
command: gh pr create --title "feat(web): A9 ? localized tap-only filter controls" --base master
--head feat/a9-tap-filters --body "$(cat <<'EOF'
```

A9 ? tap-only localized filters (FRIENDLY_AND_MULTILINGUAL.md §6.5 / §7)

The query bar **parses English** (a documented limit). A Kannada/Hindi/? user who can't type English had no way to filter without help. A9 adds **localized tap controls** alongside the bar ? the §6.5 bridge until NL parsing exists.

What changed

- **New `TapFilters`** ? localized **Sort** (longest/shortest) + **Length** (over 10s/20s) + **Count** (top 5/10) controls (244px, A4). Each builds the **same `ParsedQuery`** the parser would for its English equivalent ? directly, no English round-trip ? and routes through the shared `onApply` path. A test pins **tap == type** for **every** control (the emitted query deep-equals `parseQuery(english)``).
- **`QueryBar`**: the English example chips now render **only in English** ? for a non-English user they were an untappable wall stacked above the localized tap row (the review's main finding).
- **Catalogs (en/kn/hi/es/da/id)**: a `tap.*`` block (self-contained "Tap to filter:" + 6 control labels).

Verification

- **173 web tests** green: `TapFilters.test.tsx`` (tap==type table + **every-control emit** check + localized labels + 244px); `tsc`` + `vite build`` clean; key-parity + no-bare-strings + the new placeholder-parity guard all hold.
- **Browser-verified**: en shows the type-examples **and** the tap controls; **kn** shows only the localized tap controls (English row hidden) with a self-contained title; tapping "Over 10 seconds" filters to 3 rallies (tap == type).
- **Reviewed by a 2-lens adversarial workflow** ? correctness **clean** (tap==type traced for all 6). Folded all UX/nit findings: the non-en declutter, the self-contained "Tap to filter:" copy, logging `raw`` as the English equivalent (cleaner trace), and looping the emit-equals-parser test over every control.

Honesty notes

- kn/hi/es/da/id `tap.*`` are AI-drafted (native review pending).
- The free-text query bar stays English-grammar; A9 is the tap bridge, not NL parsing.

? Generated with [Claude Code](https://claude.com/claude-code)

EOF

```
)" 2>&1 | tail -2
[/external_agent_tool_call]
```

821. assistant (2026-07-02T06:11:33.527Z)


```
[external_agent_tool_result]
https://github.com/avidullu/khelsutra/pull/68
[/external_agent_tool_result]
```

822. assistant (2026-07-02T06:11:33.527Z)

PR [#68](https://github.com/avidullu/khelsutra/pull/68) created. Marking A9 ? in the tracker:

823. assistant (2026-07-02T06:11:33.527Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\docs\FRIENDLY_AND_MULTILINGUAL.md
[/external_agent_tool_call]
```

824. assistant (2026-07-02T06:11:33.527Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\docs\FRIENDLY_AND_MULTILINGUAL.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

825. assistant (2026-07-02T06:11:33.527Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\docs\FRIENDLY_AND_MULTILINGUAL.md
[/external_agent_tool_call]
```

826. assistant (2026-07-02T06:11:33.527Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\docs\FRIENDLY_AND_MULTILINGUAL.md has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

827. assistant (2026-07-02T06:11:33.527Z)

```
[external_agent_tool_call: Bash]
description: Commit tracker, push, poll A9 CI
command: git add docs/FRIENDLY_AND_MULTILINGUAL.md && git commit -q -m "docs(a9): mark A9 (#68)
done in the $7 tracker + changelog
```

```
Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>" && git push -q 2>&1 | tail -1 && echo
"pushed" && sleep 25 && for i in $(seq 1 8); do s=$(gh pr checks 68 2>&1 | grep -E "^web\b" |
awk '{print $2}'); c=$(gh pr checks 68 2>&1 | grep -iE "Workers Builds" | awk '{print $4}');
echo "poll $i: web=$s cf=$c"; if { [ "$s" = "pass" ] || [ "$s" = "fail" ]; } && { [ "$c" =
"pass" ] || [ "$c" = "fail" ]; }; then break; fi; sleep 15; done && echo "=== final ===" && gh
pr checks 68 2>&1 | head
[/external_agent_tool_call]
```

828. assistant (2026-07-02T06:11:33.527Z)

```
[external_agent_tool_result]
pushed
poll 1: web=pending cf=pass
poll 2: web=pass cf=pass
=== final ===
Workers Builds: hidden-king-khelsutra pass 0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-
khelsutra/production/builds/8691e772-3870-4d18-bef2-37ced20868df
site pass 10s
https://github.com/avidullu/khelsutra/actions/runs/27911948093/job/82590475940
web pass 28s
https://github.com/avidullu/khelsutra/actions/runs/27911948093/job/82590475918
[/external_agent_tool_result]
```

829. assistant (2026-07-02T06:11:33.527Z)

All green. Merging A9:

830. assistant (2026-07-02T06:11:33.527Z)

```
[external_agent_tool_call: Bash]
description: Squash-merge A9, sync master
command: gh pr merge 68 --squash --delete-branch 2>&1 | tail -2 && git checkout master -q && git
pull --ff-only -q && git log --oneline -3
[/external_agent_tool_call]
```

831. assistant (2026-07-02T06:11:33.527Z)

```
[external_agent_tool_result]
  create mode 100644 web/src/components/TapFilters.test.tsx
  create mode 100644 web/src/components/TapFilters.tsx
46e0244 feat(web): A9 ? localized tap-only filter controls (#68)
a2292ef feat(web): MULTILINGUAL Phase A ? Spanish/Danish/Indonesian SPA catalogs (#67)
ca619fb feat(web): T4 ? in-UI file upload (the runnable-tool keystone) (#66)
[/external_agent_tool_result]
```

832. assistant (2026-07-02T06:11:33.528Z)

A9 merged (#68 ? `46e0244`) ? **6 PRs this session** (A4, A5, A8, T4, MULTILINGUAL Phase A, A9), web tests 103?173, plus 2 new invariant guards (placeholder-parity, tap==type) and a latent kn/hi bug fixed.

Let me update task state and refresh the handoff before deciding the next item.

833. assistant (2026-07-02T06:11:33.528Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"9","status":"completed"}
[/external_agent_tool_call]
```

834. assistant (2026-07-02T06:11:33.528Z)

```
[external_agent_tool_result]
Updated task #9 status
[/external_agent_tool_result]
```

835. assistant (2026-07-02T06:11:33.528Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\next-session-ux-localization.md
[/external_agent_tool_call]
```

836. assistant (2026-07-02T06:11:33.528Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\next-session-ux-localization.md has been updated successfully. (file state is
current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

837. assistant (2026-07-02T06:11:33.528Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\next-session-ux-localization.md
[/external_agent_tool_call]
```

838. assistant (2026-07-02T06:11:33.528Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\next-session-ux-localization.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

839. assistant (2026-07-02T06:11:33.528Z)

Now let me decide the next step thoughtfully. I'll update the current-state checkpoint to reflect all 6 PRs first.

840. assistant (2026-07-02T06:11:33.528Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md
[/external_agent_tool_call]
```

841. assistant (2026-07-02T06:11:33.528Z)

```
[external_agent_tool_result: error]
<tool_use_error>File has been modified since read, either by the user or by a linter. Read it again before attempting to write it.</tool_use_error>
[/external_agent_tool_result]
```

842. assistant (2026-07-02T06:11:33.528Z)

The concurrent session modified `current-state.md`. Let me re-read my checkpoint block and re-apply:

843. assistant (2026-07-02T06:11:33.528Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"UX A-items \\+ North-Star
UPLOAD", "path":"C:\\Users\\avidu\\.claude\\projects\\C--Users-avidu-Projects-badminton-highlight-indexer\\memory\\current-state.md", "output_mode":"content", "-n":true, "head_limit":3}
[/external_agent_tool_call]
```

844. assistant (2026-07-02T06:11:33.528Z)

```
[external_agent_tool_result]
16:[Omitted long matching line]
[/external_agent_tool_result]
```

845. assistant (2026-07-02T06:11:33.528Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md
[/external_agent_tool_call]
```

846. assistant (2026-07-02T06:11:33.529Z)

```
[external_agent_tool_result]
14      **Checkpoint:** 2026-06-21 (**T11 retention sweep ? engine PR OPEN, all-green, coverage verified**). The owner-spun T11 chip (`task_2480b9e2`) is now BUILT in the engine. **Extended `tools/cleanup_caches.py`** (not hand-rolled, per [[cache-hygiene-policy]]) with a pre-alpha **upload retention sweep**: a new `\"uploads\"` scan location + `classify` (`uploaded source` / `partial upload` for the orphaned `partial-<id>` files) over `security.upload_dir` (default `output/uploads`); `plan_deletions` gains an **independent `uploads_retention_days` window** (privacy) orthogonal to the cache `--older-than` (disk); `scan_windows` now **skips the nested
```

uploads dir** (was silently kept whole as "other dir" ? no retention); loud structured output (retention-window banner + per-category subtotals + per-item PURGED/FAILED + a ``-json` plan for a cron/job wrapper). **Dry-run by default; reels/DB/CSVs stay durable.** **Owner gate (AskUserQuestion): uploads 7d default; reels kept durable.** ? **engine

[#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) OPEN, ALL 11 CI checks GREEN** (ruff/mypy/full-suite/golden cross-OSx3/wheel-smoke/license-scan/syntax+import); +6 unit tests; **coverage no-regression VERIFIED** (branch **85.30%** vs `origin/master` baseline **85.06%** measured in an isolated worktree; same **9748** measured stmts, branch missed **1433 ?** baseline **1456** ? the +0.24% is obsreport reporting tests in my local lane, NOT my diff; the diff touches **ONLY tools/+tests/** and the tool imports **zero** backend/training). Branch `fix/t11-retention-sweep`, cut from `origin/master` in the shared checkout (serving session's branch untouched; explicit-path staging). **? Awaiting owner merge decision** (feature-ish; not auto-merged despite green CI). The khelsutra-side \$9 T11 doc lives in the sibling repo (not editable from this checkout). [[cache-hygiene-policy]] [[ui-driven-cloud-serving-gap]] [[gotcha-shared-checkout-branch-collisions]]

15

16 **Checkpoint:** 2026-06-21 (**UX A-items + North-Star UPLOAD ? 4 khelsutra PRs merged**). The UX/localization track (separate session from the serving/alpha-shareable one below). Shipped the SPA UX A-items + the runnable-tool keystone. **4 PRs merged ? all green CI, browser-verified, each adversarially reviewed via a workflow:**

[#61](https://github.com/avidullu/khelsutra/pull/61) **A4** (?44px mobile tap targets + tooltip-text?visible: the "not available yet" reason + per-card shot confidence) .

[#62](https://github.com/avidullu/khelsutra/pull/62) **A5** (ONE shared `lib/errorText.ts` + `errors.\*` namespace across build+load+ingest ? no more raw `ApiError`/`HTTP 500`; logs gain code+detail) .

[#63](https://github.com/avidullu/khelsutra/pull/63) **A8** (background-processing reassurance + icon+text `StatusChip`, a "Stitching your reel?" building card, ?? status icons, aria-live) .

[#66](https://github.com/avidullu/khelsutra/pull/66) T4 ? in-UI file UPLOAD (NORTH-STAR keystone) (file picker ? chunked `/api/upload/\*` via new `api/client.ts` helpers + `hooks/useUpload.ts` ? server path ? `/api/process` ? existing poll?load?compile?download; **153 web tests**; rebased onto the sibling's #64/#65 ? `apiBase()`+upload coexist cleanly). **Owner decisions (AskUserQuestion):** build T4 first; **B3 catalog-home = hoist to repo-root `/locales/` (LOCKED)**; record/spin-off the rest. **Spun off as owner-launchable chips:** B3 (`task\_b3060842`), MULTILINGUAL Phase A es/da/id (`task\_e15e9643`), T11 retention (`task\_2480b9e2`) ? ? B3+MULTILINGUAL both edit catalogs, don't run together. **Remaining:** A9 (tap-only filters), A3 (recording-edu kn/hi), native review of AI-drafted kn/hi(+es/da/id); **T2 = owner stand-up only** (sibling shipped the deploy kit #65) + the hosted-mode `allowed\_video\_root` ? `upload\_dir` caveat (from the T4 review). ? Caught+fixed a slip: started T4 edits on `master`, moved to a feature branch BEFORE any commit. ? Track handoff = [[next-session-ux-localizat...
[/external_agent_tool_result]

847. assistant (2026-07-02T06:11:33.529Z)

The concurrent session already built **T11** (engine

[#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) open) from the chip I spawned ? so T11 is handled. Let me update my checkpoint block to reflect all 6 PRs + that T11/MULTILINGUAL are resolved:

848. assistant (2026-07-02T06:11:33.529Z)

[external_agent_tool_call: Edit]

file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md

[/external_agent_tool_call]

849. assistant (2026-07-02T06:11:33.529Z)

[external_agent_tool_result]

The file C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

[/external_agent_tool_result]

850. assistant (2026-07-02T06:11:33.529Z)

The chips are converging nicely ? **T11** was picked up by the concurrent session (engine [#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299), open), and I did MULTILINGUAL myself (#67). Updating task state:

851. assistant (2026-07-02T06:11:33.529Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"7","status":"completed"}
[/external_agent_tool_call]
```

852. assistant (2026-07-02T06:11:33.529Z)

```
[external_agent_tool_result]
Updated task #7 status
[/external_agent_tool_result]
```

853. assistant (2026-07-02T06:11:33.529Z)

Status: 6 PRs merged this session, all green CI, each browser-verified + adversarially reviewed.

```
| PR | What | Coverage |
|----|-----|-----|
| [#61](https://github.com/avidullu/khelsutra/pull/61) A4 | ?44px mobile tap targets + tooltip-
text?visible | +a4.touch |
| [#62](https://github.com/avidullu/khelsutra/pull/62) A5 | one shared `errorText`/`errors.*`
across build+load+ingest (no raw `ApiError`/`HTTP 500`) | +errorText, +build-error |
| [#63](https://github.com/avidullu/khelsutra/pull/63) A8 | reassurance + icon+text `StatusChip`
+ building card + ?? + aria-live | +a8.status |
| [#66](https://github.com/avidullu/khelsutra/pull/66) T4 (North Star) | in-UI file upload ?
chunked `/api/upload/*` ? `/api/process` ? existing back-half | +upload, +useUpload |
| [#67](https://github.com/avidullu/khelsutra/pull/67) MULTILINGUAL A | es/da/id ? 6
languages*; es 3-form plural; placeholder-parity guard (caught+fixed a latent kn/hi
`{videoId}` bug) | +5 |
| [#68](https://github.com/avidullu/khelsutra/pull/68) A9 | localized tap-only filters; English
chips now en-only; tap==type pinned | +TapFilters |
```

Web tests 103 ? 173, plus two new invariant guards (placeholder-parity across all 6 locales, tap==type). **Workstream A** is done (A1/A4/A5/A8/A9 ?). I caught and fixed one process slip mid-stream (T4 edits started on `master` ? moved to a feature branch before any commit).

The queue has converged:

- **T11** ? picked up by the concurrent session ? engine [#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) (open, awaiting your merge).
- **T2** ? owner stand-up only (the sibling already shipped the deploy kit #65: CF Pages + `api.\*` tunnel + CF Access). Needs: CF Access setup + **rotate `GEMINI\_API\_KEY`** + deploy. Caveat from the T4 review: set the engine's `allowed\_video\_root` to a parent of `upload\_dir`.

The one substantial item left is B3 (site/ trilingual), now **fully unblocked** (all 6 catalogs ready to hoist; approach locked = hoist to repo-root `/locales/`). It's architectural ? SPA import repoint + a new site build step + Vite out-of-root import config ? so I'd do it as **two clean PRs** (PR-1 catalog hoist + drift-guard; PR-2 vanilla shim + switcher + `data-i18n` annotation) rather than one risky change.

I'll start **B3 PR-1 (the hoist)** next on that basis ? say the word if you'd rather I hold B3 for a fresh focused session (the chip is ready) or prioritize something else.

854. user (2026-07-02T06:11:33.529Z)

when we store the video do we also record the locale in which the site was rendered in the database?

855. assistant (2026-07-02T06:11:33.529Z)

Good question ? and it's a factual one I should verify against the actual schema and API rather than answer from memory. Let me check the SPA's outgoing calls, the engine's `/api/process/upload` endpoints, and the DB schema.

856. assistant (2026-07-02T06:11:33.529Z)

```
[external_agent_tool_call: Grep]
input: {"pattern": "locale|lang|language|accept-
language|i18n", "path": "C:\\Users\\avidu\\Projects\\badminton-highlight-
indexer\\backend", "output_mode": "files_with_matches", "-i": true}
[/external_agent_tool_call]
```

857. assistant (2026-07-02T06:11:33.529Z)

```
[external_agent_tool_result]
Found 3 files
backend\reporting\spec.yaml
backend\assets\fonts\LICENSE-Apache-2.0.txt
backend\api\static\index.html
[/external_agent_tool_result]
```

858. assistant (2026-07-02T06:11:33.530Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"locale|lang(?:uage_segment)|i18n","path":"C:\\Users\\avidu\\Projects\\khelsut
ra\\web\\src\\api","output_mode":"content","-i":true}
[/external_agent_tool_call]
```

859. assistant (2026-07-02T06:11:33.530Z)

```
[external_agent_tool_result]
No matches found
[/external_agent_tool_result]
```

860. assistant (2026-07-02T06:11:33.530Z)

The SPA's API client sends `**no**` locale, and in the engine ``backend/`` the only hits are an unrelated font license, the dead ``backend/api/static/index.html``, and ``backend/reporting/spec.yaml``. Let me confirm the DB schema (videos/jobs columns) and check what the ``spec.yaml`` hit is:

861. assistant (2026-07-02T06:11:33.530Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"locale|language|lang","path":"C:\\Users\\avidu\\Projects\\badminton-
highlight-
indexer\\backend\\reporting\\spec.yaml","output_mode":"content","-i":true,"-n":true,"-C":1}
[/external_agent_tool_call]
```

862. assistant (2026-07-02T06:11:33.530Z)

```
[external_agent_tool_result]
1-# Observability report spec for the badminton-highlight-indexer.
2-# The report ENGINE (config language, renderers) lives in the shared `obsreport`
3-# library; this file is the indexer-specific layout + footgun rules. Every
[/external_agent_tool_result]
```

863. assistant (2026-07-02T06:11:33.530Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"CREATE TABLE|ADD COLUMN|c\\.\\.execute\\(|columns|videos|user_version","path":"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\backend\\infrastructure\\database.py","output_mode":"content","-n":true,"head_limit":60}
```

[/external_agent_tool_call]

864. assistant (2026-07-02T06:11:33.530Z)

[external_agent_tool_result]

```
65:         """Apply a bare ``ALTER TABLE ... ADD COLUMN`` migration step re-runably.
69:         ``user_version`` stays un-bumped ? and the next open would re-run the same
82:         """Initializes tables for videos, play_segments, and events."""
88:         # Versioned migrations live here. PRAGMA user_version is the schema
91:         version = cursor.execute("PRAGMA user_version").fetchone()[0]
110:         # future: if version < 7: cursor.execute("ALTER TABLE ..."); PRAGMA user_version
= 7
115:         """Create the videos / play_segments / events base tables (idempotent)."""
118:         CREATE TABLE IF NOT EXISTS videos (
129:         CREATE TABLE IF NOT EXISTS play_segments (
138:             FOREIGN KEY (video_id) REFERENCES videos(id) ON DELETE CASCADE
144:         CREATE TABLE IF NOT EXISTS events (
157:         # Common fields are columns; detector-specific metrics go in `stats` JSON,
160:         CREATE TABLE IF NOT EXISTS candidate_segments (
175:             FOREIGN KEY (video_id) REFERENCES videos(id) ON DELETE CASCADE,
183:         cursor.execute("PRAGMA user_version = 1")
193:         CREATE TABLE IF NOT EXISTS jobs (
210:         cursor.execute("PRAGMA user_version = 2")
218:         # ADD COLUMNS are applied idempotently so an interrupted upgrade can re-run them.
219:         self._add_column_idempotent(cursor, "ALTER TABLE jobs ADD COLUMN policy TEXT")
220:         self._add_column_idempotent(cursor, "ALTER TABLE jobs ADD COLUMN next_poll_at
TIMESTAMP")
222:         cursor.execute("PRAGMA user_version = 3")
226:         # `user_id` to videos + candidate_segments so a future per-user / multi-tenant
230:         # the ADD COLUMNS are applied idempotently so an interrupted upgrade can re-run
them.
232:         self._add_column_idempotent(cursor, "ALTER TABLE videos ADD COLUMN user_id TEXT")
233:         self._add_column_idempotent(cursor, "ALTER TABLE candidate_segments ADD COLUMN
user_id TEXT")
237:         "CREATE INDEX IF NOT EXISTS idx_videos_user ON videos(user_id) "
242:         cursor.execute("PRAGMA user_version = 4")
259:         #   n_videos_at_fit      ? held-out-user corpus size when fit (min_n trust floor
input)
262:         CREATE TABLE IF NOT EXISTS user_models (
271:             n_videos_at_fit INTEGER NOT NULL DEFAULT 0,
280:         CREATE UNIQUE INDEX IF NOT EXISTS idx_user_models_user_version
290:         cursor.execute("PRAGMA user_version = 5")
293:         # v6: per-job COST LEDGER (COMPUTE_DECOUPLED_SERVING M8, D8). NULLABLE cost columns
on
297:         # configurable FX rate. ALTER (not recreate) so v1-v5 job rows survive; ADD COLUMNS
are
311:         self._add_column_idempotent(cursor, f"ALTER TABLE jobs ADD COLUMN {col} {typ}")
318:         CREATE TABLE IF NOT EXISTS pricebook (
337:         cursor.execute("PRAGMA user_version = 6")
350:         "INSERT INTO videos (id, filepath, status, validation_results) VALUES (?, ?, ?,
?) "
472:         row = conn.cursor().execute("SELECT * FROM videos WHERE id = ?",
(video_id,)).fetchone()
531:         conn.cursor().execute("DELETE FROM videos WHERE id = ?", (video_id,))
781:         # JSON-serialised dicts; the rest are scalar columns.
789:             n_videos_at_fit: int = 0,
818:             n_videos_at_fit, is_active)
826:             n_videos_at_fit=excluded.n_videos_at_fit,
832:             n_videos_at_fit, 1 if is_active else 0))
[/external_agent_tool_result]
```

865. assistant (2026-07-02T06:11:33.530Z)

[external_agent_tool_call: Read]

file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\infrastructure\database.py

[/external_agent_tool_call]

866. assistant (2026-07-02T06:11:33.530Z)

```
[external_agent_tool_result]
115     """Create the videos / play_segments / events base tables (idempotent)."""
116     # Videos Table
117     cursor.execute("""
118         CREATE TABLE IF NOT EXISTS videos (
119             id TEXT PRIMARY KEY,
120             filepath TEXT NOT NULL,
121             processed_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
122             status TEXT NOT NULL,
123             validation_results TEXT
124         )
125     """)
126
127     # Play Segments Table (Extensible metadata JSON)
128     cursor.execute("""
129         CREATE TABLE IF NOT EXISTS play_segments (
130             id INTEGER PRIMARY KEY AUTOINCREMENT,
131             video_id TEXT NOT NULL,
132             start_time REAL NOT NULL,
133             end_time REAL NOT NULL,
134             duration REAL NOT NULL,
135             server TEXT,
136             receiver TEXT,
137             metadata TEXT,
138             FOREIGN KEY (video_id) REFERENCES videos(id) ON DELETE CASCADE
139         )
140     """)
141
142     # Events Table
143     cursor.execute("""
144         CREATE TABLE IF NOT EXISTS events (
[/external_agent_tool_result]
```

867. assistant (2026-07-02T06:11:33.530Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\infrastructure\database.py
[/external_agent_tool_call]
```

868. assistant (2026-07-02T06:11:33.530Z)

```
[external_agent_tool_result]
186     # v2: async job model (DEFERRED_HARDENING_PLAN #16). One row per submitted
187     # /api/process request. `status` is the EXECUTION state of the job
188     # (queued|running|done|failed); the pipeline's own verdict (e.g. a video
189     # that failed validation) lives inside `result` JSON as result["status"].
190     # `result` is the full sync-era response dict (incl. report paths), so the
191     # observability report is the job's artifact, per the plan.
192     cursor.execute("""
193         CREATE TABLE IF NOT EXISTS jobs (
194             id TEXT PRIMARY KEY,
195             video_path TEXT NOT NULL,
196             video_id TEXT,
197             segmenter TEXT,
198             player_pool TEXT,
199             max_frames INTEGER,
200             status TEXT NOT NULL DEFAULT 'queued',
201             progress TEXT,
202             error TEXT,
203             result TEXT,
```



```

204             created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
205             started_at TIMESTAMP,
206             finished_at TIMESTAMP
207         )
208     """)
209     cursor.execute("CREATE INDEX IF NOT EXISTS idx_jobs_status ON jobs(status)")
[/external_agent_tool_result]

```

869. assistant (2026-07-02T06:11:33.530Z)

Verified against the schema and the API ? answer is ****no****.

The site's rendered locale is ****purely a browser-side concern**** and never reaches the engine or the database:

****What actually gets stored when a video is processed:****

- `videos` table ([database.py:118](backend/infrastructure/database.py:118)): `id, filepath, processed\_at, status, validation\_results` (+ `user\_id` from the v4 migration). No locale.
- `jobs` table ([database.py:193](backend/infrastructure/database.py:193)), one row per `/api/process`: `id, video\_path, video\_id, segmenter, player\_pool, max\_frames, status, progress, error, result, created\_at/started\_at/finished\_at` (+ `policy`, `next\_poll\_at`, and the v6 cost columns). No locale.
- `play\_segments` / `events` / `candidate\_segments` ? no locale either.

****Where the locale lives instead:**** only in the browser ? the i18n detector persists the choice to `localStorage["khelsutra.lang"]` (web/src/i18n/index.ts:24). The SPA's outgoing calls don't carry it: `processVideo` sends only `{video\_path}` and `uploadInit` sends only `{filename, total\_size\_bytes}` (web/src/api/client.ts). The engine does no `Accept-Language` handling. (The only `language` hit in the backend is the `obsreport` config-DSL in `reporting/spec.yaml` ? unrelated, and the dead `backend/api/static/`.)

So today a Kannada-rendered session and an English-rendered session that process the same file produce identical DB rows.

****Whether that's a gap depends on what you want it for**** ? and there are two real reasons it might matter:

1. ****Localized output**** ? a Kannada/Hindi ****reel watermark**** or a localized report. The engine's `WatermarkConfig.font\_path` already takes a path (FRIENDLY \$4 flagged this), but the engine would need to ***know*** the locale at process time to pick the language/font. Recording it is the natural way.
2. ****Analytics**** ? which language markets are actually using the tool (a PMF signal).

If you want it, it's a small, clean, additive change: add an optional `locale` to the `/api/process` (and upload-init) request, have the SPA pass `i18n.resolvedLanguage`, and store it as a nullable column on `videos` (or `jobs`) via a v7 migration ? additive-NULL, default-OFF behavior, no impact on existing rows.

Want me to spec/build that (it'd touch the engine, so I'd coordinate with the concurrent serving session and likely fold it into the same v7 migration they're planning), or were you just checking the current state?

870. user (2026-07-02T06:11:33.531Z)

yea you very rightly identified the top reasons (watermarking and PMF) for the locale inclusion. Can you spec and build it and also do the work of localizing the watermarks in all the languages that we support? Would be good to also ensure that some basic analytics (how many spanish users etc.) are possible in the future

871. assistant (2026-07-02T06:11:33.531Z)

This is a meaty cross-repo feature, and the engine side has a real landmine: the concurrent serving session is ****planning a v7 migration**** (M9-consent + workspace_uri), so I must not blindly author a colliding `\_migrate\_to\_v7`. Let me investigate the watermark/compile/migration

surface against `origin/master` (read-only, not touching the concurrent session's working tree) before I spec anything.

872. assistant (2026-07-02T06:11:33.531Z)

```
[external_agent_tool_call: mcp_ccd_session__mark_chapter]
input: {"title":"Locale capture + localized watermarks","summary":"Owner ask: capture UI locale
on stored videos (PMF analytics) + localize the reel watermark in all 6 languages. Cross-repo;
engine v7-migration coordination needed."}
[/external_agent_tool_call]
```

873. assistant (2026-07-02T06:11:33.531Z)

```
[external_agent_tool_result]
Chapter marked: "Locale capture + localized watermarks". Continue your current work.
[/external_agent_tool_result]
```

874. assistant (2026-07-02T06:11:33.531Z)

```
[external_agent_tool_call: Bash]
description: Engine git state + migration head on origin/master
command: echo "=== branch (concurrent session's) ===" && git branch --show-current && git fetch
origin -q && echo "origin/master: $(git rev-parse --short origin/master)" && echo "" && echo
"=== migration head on origin/master (database.py user_version) ===" && git show
origin/master:backend/infrastructure/database.py 2>/dev/null | grep -nE "user_version =
[0-9]|_migrate_to_v|def _migrate|if version <" | head -30
[/external_agent_tool_call]
```

875. assistant (2026-07-02T06:11:33.531Z)

```
[external_agent_tool_result]
=== branch (concurrent session's) ===
serving/exposure-safety-preflight
origin/master: 4a06b8b

=== migration head on origin/master (database.py user_version) ===
89:         # contract ? bump it for every change and add an ordered `if version < N`
93:         if version < 1:
94:             self._migrate_to_v1(cursor)
96:         if version < 2:
97:             self._migrate_to_v2(cursor)
99:         if version < 3:
100:             self._migrate_to_v3(cursor)
102:         if version < 4:
103:             self._migrate_to_v4(cursor)
105:         if version < 5:
106:             self._migrate_to_v5(cursor)
108:         if version < 6:
109:             self._migrate_to_v6(cursor)
110:         # future: if version < 7: cursor.execute("ALTER TABLE ..."); PRAGMA user_version
= 7
155:     def _migrate_to_v1(self, cursor):
183:         cursor.execute("PRAGMA user_version = 1")
185:     def _migrate_to_v2(self, cursor):
210:         cursor.execute("PRAGMA user_version = 2")
212:     def _migrate_to_v3(self, cursor):
222:         cursor.execute("PRAGMA user_version = 3")
224:     def _migrate_to_v4(self, cursor):
242:         cursor.execute("PRAGMA user_version = 4")
244:     def _migrate_to_v5(self, cursor):
290:         cursor.execute("PRAGMA user_version = 5")
292:     def _migrate_to_v6(self, cursor):
337:         cursor.execute("PRAGMA user_version = 6")
[/external_agent_tool_result]
```

876. assistant (2026-07-02T06:11:33.531Z)

v6 is the migration head on `origin/master`; v7 is free (the serving session's v7 is planned, not yet merged). The engine checkout is on the concurrent session's branch (`serving/exposure-safety-preflight`) ? I'll do any engine work in an isolated worktree.

Now let me map the watermark + compile + font surface (the core of the build):

877. assistant (2026-07-02T06:11:33.531Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"[Ww]atermark|WatermarkConfig|font_path|draw.*text|VideoCompiler","path":"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\backend","output_mode":"files_with_matches"}
[/external_agent_tool_call]
```

878. assistant (2026-07-02T06:11:33.531Z)

```
[external_agent_tool_result]
Found 6 files
backend\\main.py
backend\\infrastructure\\database.py
backend\\config\\models.py
backend\\pipeline\\stitcher\\compiler.py
backend\\assets\\fonts\\README.md
backend\\tools\\rally_slicer.py
[/external_agent_tool_result]
```

879. assistant (2026-07-02T06:11:33.531Z)

```
[external_agent_tool_call: Glob]
input: {"pattern":"backend/assets/fonts/**"}
[/external_agent_tool_call]
```

880. assistant (2026-07-02T06:11:33.531Z)

```
[external_agent_tool_result]
backend\\assets\\fonts\\LICENSE-Apache-2.0.txt
backend\\assets\\fonts\\README.md
backend\\assets\\fonts\\RobotoSlab-Regular.ttf
[/external_agent_tool_result]
```

881. assistant (2026-07-02T06:11:33.531Z)

Critical finding: the engine bundles **only** `RobotoSlab-Regular.ttf` (Latin) ? so kn/hi watermarks will tofu without bundling the OFL Noto fonts. Let me read the watermark drawing code, the `WatermarkConfig`, and how the compile flow uses it:

882. assistant (2026-07-02T06:11:33.531Z)

```
[external_agent_tool_call: Read]
file: C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\backend\\pipeline\\stitcher\\compiler.py
[/external_agent_tool_call]
```

883. assistant (2026-07-02T06:11:33.531Z)

```
[external_agent_tool_result]
1      import os
2      import subprocess
3      import tempfile
4      import logging
5      from pathlib import Path
6      from typing import Any, Dict, List, Optional, Tuple, Union
7
```

```

8     logger = logging.getLogger(__name__)
9
10    # Bundled, commercially-clean default watermark font (Apache-2.0). Shipped so the
overlay
11    # renders out-of-the-box without a system fontconfig setup (a common gap on Windows,
where
12    # `font='Sans'` fails). Overridable via stitching.watermark.font_path. This module lives
at
13    # backend/pipeline/stitcher/, so parents[2] == backend/.
14    _BUNDLED_FONT = Path(__file__).resolve().parents[2] / "assets" / "fonts" / "RobotoSlab-
Regular.ttf"
15
16    class VideoCompiler:
17        def __init__(self, output_dir: str, end_padding: float = 1.0, begin_padding: float =
0.5,
18                    watermark: Optional[Any] = None):
19            self.output_dir = output_dir
20            self.end_padding = end_padding
21            self.begin_padding = begin_padding
22            # Optional branding overlay. Accepts a WatermarkConfig model or a plain dict;
23            # normalized to a dict (or None) so this module stays config-package-agnostic.
24            self.watermark = self._normalize_watermark(watermark)
25            os.makedirs(output_dir, exist_ok=True)
26
27        @staticmethod
28        def _normalize_watermark(watermark: Optional[Any]) -> Optional[Dict[str, Any]]:
29            if watermark is None:
30                return None
31            if hasattr(watermark, "model_dump"):
32                return watermark.model_dump()
33            if isinstance(watermark, dict):
34                return dict(watermark)
35            return None
36
37        @staticmethod
38        def _ff_quote(path: str) -> str:
39            """Make a filesystem path safe to embed inside single quotes in an ffmpeg
40            filtergraph operand (drawtext fontfile/textfile, movie= source): forward
41            slashes (Windows-friendly), escaped single quotes, AND escaped colons.
42
43            The colon MUST be escaped even inside single quotes: ffmpeg's filtergraph
44            option parser treats ':' as the option separator, so a Windows drive-letter
45            path like ``C:/dir/font.ttf`` otherwise truncates the operand at ``C`` and
46            the encode fails with "Invalid argument". This silently disabled the
47            on-by-default bundled-font watermark (and absolute logo paths) on Windows."""
48            return path.replace("\\", "/").replace("'", r"\'").replace(":", r"\:")
49
50        @staticmethod
51        def _parse_time(val: Union[int, float, str]) -> float:
52            """Normalizes a timestamp value to float seconds.
53            Handles: float/int (pass-through), plain numeric strings ("12.4"),
54            and colon-separated strings ("MM:SS", "HH:MM:SS", "H:MM:SS.ms").
55            This mirrors the parse_time logic used in gemini.py to ensure
56            the stitcher can always consume whatever format the indexer produced."""
57            if isinstance(val, (int, float)):
58                return float(val)
59            if isinstance(val, str):
60                val = val.strip()
61                if ":" in val:
62                    # Handle MM:SS, HH:MM:SS, etc.
63                    parts = val.split(":")
64                    parts.reverse()
65                    total = 0.0
66                    for i, p in enumerate(parts):
67                        try:

```

```

68             total += float(p) * (60 ** i)
69         except ValueError:
70             pass
71         return total
72     try:
73         return float(val)
74     except ValueError:
75         logger.warning(f"Could not parse timestamp value '{val}' as a number.
Defaulting to 0.0.")
76         return 0.0
77         logger.warning(f"Unexpected timestamp type {type(val).__name__} for value
'{val}'. Defaulting to 0.0.")
78         return 0.0
79
80     @staticmethod
81     def _merge_overlapping(segments: List[Tuple[float, float]]) -> List[Tuple[float,
float]]:
82         """Normalize, sort...
[/external_agent_tool_result]

```

884. assistant (2026-07-02T06:11:33.532Z)

Now I understand the watermark mechanism (ffmpeg `drawtext` with `text` + `font\_path`, defaulting to bundled RobotoSlab). To localize, I override `text` + `font\_path` per locale at compile time. Let me see `WatermarkConfig` and how `/api/compile` constructs the compiler:

885. assistant (2026-07-02T06:11:33.532Z)

```

[external_agent_tool_call: Grep]
input: {"pattern":"class WatermarkConfig|class
StitchingConfig|watermark|font_path|text:", "path":"C:\\Users\\avidu\\Projects\\badminton-
highlight-indexer\\backend\\config\\models.py", "output_mode":"content", "-n":true, "-C":1}
[/external_agent_tool_call]

```

886. assistant (2026-07-02T06:11:33.532Z)

```

[external_agent_tool_result]
237-
238:class WatermarkConfig(BaseModel):
239-     """Branding overlay burned into each highlight clip during the per-clip re-encode
240-     (backend\\pipeline\\stitcher\\compiler.py). **On by default** ? set `enabled: false`
241-     (under `stitching.watermark` in config.json) to turn it off. The text is fully
242-     configurable. The concat stage stays stream-copy (-c copy)."""
243-     enabled: bool = True
244-     text: str = "Generated by Khelsutra.guru"
245-     logo_path: str = "" # optional transparent PNG overlay; "" = no logo
246-     font_path: str = "" # optional .ttf; "" = use the fontconfig `font` family below
247-     font: str = "Sans" # fontconfig family used when font_path is empty
248-     font_size_ratio: float = Field(default=0.035, gt=0.0, le=0.5) # text height as a
fraction of frame height
--
254-
255:class StitchingConfig(BaseModel):
256-     begin_padding: float = 0.5
--
267-     min_rally_duration: float = Field(default=0.0, ge=0.0)
268-     watermark: WatermarkConfig = Field(default_factory=WatermarkConfig)
269-
[/external_agent_tool_result]

```

887. assistant (2026-07-02T06:11:33.532Z)

```

[external_agent_tool_call: Grep]
input: {"pattern":"VideoCompiler|watermark|def compile|video_path.*=|class ProcessRequest|class

```

```
CompileRequest|locale","path":"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\backend\\main.py","output_mode":"content","-n":true,"head_limit":50}[/external_agent_tool_call]
```

888. assistant (2026-07-02T06:11:33.532Z)

```
[external_agent_tool_result]
35:from backend.pipeline.stitcher.compiler import VideoCompiler
127:class ProcessRequest(BaseModel):
140:class CompileRequest(BaseModel):
148:     On a non-empty run, drop the PRIOR rows (id <= the pre-run watermark) so only the
149:     fresh segments remain. On an empty run, drop the NEW partial rows (id > watermark) and
247:         play_wm, cand_wm = db_instance.segment_watermarks(video_id)
287:         db=db_instance, video_id=video_id, video_path=req.video_path,
config=global_config,
315:         validate_input_path(req.video_path, allowed_root=allowed_root)
442:def compile_highlights(req: CompileRequest):
487:     # Run compiler with the configured stitching paddings + optional branding watermark
489:     compiler = VideoCompiler(output_dir, begin_padding=stitching.begin_padding,
490:                             end_padding=stitching.end_padding,
watermark=stitching.watermark)
646:         # clip carries the same configured stitching paddings + watermark as a real
647:         # highlight (watermark default-on).
649:         compiler = VideoCompiler(out_dir, begin_padding=stitching.begin_padding,
650:                                 end_padding=stitching.end_padding,
watermark=stitching.watermark)
740:         play_wm, cand_wm = db_instance.segment_watermarks(video_id)
757:         db=db_instance, video_id=video_id, video_path=args.video_path,
config=global_config,
[/external_agent_tool_result]
```

889. assistant (2026-07-02T06:11:33.532Z)

Let me read the request models and the compile endpoint body to see exactly where to thread the locale:

890. assistant (2026-07-02T06:11:33.532Z)

```
[external_agent_tool_call: Read]
file: C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\backend\\main.py
[/external_agent_tool_call]
```

891. assistant (2026-07-02T06:11:33.532Z)

```
[external_agent_tool_result]
127     class ProcessRequest(BaseModel):
128         video_path: str
129         player_pool: Optional[List[str]] = None
130         max_frames: Optional[int] = None
131         segmenter: Optional[str] = None
132
133     class QueryRequest(BaseModel):
134         video_id: str
135         server: Optional[str] = None
136         receiver: Optional[str] = None
137         duration_min: Optional[float] = None
138         event_type: Optional[str] = None
139
140     class CompileRequest(BaseModel):
141         video_id: str
142         segment_ids: List[int]
143         output_filename: str
144
145     def _finalize_reingest(video_id: str, segments, play_wm: int, cand_wm: int) -> None:
```

```

146         """Commit a (re)segmentation with destroy-AFTER-confirm semantics.
147
148         On a non-empty run, drop the PRIOR rows (id <= the pre-run watermark) so only the
149         fresh segments remain. On an empty run, drop the NEW partial rows (id > watermark)
150         and
151         keep the prior good results intact ? a failed/empty re-run never destroys prior
152         data.
153         """
154         if segments:
155             db_instance.prune_segments(video_id, play_upto=play_wm, cand_upto=cand_wm)
156         else:
157             db_instance.prune_segments(video_id, play_after=play_wm, cand_after=cand_wm)
158
159     # --- API Endpoints ---
160     @app.get("/api/config")
161     def get_config():
162         return global_config
163 [/external_agent_tool_result]

```

892. assistant (2026-07-02T06:11:33.532Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py
[/external_agent_tool_call]

```

893. assistant (2026-07-02T06:11:33.532Z)

```

[external_agent_tool_result]
442     def compile_highlights(req: CompileRequest):
443         video = db_instance.get_video(req.video_id)
444         if not video:
445             raise HTTPException(status_code=404, detail="Indexed video record not found.")
446
447         # Retrieve matching play segments from db
448         all_segments = db_instance.query_play_segments(req.video_id, {})
449         matched_segments = [s for s in all_segments if s["id"] in req.segment_ids]
450
451         if not matched_segments:
452             raise HTTPException(status_code=400, detail="No matching play segments found to
453             stitch.")
454
455         # Reject path traversal / absolute output names (os.path.join would otherwise let an
456         # absolute or '..'-laden output_filename write anywhere on disk).
457         try:
458             out_name = safe_output_filename(req.output_filename)
459         except ValueError as e:
460             raise HTTPException(status_code=400, detail=str(e)) from e
461
462         # The configured shot-count floor (stitching.min_shot_count) applies on the API
463         # path too ? parity with scripts/run_process_and_stitch.py. Segments WITHOUT
464         shot_count
465         # metadata are exempt: the floor filters known counts only, so explicitly
466         # requested manual/legacy segments can't silently vanish under the default floor.
467         stitching = getattr(global_config, "stitching", None) or StitchingConfig()
468         kept = []
469         for s in matched_segments:
470             meta = s.get("metadata") or {}
471             sc = meta.get("shot_count") if isinstance(meta, dict) else None
472             try:
473                 if sc is not None and int(sc) < stitching.min_shot_count:
474                     continue
475             except (TypeError, ValueError):
476                 pass # unparseable count -> treat as unknown, keep
477             kept.append(s)

```

```

476         if len(kept) < len(matched_segments):
477             logger.info(f"[compile] stitching.min_shot_count={stitching.min_shot_count}:
dropped "
478                         f"{len(matched_segments) - len(kept)}/{len(matched_segments)}
segments below the floor.")
479         if not kept:
480             raise HTTPException(status_code=400,
481                                 detail=f"All {len(matched_segments)} matching segments fall
below "
482                                     f"stitching.min_shot_count={stitching.min_shot_count}.")
483
484         # Extract start/end time pairs
485         time_pairs = [(s["start_time"], s["end_time"]) for s in kept]
486
487         # Run compiler with the configured stitching paddings + optional branding watermark
488         output_dir = getattr(getattr(global_config, "output", None), "output_dir", "output")
or "output"
489         compiler = VideoCompiler(output_dir, begin_padding=stitching.begin_padding,
490                                 end_padding=stitching.end_padding,
watermark=stitching.watermark)
491         try:
492             # The stored `filepath` is the SOURCE ref ? a bucket/Drive URI (gs://, s3://,
gdrive://) for a
493             # video ingested via /api/process from a URI, or a plain local path. Resolve it
to a LOCAL file
494             # the stitcher can read: identity for a local path (byte-identical to before),
fetch-to-scratch
495             # for a bucket URI (the same seam the process path uses, _execute_processing
main.py:159).
496             # Without this, compiling a bucket-ingested video 500s ? ffmpeg can't open a
gs:// path.
[/external_agent_tool_result]

```

894. assistant (2026-07-02T06:11:33.532Z)

I now have the complete engine picture. Key facts driving the design:

- ****Watermark**** = ffmpeg `drawtext` with `text` + `font\_path` (defaults to bundled ****RobotoSlab**, Latin-only**), built at `/api/compile` from `global\_config.stitching.watermark`.
- ****`CompileRequest`**** (video_id, segment_ids, output_filename) and ****`ProcessRequest`**** (video_path, ?) ? neither carries locale.
- ****DB****: migration head ****v6**** on origin/master; v7 is free but the serving session plans its own v7 ? real collision risk.
- ****Fonts****: only RobotoSlab is bundled ? es/da/id work as-is, but ****kn/hi** need Noto Sans Kannada/Devanagari **`.ttf`** bundled.

The font availability decides the kn/hi approach. Let me check whether I can source the Noto **`.ttf`** (from @fontsource in node_modules, or a download):

895. assistant (2026-07-02T06:11:33.532Z)

```

[external_agent_tool_call: Bash]
description: Check @fontsource ttf availability + network font fetch
command: echo "=== @fontsource files (kannada) ===" && ls
/c/Users/avidu/Projects/khelsutra/web/node_modules/@fontsource-variable/noto-sans-kannada/files/
2>/dev/null | head -20 && echo "" && echo "=== any .ttf in @fontsource kannada/devanagari? ==="
&& find /c/Users/avidu/Projects/khelsutra/web/node_modules/@fontsource-variable/noto-sans-
kannada /c/Users/avidu/Projects/khelsutra/web/node_modules/@fontsource-variable/noto-sans-
devanagari -name '*.ttf' 2>/dev/null | head && echo "(end ttf search)" && echo "" && echo "===
network test: can we fetch a Noto ttf? ===" && curl -sL -o /tmp/notok.ttf "https://raw.githubuse
rcontent.com/google/fonts/main/ofl/notosanskannada/NotoSansKannada%5Bwght%5D.ttf" 2>&1 | tail -2
&& ls -la /tmp/notok.ttf 2>/dev/null && file /tmp/notok.ttf 2>/dev/null
[/external_agent_tool_call]

```


896. assistant (2026-07-02T06:11:33.532Z)

```
[external_agent_tool_result]
=== @fontsource files (kannada) ===
noto-sans-kannada-kannada-standard-normal.woff2
noto-sans-kannada-kannada-wdth-normal.woff2
noto-sans-kannada-kannada-wght-normal.woff2
noto-sans-kannada-latin-ext-standard-normal.woff2
noto-sans-kannada-latin-ext-wdth-normal.woff2
noto-sans-kannada-latin-ext-wght-normal.woff2
noto-sans-kannada-latin-standard-normal.woff2
noto-sans-kannada-latin-wdth-normal.woff2
noto-sans-kannada-latin-wght-normal.woff2

=== any .ttf in @fontsource kannada/devanagari? ===
(end ttf search)

=== network test: can we fetch a Noto ttf? ===
-rw-r--r-- 1 avidu 197609 14 Jun 21 23:34 /tmp/notok.ttf
/tmp/notok.ttf: ASCII text, with no line terminators
[/external_agent_tool_result]
```

897. assistant (2026-07-02T06:11:33.532Z)

@fontsource ships only woff2 (no good for ffmpeg), and the first fetch 404'd (wrong path). Let me check that response and try the correct google/fonts variable-`.ttf` paths:

898. assistant (2026-07-02T06:11:33.532Z)

```
[external_agent_tool_call: Bash]
description: Inspect 404 + try correct font URLs
command: echo "=== what was the 14-byte response ===" && cat /tmp/notok.ttf; echo ""; echo "===
try correct google/fonts variable ttf paths ==="
for url in \
  "https://raw.githubusercontent.com/google/fonts/main/ofl/notosanskannada/NotoSansKannada%5Bwdt
h%2Cwght%5D.ttf" \
  "https://github.com/google/fonts/raw/main/ofl/notosanskannada/NotoSansKannada%5Bwdth,wght%5D.t
tf" ; do
  curl -sL -o /tmp/nk.ttf "$url"; sz=$(stat -c%s /tmp/nk.ttf 2>/dev/null); ft=$(file -b
/tmp/nk.ttf 2>/dev/null); echo "size=$sz type=$ft <- $url";
done
[/external_agent_tool_call]
```

899. assistant (2026-07-02T06:11:33.533Z)

```
[external_agent_tool_result]
=== what was the 14-byte response ===
404: Not Found
=== try correct google/fonts variable ttf paths ===
size=641372 type=TrueType Font data, 21 tables, 1st "GDEF", 58 names, Macintosh, type 295
string, ThinNotoSansKannada-ThinExtraLightNotoSansKannada-ExtraLightLightNotoSansKannada-
LightRegularNot <- https://raw.githubusercontent.com/google/fonts/main/ofl/notosanskannada/Noto
SansKannada%5Bwdth%2Cwght%5D.ttf
size=641372 type=TrueType Font data, 21 tables, 1st "GDEF", 58 names, Macintosh, type 295
string, ThinNotoSansKannada-ThinExtraLightNotoSansKannada-ExtraLightLightNotoSansKannada-
LightRegularNot <-
https://github.com/google/fonts/raw/main/ofl/notosanskannada/NotoSansKannada%5Bwdth,wght%5D.ttf
[/external_agent_tool_result]
```

900. assistant (2026-07-02T06:11:33.533Z)

The correct URL works ? I fetched a valid **641 KB Noto Sans Kannada variable `.ttf` (OFL)**. So kn/hi watermarks are fully achievable (fetch Noto Kannada + Devanagari from google/fonts, bundle as OFL media per the locked D7 carve-out).

Here's the spec, then I'll build it.

Spec ? locale capture + localized watermarks

****Architecture (deliberately decoupled, to ship the watermark without waiting on a DB migration):****

```
| Concern | How | Repo | Migration? |
|---|---|---|---|
| **Localized watermark** | `CompileRequest.locale` ? a `localize_watermark(cfg, locale)` helper overrides `text` + `font_path` per locale ? ffmpeg `drawtext`. en/es/da/id use RobotoSlab; kn/hi use bundled Noto. | engine | **No** |
| **SPA sends locale** | pass `i18n.resolvedLanguage` to `/api/compile` (watermark) **and** `/api/process` (analytics) | khelsutra | ? |
| **PMF analytics** | `ProcessRequest.locale` ? `jobs.locale` column ? `SELECT locale, COUNT(*) FROM jobs GROUP BY locale` | engine | **v7 (the collision-risk piece)** |
```

****3 PRs, lowest-risk first:**** PR-1 watermark (engine, no migration) ? PR-2 SPA (no engine collision) ? PR-3 analytics v7 migration (engine, ****done last + in an isolated worktree + re-verified against `origin/master`**** so it can't collide with the serving session's planned v7 ? renumber to v8 if theirs lands first). Sending `locale` to `/api/process` before PR-3 is harmless (Pydantic ignores unknown fields).

****Watermark fallback (honest):**** unknown/missing locale ? the existing default text+font (en behavior unchanged). If a Noto font is somehow absent, the compiler already retries without the watermark rather than failing the reel.

Let me build. Setting up tasks + an isolated engine worktree for PR-1, and fetching the fonts:

901. assistant (2026-07-02T06:11:33.533Z)

```
[external_agent_tool_call: TaskCreate]
description: CompileRequest.locale ? localize_watermark() overrides text+font_path per locale; bundle OFL Noto Sans Kannada/Devanagari for kn/hi; en/es/da/id use RobotoSlab. No DB migration. Verify ffmpeg renders kn/hi. Isolated worktree off origin/master.
[/external_agent_tool_call]
```

902. assistant (2026-07-02T06:11:33.533Z)

```
[external_agent_tool_result]
Task #10 created successfully: LW1 ? engine: localized reel watermark (locale ? text+font) + Noto fonts
[/external_agent_tool_result]
```

903. assistant (2026-07-02T06:11:33.533Z)

```
[external_agent_tool_call: TaskCreate]
description: web/src/api/client.ts compileReel + processVideo send i18n.resolvedLanguage; App passes it. Small khelsutra PR.
[/external_agent_tool_call]
```

904. assistant (2026-07-02T06:11:33.533Z)

```
[external_agent_tool_result]
Task #11 created successfully: LW2 ? SPA: pass UI locale to /api/compile (+ /api/process)
[/external_agent_tool_result]
```

905. assistant (2026-07-02T06:11:33.533Z)

```
[external_agent_tool_call: TaskCreate]
description: ProcessRequest.locale ? jobs.locale (v7 nullable column) + a count-by-locale helper. DONE LAST, isolated worktree, re-verify origin/master for a competing v7 (renumber to v8 if needed). Coordinate with serving session.
[/external_agent_tool_call]
```

906. assistant (2026-07-02T06:11:33.533Z)

```
[external_agent_tool_result]
Task #12 created successfully: LW3 ? engine: jobs.locale v7 migration + PMF analytics query
[/external_agent_tool_result]
```

907. assistant (2026-07-02T06:11:33.533Z)

Now the isolated worktree for the engine watermark PR + fetch both Noto fonts:

908. assistant (2026-07-02T06:11:33.533Z)

```
[external_agent_tool_call: Bash]
description: Create engine worktree + fetch Noto Kannada/Devanagari fonts
command: git fetch origin -q && WT=/c/Users/avidu/Projects/bhi-loc-watermark && git worktree add
"$WT" -b feat/localized-watermark origin/master 2>&1 | tail -3 && echo "--- worktree HEAD ---"
&& git -C "$WT" log --oneline -1 && echo "--- fetch Noto fonts (OFL) into the worktree ---" &&
FD="$WT/backend/assets/fonts" && curl -sL -o "$FD/NotoSansKannada-Variable.ttf" "https://github.
com/google/fonts/raw/main/ofl/notosanskannada/NotoSansKannada%5Bwdth,wght%5D.ttf" && curl -sL -o
"$FD/NotoSansDevanagari-Variable.ttf" "https://github.com/google/fonts/raw/main/ofl/notosansdeva
nagari/NotoSansDevanagari%5Bwdth,wght%5D.ttf" && ls -la "$FD" && echo "--- verify both are valid
TrueType ---" && file "$FD/NotoSansKannada-Variable.ttf" "$FD/NotoSansDevanagari-Variable.ttf"
[/external_agent_tool_call]
```

909. assistant (2026-07-02T06:11:33.533Z)

```
[external_agent_tool_result]
Preparing worktree (new branch 'feat/localized-watermark')
branch 'feat/localized-watermark' set up to track 'origin/master'.
HEAD is now at 4a06b8b feat(pkg): P1b ? bundle the KhelSutra SPA single-origin into the engine
wheel (default-OFF) (#302)
--- worktree HEAD ---
4a06b8b feat(pkg): P1b ? bundle the KhelSutra SPA single-origin into the engine wheel (default-
OFF) (#302)
--- fetch Noto fonts (OFL) into the worktree ---
total 1528
drwxr-xr-x 1 avidu 197609      0 Jun 21 23:36 .
drwxr-xr-x 1 avidu 197609      0 Jun 21 23:36 ..
-rw-r--r-- 1 avidu 197609 11560 Jun 21 23:36 LICENSE-Apache-2.0.txt
-rw-r--r-- 1 avidu 197609 647144 Jun 21 23:36 NotoSansDevanagari-Variable.ttf
-rw-r--r-- 1 avidu 197609 641372 Jun 21 23:36 NotoSansKannada-Variable.ttf
-rw-r--r-- 1 avidu 197609 1099 Jun 21 23:36 README.md
-rw-r--r-- 1 avidu 197609 251880 Jun 21 23:36 RobotoSlab-Regular.ttf
--- verify both are valid TrueType ---
/c/Users/avidu/Projects/bhi-loc-watermark/backend/assets/fonts/NotoSansKannada-Variable.ttf:
TrueType Font data, 21 tables, 1st "GDEF", 58 names, Macintosh, type 295 string,
ThinNotoSansKannada-ThinExtraLightNotoSansKannada-ExtraLightLightNotoSansKannada-LightRegularNot
/c/Users/avidu/Projects/bhi-loc-watermark/backend/assets/fonts/NotoSansDevanagari-Variable.ttf:
TrueType Font data, 20 tables, 1st "GDEF", 44 names, Macintosh, type 299 string,
ThinExtraLightLightRegularMediumSemiBoldBoldExtraBoldBlack
[/external_agent_tool_result]
```

910. assistant (2026-07-02T06:11:33.533Z)

Both Noto fonts fetched (valid TrueType, OFL) into the worktree (now on the latest
`origin/master` `4a06b8b`). Now the critical check ? does ffmpeg `drawtext` actually render
Kannada/Devanagari with these fonts?

911. assistant (2026-07-02T06:11:33.533Z)

```
[external_agent_tool_call: Bash]
description: Verify ffmpeg renders Kannada/Devanagari/Spanish watermarks
command: cd /c/Users/avidu/Projects/bhi-loc-watermark/backend/assets/fonts && echo "=== Kannada
render test ===" && ffmpeg -y -f lavfi -i color=c=black:s=700x200:d=1 -vf
```

```
"drawtext=fontfile='NotoSansKannada-Variable.ttf':text='KheISutra.guru ????'
?????????':fontcolor=white;fontsize=36:x=20:y=80" -frames:v 1 /tmp/wm_kn.png 2>&1 | tail -3;
echo "exit=$? size=$(stat -c%s /tmp/wm_kn.png 2>/dev/null)"; echo "=== Devanagari render test
===" && ffmpeg -y -f lavfi -i color=c=black:s=700x200:d=1 -vf
"drawtext=fontfile='NotoSansDevanagari-Variable.ttf':text='KheISutra.guru ?? ?????'
???':fontcolor=white;fontsize=36:x=20:y=80" -frames:v 1 /tmp/wm_hi.png 2>&1 | tail -3; echo
"exit=$? size=$(stat -c%s /tmp/wm_hi.png 2>/dev/null)"; echo "=== Spanish via RobotoSlab (Latin
diacritics) ===" && ffmpeg -y -f lavfi -i color=c=black:s=700x200:d=1 -vf
"drawtext=fontfile='RobotoSlab-Regular.ttf':text='Generado con
KheISutra.guru':fontcolor=white;fontsize=36:x=20:y=80" -frames:v 1 /tmp/wm_es.png 2>&1 | tail
-2; echo "exit=$? size=$(stat -c%s /tmp/wm_es.png 2>/dev/null)"
[/external_agent_tool_call]
```

912. assistant (2026-07-02T06:11:33.533Z)

```
[external_agent_tool_result]
=== Kannada render test ===
[image2 @ 0000015c9df3ee00] Use a pattern such as %03d for an image sequence or use the -update
option (with -frames:v 1 if needed) to write a single image.
[out#0/image2 @ 0000015c9dec5bc0] video:13KiB audio:0KiB subtitle:0KiB other streams:0KiB global
headers:0KiB muxing overhead: unknown
frame= 1 fps=0.0 q=-0.0 Lsize=N/A time=00:00:00.04 bitrate=N/A speed=3.06x elapsed=0:00:00.01
exit=0 size=13526
=== Devanagari render test ===
[image2 @ 000001fdd913e600] Use a pattern such as %03d for an image sequence or use the -update
option (with -frames:v 1 if needed) to write a single image.
[out#0/image2 @ 000001fdd90c5f00] video:8KiB audio:0KiB subtitle:0KiB other streams:0KiB global
headers:0KiB muxing overhead: unknown
frame= 1 fps=0.0 q=-0.0 Lsize=N/A time=00:00:00.04 bitrate=N/A speed=3.66x elapsed=0:00:00.01
exit=0 size=8487
=== Spanish via RobotoSlab (Latin diacritics) ===
[out#0/image2 @ 000001b7dc125a00] video:10KiB audio:0KiB subtitle:0KiB other streams:0KiB global
headers:0KiB muxing overhead: unknown
frame= 1 fps=0.0 q=-0.0 Lsize=N/A time=00:00:00.04 bitrate=N/A speed=3.76x elapsed=0:00:00.01
exit=0 size=9952
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

913. assistant (2026-07-02T06:11:33.533Z)

All three render (exit 0, real output). Let me visually confirm the glyphs aren't tofu by viewing the rendered frames:

914. assistant (2026-07-02T06:11:33.534Z)

```
[external_agent_tool_call: Read]
file: /tmp/wm_kn.png
[/external_agent_tool_call]
```

915. assistant (2026-07-02T06:11:33.534Z)

```
[external_agent_tool_result: error]
File does not exist. Note: your current working directory is C:\Users\avidu\Projects\badminton-
highlight-indexer.
[/external_agent_tool_result]
```

916. assistant (2026-07-02T06:11:33.534Z)

```
[external_agent_tool_call: Bash]
description: Copy render test PNGs to a readable path
command: cp /tmp/wm_kn.png /tmp/wm_hi.png /tmp/wm_es.png /c/Users/avidu/Projects/bhi-loc-
watermark/ 2>&1 && ls -la /c/Users/avidu/Projects/bhi-loc-watermark/*.png
[/external_agent_tool_call]
```

917. assistant (2026-07-02T06:11:33.534Z)

```
[external_agent_tool_result]
-rw-r--r-- 1 avidu 197609 9952 Jun 21 23:37 /c/Users/avidu/Projects/bhi-loc-watermark/wm_es.png
-rw-r--r-- 1 avidu 197609 8487 Jun 21 23:37 /c/Users/avidu/Projects/bhi-loc-watermark/wm_hi.png
-rw-r--r-- 1 avidu 197609 13526 Jun 21 23:37 /c/Users/avidu/Projects/bhi-loc-watermark/wm_kn.png
[/external_agent_tool_result]
```

918. assistant (2026-07-02T06:11:33.534Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-loc-watermark\wm_kn.png
[/external_agent_tool_call]
```

919. assistant (2026-07-02T06:11:33.534Z)

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

920. assistant (2026-07-02T06:11:33.534Z)

Kannada renders perfectly ? Latin brand + Kannada glyphs, no tofu. Let me confirm Devanagari:

921. assistant (2026-07-02T06:11:33.534Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-loc-watermark\wm_hi.png
[/external_agent_tool_call]
```

922. assistant (2026-07-02T06:11:33.534Z)

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

923. assistant (2026-07-02T06:11:33.534Z)

Both render flawlessly ? Kannada and Devanagari glyphs (no tofu), with the Latin "KhelSutra.guru" brand intact. The approach is proven end-to-end. Building LW1 now. First clean up the test artifacts and read the fonts README + fetch the OFL license:

924. assistant (2026-07-02T06:11:33.534Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"10","status":"in_progress"}
[/external_agent_tool_call]
```

925. assistant (2026-07-02T06:11:33.534Z)

```
[external_agent_tool_result]
Updated task #10 status
[/external_agent_tool_result]
```

926. assistant (2026-07-02T06:11:33.534Z)

```
[external_agent_tool_call: Bash]
description: Clean test pngs, read README, fetch OFL license
command: WT=/c/Users/avidu/Projects/bhi-loc-watermark && rm -f "$WT"/*.png && echo "removed test pngs" && echo "=== current fonts README ===" && cat "$WT/backend/assets/fonts/README.md" && echo "" && echo "=== fetch OFL-1.1 license text ===" && curl -sL -o "$WT/backend/assets/fonts/LICENSE-OFL-1.1.txt" "https://github.com/google/fonts/raw/main/ofl/notosanskannada/OFL.txt" && head -3 "$WT/backend/assets/fonts/LICENSE-OFL-1.1.txt" && echo "... ($(wc -l < "$WT/backend/assets/fonts/LICENSE-OFL-1.1.txt") lines)"
```

```
[/external_agent_tool_call]
```

927. assistant (2026-07-02T06:11:33.534Z)

```
[external_agent_tool_result]
removed test pngs
=== current fonts README ===
```

Bundled fonts

Fonts shipped with the highlight **watermark** (``stitching.watermark``). All bundled fonts are **Apache-2.0** ? commercial-clean per the project licensing guardrail (MIT/Apache/BSD only), so they can be redistributed with the tool and self-hosted derivatives.

```
| File | Font | License | Source |
|-----|-----|-----|-----|
| `RobotoSlab-Regular.ttf` | Roboto Slab | Apache-2.0 (see `LICENSE-Apache-2.0.txt`) |
https://github.com/google/fonts/tree/main/apache/robotoslab |
```

``RobotoSlab-Regular.ttf`` is the **default** watermark font: ``stitching.watermark.font_path`` defaults to it so the overlay renders **without a system fontconfig setup** (a common gap on Windows, where ``font='Sans'`` fails). To use a different typeface ? your own, or another bundled one ? set ``stitching.watermark.font_path`` to its path in ``config.json``.

To add more bundled options, drop the ``*.ttf`` here, keep it **MIT/Apache/BSD-licensed**, and add a row to the table above (with its license file).

Roboto Slab is © Google, licensed under the Apache License, Version 2.0.

```
=== fetch OFL-1.1 license text ===
Copyright 2022 The Noto Project Authors (https://github.com/notofonts/kannada)
```

```
This Font Software is licensed under the SIL Open Font License, Version 1.1.
... (93 lines)
[/external_agent_tool_result]
```

928. assistant (2026-07-02T06:11:33.534Z)

OFL license fetched (SIL OFL 1.1). Now build the engine pieces. First the localization helper:

929. assistant (2026-07-02T06:11:33.535Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\bhi-loc-watermark\backend\pipeline\stitcher\watermark_i18n.py
[/external_agent_tool_call]
```

930. assistant (2026-07-02T06:11:33.535Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\bhi-loc-
watermark\backend\pipeline\stitcher\watermark_i18n.py (file state is current in your context ?
no need to Read it back)
[/external_agent_tool_result]
```

931. assistant (2026-07-02T06:11:33.535Z)

Now wire it into ``/api/compile`` ? add ``locale`` to ``CompileRequest`` and apply the helper:

932. assistant (2026-07-02T06:11:33.535Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-loc-watermark\backend\main.py
[/external_agent_tool_call]
```

933. assistant (2026-07-02T06:11:33.535Z)

```
[external_agent_tool_result: error]
<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>
[/external_agent_tool_result]
```

934. assistant (2026-07-02T06:11:33.535Z)

Right ? the worktree is off `origin/master`, which differs from the concurrent branch I read earlier. Let me read the worktree's actual `main.py` sections:

935. assistant (2026-07-02T06:11:33.535Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"class CompileRequest|class ProcessRequest|def compile_highlights|watermark=stitching.watermark|VideoCompiler\\(", "path":"C:\\Users\\avidu\\Projects\\bhi-loc-watermark\\backend\\main.py", "output_mode":"content", "-n":true}
[/external_agent_tool_call]
```

936. assistant (2026-07-02T06:11:33.535Z)

```
[external_agent_tool_result]
202:class ProcessRequest(BaseModel):
215:class CompileRequest(BaseModel):
642:def compile_highlights(req: CompileRequest):
689:     compiler = VideoCompiler(output_dir, begin_padding=stitching.begin_padding,
690:                               end_padding=stitching.end_padding,
watermark=stitching.watermark)
852:     compiler = VideoCompiler(out_dir, begin_padding=stitching.begin_padding,
853:                               end_padding=stitching.end_padding,
watermark=stitching.watermark)
[/external_agent_tool_result]
```

937. assistant (2026-07-02T06:11:33.535Z)

```
[external_agent_tool_call: Read]
file: C:\\Users\\avidu\\Projects\\bhi-loc-watermark\\backend\\main.py
[/external_agent_tool_call]
```

938. assistant (2026-07-02T06:11:33.535Z)

```
[external_agent_tool_result]
202     class ProcessRequest(BaseModel):
203         video_path: str
204         player_pool: Optional[List[str]] = None
205         max_frames: Optional[int] = None
206         segmenter: Optional[str] = None
207
208     class QueryRequest(BaseModel):
209         video_id: str
210         server: Optional[str] = None
211         receiver: Optional[str] = None
212         duration_min: Optional[float] = None
213         event_type: Optional[str] = None
214
215     class CompileRequest(BaseModel):
216         video_id: str
217         segment_ids: List[int]
218         output_filename: str
219
220     def _finalize_reingest(video_id: str, segments, play_wm: int, cand_wm: int) -> None:
221         """Commit a (re)segmentation with destroy-AFTER-confirm semantics.
222
223         On a non-empty run, drop the PRIOR rows (id <= the pre-run watermark) so only the
```

```
[/external_agent_tool_result]
```

939. assistant (2026-07-02T06:11:33.535Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-loc-watermark\backend\main.py
[/external_agent_tool_call]
```

940. assistant (2026-07-02T06:11:33.535Z)

```
[external_agent_tool_result]
684         # Extract start/end time pairs
685         time_pairs = [(s["start_time"], s["end_time"]) for s in kept]
686
687         # Run compiler with the configured stitching paddings + optional branding watermark
688         output_dir = getattr(getattr(global_config, "output", None), "output_dir", "output")
689         or "output"
690         compiler = VideoCompiler(output_dir, begin_padding=stitching.begin_padding,
691                                 end_padding=stitching.end_padding,
692                                 watermark=stitching.watermark)
693         try:
694             # The stored `filepath` is the SOURCE ref ? a bucket/Drive URI (gs://, s3://,
695             # gdrive://) for a
696             # video ingested via /api/process from a URI, or a plain local path. Resolve it
697             # to a LOCAL file
698         except:
699             pass
700
701     [/external_agent_tool_result]
```

941. assistant (2026-07-02T06:11:33.535Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-loc-watermark\backend\main.py
[/external_agent_tool_call]
```

942. assistant (2026-07-02T06:11:33.535Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-loc-watermark\backend\main.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

943. assistant (2026-07-02T06:11:33.536Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-loc-watermark\backend\main.py
[/external_agent_tool_call]
```

944. assistant (2026-07-02T06:11:33.536Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-loc-watermark\backend\main.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

945. assistant (2026-07-02T06:11:33.536Z)

Now add the import. Let me find the `VideoCompiler` import line in the worktree:

946. assistant (2026-07-02T06:11:33.536Z)

```
[external_agent_tool_call: Grep]
input: {"pattern":"from backend.pipeline.stitcher","path":"C:\\Users\\avidu\\Projects\\bhi-loc-
watermark\\backend\\main.py","output_mode":"content","-n":true}
[/external_agent_tool_call]
```


947. assistant (2026-07-02T06:11:33.536Z)

```
[external_agent_tool_result]
39:from backend.pipeline.stitcher.compiler import VideoCompiler
[/external_agent_tool_result]
```

948. assistant (2026-07-02T06:11:33.536Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-loc-watermark\backend\main.py
[/external_agent_tool_call]
```

949. assistant (2026-07-02T06:11:33.536Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-loc-watermark\backend\main.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

950. assistant (2026-07-02T06:11:33.536Z)

Now update the fonts README to attribute the OFL Noto fonts + the D7 carve-out:

951. assistant (2026-07-02T06:11:33.536Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\bhi-loc-watermark\backend\assets\fonts\README.md
[/external_agent_tool_call]
```

952. assistant (2026-07-02T06:11:33.536Z)

```
[external_agent_tool_result: error]
<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>
[/external_agent_tool_result]
```

953. assistant (2026-07-02T06:11:33.537Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-loc-watermark\backend\assets\fonts\README.md
[/external_agent_tool_call]
```

954. assistant (2026-07-02T06:11:33.537Z)

```
[external_agent_tool_result]
1      # Bundled fonts
2
3      Fonts shipped with the highlight **watermark** (`stitching.watermark`). All bundled
fonts
4      are **Apache-2.0** ? commercial-clean per the project licensing guardrail
(MIT/Apache/BSD
5      only), so they can be redistributed with the tool and self-hosted derivatives.
6
7      | File | Font | License | Source |
8      |-----|-----|-----|-----|
9      | `RobotoSlab-Regular.ttf` | Roboto Slab | Apache-2.0 (see `LICENSE-Apache-2.0.txt`) |
https://github.com/google/fonts/tree/main/apache/robotoslab |
10
11     `RobotoSlab-Regular.ttf` is the **default** watermark font:
`stitching.watermark.font_path`
12     defaults to it so the overlay renders **without a system fontconfig setup** (a common
gap
13     on Windows, where `font='Sans'` fails). To use a different typeface ? your own, or
another
14     bundled one ? set `stitching.watermark.font_path` to its path in `config.json`.
```

```
15
16     To add more bundled options, drop the `.ttf` here, keep it **MIT/Apache/BSD-licensed**,
and
17     add a row to the table above (with its license file).
18
19     *Roboto Slab is © Google, licensed under the Apache License, Version 2.0.*
20
[/external_agent_tool_result]
```

955. assistant (2026-07-02T06:11:33.537Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\bhi-loc-watermark\backend\assets\fonts\README.md
[/external_agent_tool_call]
```

956. assistant (2026-07-02T06:11:33.537Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-loc-watermark\backend\assets\fonts\README.md has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

957. assistant (2026-07-02T06:11:33.537Z)

Now the test for the helper. Let me check the engine test conventions, then write it:

958. assistant (2026-07-02T06:11:33.538Z)

```
[external_agent_tool_call: Bash]
description: Find existing watermark/compile tests
command: cd /c/Users/avidu/Projects/bhi-loc-watermark && ls tests/ | grep -iE
"watermark|compile|stitch|api_endpoint" && echo "---" && grep -rln
"api/compile\|compile_highlights\|/api/compile" tests/ | head
[/external_agent_tool_call]
```

959. assistant (2026-07-02T06:11:33.538Z)

```
[external_agent_tool_result]
test_api_endpoints.py
test_compiler.py
test_watermark.py
---
tests/test_api_endpoints.py
tests/test_compiler.py
tests/test_config.py
tests/test_generate_highlights.py
tests/test_import_smoke.py
tests/test_watermark.py
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

960. assistant (2026-07-02T06:11:33.538Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-loc-watermark\tests\test_watermark.py
[/external_agent_tool_call]
```

961. assistant (2026-07-02T06:11:33.538Z)

```
[external_agent_tool_result]
1     """Tests for the configurable highlight watermark (stitching.watermark).
2
3     Covers the config defaults, the filtergraph builder (text + logo, positions, font,
4     the configurable string), and the end-to-end wiring through compile_highlights with
```

```

5     ffmpeg/ffprobe mocked ? including the graceful fallback when the watermark encode fails.
6     """
7     import os
8     import subprocess
9     from pathlib import Path
10    from unittest.mock import MagicMock, patch
11
12    import pytest
13    from pydantic import ValidationError
14
15    from backend.config import AppConfig
16    from backend.config.models import WatermarkConfig
17    from backend.pipeline.stitcher.compiler import VideoCompiler
18
19
20    # --- config -----
21
22    def test_watermark_config_defaults_on_with_brand_text():
23        wm = AppConfig().stitching.watermark
24        assert wm.enabled is True # default-on, with an off switch
25        assert wm.text == "Generated by Khelsutra.guru"
26        assert wm.position == "bottom-right"
27
28
29    def test_watermark_can_be_disabled_via_config():
30        cfg = AppConfig.model_validate({"stitching": {"watermark": {"enabled": False}}})
31        assert cfg.stitching.watermark.enabled is False
32
33
34    def test_font_size_ratio_and_opacity_bounds_enforced():
35        with pytest.raises(ValidationError):
36            WatermarkConfig(opacity=1.5)
37        with pytest.raises(ValidationError):
38            WatermarkConfig(font_size_ratio=0.0)
39
40
41    # --- filtergraph builder -----
42
43    def test_disabled_or_none_builds_no_filter(tmp_path):
44        assert VideoCompiler(str(tmp_path / "a"),
45watermark=None)._watermark_filter(str(tmp_path)) is None
46        off = VideoCompiler(str(tmp_path / "b"), watermark={"enabled": False})
47        assert off._watermark_filter(str(tmp_path)) is None
48
49    def test_text_filter_writes_configurable_string_and_builds_drawtext(tmp_path):
50        custom = "My Academy 2026 ? Khelsutra.guru"
51        comp = VideoCompiler(str(tmp_path / "o"), watermark={"enabled": True, "text":
custom})
52        graph = comp._watermark_filter(str(tmp_path))
53        assert graph is not None and graph.startswith("drawtext=")
54        # The arbitrary, configurable string is written verbatim to the textfile (no
escaping pain).
55        assert (tmp_path / "watermark.txt").read_text(encoding="utf-8") == custom
56        assert "textfile=" in graph
57        assert "fontcolor=white@0.85" in graph
58        assert "fontsize=h/29" in graph # round(1/0.035) == 29
59        assert "x=w-tw-24:y=h-th-24" in graph # bottom-right default, margin 24
60        assert "box=1" in graph
61
62
63    def test_position_and_margin_map_to_corner(tmp_path):
64        comp = VideoCompiler(str(tmp_path / "o"),
65watermark={"enabled": True, "text": "X", "position": "top-
left", "margin": 12})

```

```

66         graph = comp._watermark_filter(str(tmp_path))
67         assert "x=12:y=12" in graph
68
69
70     def test_explicit_font_path_uses_fontfile(tmp_path):
71         comp = VideoCompiler(str(tmp_path / "o1"),
72                               watermark={"enabled": True, "text": "X", "font_path":
"/fonts/Brand.ttf"})
73         g = comp._watermark_filter(str(tmp_path))
74         assert "fontfile='/fonts/Brand.ttf'" in g
75
76
77     def test_default_uses_bundled_apache_font(tmp_path):
78         # No font_path ? the bundled Apache-2.0 Roboto Slab (renders without fontconfig).
79         comp = VideoCompiler(str(tmp_path / "o2"), watermark={"enabled": True, "text": "X"})
80         g = comp._watermark_filter(str(tmp_path))
81         assert "fontfile=" in g and "RobotoSlab-Regular.ttf" in g
82         assert "font=" not in g # not the fontconfig path
83
84
85     def test_bundled_font_actually_ships_and_is_apache():
86         import backend.pipeline.stitcher.compiler as cmod
87         assert cmod._BUNDLED_FONT.exists(), "bundled default watermark font must ship in the
repo"
88         with open(cmod._BUNDLED_FONT, "r...
[/external_agent_tool_result]

```

962. assistant (2026-07-02T06:11:33.538Z)

Good ? clear conventions. Now the test for the i18n helper (mirroring the style, including a filtergraph integration check):

963. assistant (2026-07-02T06:11:33.538Z)

```

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\bhi-loc-watermark\tests\test_watermark_i18n.py
[/external_agent_tool_call]

```

964. assistant (2026-07-02T06:11:33.539Z)

```

[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\bhi-loc-
watermark\tests\test_watermark_i18n.py (file state is current in your context ? no need to Read
it back)
[/external_agent_tool_result]

```

965. assistant (2026-07-02T06:11:33.539Z)

Now run the engine gate on the worktree ? the new test, related tests, ruff, mypy:

966. assistant (2026-07-02T06:11:33.539Z)

```

[external_agent_tool_call: Bash]
description: Run new + related watermark/compiler tests
command: cd /c/Users/avidu/Projects/bhi-loc-watermark && python -m pytest
tests/test_watermark_i18n.py tests/test_watermark.py tests/test_compiler.py -q 2>&1 | tail -20
[/external_agent_tool_call]

```

967. assistant (2026-07-02T06:11:33.539Z)

```

[external_agent_tool_result]
..... [100%]
31 passed in 0.59s
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer

```

```
[/external_agent_tool_result]
```

968. assistant (2026-07-02T06:11:33.539Z)

31 passed (incl. my 8 new tests). Now ruff + mypy on the changes + the API-endpoint tests:

969. assistant (2026-07-02T06:11:33.539Z)

```
[external_agent_tool_call: Bash]
description: Ruff, mypy, api endpoint tests
command: cd /c/Users/avidu/Projects/bhi-loc-watermark && echo "=== ruff ===" && python -m ruff
check backend/pipeline/stitcher/watermark_i18n.py backend/main.py tests/test_watermark_i18n.py
2>&1 | tail -15 && echo "ruff done" && echo "=== mypy ===" && python -m mypy
backend/pipeline/stitcher/watermark_i18n.py 2>&1 | tail -15 && echo "=== api endpoints test
(compile path) ===" && python -m pytest tests/test_api_endpoints.py -q 2>&1 | tail -8
[/external_agent_tool_call]
```

970. assistant (2026-07-02T06:11:33.539Z)

```
[external_agent_tool_result]
=== ruff ===
All checks passed!
ruff done
=== mypy ===
Success: no issues found in 1 source file
=== api endpoints test (compile path) ===
..... [100%]
38 passed in 12.08s
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

971. assistant (2026-07-02T06:11:33.539Z)

ruff + mypy clean; API endpoints 38 pass. Now the full engine suite (the owner's pre-merge bar):

972. assistant (2026-07-02T06:11:33.539Z)

```
[external_agent_tool_call: Bash]
description: Run full engine test suite
command: cd /c/Users/avidu/Projects/bhi-loc-watermark && timeout 540 python run_all_tests.py
2>&1 | tail -25
[/external_agent_tool_call]
```

973. assistant (2026-07-02T06:11:33.539Z)

```
[external_agent_tool_result]
obsreport (optional reporting dep): available
=====
.....SS..... [ 5%]
..... [ 11%]
..... [ 16%]
..... [ 22%]
..... [ 27%]
..... [ 33%]
..... [ 38%]
..... [ 44%]
..... [ 49%]
..... [ 55%]
..... [ 60%]
..... [ 66%]
..... [ 71%]
.....SSSSS..... [ 77%]
..... [ 82%]
..... [ 88%]
```

```
..... [ 93%]
..... [ 99%]
..... [100%]
1302 passed, 7 skipped in 97.57s (0:01:37)
=====
RESULT: PASS - all tests green
=====
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

974. assistant (2026-07-02T06:11:33.539Z)

Full engine suite: **1302 passed, 7 skipped** ? green. ruff + mypy clean, ffmpeg renders kn/hi (visually verified). LW1 is complete. Committing in the worktree:

975. assistant (2026-07-02T06:11:33.539Z)

```
[external_agent_tool_call: Bash]
description: Stage LW1, verify clean base
command: cd /c/Users/avidu/Projects/bhi-loc-watermark && git fetch origin -q && echo "branch:
$(git branch --show-current) origin/master: $(git rev-parse --short origin/master)" && git add
backend/pipeline/stitcher/watermark_i18n.py backend/main.py
backend/assets/fonts/NotoSansKannada-Variable.ttf backend/assets/fonts/NotoSansDevanagari-
Variable.ttf backend/assets/fonts/LICENSE-OFL-1.1.txt backend/assets/fonts/README.md
tests/test_watermark_i18n.py && echo "--- staged ---" && git diff --cached --name-only && echo
"--- base (only my work) ---" && git log --oneline origin/master..HEAD
[/external_agent_tool_call]
```

976. assistant (2026-07-02T06:11:33.540Z)

```
[external_agent_tool_result]
branch: feat/localized-watermark origin/master: ca34787
warning: in the working copy of 'backend/assets/fonts/README.md', LF will be replaced by CRLF
the next time Git touches it
warning: in the working copy of 'backend/assets/fonts/LICENSE-OFL-1.1.txt', LF will be replaced
by CRLF the next time Git touches it
warning: in the working copy of 'backend/pipeline/stitcher/watermark_i18n.py', LF will be
replaced by CRLF the next time Git touches it
warning: in the working copy of 'tests/test_watermark_i18n.py', LF will be replaced by CRLF the
next time Git touches it
--- staged ---
backend/assets/fonts/LICENSE-OFL-1.1.txt
backend/assets/fonts/NotoSansDevanagari-Variable.ttf
backend/assets/fonts/NotoSansKannada-Variable.ttf
backend/assets/fonts/README.md
backend/main.py
backend/pipeline/stitcher/watermark_i18n.py
tests/test_watermark_i18n.py
--- base (only my work) ---
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

977. assistant (2026-07-02T06:11:33.540Z)

`origin/master` advanced to `ca34787` (serving session merging fast) ? my branch is on `4a06b8b`. Let me commit, then rebase onto the new master and check for conflicts:

978. assistant (2026-07-02T06:11:33.540Z)

```
[external_agent_tool_call: Bash]
description: Commit LW1, inspect origin/master delta
command: cd /c/Users/avidu/Projects/bhi-loc-watermark && git commit -q -F - <<'EOF'
feat(stitch): localize the reel watermark to the UI locale (+ OFL Noto fonts)
```

The reel's branding watermark now renders in the language the SPA was used in, so a Kannada/Hindi/Spanish/? user's highlight reel is watermarked in their language.

- New backend/pipeline/stitcher/watermark_i18n.py: localize_watermark(cfg, locale) overrides the watermark text (+ a script font for non-Latin) per locale. en/es/da/id keep RobotoSlab; kn ? Noto Sans Kannada, hi ? Noto Sans Devanagari. Unknown/missing/en locale ? the configured default UNCHANGED (no behavior change for existing reels). The brand token "KheI Sutra.guru" stays Latin.
- CompileRequest gains an optional `locale`; /api/compile localizes stitching.watermark before the compile. Backward-compatible (absent locale = today's behavior).
- Bundle Noto Sans Kannada + Devanagari (OFL-1.1) in backend/assets/fonts/ ? the D7 OFL-for-media carve-out (the same faces the SPA self-hosts) + LICENSE-OFL-1.1.txt + README attribution.

Graceful: a missing script font falls back to RobotoSlab (no crash), and the compiler already retries the encode without the watermark if drawtext can't render ? a font gap never nukes the reel.

Tests: tests/test_watermark_i18n.py (pass-through, Latin text-only, kn/hi text+Noto font, BCP-47 base-subtag normalization, no input mutation, fonts ship + valid sfnt + OFL, and a localized watermark flowing into the drawtext filtergraph). Verified ffmpeg actually renders Kannada + Devanagari (no tofu) end to end. Full suite 1302 passed / 7 skipped; ruff + mypy clean.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>

EOF

```
git log --oneline -1 && echo "=== what did origin/master add since my base? ===" && git diff --stat 4a06b8b origin/master | tail -15
[/external_agent_tool_call]
```

979. assistant (2026-07-02T06:11:33.540Z)

```
[external_agent_tool_result]
1f372cf feat(stitch): localize the reel watermark to the UI locale (+ OFL Noto fonts)
=== what did origin/master add since my base? ===
 backend/main.py | 96 ++++++
 tests/test_app_launcher.py | 128 ++++++
 2 files changed, 213 insertions(+), 11 deletions(-)
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

980. assistant (2026-07-02T06:11:33.540Z)

`origin/master` also changed `main.py` (+96, an app launcher) ? possible overlap with my `main.py` edits. Rebasing onto the new master:

981. assistant (2026-07-02T06:11:33.540Z)

```
[external_agent_tool_call: Bash]
description: Rebase LW1 onto new origin/master
command: cd /c/Users/avidu/Projects/bhi-loc-watermark && git rebase origin/master 2>&1 | tail -15
[/external_agent_tool_call]
```

982. assistant (2026-07-02T06:11:33.540Z)

```
[external_agent_tool_result]
Rebasing (1/1)
```

Successfully rebased and updated refs/heads/feat/localized-watermark.
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]

983. assistant (2026-07-02T06:11:33.540Z)

Rebase merged cleanly ? my `localize\_watermark` import + `CompileRequest.locale` + the compile-endpoint change coexist with the serving session's new launcher. Let me re-verify on the rebased base (the new tests included) + ruff/mypy:

984. assistant (2026-07-02T06:11:33.540Z)

```
[external_agent_tool_call: Bash]
description: Re-verify after rebase + confirm wiring intact
command: cd /c/Users/avidu/Projects/bhi-loc-watermark && python -m pytest
tests/test_watermark_i18n.py tests/test_watermark.py tests/test_api_endpoints.py
tests/test_import_smoke.py tests/test_app_launcher.py -q 2>&1 | tail -6 && echo "=== ruff
main.py + new module ===" && python -m ruff check backend/main.py
backend/pipeline/stitcher/watermark_i18n.py 2>&1 | tail -3 && echo "=== mypy ===" && python -m
mypy backend/pipeline/stitcher/watermark_i18n.py 2>&1 | tail -3 && echo "=== compile endpoint
intact? ===" && grep -n "localize_watermark" backend/main.py
[/external_agent_tool_call]
```

985. assistant (2026-07-02T06:11:33.540Z)

```
[external_agent_tool_result]
..... [ 96%]
... [100%]
75 passed in 33.10s
=== ruff main.py + new module ===
All checks passed!
=== mypy ===
Success: no issues found in 1 source file
=== compile endpoint intact? ===
43:from backend.pipeline.stitcher.watermark_i18n import localize_watermark
749:     localized_watermark = localize_watermark(stitching.watermark, req.locale)
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

986. assistant (2026-07-02T06:11:33.540Z)

Post-rebase: 75 tests pass (incl. the serving session's new launcher tests), ruff + mypy clean, wiring intact. Pushing LW1 + opening the PR:

987. assistant (2026-07-02T06:11:33.540Z)

```
[external_agent_tool_call: Bash]
description: Push LW1 engine branch
command: cd /c/Users/avidu/Projects/bhi-loc-watermark && git log --oneline origin/master..HEAD
&& echo "--- remote ---" && git remote get-url origin && git push -u origin feat/localized-
watermark 2>&1 | tail -4
[/external_agent_tool_call]
```

988. assistant (2026-07-02T06:11:33.541Z)

```
[external_agent_tool_result]
5132e14 feat(stitch): localize the reel watermark to the UI locale (+ OFL Noto fonts)
--- remote ---
https://github.com/Khelsutra/badminton-highlight-indexer.git
remote:
branch 'feat/localized-watermark' set up to track 'origin/feat/localized-watermark'.
To https://github.com/Khelsutra/badminton-highlight-indexer.git
* [new branch]      feat/localized-watermark -> feat/localized-watermark
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

989. assistant (2026-07-02T06:11:33.541Z)


```
[external_agent_tool_call: Bash]
description: Create LW1 engine PR
command: cd /c/Users/avidu/Projects/bhi-loc-watermark && gh pr create --repo
Khelsutra/badminton-highlight-indexer --base master --head feat/localized-watermark --title
"feat(stitch): localize the reel watermark to the UI locale (+ OFL Noto fonts)" --body "$(cat
<<'EOF'
```

Localized reel watermark

The reel's branding watermark now renders in the language the SPA was used in ? a Kannada/Hindi/Spanish/? user's highlight reel is watermarked in their language, with a script-appropriate font.

What changed

```
- **New `backend/pipeline/stitcher/watermark_i18n.py` ? `localize_watermark(cfg, locale)`
overrides the watermark `text` (and a script `font_path` for non-Latin) per locale: en/es/da/id
keep RobotoSlab; **kn ? Noto Sans Kannada, hi ? Noto Sans Devanagari**. The brand token
"KhelSutra.guru" stays Latin. **Unknown/missing/`en` locale ? the configured default UNCHANGED**
(no behavior change for existing reels).
- **`CompileRequest` gains an optional `locale`; `/api/compile` localizes
`stitching.watermark` before compiling. Backward-compatible ? absent locale = today's behavior.
- **Bundled Noto Sans Kannada + Devanagari (OFL-1.1)** in `backend/assets/fonts/` ? the **D7
OFL-for-media carve-out** (the same faces the SPA self-hosts) + `LICENSE-OFL-1.1.txt` + README
attribution.
```

Graceful by construction

A missing script font falls back to RobotoSlab (no crash), and the compiler already retries the encode **without** the watermark if `drawtext` can't render ? a font gap can never nuke the reel.

Verification

```
- **Full suite: 1302 passed / 7 skipped** ; `ruff` + `mypy` clean (rebased onto latest `master`,
re-verified ? 75 targeted tests incl. the new app-launcher tests green).
- New `tests/test_watermark_i18n.py`: pass-through (en/unknown/None), Latin text-only override,
kn/hi text + bundled Noto font, BCP-47 base-subtag normalization, no input mutation, fonts ship
+ valid sfnt + OFL, and a localized watermark flowing into the `drawtext` filtergraph.
- **Rendered with real ffmpeg** ? Kannada (`KhelSutra.guru` ???) and Devanagari
(`KhelSutra.guru` ???) draw correctly (no tofu), Latin brand intact.
```

Notes

```
- Localized watermark strings are AI-drafted ? **native-speaker review pending** (same posture
as the SPA catalogs).
- The SPA must pass `locale` to `/api/compile` to activate this (separate khelsutra PR); this is
inert + backward-compatible until then.
- Built in an isolated worktree off `origin/master`; the concurrent serving session's working
tree was untouched.
```

```
? Generated with [Claude Code](https://claude.com/claude-code)
EOF
)" 2>&1 | tail -2
[/external_agent_tool_call]
```

990. assistant (2026-07-02T06:11:33.541Z)

```
[external_agent_tool_result]
https://github.com/Khelsutra/badminton-highlight-indexer/pull/304
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

991. assistant (2026-07-02T06:11:33.541Z)

LW1 PR **[#304](https://github.com/Khelsutra/badminton-highlight-indexer/pull/304)** is up. The OFL fonts could trip the engine's **license-scan** CI check (it may only allow MIT/Apache/BSD). Let me check CI status, and build LW2 (SPA passthrough) while it runs.

992. assistant (2026-07-02T06:11:33.541Z)

```
[external_agent_tool_call: Bash]
description: Check LW1 CI (esp. license-scan)
command: cd /c/Users/avidu/Projects/bhi-loc-watermark && sleep 20 && gh pr checks 304 --repo
Khelsutra/badminton-highlight-indexer 2>&1 | head -20
[/external_agent_tool_call]
```

993. assistant (2026-07-02T06:11:33.541Z)

```
[external_agent_tool_result]
Golden fixtures cross-OS (windows-latest)      pass      52s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27913265375/job/82593977777
Ruff lint          pass      10s      https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27913265375/job/82593977849
Syntax sweep + import smoke (no GPU stack)      pass      20s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27913265375/job/82593977825
Built wheel smoke (install + launch)      pending 0      https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27913265375/job/82593977755
Mypy (lenient baseline) pass      31s      https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27913265375/job/82593977808
Full test suite (offline, mocked)      pending 0      https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27913265375/job/82593977761
Dependency license scan (no copyleft in runtime deps) pass      32s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27913265375/job/82593977782
Golden fixtures cross-OS (macos-latest) pass      17s      https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27913265375/job/82593977794
Golden fixtures cross-OS (ubuntu-latest)      pass      25s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27913265375/job/82593977785
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

994. assistant (2026-07-02T06:11:33.541Z)

License-scan passed (it scans dependency licenses, not font media). Most checks green; wheel-smoke + full-suite pending. Let me verify the new fonts are included in the wheel package-data (else the packaged kn/hi watermark would break), then build LW2 while CI finishes:

995. assistant (2026-07-02T06:11:33.541Z)

```
[external_agent_tool_call: Bash]
description: Check wheel packaging includes the fonts
command: cd /c/Users/avidu/Projects/bhi-loc-watermark && echo "=== package-data / font inclusion
=== " && grep -rNE "ttf|assets|fonts|package.?data|package_data|force-include|MANIFEST"
pyproject.toml MANIFEST.in setup.py setup.cfg 2>/dev/null | grep -iE
"ttf|font|assets|package.?data|include" | head -20
[/external_agent_tool_call]
```

996. assistant (2026-07-02T06:11:33.541Z)

```
[external_agent_tool_result]
=== package-data / font inclusion ===
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

997. assistant (2026-07-02T06:11:33.541Z)

```
[external_agent_tool_call: Bash]
description: Inspect pyproject packaging + tracked fonts
command: cd /c/Users/avidu/Projects/bhi-loc-watermark && echo "=== pyproject build + package
```

```
config ===" && grep -nE "build-  
backend|\[tool\.(setuptools|hatch|poetry)|include|packages|artifacts|force-  
include|\.ttf|assets|fonts|spa" pyproject.toml | head -40 && echo "" && echo "=== is RobotoSlab  
tracked + how (other .ttf already shipped) ===" && git ls-files backend/assets/fonts/  
[/external_agent_tool_call]
```

998. assistant (2026-07-02T06:11:33.541Z)

```
[external_agent_tool_result]  
=== pyproject build + package config ===  
5:# reason), so the wheel target packages are listed EXPLICITLY below ? hatchling's  
14:build-backend = "hatchling.build"  
88:[tool.hatch.metadata]  
93:[tool.hatch.build.targets.wheel]  
94:# `backend` is a namespace package (no __init__.py); list both import roots  
97:packages = ["backend", "training"]
```

```
=== is RobotoSlab tracked + how (other .ttf already shipped) ===  
backend/assets/fonts/LICENSE-Apache-2.0.txt  
backend/assets/fonts/LICENSE-OFL-1.1.txt  
backend/assets/fonts/NotoSansDevanagari-Variable.ttf  
backend/assets/fonts/NotoSansKannada-Variable.ttf  
backend/assets/fonts/README.md  
backend/assets/fonts/RobotoSlab-Regular.ttf  
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer  
[/external_agent_tool_result]
```

999. assistant (2026-07-02T06:11:33.541Z)

Hatchling packages all files under `backend/` by default (RobotoSlab already ships this way), so the new Noto `.ttf` are auto-included in the wheel ? no packaging change needed, and they're tracked. Let me let LW1 CI finish while I build **LW2** (the SPA passthrough). Cutting the khelsutra branch:

1000. assistant (2026-07-02T06:11:33.542Z)

```
[external_agent_tool_call: TaskUpdate]  
input: {"taskId":"11","status":"in_progress"}  
[/external_agent_tool_call]
```

1001. assistant (2026-07-02T06:11:33.542Z)

```
[external_agent_tool_result]  
Updated task #11 status  
[/external_agent_tool_result]
```

1002. assistant (2026-07-02T06:11:33.542Z)

```
[external_agent_tool_call: Bash]  
description: Cut LW2 branch, locate client fns  
command: cd /c/Users/avidu/Projects/khelsutra && git fetch origin -q && git checkout -b  
feat/lw2-locale-passthrough origin/master 2>&1 | tail -2 && git log --oneline -1 && echo "===  
current compileReel + processVideo signatures ===" && grep -nE "export async function  
compileReel|export async function processVideo" web/src/api/client.ts  
[/external_agent_tool_call]
```

1003. assistant (2026-07-02T06:11:33.542Z)

```
[external_agent_tool_result]  
branch 'feat/lw2-locale-passthrough' set up to track 'origin/master'.  
Switched to a new branch 'feat/lw2-locale-passthrough'  
1b1fa7b feat(web+site): B3 ? trilingual marketing site via shared t()/data-i18n shim (D-CATALOG-  
HOME) (#70)  
=== current compileReel + processVideo signatures ===
```

```
84:export async function processVideo(videoPath: string, cid?: string): Promise<ProcessResponse>
{
108:export async function compileReel(
[/external_agent_tool_result]
```

1004. assistant (2026-07-02T06:11:33.542Z)

`origin/master` shows **B3** is already done** (#70 ? "trilingual marketing site via shared t()/data-i18n shim", the catalog hoist) ? the concurrent session/owner picked up that chip. Good, one less item. (My LW2 doesn't touch catalogs, so the hoist doesn't affect it.) Let me read the client functions to edit:

1005. assistant (2026-07-02T06:11:33.542Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khehsutra\web\src\api\client.ts
[/external_agent_tool_call]
```

1006. assistant (2026-07-02T06:11:33.542Z)

```
[external_agent_tool_result]
80
81      /** Submit a video (a `gs://`/`gcs://`/`s3://` bucket URI, or a path on the engine host)
for
82      * processing. POST /api/process {video_path} ? 202 {job_id, ...} [main.py:260]. Fast-
fails 400
83      * (unsupported scheme / out-of-allowed-root) or 404 (missing LOCAL file) ? surfaced as
ApiError. */
84      export async function processVideo(videoPath: string, cid?: string):
Promise<ProcessResponse> {
85          return request<ProcessResponse>("/process", {
86              method: "POST",
87              body: JSON.stringify({ video_path: videoPath }),
88              cid,
89          });
90      }
91
92      /** Poll a submitted job. GET /api/jobs/{job_id} [main.py:303]. 404 ? unknown job_id. */
93      export async function getJob(jobId: string, cid?: string): Promise<JobStatus> {
94          return request<JobStatus>(`/jobs/${encodeURIComponent(jobId)}`, { method: "GET", cid
});
95      }
96
97      /** List a video's detected rallies. POST /api/query {video_id} ? {play_segments}
[main.py:331]. */
98      export async function listRallies(videoId: string, cid?: string):
Promise<RallySegment[]> {
99          const data = await request<QueryResponse>("/query", {
100              method: "POST",
101              body: JSON.stringify({ video_id: videoId }),
102              cid,
103          });
104          return data.play_segments ?? [];
105      }
106
107      /** Stitch selected rallies into one reel. POST /api/compile {video_id, segment_ids,
output_filename} [main.py:342]. */
108      export async function compileReel(
109          videoId: string,
110          segmentIds: number[],
111          outputFilename: string,
112          cid?: string,
113      ): Promise<CompileResponse> {
114          return request<CompileResponse>("/compile", {
```

```
115         method: "POST",
116         body: JSON.stringify({ video_id: videoId, segment_ids: segmentIds, output_filename:
outputFilename }),
117         cid,
118     });
119 }
[/external_agent_tool_result]
```

1007. assistant (2026-07-02T06:11:33.542Z)

LW2 ? thread locale through the client (optional, so existing callers are unaffected;
`JSON.stringify` drops `undefined`):

1008. assistant (2026-07-02T06:11:33.542Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\api\client.ts
[/external_agent_tool_call]
```

1009. assistant (2026-07-02T06:11:33.542Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\api\client.ts has been updated successfully.
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

1010. assistant (2026-07-02T06:11:33.542Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\api\client.ts
[/external_agent_tool_call]
```

1011. assistant (2026-07-02T06:11:33.542Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\api\client.ts has been updated successfully.
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

1012. assistant (2026-07-02T06:11:33.542Z)

Now thread locale through the ingest hook and the components. Let me read the current
`useIngestJob.submit` + `IngestPanel` submit calls + `App.onBuild`:

1013. assistant (2026-07-02T06:11:33.542Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\web\src\hooks\useIngestJob.ts
[/external_agent_tool_call]
```

1014. assistant (2026-07-02T06:11:33.542Z)

```
[external_agent_tool_result]
51     const submit = useCallback(
52       (rawUri: string) => {
53         const uri = rawUri.trim();
54         if (!uri) {
55           setPhase({ status: "error", error: { code: "invalidUri" } });
56           return;
57         }
58
59         stop(); // supersede any prior run
60         const run: Run = { cancelled: false };
```

```

61         runRef.current = run;
62
63         const cid = newCorrelationId();
64         const startedAt = Date.now();
65         setPhase({ status: "submitting" });
66         log("info", "ingest.submit_start", { cid, uri });
67
68         const fail = (error: IngestError, event: string) => {
69             if (run.cancelled) return;
[/external_agent_tool_result]

```

1015. assistant (2026-07-02T06:11:33.543Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\web\src\hooks\useIngestJob.ts
[/external_agent_tool_call]

```

1016. assistant (2026-07-02T06:11:33.543Z)

```

[external_agent_tool_result]
116         void (async () => {
117             let jobId: string;
118             try {
119                 const resp = await processVideo(uri, cid);
120                 jobId = resp.job_id;
121             } catch (err) {
122                 fail(classifyError(err), "ingest.submit_error");
123                 return;
124             }
125             if (run.cancelled) return;
126             log("info", "ingest.queued", { cid, uri, job_id: jobId });
127             setPhase({ status: "polling", progress: null });
128             poll(jobId, POLL_MIN_MS);
129         })();
130     },
131     [stop],
[/external_agent_tool_result]

```

1017. assistant (2026-07-02T06:11:33.543Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\hooks\useIngestJob.ts
[/external_agent_tool_call]

```

1018. assistant (2026-07-02T06:11:33.543Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\hooks\useIngestJob.ts has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

1019. assistant (2026-07-02T06:11:33.543Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\hooks\useIngestJob.ts
[/external_agent_tool_call]

```

1020. assistant (2026-07-02T06:11:33.543Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\hooks\useIngestJob.ts has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

1021. assistant (2026-07-02T06:11:33.543Z)

Now `IngestPanel` ? pass the active locale on both submit paths. Let me read the current relevant lines:

1022. assistant (2026-07-02T06:11:33.543Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx
[/external_agent_tool_call]
```

1023. assistant (2026-07-02T06:11:33.543Z)

```
[external_agent_tool_result]
19
20   export function IngestPanel({ onLoaded }: Props) {
21     const { t } = useTranslation();
22     const [uri, setUri] = useState("");
23     const { phase, submit, cancel: cancelIngest, reset: resetIngest, busy: ingestBusy } =
useIngestJob(onLoaded);
24     // On upload completion, feed the returned server PATH straight into the engine
processing back-half.
25     const upload = useUpload((path) => submit(path));
26     const busy = ingestBusy || upload.busy;
27
28     const onSubmitUri = (e: FormEvent) => {
29       e.preventDefault();
30       submit(uri);
31     };
32     const onFile = (e: ChangeEvent<HTMLInputElement>) => {
33       const file = e.target.files?.[0];
34       if (file) upload.start(file);
35       e.target.value = ""; // let re-picking the same file fire change again
36     };
37     const cancelAll = () => {
38       upload.cancel();
39       cancelIngest();
40     };
41     const resetAll = () => {
42       upload.reset();
43       resetIngest();
44     };
45
46     // Shared mapping with the build flow (lib/errorText): an `engine` failure carries the
engine's own
47     // message (a 400 scheme/path/extension detail, a worker error); every other code maps
to a friendly
48     // localized line. Surface whichever half failed (upload or engine) ? they're mutually
exclusive.
[/external_agent_tool_result]
```

1024. assistant (2026-07-02T06:11:33.543Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx
[/external_agent_tool_call]
```

1025. assistant (2026-07-02T06:11:33.543Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\IngestPanel.tsx has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

1026. assistant (2026-07-02T06:11:33.543Z)

Now `App.onBuild` ? pass the locale to `compileReel`:

1027. assistant (2026-07-02T06:11:33.543Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\App.tsx
[/external_agent_tool_call]
```

1028. assistant (2026-07-02T06:11:33.543Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\App.tsx has been updated successfully. (file
state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

1029. assistant (2026-07-02T06:11:33.543Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\App.tsx
[/external_agent_tool_call]
```

1030. assistant (2026-07-02T06:11:33.544Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\App.tsx has been updated successfully. (file
state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

1031. assistant (2026-07-02T06:11:33.544Z)

Now tests for the passthrough. A client-level test (locale included when present, omitted when absent) + an App-integration check. First the client test:

1032. assistant (2026-07-02T06:11:33.544Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\khelsutra\web\src\api\locale.test.ts
[/external_agent_tool_call]
```

1033. assistant (2026-07-02T06:11:33.544Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\khelsutra\web\src\api\locale.test.ts (file
state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

1034. assistant (2026-07-02T06:11:33.544Z)

Now an App-integration test proving the active locale reaches `/api/compile`. Let me add it to the existing build-error harness:

1035. assistant (2026-07-02T06:11:33.544Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\build-error.test.tsx
[/external_agent_tool_call]
```

1036. assistant (2026-07-02T06:11:33.544Z)

```
[external_agent_tool_result]
73         const u = String(url);
```



```

74         const method = init?.method ?? "GET";
75         if (u === "/api/query" && method === "POST") throw new Error("offline"); // engine
down on load
76         if (u === "/api/config") return httpErr(500, "");
77         throw new Error(`unexpected fetch ${method} ${u}`);
78     }) as unknown as typeof fetch);
79     render(<App />);
80     const banner = await screen.findByTestId("mode-banner");
81     await waitFor(() => expect(banner.textContent).toMatch(/can't reach khelsutra/i));
82     expect(banner.textContent).not.toMatch(/ApiError|Network error calling/);
83     });
84     });
85
[/external_agent_tool_result]

```

1037. assistant (2026-07-02T06:11:33.544Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\khelsutra\web\src\components\build-error.test.tsx
[/external_agent_tool_call]

```

1038. assistant (2026-07-02T06:11:33.544Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\khelsutra\web\src\components\build-error.test.tsx has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

1039. assistant (2026-07-02T06:11:33.544Z)

Now run the web suite + tsc + build for LW2:

1040. assistant (2026-07-02T06:11:33.544Z)

```

[external_agent_tool_call: Bash]
description: Run web suite + tsc + build for LW2
command: cd /c/Users/avidu/Projects/khelsutra/web && npm test --silent 2>/dev/null | grep -E
"Test Files|Tests |FAIL|?|x|? src" | tail -10 && echo "=== tsc ===" && npx tsc --noEmit && echo
"tsc OK" && echo "=== build ===" && npm run build --silent 2>&1 | tail -1
[/external_agent_tool_call]

```

1041. assistant (2026-07-02T06:11:33.544Z)

```

[external_agent_tool_result]
Test Files  20 passed (20)
Tests      177 passed (177)
=== tsc ===
tsc OK
=== build ===
? built in 1.36s
[/external_agent_tool_result]

```

1042. assistant (2026-07-02T06:11:33.545Z)

177 tests (+4), tsc + build clean. LW2 has no visible UI change (data passthrough, fully test-covered). Committing:

1043. assistant (2026-07-02T06:11:33.545Z)

```

[external_agent_tool_call: Bash]
description: Stage, commit, push LW2
command: cd /c/Users/avidu/Projects/khelsutra && echo "branch: $(git branch --show-current)" &&
git add web/src/api/client.ts web/src/api/locale.test.ts web/src/hooks/useIngestJob.ts
web/src/components/IngestPanel.tsx web/src/components/build-error.test.tsx web/src/App.tsx &&

```

```
git diff --cached --name-only && echo "--- base (only my work) ---" && git log --oneline
origin/master..HEAD && git commit -q -F - <<'EOF'
feat(web): LW2 ? send the UI locale to /api/compile (watermark) + /api/process (analytics)
```

The SPA now records the language it was used in on the two engine calls that need it:

- compileReel + processVideo (api/client.ts) take an optional `locale`, included in the body and omitted when absent (JSON.stringify drops undefined ? full back-compat).
- IngestPanel passes i18n.resolvedLanguage on both ingest paths (URI + file upload), threaded through useIngestJob.submit(uri, locale) ? processVideo.
- App.onBuild passes i18n.resolvedLanguage to compileReel ? the engine localizes the reel's branding watermark (engine LW1) to the user's language.

Tests: api/locale.test.ts (locale included when present, omitted when absent, on both calls) + a build-error.test App-integration check (the active locale reaches /api/compile). 177 web tests green; tsc + build clean. Inert/back-compat until the engine LW1 (watermark) + LW3 (analytics) land.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>

EOF

```
git log --oneline -1 && git push -u origin feat/lw2-locale-passthrough 2>&1 | tail -2
[/external_agent_tool_call]
```

1044. assistant (2026-07-02T06:11:33.545Z)

```
[external_agent_tool_result]
branch: feat/lw2-locale-passthrough
web/src/App.tsx
web/src/api/client.ts
web/src/api/locale.test.ts
web/src/components/IngestPanel.tsx
web/src/components/build-error.test.tsx
web/src/hooks/useIngestJob.ts
--- base (only my work) ---
67a386c feat(web): LW2 ? send the UI locale to /api/compile (watermark) + /api/process
(analytics)
To https://github.com/avidullu/khelsutra.git
* [new branch]      feat/lw2-locale-passthrough -> feat/lw2-locale-passthrough
[/external_agent_tool_result]
```

1045. assistant (2026-07-02T06:11:33.545Z)

```
[external_agent_tool_call: Bash]
description: Create LW2 PR
command: gh pr create --base master --head feat/lw2-locale-passthrough --title "feat(web): LW2 ?
send the UI locale to /api/compile + /api/process" --body "$(cat <<'EOF'
```

LW2 ? record the UI locale on the engine calls that need it

Pairs with the engine watermark localization. The SPA now sends the language it was rendered in on the two calls that use it:

- **`/api/compile`** ? localizes the **reel branding watermark** (engine LW1).
- **`/api/process`** ? records the locale for **PMF analytics** (which markets use the tool; engine LW3).

What changed

- `compileReel` + `processVideo` (`api/client.ts`) take an optional `locale`, included in the body and **omitted when absent** (`JSON.stringify` drops `undefined` ? full back-compat).
- `IngestPanel` passes `i18n.resolvedLanguage` on **both** ingest paths (URI + file upload), threaded through `useIngestJob.submit(uri, locale)` ? `processVideo`.
- `App.onBuild` passes `i18n.resolvedLanguage` to `compileReel`.

Verification

- **177 web tests** green (`api/locale.test.ts`: locale included when present / omitted when absent on both calls; a `build-error.test` App-integration check that the active locale reaches

```
`/api/compile`); `tsc` + `vite build` clean.  
- No visible UI change (data passthrough) ? fully test-covered.  
- **Inert + backward-compatible** until the engine PRs land: the engine ignores an unknown  
`/api/process` field (Pydantic) and falls back to the default watermark for an absent/unknown  
compile locale.  
  
Engine side: watermark = [Khelsutra/badminton-highlight-indexer#304](https://github.com/Khelsutra/badminton-highlight-indexer/pull/304) (LW1); analytics  
migration = LW3 (next).  
  
? Generated with [Claude Code](https://claude.com/claude-code)  
EOF  
)" 2>&1 | tail -2  
[/external_agent_tool_call]
```

1046. assistant (2026-07-02T06:11:33.545Z)

```
[external_agent_tool_result]  
https://github.com/avidullu/khelsutra/pull/73  
[/external_agent_tool_result]
```

1047. assistant (2026-07-02T06:11:33.545Z)

LW2 PR **[#73](https://github.com/avidullu/khelsutra/pull/73)** up. Let me check LW1 (#304) CI ?
the full-suite + wheel-smoke were pending ? and LW2 CI:

1048. assistant (2026-07-02T06:11:33.545Z)

```
[external_agent_tool_call: Bash]  
description: Check LW1 engine CI status  
command: echo "=== LW1 #304 (engine) ===" && gh pr checks 304 --repo Khelsutra/badminton-  
highlight-indexer 2>&1 | awk '{print $1"\t"$2}' | head -20  
[/external_agent_tool_call]
```

1049. assistant (2026-07-02T06:11:33.545Z)

```
[external_agent_tool_result]  
=== LW1 #304 (engine) ===  
Built    wheel  
Dependency    license  
Full    test  
Golden    fixtures  
Golden    fixtures  
Golden    fixtures  
Mypy      (lenient  
Ruff      lint  
Syntax    sweep  
[/external_agent_tool_result]
```

1050. assistant (2026-07-02T06:11:33.545Z)

```
[external_agent_tool_call: Bash]  
description: Clean CI status for LW1 + LW2  
command: echo "=== LW1 #304 ===" && gh pr checks 304 --repo Khelsutra/badminton-highlight-  
indexer --json name,state -q '.[ ] | "\(.state)\t\(.name)'" 2>&1 && echo "" && echo "=== LW2 #73  
=== " && sleep 5 && gh pr checks 73 --json name,state -q '.[ ] | "\(.state)\t\(.name)'" 2>&1  
[/external_agent_tool_call]
```

1051. assistant (2026-07-02T06:11:33.545Z)

```
[external_agent_tool_result]  
=== LW1 #304 ===  
SUCCESS Ruff lint  
SUCCESS Syntax sweep + import smoke (no GPU stack)
```

```
SUCCESS Golden fixtures cross-OS (windows-latest)
SUCCESS Built wheel smoke (install + launch)
SUCCESS Mypy (lenient baseline)
SUCCESS Full test suite (offline, mocked)
SUCCESS Golden fixtures cross-OS (ubuntu-latest)
SUCCESS Golden fixtures cross-OS (macos-latest)
SUCCESS Dependency license scan (no copyleft in runtime deps)
```

```
=== LW2 #73 ===
```

```
SUCCESS Workers Builds: hidden-king-khelsutra
SUCCESS site
IN_PROGRESS      web
[/external_agent_tool_result]
```

1052. assistant (2026-07-02T06:11:33.545Z)

****LW1 #304: all 9 engine checks green**** (full suite cross-OS, wheel-smoke, license-scan included). LW2 web check still running. Per "keep merging green CI PRs" and my merge authority, merging LW1 (owner-requested, fully green, backward-compatible):

1053. assistant (2026-07-02T06:11:33.545Z)

```
[external_agent_tool_call: Bash]
description: Squash-merge LW1 engine PR
command: cd /c/Users/avidu/Projects/bhi-loc-watermark && gh pr merge 304 --repo
Khelsutra/badminton-highlight-indexer --squash --delete-branch 2>&1 | tail -4
[/external_agent_tool_call]
```

1054. assistant (2026-07-02T06:11:33.545Z)

```
[external_agent_tool_result]
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

1055. assistant (2026-07-02T06:11:33.546Z)

```
[external_agent_tool_call: Bash]
description: Confirm LW1 merged + remove worktree
command: gh pr view 304 --repo Khelsutra/badminton-highlight-indexer --json state,mergedAt -q
'"state=\.state) merged=\.mergedAt"' 2>&1 && echo "--- clean up the worktree (merged) ---" &&
git worktree remove /c/Users/avidu/Projects/bhi-loc-watermark --force 2>&1 | tail -2 && echo
"worktree removed" && git worktree list
[/external_agent_tool_call]
```

1056. assistant (2026-07-02T06:11:33.546Z)

```
[external_agent_tool_result]
state=MERGED merged=2026-06-21T18:24:57Z
--- clean up the worktree (merged) ---
worktree removed
C:/Users/avidu/Projects/badminton-highlight-indexer
6fdd1d4 [serving/exposure-safety-preflight]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/agent-a771ff1d85e18f2ed
7e0881a [perf/wasb-fast-frames]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-execpolicy
5cb7bbf [bhi-execpolicy]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/boxhill17
8736e2d [run/boxhill17-highlights]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gap-doc
f04f3f3 [docs/rally-gap-closure]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/hardchop
614325d [feat/w1-hard-cap-fallback]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/pr-stitch
a6adcc5 [fix/runner-stitch-watermark]
```

```
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/recdoc
9fd4215 [docs/recording-single-match]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/resweep
fe34013 [docs/lovo-resweep-n13]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing
a3b209a [master]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/w2-safeguard
5ba6955 [feat/w2-never-empty-safeguard]
C:/Users/avidu/Projects/bhi-wt-p2a
ca34787 [feat/packaging-p2a-ffmpeg]
[/external_agent_tool_result]
```

1057. assistant (2026-07-02T06:11:33.546Z)

LW1 merged (engine PR #304) + worktree cleaned up. Marking it done; checking LW2 CI:

1058. assistant (2026-07-02T06:11:33.546Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"10","status":"completed"}
[/external_agent_tool_call]
```

1059. assistant (2026-07-02T06:11:33.546Z)

```
[external_agent_tool_result]
Updated task #10 status
[/external_agent_tool_result]
```

1060. assistant (2026-07-02T06:11:33.546Z)

```
[external_agent_tool_call: Bash]
description: Poll LW2 web CI
command: cd /c/Users/avidu/Projects/khelsutra && for i in $(seq 1 8); do s=$(gh pr checks 73
--json name,state -q '.[ ] | select(.name=="web") | .state' 2>/dev/null); echo "poll $i: web=$s";
if [ "$s" = "SUCCESS" ] || [ "$s" = "FAILURE" ]; then break; fi; sleep 15; done && gh pr checks
73 --json name,state -q '.[ ] | "\(.state)\t\(.name)'"
[/external_agent_tool_call]
```

1061. assistant (2026-07-02T06:11:33.546Z)

```
[external_agent_tool_result]
poll 1: web=SUCCESS
SUCCESS Workers Builds: hidden-king-khelsutra
SUCCESS site
SUCCESS web
[/external_agent_tool_result]
```

1062. assistant (2026-07-02T06:11:33.546Z)

LW2 all green. Merging:

1063. assistant (2026-07-02T06:11:33.546Z)

```
[external_agent_tool_call: Bash]
description: Merge LW2, sync khelsutra master
command: gh pr merge 73 --squash --delete-branch 2>&1 | tail -2 && git checkout master -q && git
pull --ff-only -q && git log --oneline -1
[/external_agent_tool_call]
```

1064. assistant (2026-07-02T06:11:33.546Z)

```
[external_agent_tool_result]
fatal: unable to access 'https://github.com/avidullu/khelsutra.git/': Failed to connect to
```

```
github.com port 443 after 21053 ms: Couldn't connect to server
! warning: not possible to fast-forward to: "master"
c1848df feat(web): LW2 ? send the UI locale to /api/compile (watermark) + /api/process
(analytics) (#73)
[/external_agent_tool_result]
```

1065. assistant (2026-07-02T06:11:33.546Z)

The merge went through on GitHub, but a transient network blip hit the follow-up `git pull`. Let me retry and verify state:

1066. assistant (2026-07-02T06:11:33.546Z)

```
[external_agent_tool_call: Bash]
description: Retry fetch/pull khelsutra master
command: sleep 5 && git fetch origin -q 2>&1 | tail -2 && echo "fetch ok" && echo "local master:
$(git rev-parse --short master)  origin/master: $(git rev-parse --short origin/master)" && git
pull --ff-only -q 2>&1 | tail -2; echo "after pull: local master = $(git rev-parse --short
master)" && git log --oneline -2
[/external_agent_tool_call]
```

1067. assistant (2026-07-02T06:11:33.546Z)

```
[external_agent_tool_result]
fetch ok
local master: c1848df  origin/master: c1848df
after pull: local master = c1848df
c1848df feat(web): LW2 ? send the UI locale to /api/compile (watermark) + /api/process
(analytics) (#73)
1b1fa7b feat(web+site): B3 ? trilingual marketing site via shared t()/data-i18n shim (D-CATALOG-
HOME) (#70)
[/external_agent_tool_result]
```

1068. assistant (2026-07-02T06:11:33.547Z)

LW2 merged (#73 ? `c1848df`), khelsutra master synced. **LW1 + LW2 done.** Now **LW3** ? the analytics v7 migration, the collision-risk piece. First re-check the engine's current migration head (the serving session may have taken v7):

1069. assistant (2026-07-02T06:11:33.547Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"11","status":"completed"}
[/external_agent_tool_call]
```

1070. assistant (2026-07-02T06:11:33.547Z)

```
[external_agent_tool_result]
Updated task #11 status
[/external_agent_tool_result]
```

1071. assistant (2026-07-02T06:11:33.547Z)

```
[external_agent_tool_call: Bash]
description: Re-check engine migration head for a competing v7
command: cd /c/Users/avidu/Projects/badminton-highlight-indexer && git fetch origin -q && echo
"engine origin/master: $(git rev-parse --short origin/master)" && echo "=== migration head on
origin/master ===" && git show origin/master:backend/infrastructure/database.py | grep -nE "if
version < [0-9]|PRAGMA user_version = [0-9]|def _migrate_to_v[0-9]" | tail -12
[/external_agent_tool_call]
```

1072. assistant (2026-07-02T06:11:33.547Z)

```
[external_agent_tool_result]
engine origin/master: e99d0c3
=== migration head on origin/master ===
155:     def _migrate_to_v1(self, cursor):
183:         cursor.execute("PRAGMA user_version = 1")
185:     def _migrate_to_v2(self, cursor):
210:         cursor.execute("PRAGMA user_version = 2")
212:     def _migrate_to_v3(self, cursor):
222:         cursor.execute("PRAGMA user_version = 3")
224:     def _migrate_to_v4(self, cursor):
242:         cursor.execute("PRAGMA user_version = 4")
244:     def _migrate_to_v5(self, cursor):
290:         cursor.execute("PRAGMA user_version = 5")
292:     def _migrate_to_v6(self, cursor):
337:         cursor.execute("PRAGMA user_version = 6")
[/external_agent_tool_result]
```

1073. assistant (2026-07-02T06:11:33.547Z)

Migration head is **still v6** on `origin/master` (`e99d0c3`) ? v7 is still free (the serving session hasn't merged theirs). I'll take v7 for `jobs.locale`, in an isolated worktree, and re-verify before merge. Setting up + reading the relevant code:

1074. assistant (2026-07-02T06:11:33.547Z)

```
[external_agent_tool_call: Bash]
description: Create LW3 worktree + locate create_job
command: WT=/c/Users/avidu/Projects/bhi-loc-analytics && git worktree add "$WT" -b feat/locale-
analytics origin/master 2>&1 | tail -2 && echo "--- create_job signature + jobs insert ---" &&
grep -nE "def create_job|INSERT INTO jobs|def _add_column_idempotent|def count_|GROUP BY"
"$WT/backend/infrastructure/database.py" | head
[/external_agent_tool_call]
```

1075. assistant (2026-07-02T06:11:33.547Z)

```
[external_agent_tool_result]
branch 'feat/locale-analytics' set up to track 'origin/master'.
HEAD is now at e99d0c3 feat(stitch): localize the reel watermark to the UI locale (+ OFL Noto
fonts) (#304)
--- create_job signature + jobs insert ---
64:     def _add_column_idempotent(cursor: sqlite3.Cursor, ddl: str) -> None:
557:     def create_job(self, job_id: str, video_path: str, segmenter: Optional[str] = None,
566:         "INSERT INTO jobs (id, video_path, segmenter, player_pool, max_frames, status,
policy, "
[/external_agent_tool_result]
```

1076. assistant (2026-07-02T06:11:33.547Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-loc-analytics\backend\infrastructure\database.py
[/external_agent_tool_call]
```

1077. assistant (2026-07-02T06:11:33.547Z)

```
[external_agent_tool_result]
86         self._create_base_tables(cursor)
87
88         # Versioned migrations live here. PRAGMA user_version is the schema
89         # contract ? bump it for every change and add an ordered `if version < N`
90         # block below. Tests assert the expected version, so accidental drift fails
CI.
91         version = cursor.execute("PRAGMA user_version").fetchone()[0]
92
93         if version < 1:
```

```

94         self._migrate_to_v1(cursor)
95
96     if version < 2:
97         self._migrate_to_v2(cursor)
98
99     if version < 3:
100         self._migrate_to_v3(cursor)
101
102     if version < 4:
103         self._migrate_to_v4(cursor)
104
105     if version < 5:
106         self._migrate_to_v5(cursor)
107
108     if version < 6:
109         self._migrate_to_v6(cursor)
110     # future: if version < 7: cursor.execute("ALTER TABLE ..."); PRAGMA
user_version = 7
111
[/external_agent_tool_result]

```

1078. assistant (2026-07-02T06:11:33.547Z)

```

[external_agent_tool_call: Read]
input: {"__unparsedToolInput":{"raw":{"file_path": "C:\\\\Users\\\\avidu\\\\Projects\\\\bhi-
loc-analytics\\\\backend\\\\infrastructure\\\\database.py", "offset": 557, , "limit":
28},"len":130}}
[/external_agent_tool_call]

```

1079. assistant (2026-07-02T06:11:33.547Z)

```

[external_agent_tool_result: error]
<tool_use_error>InputValidationError: Read was called with input that could not be parsed as
JSON.
You sent (first 130 of 130 bytes): {"file_path": "C:\\Users\\avidu\\Projects\\bhi-loc-
analytics\\backend\\infrastructure\\database.py", "offset": 557, , "limit": 28}
Common causes: unescaped backslashes in file paths (use / or \\), unescaped control characters,
or truncated output. Retry with valid JSON.</tool_use_error>
[/external_agent_tool_result]

```

1080. assistant (2026-07-02T06:11:33.547Z)

```

[external_agent_tool_call: Read]
file: C:\\Users\\avidu\\Projects\\bhi-loc-analytics\\backend\\infrastructure\\database.py
[/external_agent_tool_call]

```

1081. assistant (2026-07-02T06:11:33.547Z)

```

[external_agent_tool_result]
557     def create_job(self, job_id: str, video_path: str, segmenter: Optional[str] = None,
558                     player_pool: Optional[List[str]] = None,
559                     max_frames: Optional[int] = None,
560                     policy: Optional[Dict[str, Any]] = None,
561                     owner_id: Optional[str] = None) -> bool:
562         """Insert a new job in state 'queued'. ``policy`` is an optional JSON
ExecutionPolicy
563         (EXECUTION_POLICY.md) the worker honours; None ? the worker's default policy.
``owner_id``
564         (M9, D9) is the verified tenant from edge auth ? NULL for the local single-user
(default)."""
565         return self._execute_write(
566             "INSERT INTO jobs (id, video_path, segmenter, player_pool, max_frames,
status, policy, "
567             "owner_id) VALUES (?, ?, ?, ?, ?, 'queued', ?, ?)",

```



```

568         (job_id, video_path, segmenter,
569         json.dumps(player_pool) if player_pool is not None else None,
570         max_frames, json.dumps(policy) if policy is not None else None, owner_id),
571         f"Failed to create job {job_id}",
572     )
573
574     def mark_job_running(self, job_id: str) -> bool:
575         return self._execute_write(
576             "UPDATE jobs SET status='running', started_at=CURRENT_TIMESTAMP, "
577             "progress='starting' WHERE id = ?", (job_id,),
578             f"Failed to mark job {job_id} running",
579         )
580
581     def set_job_progress(self, job_id: str, progress: str) -> bool:
582         return self._execute_write(
583             "UPDATE jobs SET progress=? WHERE id = ?", (progress, job_id),
584             f"Failed to set progress for job {job_id}",
585         )
586
[/external_agent_tool_result]

```

1082. assistant (2026-07-02T06:11:33.548Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-loc-analytics\backend\infrastructure\database.py
[/external_agent_tool_call]

```

1083. assistant (2026-07-02T06:11:33.548Z)

```

[external_agent_tool_result]
64     def _add_column_idempotent(cursor: sqlite3.Cursor, ddl: str) -> None:
65         """Apply a bare ``ALTER TABLE ... ADD COLUMN`` migration step re-runnably.
66
67         A DDL ``ALTER`` auto-commits independently of the surrounding ``with conn``
68         transaction, so an interrupted v3/v4 block can leave the column added while
69         ``user_version`` stays un-bumped ? and the next open would re-run the same
70         ``ALTER`` and raise ``OperationalError: duplicate column name``, bricking every
71         future ``Database(...)`` construction (``__init_db`` runs in ``__init__``).
Swallow
72         ONLY that specific error so the ladder stays crash-safe, matching the re-
runnable
73         ``CREATE ... IF NOT EXISTS`` v1/v2/v5 blocks.
74         """
75         try:
76             cursor.execute(ddl)
77         except sqlite3.OperationalError as exc:
78             if "duplicate column name" not in str(exc).lower():
79                 raise
80
81     def _init_db(self):
82         """Initializes tables for videos, play_segments, and events."""
83         with self._connect() as conn:
84             cursor = conn.cursor()
85
[/external_agent_tool_result]

```

1084. assistant (2026-07-02T06:11:33.548Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-loc-analytics\backend\infrastructure\database.py
[/external_agent_tool_call]

```

1085. assistant (2026-07-02T06:11:33.548Z)

```

[external_agent_tool_result]
292     def _migrate_to_v6(self, cursor):
293         # v6: per-job COST LEDGER (COMPUTE_DECOUPLED_SERVING M8, D8). NULLABLE cost
columns on
294         # `jobs` (NULL = no cost recorded = byte-identical to today; nothing records
until
295         # billing.enabled + a real pricebook) + a VERSIONED `pricebook` table. Cost is
stored in
296         # USD (matches GCP billing = one source of truth); INR is a display conversion
at a
297         # configurable FX rate. ALTER (not recreate) so v1-v5 job rows survive; ADD
COLUMNS are
298         # idempotent so an interrupted upgrade can re-run them. Additive ONLY ? no
served change.
299         for col, typ in (
300             ("owner_id", "TEXT"),          # who pays (the non-owner user / tenant)
301             ("compute_backend", "TEXT"),    # local_gpu | cloud_run | ...
302             ("gpu_type", "TEXT"),           # L4 | T4 | ... (selects the per-second
GPU rate)
303             ("gpu_active_seconds", "REAL"),
304             ("wall_seconds", "REAL"),
305             ("bytes_out", "INTEGER"),        # egress
306             ("gemini_cost_usd", "REAL"),     # provider-reported, already USD
307             ("pricebook_version", "TEXT"),
308             ("cost_usd", "REAL"),            # computed total (USD = source of truth)
309             ("cost_breakdown_json", "TEXT"), # {gpu_seconds: .., egress_gb: ..,
gemini_usd: ..}
310         ):
311             self._add_column_idempotent(cursor, f"ALTER TABLE jobs ADD COLUMN {col}
{typ}")
312         # Partial index keeps the NULL=no-cost fast path churn-free while making per-
owner billing cheap.
313         cursor.execute("CREATE INDEX IF NOT EXISTS idx_jobs_owner ON jobs(owner_id) "
314                        "WHERE owner_id IS NOT NULL")
315         # Versioned pricebook: one row per (version, resource, gpu_type). Rates change +
differ by
316         # gpu_type ? re-version (never mutate a shipped version, so a recorded cost
stays auditable).
317         cursor.execute("""
318             CREATE TABLE IF NOT EXISTS pricebook (
319                 version TEXT NOT NULL,
320                 resource TEXT NOT NULL,
321                 gpu_type TEXT NOT NULL DEFAULT '',
322                 unit_price_usd REAL NOT NULL DEFAULT 0.0,
323                 currency TEXT NOT NULL DEFAULT 'USD',
324                 effective_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
325             )
326         """)
327         cursor.execute("CREATE UNIQUE INDEX IF NOT EXISTS idx_pricebook_key "
328                        "ON pricebook(version, resource, gpu_type)")
329         # Seed a PLACEHOLDER version with every rate 0.0 ? cost_usd computes to 0
(nothing
330         # user-facing changes). The owner INSERTs a real, calibrated version (real GCP
$/GPU-s,
331         # egress $/GB) and flips billing.pricebook_version to it ? never overwrite this
seed.
332         for resource, gpu_type in (("gpu_seconds", "L4"), ("gpu_seconds", "T4"),
333                                     ("wall_seconds", ""), ("egress_gb", "")):
334             cursor.execute(
335                 "INSERT OR IGNORE INTO pricebook (version, resource, gpu_type,
unit_price_usd, "
336                 "currency) VALUES ('placeholder-v0', ?, ?, 0.0, 'USD')", (resource,
gpu_type))
337         cursor.execute("PRAGMA user_version = 6")
[/external_agent_tool_result]

```

1086. assistant (2026-07-02T06:11:33.548Z)

Migration pattern is clear. Let me find where `/api/process` calls `create\_job` and the test that pins the schema version:

1087. assistant (2026-07-02T06:11:33.548Z)

```
[external_agent_tool_call: Bash]
description: Find create_job caller + version-pinning test
command: cd /c/Users/avidu/Projects/bhi-loc-analytics && echo "=== create_job caller in main.py
=== " && grep -nE "create_job\(" backend/main.py && echo "=== schema-version assertions in tests
=== " && grep -rnE "user_version|PRAGMA user_version|== 6|version.*6" tests/ | grep -iE "version"
| head
[/external_agent_tool_call]
```

1088. assistant (2026-07-02T06:11:33.548Z)

```
[external_agent_tool_result]
=== create_job caller in main.py ===
595:     if not db_instance.create_job(job_id, req.video_path, req.segmenter,
=== schema-version assertions in tests ===
tests/test_async_jobs.py:10:- migration: fresh DB lands at user_version 2 with a jobs table; v1
DBs upgrade in place
tests/test_async_jobs.py:83:         assert conn.execute("PRAGMA user_version").fetchone()[0] ==
6
tests/test_async_jobs.py:101:     conn.execute("PRAGMA user_version = 1")
tests/test_async_jobs.py:107:         assert c.execute("PRAGMA user_version").fetchone()[0] == 6
tests/test_billing.py:39:         ver = conn.execute("PRAGMA user_version").fetchone()[0]
tests/test_billing.py:50:     """Re-opening the DB re-runs the ladder guard (version already 6 ?
no-op, no duplicate seed)."""
tests/test_database.py:86:     version = conn.execute("PRAGMA user_version").fetchone()[0]
tests/test_database.py:89:     assert version == 6
tests/test_database.py:108:     version = conn.execute("PRAGMA user_version").fetchone()[0]
tests/test_database.py:114:     assert version == 6         # v6 = per-job cost ledger
(COMPUTE_DECOUPLED_SERVING M8)
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

1089. assistant (2026-07-02T06:11:33.548Z)

The schema-version is pinned to `== 6` in `test\_async\_jobs.py` + `test\_database.py` (the intentional-drift guard) ? I'll bump those to 7. Now building LW3. Marking in-progress and adding the v7 migration:

1090. assistant (2026-07-02T06:11:33.548Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"12","status":"in_progress"}
[/external_agent_tool_call]
```

1091. assistant (2026-07-02T06:11:33.548Z)

```
[external_agent_tool_result]
Updated task #12 status
[/external_agent_tool_result]
```

1092. assistant (2026-07-02T06:11:33.548Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-loc-analytics\backend\infrastructure\database.py
[/external_agent_tool_call]
```

1093. assistant (2026-07-02T06:11:33.548Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-loc-analytics\backend\infrastructure\database.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

1094. assistant (2026-07-02T06:11:33.548Z)

Now add the `\_migrate\_to\_v7` method after `\_migrate\_to\_v6`:

1095. assistant (2026-07-02T06:11:33.549Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-loc-analytics\backend\infrastructure\database.py
[/external_agent_tool_call]
```

1096. assistant (2026-07-02T06:11:33.549Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-loc-analytics\backend\infrastructure\database.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

1097. assistant (2026-07-02T06:11:33.549Z)

Now add `locale` to `create\_job` + a `count\_jobs\_by\_locale` analytics helper:

1098. assistant (2026-07-02T06:11:33.549Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-loc-analytics\backend\infrastructure\database.py
[/external_agent_tool_call]
```

1099. assistant (2026-07-02T06:11:33.549Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-loc-analytics\backend\infrastructure\database.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

1100. assistant (2026-07-02T06:11:33.549Z)

Now `ProcessRequest.locale` + pass it to `create\_job` in the worktree's main.py:

1101. assistant (2026-07-02T06:11:33.549Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-loc-analytics\backend\main.py
[/external_agent_tool_call]
```

1102. assistant (2026-07-02T06:11:33.549Z)

```
[external_agent_tool_result]
592         )
593
594         job_id = uuid.uuid4().hex
595         if not db_instance.create_job(job_id, req.video_path, req.segmenter,
596                                     req.player_pool, req.max_frames, owner_id=owner_id):
597             raise HTTPException(status_code=500, detail="Failed to persist job record.")
598         _get_executor().submit(_drain_due_jobs)
599         return JSONResponse(
[/external_agent_tool_result]
```

1103. assistant (2026-07-02T06:11:33.549Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-loc-analytics\backend\main.py
[/external_agent_tool_call]
```

1104. assistant (2026-07-02T06:11:33.549Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-loc-analytics\backend\main.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

1105. assistant (2026-07-02T06:11:33.549Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-loc-analytics\backend\main.py
[/external_agent_tool_call]
```

1106. assistant (2026-07-02T06:11:33.549Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-loc-analytics\backend\main.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

1107. assistant (2026-07-02T06:11:33.549Z)

Now fix the schema-version assertions (the intentional-drift guard). Let me find every one:

1108. assistant (2026-07-02T06:11:33.550Z)

```
[external_agent_tool_call: Bash]
description: Find all schema-version assertions to bump
command: cd /c/Users/avidu/Projects/bhi-loc-analytics && grep -rNE "user_version" tests/ | grep
-E "==" ?6|>= ?6|fetchone(\)\[0\] == 6" && echo "--- billing line 39 context ---" && sed -n
'36,52p' tests/test_billing.py
[/external_agent_tool_call]
```

1109. assistant (2026-07-02T06:11:33.550Z)

```
[external_agent_tool_result]
tests/test_async_jobs.py:83:         assert conn.execute("PRAGMA user_version").fetchone()[0] ==
6
tests/test_async_jobs.py:107:        assert c.execute("PRAGMA user_version").fetchone()[0] == 6
tests/test_database.py:237:        assert conn.execute("PRAGMA user_version").fetchone()[0] == 6
tests/test_database.py:275:        assert conn.execute("PRAGMA user_version").fetchone()[0] == 6
tests/test_job_worker_loop.py:275:        assert conn.execute("PRAGMA
user_version").fetchone()[0] == 6
tests/test_job_worker_loop.py:292:        assert c.execute("PRAGMA user_version").fetchone()[0]
== 6
--- billing line 39 context ---
def test_v6_migration_adds_cost_columns_and_seeds_placeholder(tmp_path):
    db = Database(str(tmp_path / "m.db"))
    with db._connect() as conn:
        ver = conn.execute("PRAGMA user_version").fetchone()[0]
        cols = {r[1] for r in conn.execute("PRAGMA table_info(jobs)").fetchall()}
        seeded = conn.execute(
            "SELECT unit_price_usd FROM pricebook WHERE version='placeholder-v0'").fetchall()
        assert ver >= 6
        assert {"cost_usd", "gpu_active_seconds", "wall_seconds", "bytes_out", "owner_id",
            "pricebook_version", "cost_breakdown_json", "compute_backend", "gpu_type"} <= cols
        assert len(seeded) >= 4 and all(r[0] == 0.0 for r in seeded) # placeholder = every rate $0

def test_v6_migration_is_idempotent(tmp_path):
```

```
"""Re-opening the DB re-runs the ladder guard (version already 6 ? no-op, no duplicate
seed)."""
path = str(tmp_path / "m.db")
Database(path)
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

1110. assistant (2026-07-02T06:11:33.550Z)

`test\_billing.py` uses `>= 6` (stays valid). Six `== 6` assertions across 3 files need bumping. Let me read their context (to fix any adjacent "v6 = ..." comments too):

1111. assistant (2026-07-02T06:11:33.550Z)

```
[external_agent_tool_call: Bash]
description: Context for version assertions + comments
command: cd /c/Users/avidu/Projects/bhi-loc-analytics && for f in tests/test_async_jobs.py
tests/test_database.py tests/test_job_worker_loop.py; do echo "==== $f ====="; grep -nE
"user_version.*== 6|fresh|lands at|v6|version 6|latest" "$f" | head; done
[/external_agent_tool_call]
```

1112. assistant (2026-07-02T06:11:33.550Z)

```
[external_agent_tool_result]
===== tests/test_async_jobs.py =====
10:- migration: fresh DB lands at user_version 2 with a jobs table; v1 DBs upgrade in place
79:def test_migration_fresh_db_is_v3_with_jobs_table(tmp_path):
80:    db = Database(str(tmp_path / "fresh.db"))
82:        # v6 = per-job cost ledger (M8); the jobs table arrived at v2/v3, user_models at v5.
83:        assert conn.execute("PRAGMA user_version").fetchone()[0] == 6
107:        assert c.execute("PRAGMA user_version").fetchone()[0] == 6
===== tests/test_database.py =====
88:    # v6 = per-job cost ledger (COMPUTE_DECOUPLED_SERVING M8). Bump with each migration.
113:    assert "pricebook" in tables # v6 (per-job cost ledger, M8)
114:    assert version == 6 # v6 = per-job cost ledger (COMPUTE_DECOUPLED_SERVING M8)
213:def test_migration_v3_db_upgrades_to_v6_preserving_rows(tmp_path):
214:    """A real v3 DB (jobs + candidates already populated) upgrades cleanly to v6: the new
217:    db = Database(str(db_path)) # fresh DB is born at the current version; simulate a v3
state
234:    # Re-open: the migration ladder must carry the v3 DB up to v6 without data loss.
237:    assert conn.execute("PRAGMA user_version").fetchone()[0] == 6
241:    assert "pricebook" in tables # v6 (cost ledger)
244:    assert "cost_usd" in _table_cols(db_path, "jobs") # v6 cost columns back-filled (NULL)
===== tests/test_job_worker_loop.py =====
261:    db = Database(str(tmp_path / "fresh.db"))
271:def test_fresh_db_is_v6(tmp_path):
275:    assert conn.execute("PRAGMA user_version").fetchone()[0] == 6
280:    # migration ladder up to the current head (v6) without error ? this exercises the v0?v6
292:    assert c.execute("PRAGMA user_version").fetchone()[0] == 6
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

1113. assistant (2026-07-02T06:11:33.550Z)

Six `== 6` assertions (+ one `assert version == 6` in test_database.py:114) plus some "v6 = head" comments/names. Let me read all three test sections to bump them accurately:

1114. assistant (2026-07-02T06:11:33.550Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_async_jobs.py
[/external_agent_tool_call]
```

1115. assistant (2026-07-02T06:11:33.550Z)

```

[external_agent_tool_result]
79     def test_migration_fresh_db_is_v3_with_jobs_table(tmp_path):
80         db = Database(str(tmp_path / "fresh.db"))
81         with db._get_connection() as conn:
82             # v6 = per-job cost ledger (M8); the jobs table arrived at v2/v3, user_models at
v5.
83             assert conn.execute("PRAGMA user_version").fetchone()[0] == 6
84             cols = {r["name"] for r in conn.execute("PRAGMA table_info(jobs)").fetchall()}
85             assert {"id", "video_path", "video_id", "status", "progress", "error", "result"} <=
cols
86             assert {"policy", "next_poll_at"} <= cols                # v3 self-scheduling columns
87
88
89     def test_migration_upgrades_v1_db_in_place(tmp_path):
90         path = str(tmp_path / "v1.db")
91         conn = sqlite3.connect(path)
92         conn.execute(
93             "CREATE TABLE videos (id TEXT PRIMARY KEY, filepath TEXT NOT NULL, "
94             "processed_at TIMESTAMP, status TEXT NOT NULL, validation_results TEXT)")
95         # A genuine v1 DB also has candidate_segments (created in the version<1 block);
include it
96         # so the v4 personalization ALTER (which adds user_id to that table) has a table to
alter.
97         conn.execute(
98             "CREATE TABLE candidate_segments (id INTEGER PRIMARY KEY AUTOINCREMENT, "
99             "video_id TEXT NOT NULL, detector TEXT NOT NULL, start_time REAL NOT NULL, "
100             "end_time REAL NOT NULL, duration REAL NOT NULL, stats TEXT)")
101         conn.execute("PRAGMA user_version = 1")
102         conn.commit()
103         conn.close()
104
105         db = Database(path) # ladder should take it 1 ? 6
106         with db._get_connection() as c:
107             assert c.execute("PRAGMA user_version").fetchone()[0] == 6
108             assert c.execute(
[/external_agent_tool_result]

```

1116. assistant (2026-07-02T06:11:33.550Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_database.py
[/external_agent_tool_call]

```

1117. assistant (2026-07-02T06:11:33.550Z)

```

[external_agent_tool_result]
84         Database(str(db_path))
85         conn = sqlite3.connect(str(db_path))
86         version = conn.execute("PRAGMA user_version").fetchone()[0]
87         conn.close()
88         # v6 = per-job cost ledger (COMPUTE_DECOUPLED_SERVING M8). Bump with each migration.
89         assert version == 6
90
91
92     def test_init_db_idempotent_and_upgrades_old_db(tmp_path):
93         db_path = tmp_path / "legacy.db"
94         # Simulate a pre-v1 database that only has the old tables and no version.
95         conn = sqlite3.connect(str(db_path))
96         conn.execute("CREATE TABLE videos (id TEXT PRIMARY KEY, filepath TEXT, status
TEXT)")
97         conn.commit()
98         conn.close()
99
100         # Opening with the new Database should add candidate_segments without error...
101         Database(str(db_path))

```

```

102     # ...and running init again must be a no-op (idempotent).
103     Database(str(db_path))
104
105     conn = sqlite3.connect(str(db_path))
106     tables = [r[0] for r in conn.execute(
107         "SELECT name FROM sqlite_master WHERE type='table'").fetchall()]
108     version = conn.execute("PRAGMA user_version").fetchone()[0]
109     conn.close()
110     assert "candidate_segments" in tables
111     assert "jobs" in tables # v2 (async job model, #16)
112     assert "user_models" in tables # v5 (per-user model store)
113     assert "pricebook" in tables # v6 (per-job cost ledger, M8)
114     assert version == 6 # v6 = per-job cost ledger (COMPUTE_DECOUPLED_SERVING M8)
115
[/external_agent_tool_result]

```

1118. assistant (2026-07-02T06:11:33.550Z)

Now bump the assertions (+ note v7 in the comments). test_async_jobs.py:

1119. assistant (2026-07-02T06:11:33.550Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_async_jobs.py
[/external_agent_tool_call]

```

1120. assistant (2026-07-02T06:11:33.551Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_async_jobs.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

1121. assistant (2026-07-02T06:11:33.551Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_async_jobs.py
[/external_agent_tool_call]

```

1122. assistant (2026-07-02T06:11:33.551Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_async_jobs.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

1123. assistant (2026-07-02T06:11:33.551Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_database.py
[/external_agent_tool_call]

```

1124. assistant (2026-07-02T06:11:33.551Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_database.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

1125. assistant (2026-07-02T06:11:33.551Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_database.py

```



```
[/external_agent_tool_call]
```

1126. assistant (2026-07-02T06:11:33.551Z)

```
[external_agent_tool_result]
```

The file C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_database.py has been updated successfully. (file state is current in your context ? no need to Read it back)

```
[/external_agent_tool_result]
```

1127. assistant (2026-07-02T06:11:33.551Z)

Now the v3-upgrade test + the worker-loop test. Reading their context:

1128. assistant (2026-07-02T06:11:33.551Z)

```
[external_agent_tool_call: Read]
```

file: C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_database.py

```
[/external_agent_tool_call]
```

1129. assistant (2026-07-02T06:11:33.551Z)

```
[external_agent_tool_result]
```

```
210         assert len(db.query_candidate_segments("vid1")) == 1
211
212
213     def test_migration_v3_db_upgrades_to_v6_preserving_rows(tmp_path):
214         """A real v3 DB (jobs + candidates already populated) upgrades cleanly to v6: the
new
215         columns/tables appear, user_id back-fills to NULL, and prior rows survive the
ALTERs."""
216         db_path = tmp_path / "legacy_v3.db"
217         db = Database(str(db_path)) # fresh DB is born at the current version; simulate a
v3 state
218         # Seed pre-upgrade data, then force the version back to 3 and re-open to run v4+v5.
219         db.add_video("vidA", "a.mp4", "processed", [{v: 1}])
220         db.add_candidate_segment("vidA", "wasb-hybrid", 2.0, 9.0, stats={"net_crossings":
3})
221         conn = sqlite3.connect(str(db_path))
222         conn.execute("DROP INDEX IF EXISTS idx_videos_user")
223         conn.execute("DROP INDEX IF EXISTS idx_candidates_user")
224         conn.execute("DROP TABLE IF EXISTS user_models")
225         # Recreate v3-shaped tables WITHOUT user_id by rebuilding (simplest: just rewind
version
226         # and drop the v4/v5 artifacts that a true v3 DB wouldn't have).
227         conn.execute("ALTER TABLE videos DROP COLUMN user_id")
228         conn.execute("ALTER TABLE candidate_segments DROP COLUMN user_id")
229         conn.execute("PRAGMA user_version = 3")
230         conn.commit()
231         conn.close()
232         assert "user_id" not in _table_cols(db_path, "videos")
233
234         # Re-open: the migration ladder must carry the v3 DB up to v6 without data loss.
235         db2 = Database(str(db_path))
236         conn = sqlite3.connect(str(db_path))
237         assert conn.execute("PRAGMA user_version").fetchone()[0] == 6
238         tables = {r[0] for r in conn.execute(
239             "SELECT name FROM sqlite_master WHERE type='table'").fetchall()}
240         assert "user_models" in tables # v5
241         assert "pricebook" in tables # v6 (cost ledger)
242         conn.close()
243         assert "user_id" in _table_cols(db_path, "videos")
244         assert "cost_usd" in _table_cols(db_path, "jobs") # v6 cost columns back-filled
(NULL)
245         assert "user_id" in _table_cols(db_path, "candidate_segments")
```

```
[/external_agent_tool_result]
```

1130. assistant (2026-07-02T06:11:33.551Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_job_worker_loop.py
[/external_agent_tool_call]
```

1131. assistant (2026-07-02T06:11:33.551Z)

```
[external_agent_tool_result]
271     def test_fresh_db_is_v6(tmp_path):
272         db = Database(str(tmp_path / "v.db"))
273         with db._get_connection() as conn:
274             # Head of the migration ladder (bump when database.py adds a new `if version <
N`).
275             assert conn.execute("PRAGMA user_version").fetchone()[0] == 6
276
277
278     def test_legacy_db_upgrades_to_head(tmp_path):
279         # A pre-migration legacy DB (only `videos`, user_version 0) must run the FULL
ordered
280         # migration ladder up to the current head (v6) without error ? this exercises the
v0?v6
281         # path including the v1 `candidate_segments` CREATE that the later v4 ALTER depends
on.
282         path = str(tmp_path / "legacy.db")
283         conn = sqlite3.connect(path)
284         conn.execute(
285             "CREATE TABLE videos (id TEXT PRIMARY KEY, filepath TEXT NOT NULL, "
286             "processed_at TIMESTAMP, status TEXT NOT NULL, validation_results TEXT)")
287         conn.execute("PRAGMA user_version = 0")
288         conn.commit()
289         conn.close()
290         db = Database(path)
291         with db._get_connection() as c:
292             assert c.execute("PRAGMA user_version").fetchone()[0] == 6
293
[/external_agent_tool_result]
```

1132. assistant (2026-07-02T06:11:33.551Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_database.py
[/external_agent_tool_call]
```

1133. assistant (2026-07-02T06:11:33.551Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_database.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

1134. assistant (2026-07-02T06:11:33.552Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_database.py
[/external_agent_tool_call]
```

1135. assistant (2026-07-02T06:11:33.552Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_database.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
```

```
[/external_agent_tool_result]
```

1136. assistant (2026-07-02T06:11:33.552Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_job_worker_loop.py
[/external_agent_tool_call]
```

1137. assistant (2026-07-02T06:11:33.552Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_job_worker_loop.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

1138. assistant (2026-07-02T06:11:33.552Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_job_worker_loop.py
[/external_agent_tool_call]
```

1139. assistant (2026-07-02T06:11:33.552Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_job_worker_loop.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

1140. assistant (2026-07-02T06:11:33.552Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_job_worker_loop.py
[/external_agent_tool_call]
```

1141. assistant (2026-07-02T06:11:33.552Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_job_worker_loop.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

1142. assistant (2026-07-02T06:11:33.552Z)

Now the dedicated LW3 test. Let me check the existing `/api/process` test harness to mirror the API-level assertion:

1143. assistant (2026-07-02T06:11:33.552Z)

```
[external_agent_tool_call: Bash]
description: Inspect /api/process test harness
command: cd /c/Users/avidu/Projects/bhi-loc-analytics && grep -nE "api/process|def
test.*process|create_job|get_job|TestClient|client\.post|monkeypatch.*db|db_instance"
tests/test_api_endpoints.py | head -25
[/external_agent_tool_call]
```

1144. assistant (2026-07-02T06:11:33.552Z)

```
[external_agent_tool_result]
1: """TestClient coverage for the FastAPI seam (previously zero ? the compiler SyntaxError
10: pytest.importorskip("httpx") # TestClient needs httpx; skip cleanly where absent
11: from fastapi.testclient import TestClient
31:     monkeypatch.setattr(m, "db_instance", db)
34:     client = TestClient(m.app)
```

```

155: def test_process_success_with_mocked_segmenter(env):
161:     r = client.post("/api/process", json={"video_path": str(vid_file), "segmenter": "x"})
169: def test_process_validation_failure_preserves_prior_segments(env):
176:     r = client.post("/api/process", json={"video_path": str(vid_file)})
185: def test_process_rejects_null_byte_path(env):
187:     r = client.post("/api/process", json={"video_path": "a\x00b.mp4"})
195: def test_process_uses_configured_output_dir_not_literal_output(env, tmp_path, monkeypatch):
218:     r = client.post("/api/process", json={"video_path": str(vid_file)})
250:     r = client.post("/api/process", json={"video_path": str(vid_file)})
262:     r2 = client.post("/api/process", json={"video_path": str(vid_file)})
275:     r = client.post("/api/query", json={"video_id": "vid"})
284:     r = client.post("/api/compile", json={
302:     r = client.post("/api/compile", json={
338:     r = client.post("/api/compile", json={
380:     r = client.post("/api/compile", json={
393:     r = client.post("/api/compile", json={
425:         r = client.post("/api/compile", json={
447:     rj = client.post("/api/process", json={"video_path": str(vid_file), "segmenter": "x"})
450:     rd = client.post("/api/process", json={"video_path": str(vid_file), "segmenter": "x"})
457:     rf = client.post("/api/process", json={"video_path": str(vid_file)})
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]

```

1145. assistant (2026-07-02T06:11:33.552Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_api_endpoints.py
[/external_agent_tool_call]

```

1146. assistant (2026-07-02T06:11:33.552Z)

```

[external_agent_tool_result]
20     """submit() runs the fn synchronously ? async jobs complete before the response
21     is inspected (full lifecycle coverage lives in test_async_jobs.py)."""
22
23     def submit(self, fn, *args, **kwargs):
24         fn(*args, **kwargs)
25
26
27     @pytest.fixture
28     def env(tmp_path, monkeypatch):
29         db = Database(str(tmp_path / "t.db"))
30         cfg = AppConfig()
31         monkeypatch.setattr(m, "db_instance", db)
32         monkeypatch.setattr(m, "global_config", cfg)
33         monkeypatch.setattr(m, "_get_executor", lambda: _InlineExecutor())
34         client = TestClient(m.app)
35         return client, db, tmp_path, monkeypatch
36
37
38     def _video(tmp_path, name="m.mp4"):
39         p = tmp_path / name
[/external_agent_tool_result]

```

1147. assistant (2026-07-02T06:11:33.553Z)

```

[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_api_endpoints.py
[/external_agent_tool_call]

```

1148. assistant (2026-07-02T06:11:33.553Z)

```

[external_agent_tool_result]
155     def test_process_success_with_mocked_segmenter(env):

```

```

156         """#16: submit returns 202 + job_id; the (inline) job carries the old response."""
157         client, db, tmp_path, mp = env
158         vid_file = _video(tmp_path)
159         mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {}))
160         mp.setattr(m.segmenter_registry, "get", lambda name: _FakeSeg)
161         r = client.post("/api/process", json={"video_path": str(vid_file), "segmenter":
"x"})
162         assert r.status_code == 202
163         job = client.get(f"/api/jobs/{r.json()['job_id']}").json()
164         assert job["status"] == "done"
165         assert job["result"]["status"] == "success"
166         assert len(job["result"]["play_segments"]) == 1
167
168     [/external_agent_tool_result]

```

1149. assistant (2026-07-02T06:11:33.553Z)

Now write the LW3 tests ? DB migration + create_job + analytics, plus an API-level check in the existing harness:

1150. assistant (2026-07-02T06:11:33.553Z)

```

[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_locale_analytics.py
[/external_agent_tool_call]

```

1151. assistant (2026-07-02T06:11:33.553Z)

```

[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\bhi-loc-
analytics\tests\test_locale_analytics.py (file state is current in your context ? no need to
Read it back)
[/external_agent_tool_result]

```

1152. assistant (2026-07-02T06:11:33.553Z)

Now add the API-level capture test to the existing harness:

1153. assistant (2026-07-02T06:11:33.553Z)

```

[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_api_endpoints.py
[/external_agent_tool_call]

```

1154. assistant (2026-07-02T06:11:33.553Z)

```

[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_api_endpoints.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

```

1155. assistant (2026-07-02T06:11:33.553Z)

Now run the engine gate for LW3 ? the new test + all the migration/version tests + ruff/mypy:

1156. assistant (2026-07-02T06:11:33.553Z)

```

[external_agent_tool_call: Bash]
description: Run LW3 + migration/version tests + ruff + mypy
command: cd /c/Users/avidu/Projects/bhi-loc-analytics && python -m pytest
tests/test_locale_analytics.py tests/test_database.py tests/test_async_jobs.py
tests/test_job_worker_loop.py tests/test_api_endpoints.py tests/test_billing.py -q 2>&1 | tail

```

```
-12 && echo "=== ruff ===" && python -m ruff check backend/infrastructure/database.py
backend/main.py tests/test_locale_analytics.py 2>&1 | tail -5 && echo "=== mypy ===" && python
-m mypy backend/infrastructure/database.py 2>&1 | tail -5
[/external_agent_tool_call]
```

1157. assistant (2026-07-02T06:11:33.553Z)

```
[external_agent_tool_result]
    # Re-opening re-runs v3 (jobs.policy/next_poll_at), v4
(videos/candidate_segments.user_id),
    # v5 (user_models), and v6 (jobs cost columns + pricebook) over artifacts that ALREADY
exist ?
    # must NOT raise (re-runnable ALTERS / INSERT OR IGNORE), and must reach v6.
    Database(str(db_path))
    conn = sqlite3.connect(str(db_path))
>    assert conn.execute("PRAGMA user_version").fetchone()[0] == 6
E    assert 7 == 6

tests\test_database.py:275: AssertionError
===== short test summary info =====
FAILED tests/test_database.py::test_migration_idempotent_on_interrupted_upgrade
1 failed, 114 passed in 18.93s
=== ruff ===
All checks passed!
=== mypy ===
Success: no issues found in 1 source file
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

1158. assistant (2026-07-02T06:11:33.553Z)

One more `== 6` I missed (test_database.py:275, a different test). ruff + mypy clean. Let me fix it:

1159. assistant (2026-07-02T06:11:33.554Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_database.py
[/external_agent_tool_call]
```

1160. assistant (2026-07-02T06:11:33.554Z)

```
[external_agent_tool_result]
266     conn.execute("PRAGMA user_version = 2")
267     conn.commit()
268     conn.close()
269
270     # Re-opening re-runs v3 (jobs.policy/next_poll_at), v4
(videos/candidate_segments.user_id),
271     # v5 (user_models), and v6 (jobs cost columns + pricebook) over artifacts that
ALREADY exist ?
272     # must NOT raise (re-runnable ALTERS / INSERT OR IGNORE), and must reach v6.
273     Database(str(db_path))
274     conn = sqlite3.connect(str(db_path))
275     assert conn.execute("PRAGMA user_version").fetchone()[0] == 6
276     conn.close()
277     # Columns/tables are all present after the clean re-apply.
[/external_agent_tool_result]
```

1161. assistant (2026-07-02T06:11:33.554Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_database.py
[/external_agent_tool_call]
```

1162. assistant (2026-07-02T06:11:33.554Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\Projects\bhi-loc-analytics\tests\test_database.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

1163. assistant (2026-07-02T06:11:33.554Z)

Now run the full engine suite to confirm everything green (catches any other version reference):

1164. assistant (2026-07-02T06:11:33.554Z)

```
[external_agent_tool_call: Bash]
description: Full engine suite for LW3
command: cd /c/Users/avidu/Projects/bhi-loc-analytics && timeout 540 python run_all_tests.py
2>&1 | tail -8
[/external_agent_tool_call]
```

1165. assistant (2026-07-02T06:11:33.555Z)

```
[external_agent_tool_result]
..... [ 87%]
..... [ 92%]
..... [ 98%]
..... [100%]
1315 passed, 7 skipped in 116.72s (0:01:56)
=====
RESULT: PASS - all tests green
=====
[/external_agent_tool_result]
```

1166. assistant (2026-07-02T06:11:33.555Z)

Full suite: **1315 passed, 7 skipped**, ruff + mypy clean. Before committing, re-verify
`origin/master` hasn't gained a competing v7 (the serving session is active) ? renumber to v8 if
so:

1167. assistant (2026-07-02T06:11:33.555Z)

```
[external_agent_tool_call: Bash]
description: Check for a competing v7 before committing LW3
command: cd /c/Users/avidu/Projects/bhi-loc-analytics && git fetch origin -q && echo
"origin/master: $(git rev-parse --short origin/master) my base: $(git rev-parse --short HEAD)"
&& echo "=== competing v7 on origin/master? ===" && git show
origin/master:backend/infrastructure/database.py | grep -nE "version <
7|_migrate_to_v7|user_version = 7" | head
[/external_agent_tool_call]
```

1168. assistant (2026-07-02T06:11:33.555Z)

```
[external_agent_tool_result]
origin/master: fa7ee3f my base: e99d0c3
=== competing v7 on origin/master? ===
110:          # future: if version < 7: cursor.execute("ALTER TABLE ..."); PRAGMA user_version
= 7
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

1169. assistant (2026-07-02T06:11:33.555Z)

`origin/master` advanced to `fa7ee3f` but **still no v7** (only the comment I'm replacing) ? v7
is mine to take. Committing LW3, then rebasing onto the new master:

1170. assistant (2026-07-02T06:11:33.555Z)

```
[external_agent_tool_call: Bash]
description: Commit LW3 + rebase onto origin/master
command: cd /c/Users/avidu/Projects/bhi-loc-analytics && echo "branch: $(git branch --show-current)" && git add backend/infrastructure/database.py backend/main.py
tests/test_locale_analytics.py tests/test_api_endpoints.py tests/test_async_jobs.py
tests/test_database.py tests/test_job_worker_loop.py && git diff --cached --name-only && git
commit -q -F - <<'EOF'
feat(db): v7 ? record the UI locale per job (PMF analytics) + count_jobs_by_locale

Captures the language the SPA was used in on every /api/process submission, so we can see which
language markets use the tool ? and it pairs with the localized reel watermark (#304).

- v7 migration: a NULLABLE `locale` column on `jobs` (NULL = unrecorded = byte-identical to
today;
  CLI / older clients / pre-v7 rows). Additive-only ALTER, idempotent (re-runnable).
- ProcessRequest gains an optional `locale`; /api/process stores it via create_job(locale=...).
- New Database.count_jobs_by_locale() ? {locale: count} (busiest-first; NULL ? '(unset)' bucket)
?
  the analytics primitive ("how many Spanish submissions").
- Bumped the schema-version contract tests 6 ? 7 (the intentional-drift guard) + a v6?v7 upgrade
test that back-fills locale=NULL on prior rows.

Tests: tests/test_locale_analytics.py (v7 fresh + v6?v7 upgrade + create_job storage + the
analytics
aggregation + ProcessRequest.locale) + an /api/process locale-capture test. Full suite 1315
passed /
7 skipped; ruff + mypy clean.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>
EOF
echo "committed: $(git log --oneline -1)" && echo "=== rebase onto origin/master ===" && git
rebase origin/master 2>&1 | tail -6
[/external_agent_tool_call]
```

1171. assistant (2026-07-02T06:11:33.555Z)

```
[external_agent_tool_result]
branch: feat/locale-analytics
warning: in the working copy of 'tests/test_locale_analytics.py', LF will be replaced by CRLF
the next time Git touches it
backend/infrastructure/database.py
backend/main.py
tests/test_api_endpoints.py
tests/test_async_jobs.py
tests/test_database.py
tests/test_job_worker_loop.py
tests/test_locale_analytics.py
committed: e8ce25a feat(db): v7 ? record the UI locale per job (PMF analytics) +
count_jobs_by_locale
=== rebase onto origin/master ===
Rebasing (1/1)

Successfully rebased and updated refs/heads/feat/locale-analytics.
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

1172. assistant (2026-07-02T06:11:33.555Z)

Rebased cleanly (the new master added a P2a ffmpeg-resolver import; my changes merged alongside). Re-verifying on the rebased base:

1173. assistant (2026-07-02T06:11:33.555Z)


```
[external_agent_tool_call: Bash]
description: Re-verify LW3 post-rebase
command: cd /c/Users/avidu/Projects/bhi-loc-analytics && echo "=== wiring intact ===" && grep
-nE "locale=req.locale|def _migrate_to_v7|count_jobs_by_locale|locale: Optional\[str\] = None"
backend/main.py backend/infrastructure/database.py | head && echo "=== targeted re-run ===" &&
python -m pytest tests/test_locale_analytics.py tests/test_database.py tests/test_async_jobs.py
tests/test_job_worker_loop.py tests/test_api_endpoints.py tests/test_billing.py -q 2>&1 | tail
-4 && echo "=== ruff + mypy ===" && python -m ruff check backend/main.py
backend/infrastructure/database.py 2>&1 | tail -2 && python -m mypy
backend/infrastructure/database.py 2>&1 | tail -2
[/external_agent_tool_call]
```

1174. assistant (2026-07-02T06:11:33.555Z)

```
[external_agent_tool_result]
=== wiring intact ===
backend/main.py:264:      # analytics (count_jobs_by_locale). Optional ? NULL when unrecorded (CLI
/ older clients).
backend/main.py:265:      locale: Optional[str] = None
backend/main.py:280:      locale: Optional[str] = None
backend/main.py:601:                                locale=req.locale):
backend/infrastructure/database.py:342:      def _migrate_to_v7(self, cursor):
backend/infrastructure/database.py:346:                                # count_jobs_by_locale) and pairs with the
localized reel watermark (the SPA sends the same
backend/infrastructure/database.py:574:                                locale: Optional[str] = None) -> bool:
backend/infrastructure/database.py:588:      def count_jobs_by_locale(self) -> Dict[str, int]:
=== targeted re-run ===
..... [ 62%]
..... [100%]
115 passed in 19.20s
=== ruff + mypy ===
All checks passed!
Success: no issues found in 1 source file
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

1175. assistant (2026-07-02T06:11:33.555Z)

All green post-rebase: 115 targeted tests, ruff + mypy clean, wiring intact
(`ProcessRequest.locale` + `\_migrate\_to\_v7` + `count\_jobs\_by\_locale`; the
`CompileRequest.locale` from the merged LW1 also present). Pushing + PR:

1176. assistant (2026-07-02T06:11:33.556Z)

```
[external_agent_tool_call: Bash]
description: Push LW3 branch
command: cd /c/Users/avidu/Projects/bhi-loc-analytics && git log --oneline origin/master..HEAD
&& git push -u origin feat/locale-analytics 2>&1 | tail -3
[/external_agent_tool_call]
```

1177. assistant (2026-07-02T06:11:33.556Z)

```
[external_agent_tool_result]
b6245ec feat(db): v7 ? record the UI locale per job (PMF analytics) + count_jobs_by_locale
branch 'feat/locale-analytics' set up to track 'origin/feat/locale-analytics'.
To https://github.com/Khelsutra/badminton-highlight-indexer.git
* [new branch]      feat/locale-analytics -> feat/locale-analytics
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

1178. assistant (2026-07-02T06:11:33.556Z)

```
[external_agent_tool_call: Bash]
description: Create LW3 engine PR
```

```
command: cd /c/Users/avidu/Projects/bhi-loc-analytics && gh pr create --repo
Khelsutra/badminton-highlight-indexer --base master --head feat/locale-analytics --title
"feat(db): v7 ? record the UI locale per job (PMF analytics) + count_jobs_by_locale" --body
"${cat <<'EOF'
```

v7 ? per-job UI locale (PMF analytics)

Captures the language the SPA was used in on every `/api/process`` submission, so we can answer `**"which language markets actually use the tool?"` ? and it complements the localized reel watermark ([#304](https://github.com/Khelsutra/badminton-highlight-indexer/pull/304)).

What changed

- `**v7 migration**`: a `**NULLABLE`locale`` column on ``jobs`` (``NULL`` = unrecorded = byte-identical to today: CLI / older clients / pre-v7 rows). Additive-only ``ALTER``, idempotent (re-runnable, crash-safe).
- `**`ProcessRequest`` gains an optional ``locale``; `/api/process`` stores it via ``create_job(locale=?)``.
- `**New`Database.count_jobs_by_locale()`` ? `{locale: count}`` (busiest-first; ``NULL`` ? ``(unset)``) ? the analytics primitive.

Migration coordination

Authored in an `**isolated worktree off`origin/master``; re-verified there is `**no competing v7**` before committing and again after rebasing onto the latest master (the serving session's planned consent-cols migration can take `**v8**` ? this is a small, self-contained, easily-renumbered ladder step). The concurrent session's working tree was untouched.

Verification

- `**Full suite: 1315 passed / 7 skipped**`; ``ruff`` + ``mypy`` clean.
- New ``tests/test_locale_analytics.py``: v7 fresh DB, a `**v6?v7 in-place upgrade**` that back-fills ``locale=NULL`` on prior rows, ``create_job`` storage, the ``count_jobs_by_locale`` aggregation (incl. the ``(unset)`` bucket), and ``ProcessRequest.locale``. Plus an `/api/process`` locale-capture test in ``test_api_endpoints.py``.
- Bumped the schema-version contract assertions `**6 ? 7**` across ``test_async_jobs`` / ``test_database`` / ``test_job_worker_loop`` (the intentional-drift guard).

Notes

- The SPA already sends ``locale`` to `/api/process`` (khelsutra #73), and Pydantic ignored it until this PR ? so this activates the capture with `**zero**` client change needed.
- Analytics is queryable now (``count_jobs_by_locale`` / SQL ``GROUP BY locale``); a read-only stats endpoint can come later if wanted.

? Generated with [Claude Code](https://claude.com/claude-code)
EOF

```
)" 2>&1 | tail -2
[/external_agent_tool_call]
```

1179. assistant (2026-07-02T06:11:33.556Z)

```
[external_agent_tool_result]
https://github.com/Khelsutra/badminton-highlight-indexer/pull/306
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

1180. assistant (2026-07-02T06:11:33.556Z)

LW3 PR `**[#306](https://github.com/Khelsutra/badminton-highlight-indexer/pull/306)**` up. Polling its CI:

1181. assistant (2026-07-02T06:11:33.556Z)

```
[external_agent_tool_call: Bash]
description: Poll LW3 engine CI
command: cd /c/Users/avidu/Projects/bhi-loc-analytics && for i in $(seq 1 14); do done_n=$(gh pr
checks 306 --repo Khelsutra/badminton-highlight-indexer --json state -q
```

```
'[.[]|select(.state=="SUCCESS")]|length' 2>/dev/null); tot=$(gh pr checks 306 --repo
Khelsutra/badminton-highlight-indexer --json state -q 'length' 2>/dev/null); fail=$(gh pr checks
306 --repo Khelsutra/badminton-highlight-indexer --json state -q
'[.[]|select(.state=="FAILURE"or .state=="ERROR")]|length' 2>/dev/null); echo "poll $i:
$done_n/$tot success, $fail failed"; if [ "$fail" != "0" ] || [ "$done_n" = "$tot" ]; then
break; fi; sleep 20; done && echo "=== final ===" && gh pr checks 306 --repo
Khelsutra/badminton-highlight-indexer --json name,state -q '[.[] | "\(.state)\t\(.name)"]'
[/external_agent_tool_call]
```

1182. assistant (2026-07-02T06:11:33.556Z)

```
[external_agent_tool_result]
Command running in background with ID: bc92pnhue. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\2ba49afa-99f4-41f2-86a2-27a289d432f3\tasks\bc92pnhue.output. You will be notified when
it completes. To check interim output, use Read on that file path.
[/external_agent_tool_result]
```

1183. assistant (2026-07-02T06:11:33.556Z)

While LW3's engine CI runs (background), let me give a status ? the feature is essentially built and the first two PRs are already merged.

Locale capture + localized watermarks ? status

****Shipped end-to-end across 3 PRs (2 merged, 1 in CI):****

```
| PR | What | State |
|----|-----|-----|
| engine [#304](https://github.com/Khelsutra/badminton-highlight-indexer/pull/304) **LW1** |
Localized reel **watermark** ? `localize_watermark(cfg, locale)` overrides text + font per
locale; bundled **Noto Sans Kannada + Devanagari (OFL)**; `/api/compile` localizes it. Verified
ffmpeg renders kn/hi (no tofu). | ? **merged** (1302 suite green) |
| khelsutra [#73](https://github.com/avidullu/khelsutra/pull/73) **LW2** | SPA sends
`i18n.resolvedLanguage` to `/api/compile` (watermark) + `/api/process` (analytics). | ?
**merged** (177 web tests) |
| engine [#306](https://github.com/Khelsutra/badminton-highlight-indexer/pull/306) **LW3** |
**v7 migration**: `jobs.locale` column + `count_jobs_by_locale()` analytics; `/api/process`
records it. | ? **CI running** (1315 suite green locally) |
```

Answering your three asks directly:

1. ****Watermarking**** ? done: a Kannada/Hindi/Spanish/? user's reel is now watermarked in their language (verified the actual ffmpeg render).
2. ****Locale in the DB**** ? yes now: `jobs.locale` (v7), captured at `/api/process`.
3. ****Basic analytics possible**** ? yes: `Database.count\_jobs\_by\_locale()` ? `{locale: count}` (e.g. how many Spanish submissions); SQL `GROUP BY locale` works.

Migration safety: I confirmed ****no competing v7**** before committing and after rebasing ? the serving session's planned consent migration can take v8. All engine work was done in ****isolated worktrees**** so the concurrent session's checkout was never touched.

I'll merge LW3 once CI is green (re-checking for a competing v7 at merge time), then refresh the memory handoff. Waiting on the CI poll now.

1184. assistant (2026-07-02T06:11:33.556Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\2ba49afa-99f4-41f2-86a2-27a289d432f3\tasks\bc92pnhue.output
[/external_agent_tool_call]
```

1185. assistant (2026-07-02T06:11:33.556Z)

```
[external_agent_tool_result]
```

```
1      poll 1: 1/9 success, 0 failed
2      poll 2: 6/9 success, 0 failed
3      poll 3: 7/9 success, 0 failed
4      poll 4: 8/9 success, 0 failed
5      poll 5: 8/9 success, 0 failed
6      poll 6: 8/9 success, 0 failed
7      poll 7: 8/9 success, 0 failed
8
[/external_agent_tool_result]
```

1186. assistant (2026-07-02T06:11:33.556Z)

CI is at **8/9 green, 0 failed** ? the last check (full suite or wheel-smoke) is finishing. I'll merge LW3 as soon as the poll reports 9/9 (re-checking for a competing v7 at merge time), then refresh the handoff memory and give the final wrap. Waiting on the background poll to complete.

1187. user (2026-07-02T06:11:33.556Z)

```
<task-notification>
<task-id>bc92pnhue</task-id>
<tool-use-id>toolu_01P4rGmRom2wACbnUDHgva7T</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\2ba49afa-99f4-41f2-86a2-27a289d432f3\tasks\bc92pnhue.output</output-file>
<status>completed</status>
<summary>Background command "Poll LW3 engine CI" completed (exit code 0)</summary>
</task-notification>
```

1188. assistant (2026-07-02T06:11:33.557Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\2ba49afa-99f4-41f2-86a2-27a289d432f3\tasks\bc92pnhue.output
[/external_agent_tool_call]
```

1189. assistant (2026-07-02T06:11:33.557Z)

```
[external_agent_tool_result]
1      poll 1: 1/9 success, 0 failed
2      poll 2: 6/9 success, 0 failed
3      poll 3: 7/9 success, 0 failed
4      poll 4: 8/9 success, 0 failed
5      poll 5: 8/9 success, 0 failed
6      poll 6: 8/9 success, 0 failed
7      poll 7: 8/9 success, 0 failed
8      poll 8: 9/9 success, 0 failed
9      === final ===
10     SUCCESS Ruff lint
11     SUCCESS Mypy (lenient baseline)
12     SUCCESS Syntax sweep + import smoke (no GPU stack)
13     SUCCESS Golden fixtures cross-OS (ubuntu-latest)
14     SUCCESS Golden fixtures cross-OS (windows-latest)
15     SUCCESS Golden fixtures cross-OS (macos-latest)
16     SUCCESS Dependency license scan (no copyleft in runtime deps)
17     SUCCESS Built wheel smoke (install + launch)
18     SUCCESS Full test suite (offline, mocked)
19
[/external_agent_tool_result]
```

1190. assistant (2026-07-02T06:11:33.557Z)

All 9 LW3 checks green ? but I'll hold the merge: the owner just directed me to **verify the watermark via actual video generation for all locales first**. That's the right call ? my earlier test was a single `drawtext` frame, not a real reel through the compile pipeline. Let me do a true end-to-end render for all 6 locales.

1191. assistant (2026-07-02T06:11:33.557Z)

```
[external_agent_tool_call: mcp_ccd_session__mark_chapter]
input: {"title": "Watermark video verification (all locales)", "summary": "Owner: generate real watermark videos for all 6 locales + verify before merging LW3."}
[/external_agent_tool_call]
```

1192. assistant (2026-07-02T06:11:33.557Z)

```
[external_agent_tool_result]
Chapter marked: "Watermark video verification (all locales)". Continue your current work.
[/external_agent_tool_result]
```

1193. assistant (2026-07-02T06:11:33.557Z)

```
[external_agent_tool_call: Write]
file: C:\Users\avidu\Projects\bhi-loc-analytics\scratch_wm_verify.py
[/external_agent_tool_call]
```

1194. assistant (2026-07-02T06:11:33.557Z)

```
[external_agent_tool_result]
File created successfully at: C:\Users\avidu\Projects\bhi-loc-analytics\scratch_wm_verify.py
(file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

1195. assistant (2026-07-02T06:11:33.557Z)

```
[external_agent_tool_call: Bash]
description: Generate real watermark reels for all 6 locales
command: cd /c/Users/avidu/Projects/bhi-loc-analytics && python scratch_wm_verify.py 2>&1 | tail
-12
[/external_agent_tool_call]
```

1196. assistant (2026-07-02T06:11:33.557Z)

```
[external_agent_tool_result]
[en] reel=reel_en.mp4 (127585 B)  text='Generated by Khelsutra.guru'  crop=wm_en.png (24409 B)
[kn] reel=reel_kn.mp4 (128885 B)  text='KhelSutra.guru ????' crop=wm_kn.png (27423 B)
[hi] reel=reel_hi.mp4 (127376 B)  text='KhelSutra.guru ??'  crop=wm_hi.png (22281 B)
[es] reel=reel_es.mp4 (127911 B)  text='Generado con KhelSutra.guru'  crop=wm_es.png (24653 B)
[da] reel=reel_da.mp4 (127226 B)  text='Lavet med KhelSutra.guru'  crop=wm_da.png (22697 B)
[id] reel=reel_id.mp4 (128029 B)  text='Dibuat dengan KhelSutra.guru'  crop=wm_id.png (24878 B)
```

```
ALL REELS GENERATED: True
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

1197. assistant (2026-07-02T06:11:33.557Z)

All 6 reels generated through the **real** VideoCompiler pipeline (each ~127 KB, watermark baked in). Now let me view each cropped watermark to confirm correct rendering per locale:

1198. assistant (2026-07-02T06:11:33.557Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\Projects\bhi-loc-analytics\scratch_wm_out\wm_en.png
[/external_agent_tool_call]
```

1199. assistant (2026-07-02T06:11:33.557Z)

```
[external_agent_tool_result]
```

[/external_agent_tool_result]

1200. assistant (2026-07-02T06:11:33.557Z)

en renders correctly ("Generated by Khelsutra.guru", with the box). Now the critical kn + hi (Noto fonts) through the real pipeline:

1201. assistant (2026-07-02T06:11:33.557Z)

[external_agent_tool_call: Read]

file: C:\Users\avidu\Projects\bhi-loc-analytics\scratch_wm_out\wm_kn.png

[/external_agent_tool_call]

1202. assistant (2026-07-02T06:11:33.558Z)

[external_agent_tool_result]

[/external_agent_tool_result]

1203. assistant (2026-07-02T06:11:33.558Z)

[external_agent_tool_call: Read]

file: C:\Users\avidu\Projects\bhi-loc-analytics\scratch_wm_out\wm_hi.png

[/external_agent_tool_call]

1204. assistant (2026-07-02T06:11:33.558Z)

[external_agent_tool_result]

[/external_agent_tool_result]

1205. assistant (2026-07-02T06:11:33.558Z)

kn + hi render perfectly through the real reel pipeline (Latin brand + native script, no tofu). Now the three Latin locales (es/da/id via RobotoSlab):

1206. assistant (2026-07-02T06:11:33.558Z)

[external_agent_tool_call: Read]

file: C:\Users\avidu\Projects\bhi-loc-analytics\scratch_wm_out\wm_es.png

[/external_agent_tool_call]

1207. assistant (2026-07-02T06:11:33.558Z)

[external_agent_tool_result]

[/external_agent_tool_result]

1208. assistant (2026-07-02T06:11:33.558Z)

[external_agent_tool_call: Read]

file: C:\Users\avidu\Projects\bhi-loc-analytics\scratch_wm_out\wm_da.png

[/external_agent_tool_call]

1209. assistant (2026-07-02T06:11:33.558Z)

[external_agent_tool_result]

[/external_agent_tool_result]

1210. assistant (2026-07-02T06:11:33.558Z)

[external_agent_tool_call: Read]

file: C:\Users\avidu\Projects\bhi-loc-analytics\scratch_wm_out\wm_id.png

[/external_agent_tool_call]

1211. assistant (2026-07-02T06:11:33.558Z)

```
[external_agent_tool_result]
[/external_agent_tool_result]
```

1212. assistant (2026-07-02T06:11:33.558Z)

****All 6 locales verified via real watermark video generation**** through the actual VideoCompiler pipeline ? every one renders correctly (no tofu), with the right localized text baked into a real reel:

```
- **en** "Generated by Khelsutra.guru" . **es** "Generado con KhelSutra.guru" . **da** "Lavet med KhelSutra.guru" . **id** "Dibuat dengan KhelSutra.guru" (RobotoSlab)
- **kn** "KhelSutra.guru ??? ?????" . **hi** "KhelSutra.guru ?? ???? ??" (Noto Kannada/Devanagari)
```

This also retroactively validates the already-merged LW1 (the test ran against master's code). Now clean up scratch, re-check for a competing v7, and merge LW3:

1213. assistant (2026-07-02T06:11:33.558Z)

```
[external_agent_tool_call: Bash]
description: Clean scratch + final v7 collision check
command: cd /c/Users/avidu/Projects/bhi-loc-analytics && rm -rf scratch_wm_verify.py
scratch_wm_out && echo "scratch cleaned" && git status --short && echo "=== final competing-v7
check on origin/master ===" && git fetch origin -q && git show
origin/master:backend/infrastructure/database.py | grep -cE "def _migrate_to_v7|version < 7" &&
echo "(0/absent comment = no competing v7) origin/master=$(git rev-parse --short origin/master)"
[/external_agent_tool_call]
```

1214. assistant (2026-07-02T06:11:33.558Z)

```
[external_agent_tool_result]
scratch cleaned
=== final competing-v7 check on origin/master ===
1
(0/absent comment = no competing v7) origin/master=fa7ee3f
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

1215. assistant (2026-07-02T06:11:33.558Z)

Confirmed: the single match is just the `# future: if version < 7` comment (no real competing migration), `origin/master` unchanged at my rebase base ? mergeable. ****All the owner's conditions are met**** ? CI 9/9 green, all 6 locales verified via real watermark video. Merging LW3:

1216. assistant (2026-07-02T06:11:33.558Z)

```
[external_agent_tool_call: Bash]
description: Merge LW3 engine PR
command: cd /c/Users/avidu/Projects/bhi-loc-analytics && gh pr merge 306 --repo
Khelsutra/badminton-highlight-indexer --squash --delete-branch 2>&1 | tail -3; sleep 3; gh pr
view 306 --repo Khelsutra/badminton-highlight-indexer --json state,mergedAt -q '"state=\.state)
merged=\.mergedAt"'
[/external_agent_tool_call]
```

1217. assistant (2026-07-02T06:11:33.558Z)

```
[external_agent_tool_result]
state=MERGED merged=2026-06-21T18:44:51Z
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
[/external_agent_tool_result]
```

1218. assistant (2026-07-02T06:11:33.559Z)

LW3 merged (#306). The full feature is complete ? all 3 PRs merged + verified. Cleaning up the

worktree:

1219. assistant (2026-07-02T06:11:33.559Z)

```
[external_agent_tool_call: Bash]
description: Remove LW3 worktree, confirm engine master
command: git worktree remove /c/Users/avidu/Projects/bhi-loc-analytics --force 2>&1 | tail -2 &&
echo "worktree removed" && git fetch origin -q && echo "engine origin/master now: $(git rev-
parse --short origin/master)" && git log --oneline origin/master -3
[/external_agent_tool_call]
```

1220. assistant (2026-07-02T06:11:33.559Z)

```
[external_agent_tool_result]
worktree removed
engine origin/master now: f01dc85
f01dc85 feat(db): v7 ? record the UI locale per job (PMF analytics) + count_jobs_by_locale
(#306)
fa7ee3f feat(pkg): P2a ? single ffmpeg/ffprobe resolver (env-overridable; one fail-loud place)
(default-OFF) (#305)
e99d0c3 feat(stitch): localize the reel watermark to the UI locale (+ OFL Noto fonts) (#304)
[/external_agent_tool_result]
```

1221. assistant (2026-07-02T06:11:33.559Z)

Worktree removed. Engine `origin/master` head is now my LW3 (#306), cleanly interleaved with the serving session's work. Let me mark LW3 done and refresh the handoff memory:

1222. assistant (2026-07-02T06:11:33.559Z)

```
[external_agent_tool_call: TaskUpdate]
input: {"taskId":"12","status":"completed"}
[/external_agent_tool_call]
```

1223. assistant (2026-07-02T06:11:33.559Z)

```
[external_agent_tool_result]
Updated task #12 status
[/external_agent_tool_result]
```

1224. assistant (2026-07-02T06:11:33.559Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\next-session-ux-localization.md
[/external_agent_tool_call]
```

1225. assistant (2026-07-02T06:11:33.559Z)

```
[external_agent_tool_result]
10      # Handoff ? khelsutra UX / localization / runnable-tool (refreshed 2026-06-21)
11
12      > This is the **khelsutra (web/ SPA + site/)** product track. The **engine/serving**
track (Cloud-Run M5?M12, a CONCURRENT session) has its own `session-handoff.md` ? **don't touch
its files** (`COMPUTE_DECOUPLED_SERVING`, `backend/compute/`, engine `DOC_STATUS`/`NEXT_STEPS`).
13
14      ## 1. You are here *(refreshed 2026-06-21, end of the UX + North-Star session)*
15      The **SPA UX A-items are DONE** (A1/A4/A5/A8 merged), the **North-Star upload keystone
(T4) is SHIPPED**, the SPA is trilingual (en+kn+hi), and the bucket?reel CUJ is UI-driven +
proven. The North Star = make the tool runnable/shareable for alpha users from the UI; **a
sibling session shipped most of the T2 deploy plumbing**
([#64](https://github.com/avidullu/khelsutra/pull/64)) `VITE_API_BASE` split-origin +
[#65](https://github.com/avidullu/khelsutra/pull/65) **alpha deploy kit** = Cloudflare Pages +
`api.*` tunnel + CF Access; see `deploy/DEPLOY_ALPHA.md`) ? so T2's remaining bits are owner-
```


gated (CF Access setup + **GEMINI_API_KEY rotation** + the actual deploy).

16

17 **Merged THIS session (all green CI, each browser-verified + adversarially reviewed via a workflow):**

18 - [#61](https://github.com/avidullu/khelsutra/pull/61) **A4** ? ?44px mobile tap targets + tooltip-text?visible (the "not available yet" reason + per-card shot confidence).

19 - [#62](https://github.com/avidullu/khelsutra/pull/62) **A5** ? friendly engine-error mapping across build + load + ingest via one shared `lib/errorText.ts` + an `errors.\*` namespace (no more raw `ApiError`/`HTTP 500`).

20 - [#63](https://github.com/avidullu/khelsutra/pull/63) **A8** ? background-processing reassurance + icon+text status chips (`StatusChip`), a "Stitching your reel?" building card + ?? status icons.

21 - [#66](https://github.com/avidullu/khelsutra/pull/66) **T4 (North Star keystone)** ? in-UI file UPLOAD: file picker ? chunked `/api/upload/\*` (`api/client.ts` + `hooks/useUpload.ts`) ? server path ? `/api/process` ? existing poll?load?compile?download. 153 web tests.

22

23 - [#67](https://github.com/avidullu/khelsutra/pull/67) **MULTILINGUAL Phase A** ? es/da/id catalogs (SPA now speaks **6 languages**); es 3-form plural; a **placeholder-parity** guard that caught + fixed a latent kn/hi A1-drift (`{videoId}`).

24 - [#68](https://github.com/avidullu/khelsutra/pull/68) **A9** ? localized tap-only filter controls (`TapFilters`); English example chips now en-only; tap==type pinned.

25

26 **Workstream A is essentially DONE (A1/A4/A5/A8/A9 ?).** SPA = en/kn/hi/es/da/id. Web tests 103?173** + 2 new invariant guards (placeholder-parity, tap==type).

27

28 Prior session (context): engine #284/#289, khelsutra #55?#60, owner-merged #58. Screencast: `gs://khelsutra-rally-corpus/promo/khelsutra-bucket-cuj-screencast.mp4`.

29

30 **Spun off as chips (owner-launchable parallel sessions, recorded 2026-06-21):** B3 (`task\_b3060842`), MULTILINGUAL Phase A (`task\_e15e9643`), T11 (`task\_2480b9e2`). ? B3 + MULTILINGUAL both edit the catalogs ? don't run them simultaneously.

31

32 ## 2. Next steps (one PR per item, off `origin/master`, merge on green + rising coverage)

33 **DONE this session:** A1 (#60) · A4 (#61) · A5 (#62) · A8 (#63) · **T4 North-Star upload (#66)** · **MULTILINGUAL Phase A es/da/id (#67)** · **A9 tap-filters (#68)** · (A2 DROPPED ? social/sharing is separate scope.)*

34

35 **? TOP NEXT ? B3 (`site/` trilingual), now FULLY UNBLOCKED.** All 6 catalogs (en/kn/hi/es/da/id) exist to hoist. Chip `task\_b3060842` is ready (still pending). **DECISION LOCKED (owner): hoist catalogs to repo-root `/locales/`** + re-point the SPA imports (or a Vite `@locales` alias; note `web/src/i18n/index.ts` imports 6 catalogs now + `hygiene.test.ts` `LOCALES\_DIR`) + a tiny site prebuild copy + drift-guard. Vanilla `site/public/i18n.js` over `index/capture/own-model.html` + switcher + `site/server.js` Accept-Language; en-only content f...

[/external_agent_tool_result]

1226. assistant (2026-07-02T06:11:33.559Z)

Let me update the handoff to reflect the locale-watermark-analytics feature shipping and that B3 + T11 are done:

1227. assistant (2026-07-02T06:11:33.559Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\next-session-ux-localization.md
[/external_agent_tool_call]

1228. assistant (2026-07-02T06:11:33.559Z)

[external_agent_tool_result]
The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-

indexer\memory\next-session-ux-localization.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

1229. assistant (2026-07-02T06:11:33.559Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\next-session-ux-localization.md
[/external_agent_tool_call]

1230. assistant (2026-07-02T06:11:33.559Z)

[external_agent_tool_result]
The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\next-session-ux-localization.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

1231. assistant (2026-07-02T06:11:33.559Z)

Now a concise checkpoint in the shared current-state. Let me read its current top (it's been moving):

1232. assistant (2026-07-02T06:11:33.559Z)

[external_agent_tool_call: Read]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md
[/external_agent_tool_call]

1233. assistant (2026-07-02T06:11:33.559Z)

[external_agent_tool_result]
12 > ****Scope of this file = LIVE only:**** the current handoff + ***today's*** checkpoints + open PRs/decisions. Keep it read-first-short (?1 page). Older checkpoints are archived in **[[current-state-archive]]** ? when today's checkpoints age out (next clean-slate day), move them there, don't let them pile up here.
13
14 ****Checkpoint:**** 2026-06-21 (****T11 retention sweep ? engine PR MERGED + both follow-ups done****). The owner-spun T11 chip (``task_2480b9e2``) is now BUILT in the engine. ****Extended `tools/cleanup\_caches.py`**** (not hand-rolled, per **[[cache-hygiene-policy]]**) with a pre-alpha ****upload retention sweep****: a new ``"uploads"`` scan location + ``classify`` (``uploaded source`` / ``partial upload`` for the orphaned ``partial-<id>`` files) over ``security.upload_dir`` (default ``output/uploads``); ``plan_deletions`` gains an ****independent `uploads\_retention\_days` window**** (privacy) orthogonal to the cache ``--older-than`` (disk); ``scan_windows`` now ****skips the nested uploads dir**** (was silently kept whole as "other dir" ? no retention); loud structured output (retention-window banner + per-category subtotals + per-item PURGED/FAILED + a ``--json`` plan for a cron/job wrapper). ****Dry-run by default; reels/DB/CSVs stay durable.**** ****Owner gate (AskUserQuestion): uploads 7d default; reels kept durable.**** ? ****engine [#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) MERGED**** (``cb7394b``, all 11 CI checks green: ruff/mypy/full-suite/golden cross-OSx3/wheel-smoke/license-scan/syntax+import); +6 unit tests; ****coverage no-regression VERIFIED**** (branch ****85.30%**** vs ``origin/master`` baseline ****85.06%**** measured in an isolated worktree; same ****9748**** measured stmts, branch missed ****1433 ?**** baseline ****1456**** ? the +0.24% is obsreport reporting tests in my local lane, NOT my diff; the diff touches ****ONLY tools/+tests/**** and the tool imports ****zero**** backend/training). Branch cut from ``origin/master`` in the shared checkout, explicit-path staging; after merge the checkout was ****restored to the serving session's branch**** + feature branches deleted. ****? BOTH FOLLOW-UPS DONE (owner-directed):**** (1) automation (unattended scheduled-job wiring for hosted alpha) filed ? ****[engine #301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301)**** (enhancement; the tool + ``--json`` hook is ready, needs a cron/Cloud-Scheduler wrapper + observability + default-OFF guard + in-flight/active-job safety). (2) ****khelsutra \$9 T11 doc flipped ??? ? [khelsutra #69](https://github.com/avidullu/khelsutra/pull/69) MERGED**** (``bea8a2a``; honest framing: sweep

TOOL shipped #299, scheduling tracked #301; also updated the \$6 retention risk row + \$10 blocking-gate-#2 [window DECIDED] + \$11 DoD). Done in a dedicated worktree off `origin/master`; the UX session's `khelsutra-b3-wt` worktree was untouched. [[cache-hygiene-policy]] [[ui-driven-cloud-serving-gap]] [[gotcha-shared-checkout-branch-collisions]] [[doc-status-register]]
15
[/external_agent_tool_result]

1234. assistant (2026-07-02T06:11:33.560Z)

The serving session updated the T11 checkpoint. Let me insert my locale-watermark checkpoint above it:

1235. assistant (2026-07-02T06:11:33.560Z)

[external_agent_tool_call: Edit]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md
[/external_agent_tool_call]

1236. assistant (2026-07-02T06:11:33.560Z)

[external_agent_tool_result]
The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)
[/external_agent_tool_result]

1237. assistant (2026-07-02T06:11:33.560Z)

All three PRs merged + verified, memory refreshed. Final sanity check on the workspace:

1238. assistant (2026-07-02T06:11:33.560Z)

[external_agent_tool_call: Bash]
description: Confirm worktrees cleaned + no open PRs of mine
command: echo "=== my engine worktrees gone? ===" && git worktree list | grep -iE "bhi-loc" || echo "(none of mine remain ?)" && echo "=== my open PRs across both repos ===" && gh pr list --repo Khelsutra/badminton-highlight-indexer --author "@me" --state open 2>/dev/null | head; gh pr list --repo avidullu/khelsutra --author "@me" --state open 2>/dev/null | head; echo "(end open PRs)"
[/external_agent_tool_call]

1239. assistant (2026-07-02T06:11:33.560Z)

[external_agent_tool_result]
=== my engine worktrees gone? ===
(none of mine remain ?)
=== my open PRs across both repos ===
(end open PRs)
[/external_agent_tool_result]

1240. assistant (2026-07-02T06:11:33.560Z)

[external_agent_tool_call: TaskUpdate]
input: {"taskId":"4","status":"completed"}
[/external_agent_tool_call]

1241. assistant (2026-07-02T06:11:33.560Z)

[external_agent_tool_result]
Updated task #4 status
[/external_agent_tool_result]

1242. assistant (2026-07-02T06:11:33.560Z)

Done ? and verified exactly as you asked. ?

Locale ? watermark + analytics: shipped & verified

All three PRs are **merged**, and I confirmed the watermark by **generating** a real reel through the actual ``VideoCompiler`` pipeline for every supported locale****** and eyeballing the baked-in watermark:

```
| Locale | Watermark on the real reel | Font | Render |
|-----|-----|-----|-----|
| en | Generated by Khelsutra.guru | RobotoSlab | ? |
| es | Generado con KhelSutra.guru | RobotoSlab | ? |
| da | Lavet med KhelSutra.guru | RobotoSlab | ? |
| id | Dibuat dengan KhelSutra.guru | RobotoSlab | ? |
| kn | KhelSutra.guru ??? ???? ???? | Noto Sans Kannada | ? no tofu |
| hi | KhelSutra.guru ?? ???? ???? | Noto Sans Devanagari | ? no tofu |
```

Not just a frame test ? each was a full ``compile_highlights`` ? per-clip re-encode + ffmpeg ``drawtext`` ? real ``mp4``, then I cropped and viewed the watermark region.

****The PRs (merged on green CI + the video verification):****

- [#304](https://github.com/Khelsutra/badminton-highlight-indexer/pull/304) ****LW1**** ? localized watermark + bundled OFL Noto fonts (engine, 1302 suite green)
- [#73](https://github.com/avidullu/khelsutra/pull/73) ****LW2**** ? SPA sends the UI locale to ``/api/compile`` + ``/api/process`` (177 web tests)
- [#306](https://github.com/Khelsutra/badminton-highlight-indexer/pull/306) ****LW3**** ? v7 ``jobs.locale`` + ``count_jobs_by_locale()`` (engine, 1315 suite green, all 9 CI checks)

****Your three asks, delivered:****

1. ****Watermarking in every language**** ? done & video-verified.
2. ****Locale recorded in the DB**** ? ``jobs.locale`` (v7), captured at ``/api/process``.
3. ****Analytics possible**** ? ``Database.count_jobs_by_locale()`` ? e.g. ``{"kn": 2, "hi": 1, "es": 1, "(unset)": 1}``; a read-only stats endpoint is a trivial follow-up if you want a dashboard.

****Safety:**** all engine work in isolated worktrees (the serving session's checkout untouched); re-checked for a competing ****v7**** at commit, rebase, and merge ? none, so the engine head is now v7 and the serving session's consent migration cleanly takes v8. Scratch render artifacts cleaned up; no worktrees or open PRs of mine remain.

****One honest caveat:**** the localized watermark strings (and the kn/hi/es/da/id catalog text) are AI-drafted ? ****native-speaker review**** is still the pre-public-launch gate. I recorded that, plus the remaining T2 owner stand-up, in the handoff.

Net for the session: ****9 PRs merged**** (A4/A5/A8/T4/MULTILINGUAL-A/A9 + LW1/LW2/LW3), web tests 103?177, engine suite green at 1315, and B3 + T11 landed via the chips. Want me to add the ``GET /api/stats/locales`` endpoint, or is the queryable column enough for now?

1243. user (2026-07-02T06:11:33.560Z)

you are >80% context now, so can you get to a clean slate, update all teh documentation and give me a nie icebreaker to continue in the next session and complete the entire work. Amazing session!

1244. assistant (2026-07-02T06:11:33.560Z)

Amazing session indeed ? thank you! Let me wrap to a clean slate, true up the docs, and leave you a crisp icebreaker. First, verify git state and check what the tracked docs currently say (the sibling's B3 #70 may have already updated some):

1245. assistant (2026-07-02T06:11:33.560Z)

[external_agent_tool_call: Bash]

```
description: khelsutra clean-slate check
command: cd /c/Users/avidu/Projects/khelsutra && git checkout master -q 2>&1 | tail -1; git
fetch origin -q && git pull --ff-only -q 2>&1 | tail -2; echo "khelsutra: branch=$(git branch
--show-current) head=$(git rev-parse --short HEAD) origin=$(git rev-parse --short
origin/master)" && echo "--- status (untracked-only expected) ---" && git status --short && echo
"--- my branches still around? ---" && git branch | grep -vE "^\\? *master$" | head
[/external_agent_tool_call]
```

1246. assistant (2026-07-02T06:11:33.560Z)

```
[external_agent_tool_result]
khelsutra: branch=master head=1cd84af origin=1cd84af
--- status (untracked-only expected) ---
?? .claude/
?? site/access.db
--- my branches still around? ---
    docs/multilingual-expansion-proposal
    feat/a3-recording-l10n
+ feat/a6-access-plain-language
+ fix/kn-hi-spa-console-native-review
[/external_agent_tool_result]
```

1247. assistant (2026-07-02T06:11:33.560Z)

khelsutra is on master, clean (only pre-existing untracked files). I can also see **parallel sessions are already knocking out the remaining items** ? there are live worktree branches ``feat/a3-recording-l10n` (A3)`, ``feat/a6-access-plain-language` (A6)`, and ``fix/kn-hi-spa-console-native-review` (native review)`. So I'll leave the khelsutra tracked docs (FRIENDLY/MULTILINGUAL) to those sessions to avoid collision, and make the **memory handoff** the authoritative "continue here" record (no conflict ? separate files).

Let me add the icebreaker to the top of the track handoff and freshen the date:

1248. assistant (2026-07-02T06:11:33.561Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\next-session-ux-localization.md
[/external_agent_tool_call]
```

1249. assistant (2026-07-02T06:11:33.561Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\next-session-ux-localization.md has been updated successfully. (file state is
current in your context ? no need to Read it back)
[/external_agent_tool_result]
```

1250. assistant (2026-07-02T06:11:33.561Z)

Let me fix a typo in that block ("018" slip):

1251. assistant (2026-07-02T06:11:33.561Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\next-session-ux-localization.md
[/external_agent_tool_call]
```

1252. assistant (2026-07-02T06:11:33.561Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\next-session-ux-localization.md has been updated successfully. (file state is
```

current in your context ? no need to Read it back)
[/external_agent_tool_result]

1253. assistant (2026-07-02T06:11:33.561Z)

Now update the MEMORY.md index entry so the "? START HERE" pointer reflects the new reality (it still says "next: A4?A5?A8?B3"):

1254. assistant (2026-07-02T06:11:33.561Z)

```
[external_agent_tool_call: Read]
file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\MEMORY.md
[/external_agent_tool_call]
```

1255. assistant (2026-07-02T06:11:33.561Z)

```
[external_agent_tool_result]
1      # Memory Index
2
3      ## ? Start here
4      - [? START HERE ? khelsutra UX/l10n/runnable-tool](next-session-ux-localization.md) ?
**refreshed 2026-06-21.** CUJ done + SPA trilingual + A1 copy merged + 7-lang expansion APPROVED
(#58). **Ordered next:** A4?A5?A8?B3 (SPA UX), then the **runnable-tool NORTH STAR** (T4 in-UI
upload ? T2 droplet+Cloudflare-Access+Gemini ? T11 retention; NOT GPU/NOT M12) + ramp-up + open
owner decisions. *(The engine/serving track has its OWN handoff in [session-handoff](session-
handoff.md) ? don't conflate.)*
5      - [Current state](current-state.md) ? **live micro-checkpoint** (updated every
checkpoint): open PRs, what just landed, next steps, open decisions. **Read this first.** Now
LIVE-only (handoff + today's checkpoints); kept read-first-short.
6      - [Current-state archive](current-state-archive.md) ? chronological checkpoint history
(2026-06-06 ? 06-18, 96 blocks) split out of current-state 2026-06-19. Deep history only; NOT
read each session.
7      - [Session handoff](session-handoff.md) ? current state · next steps · open decisions ·
ramp-up kit. AUTO-refreshed at meaningful checkpoints (PR merged, key decision, phase shift,
session wrap) ? see working-agreement-checkpoints staleness tripwire.
8      - [Code map artifact](code-map-artifact.md) ? for CODE/architecture ramp-up, read
`docs/CODE_MAP.md` FIRST (dense anchor-rich map; replaces a file crawl). How to regenerate it.
9      - [File-placement conventions](file-placement-conventions.md) ? **where new files go**
(scripts/ vs backend/eval/ vs tools/ vs scratch/ vs training/ vs artifacts/ ?); **read before
committing a new file.** Authoritative table = `docs/CODE_MAP.md` §0 (PR #138); scratch/
gitignored (#136).
10
[/external_agent_tool_result]
```

1256. assistant (2026-07-02T06:11:33.561Z)

```
[external_agent_tool_call: Edit]
file: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\MEMORY.md
[/external_agent_tool_call]
```

1257. assistant (2026-07-02T06:11:33.561Z)

```
[external_agent_tool_result]
The file C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-
indexer\memory\MEMORY.md has been updated successfully. (file state is current in your context ?
no need to Read it back)
[/external_agent_tool_result]
```

1258. assistant (2026-07-02T06:11:33.561Z)

Everything's tidy. Clean slate confirmed and docs trued up. Here's the wrap.

? Session wrap ? clean slate

****Git/workspace:**** khelsutra on `master` (clean; only pre-existing untracked `.claude/`, `site/access.db`). ****No branches, worktrees, or open PRs of mine remain**** ? all engine work was done in isolated worktrees (removed), nothing touched the serving session's checkout.

****Docs updated:****

- ****Memory handoff**** ([next-session-ux-localization.md](memory/next-session-ux-localization.md)) ? added a ****\$0 icebreaker****, refreshed \$1/\$2, dated 2026-06-22.
- ****current-state.md**** ? checkpoint for the locale?watermark+analytics feature.
- ****MEMORY.md**** ? "? START HERE" pointer rewritten to the new reality.
- The khelsutra ****tracked docs**** (FRIENDLY/MULTILINGUAL \$7) I deliberately ****left to the parallel sessions**** ? there are live worktrees for `feat/a3-recording-l10n`, `feat/a6-access-plain-language`, and `fix/kn-hi-spa-console-native-review`, so a doc PR from me would collide. The memory captures the full picture for safety.

What landed this session

****9 of my PRs merged**** + 2 via chips I spawned:

- ****SPA UX/i18n:**** A4 [#61](https://github.com/avidullu/khelsutra/pull/61) · A5 [#62](https://github.com/avidullu/khelsutra/pull/62) · A8 [#63](https://github.com/avidullu/khelsutra/pull/63) · ****T4 in-UI upload**** [#66](https://github.com/avidullu/khelsutra/pull/66) · MULTILINGUAL-A (es/da/id) [#67](https://github.com/avidullu/khelsutra/pull/67) · A9 tap-filters [#68](https://github.com/avidullu/khelsutra/pull/68)
- ****Locale ? watermark + PMF analytics:**** LW1 [#304](https://github.com/Khelsutra/badminton-highlight-indexer/pull/304) · LW2 [#73](https://github.com/avidullu/khelsutra/pull/73) · LW3 [#306](https://github.com/Khelsutra/badminton-highlight-indexer/pull/306) ? ****verified via real watermark video for all 6 locales****
- ****Via spawned chips:**** B3 site-trilingual [#70](https://github.com/avidullu/khelsutra/pull/70) · T11 retention [#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299)

Web tests ****103 ? 177****, engine suite green at ****1315****, two new invariant guards (placeholder-parity, tap==type), one latent kn/hi bug caught+fixed. Every PR browser/video-verified + adversarially reviewed.

? Icebreaker for next session ? "make the alpha real"

> ? ****The runnable tool is built and fully localized.**** A non-tech Bangalore coach can, in ****6 languages****, on a phone: upload a video ? watch it process with reassurance ? tap/type to pick rallies ? download a reel ****watermarked in their own language**** ? and we capture which language they used.

>

> ****One thing stands between this and real alpha users: nothing is deployed yet.**** So ****start with T2**** (~an afternoon, your call on the gated bits): Cloudflare Pages + a `cloudflared api.` tunnel + ****Cloudflare Access**** + ****rotate `GEMINI\_API\_KEY`****, per `khelsutra/deploy/DEPLOY\_ALPHA.md`. ? set the engine's `allowed\_video\_root` to a parent of `upload\_dir` or uploads get rejected. Then run the in-UI upload?reel CUJ against the live URL and share it.

>

> ****Then to finish:**** native-speaker review of all AI-drafted strings (catalogs + watermark ? the public-launch gate) · A3/A6/A7 site polish (parallel sessions in flight) · MULTILINGUAL Phase B/C (Thai/CJK ? owner-gated).

Thank you ? this was a genuinely fun one. The tool's gone from "works on my machine in English" to "point-and-go in six languages with localized reels." See you next session to flip the switch. ?